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This Test Operating Procedure (TOP) describes the test requirements, objectives, responsibilities, and techniques/methodologies required to perform flight performance testing on unmanned aircraft systems (UAS), both in the fixed wing and vertical takeoff and landing (VTOL) configurations. Performance test methodologies that pertain to both the UAS fixed wing and VTOL configurations include pitot static, level flight, and climb and descent. Test methodologies that pertain only to the UAS fixed wing configuration include those for the conventional and augmented takeoff and landing flight events. Performance test methodologies that pertain only to the VTOL configuration include powerplant assessments, powerplant performance, hover, and vertical climb. The performance test methodologies for the specific and individual tests that fall under these primary flight regimes are detailed in this procedure and are represented as standalone sections. Each section in this document contains a complete discussion of the scope, facilities, instrumentation requirements, test conditions, procedures, and data requirements needed to conduct the aforementioned performance assessments.

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US ARMY DEVELOPMENTAL TEST COMMAND
TEST OPERATIONS PROCEDURE

Test Operations Procedure (TOP) 7-2-030 **22 July 2006**
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UAS FIXED WING AND VTOL PERFORMANCE FLIGHT TEST METHODOLOGY

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SECTION 1

INTRODUCTION

1.1 Scope

This Test Operating Procedure (TOP) contains the test approach, test article configuration, test facilities, test instrumentation, required test conditions, test procedures, data required, and presentation of data necessary to perform flight performance testing on unmanned aircraft systems (UAS), both in the fixed wing and vertical takeoff and landing (VTOL) configuration. This document is both a technical and administrative document created to aid in defining a viable and cost effective flight performance testing program to help ensure and/or verify if a particular UAS can operate within its intended mission profile.

This document compiles many of the flight performance test requirements found in References 1 and 2 and Appendices B and C. Much of the theoretical background into the various test events and test techniques can be found in the aforementioned references and appendices. Test personnel desiring additional information can refer to these references and appendices for detailed discussions into the theory of the flight performance tests contained in this document. This document was released using the requirements defined in Reference 3 and will remain under configuration control. Refer to the requirements found in Reference 3 to revise this document.

The contents of this document follow, in general, the formats defined in Reference 3. Section 1 contains a brief overview of this TOP, containing the scope of this document, the unique differences between UAS and manned aircraft, and a description of the various UAS size classifications that will be needed for test planning purposes. For completeness, the sections that contain detailed discussions into the various performance tests are organized and presented in a similar fashion. For continuity, each performance test is documented in a format that is standalone from the other tests. Individual test sections contain discussions pertaining to the scope of the test, the facilities and instrumentation required to conduct the test, the test requirements, the test procedures, the required test data, and the presentation of the test data.

1.1.1 Test Approach (UAS vs. Manned Aircraft)

When planning a performance flight test program for UAS, the test engineer should note several subtle yet distinct differences between unmanned and manned aircraft. First and foremost are the components that comprise the "system." A typical UAS will consist of an air vehicle (AV), an air vehicle operator (AVO), and a ground control station (GCS). This equipment and the operator/pilot comprise a system that presents certain challenges when testing for flight performance characteristics. Due to flight control and flight performance limitations, certain flight test procedures simply cannot be conducted with some UAS. For

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example, the AVO or pilot has the difficult task of performing flight test maneuvers without the same level of situational awareness afforded to a pilot in a manned aircraft.

Another primary difference between unmanned and manned aircraft that must be considered is the incorporation of the autonomous flight control system or autopilot. Most UAS employ a flight control system that limits the operational capability of the aircraft. A thorough understanding of the UAS flight control system, its limitations, and extended capabilities is required to successfully conduct a performance flight test program. An example of this flight control limitation is often found in the climb performance capability of the AV. Many UAS autopilot systems are optimized to climb at a given rate of ascent. This limitation, while not completely insurmountable, often precludes testing at different airspeeds to collect the data needed to generate performance parameters for maximum rate of climb, maximum climb angle, minimum time to climb, and minimum fuel burn. In most cases, performance flight testing will simply involve the verification of the flight control software that is embedded in the autopilot system. On other occasions, the flight control software can be altered to accommodate certain test conditions. While this is certainly a feasible approach, issues such as airworthiness certification should be factored into the risk mitigation plan for the given test.

As previously mentioned, the pilot or AVO plays a critical role in the operation of the AV and this role must be understood to conduct a successful performance flight test program. Most UAS flight control systems are designed such that the AVO is not required to "close the loop" while operating the AV. Many UAS are designed to be operated in a "hands off" mode where the AVO can simply operate the AV using only map displays and mouse-like controls for the input of desired waypoints. This operational mode, from a data link and flight telemeter point of view, is generally not conducive for performance flight testing. To mitigate the risk of interfering with the primary down link and up link required to operate the AV, it is recommended that a separate onboard data logger and telemetry system be incorporated for many of the performance flight test events. This onboard telemetry system is required to collect, process, and telemeter the high-fidelity data for many of the flight test events presented in this document. In subsequent sections of this TOP, specific data and instrumentation requirements will be addressed for each performance parameter.

For risk mitigation and safety purposes, a thorough understanding of the aforementioned down link and up link operation and limitations will be required to conduct a viable and safe flight test program. Many fixed wing and VTOL UAS operate without redundant data links. As such, they frequently experience loss of data, and sometimes they experience a complete loss of link. These situations require that the test agency have safety plans in place to quickly respond to these failure modes, noting that these lost link failures often occur during times of limited situational awareness of the AV and its operational status.

Finally, there are many different UAS configurations that fall within the scope of this TOP. They range from small, lightweight systems (under 55 lb) to large full-scale aircraft that can achieve gross weights greater than 1,320 lb. While certain weather conditions and

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terrain can influence the test performance of manned aircraft, particular consideration for these factors shall be given to small or micro UAS. Seemingly insignificant winds can have a dynamic impact on the flight control systems associated with small UAS. Test planning shall include the possibility of testing some small or micro UAS in closed or confined environments that are sheltered from winds and temperature variations that could adversely affect the accuracy of the test data.

1.1.2 UAS Size Classifications

The UAS size classifications presented in table 1 are consistent with those defined in the U.S. Army Aviation Engineering Directorate Standard Operating Procedure for UAS Airworthiness (see Reference 4). The size limits presented in this document are to be used for test planning purposes only. The UAS size classifications presented in Table 1 will aid the test agency in the planning, conduct, and risk mitigation of the flight test program.

The small UAS class comprises AVs weighing up to 55 lb with a maximum airspeed of 120 knots. This classification covers both fixed wing and VTOL UAS configurations. The aircraft that fall within these parameters could potentially be sensitive to even the slightest weather phenomenon, including winds and temperature inversions. Excessive winds, especially at the surface, would most likely produce poor or nonrepeatable test results. In a worst case scenario, certain weather conditions could exceed the UAS ability to maintain course, resulting in a loss of AV control and possible damage to the AV and possible injury to ground support personnel.

The light UAS class comprises AVs weighing between 55 to 1,320 lb, with a maximum airspeed of 200 knots. This classification covers both fixed wing and VTOL UAS configurations. The medium/large UAS class is defined for AVs that have maximum gross weights exceeding 1,320 lb, regardless of their configuration. However, airspeed limits greater than 200 knots are assigned to the fixed wing UAS configurations. As shown in table 1, there are no airspeed requirements defined for the medium/large VTOL UAS configuration.

Table 1. UAS Size Classifications

UAS Classification	Configuration	UAS Capability	
		Max Gross Weight (lb)	Max Airspeed (knots)
Medium/Large	Fixed Wing	>1,320	>200
Medium/Large	VTOL	>1,320	NA
Light	Fixed Wing & VTOL	55 to 1,320	120 to 200
Small/Micro/Mini	Fixed Wing & VTOL	<55	<120

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1.1.3 Test Article Configuration

The test article must be in a flight configuration that is representative of the type of mission to be flown by the AV. All rotor systems (VTOL), pumps, generators, cooling fans, telemetry systems, and all other parasitic/auxiliary devices requiring power from the engine must be operational. The test article must be equipped with the necessary auxiliary instrumentation required to collect the test data needed to evaluate the performance of the AV, particularly the accurate measurement of the engine fuel flow data. If such instrumentation is not standard to the AV, then additional instrumentation (e.g., fuel flowmeter) must be installed to obtain the required fuel flow data. As required, the test instrumentation and related recording/telemetry systems should be powered using a separate source.

1.2 Facilities and Instrumentation

Each performance flight test presented in this document contains a discussion into the general and specific test facilities and test instrumentation required for successful execution of a given test event. The test facilities required to conduct many of the performance tests presented in this document need not exceed those required for other manned performance flight testing. UAS flight testing, in general, requires but is not limited to:

- **Infrastructure (hangar facilities, meteorological services, data acquisition services, miscellaneous test and ground support equipment)**
- **Airspace (air traffic control (ATC), communication between AVO and ATC)**
- **Time Synchronization using global positioning system (GPS) Universal Time Constant (UTC)**
- **Indoor facilities for some mini/micro class UAS that might be susceptible to certain wind and/or weather conditions in order to control the environmental and weather effects during certain test events.**

Because UAS flight testing lacks pilot input, certain instrumentation can be used that will monitor various parameters that would fall under pilot observation in manned testing. Some instruments that can be added in an attempt to recoup some of the lost pilot input are video cameras, accelerometers, and acoustic sensors.

The test instrumentation required for the performance flight tests covered in this document must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. A complete list of instrumentation requirements needed to perform a given test is presented in the various performance tests covered in this document. These instrumentation requirements are presented for the various engine types covered under the scope of this document, namely gas turbine,

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reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in each performance flight test section.

1.3 Required Test Conditions

1.3.1 Test Tolerances

The general test tolerances defined in this test operating procedure should be applied to the nominal test values. Table 2 defines the test tolerances that will apply to most of the performance flight tests contained in this document. Specific test tolerances for each of the performance tests are provided in each of the test sections. In many cases, the test tolerances are suggested values, noting that many UAS will have limitations in meeting these requirements. In addition to the tolerances mentioned below in Table 2, it is important to have a stable air mass to prevent wind shear from corrupting the flight test data. Each test center will have different tolerances for stable air mass, usually a combination of wind vector (speed and direction) changes, pressure changes, and temperature changes.

Table 2. Performance Flight Test Tolerances

Test Parameter	Tolerance
Ambient Air Temperature	±1°C
Temperature Inversion	None
Pressure Altitude	±10 ft
GPS UTC Time	Standard Drift
Wind Speed	<3 knots
GPS Accuracy	Minimum number of satellites determined to provide required test accuracy.

1.3.2 Test Responsibilities

The testing agency and the appointed test conductor will have the following general responsibilities:

1. **To ensure that the appropriate procedures, as defined in the specific test plan(s), are adhered to**
2. **To make certain all activities involving the test (e.g., airspace management, safety coordination, etc.) are coordinated through him and the test engineer**
3. **To ensure the test is conducted in a safe manner in accordance with the applicable test range safety requirements, including but not limited to loss of link (LOL), flight termination requirements, and loss of control issues**

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4. **To be prepared to respond to any unplanned event that may occur during the test. An unplanned event is any action which occurs that is not the next planned step in the test plan or test procedure. Examples of these actions include, but are not limited to:**
   a. **Personal injury**
   b. **Unplanned test hardware damage**
   c. **Emergency facility shutdown**
   d. **Unplanned equipment failures**
   e. **Unplanned amendments to the test plan or test procedure**
5. **To take immediate action to prevent injury to personnel and/or damage to equipment when an unsafe condition arises**
6. **To direct all personnel within the test area during the test operation**
7. **To verify that unauthorized personnel have been cleared from the test area**
8. **To provide all equipment necessary for data acquisition, data reduction, and still and video photography**
9. **Present the test readiness of the facility during the formal test readiness reviews**

The test engineer will have the following responsibilities:

1. **To make all decisions involving the handling, configuration, and operation of the test article**
2. **To perform any pre- and post-test inspections of the test article**
3. **To provide all of the hardware associated with the test article configuration**
4. **To support/assist the testing agency to ensure that the test article handling procedures are observed**
5. **To review and approve all test procedures submitted by the testing agency**

1.3.3 Test Planning

A test plan, generated by the testing agency, must give a step-by-step account of how the testing agency has planned to conduct the required test. Attention needs to be given to the testing equipment stating all tolerances and sensitivity ranges. All test-sequencing steps must be clearly outlined and detailed enough to allow the test team to assess the procedures for completeness. All data recording plans (photographic, electronic, and paper) must be included, as well as any other plans that would present a well-defined record of all intended test practices.

Formal test readiness reviews must be conducted before the start of each test series. These reviews will include the appropriate representatives from the test team. These reviews will provide a forum to discuss any and all issues applicable to the status of the test article, procedures, reporting, and schedules. Action items assigned at these test readiness reviews must be documented in the minutes of the review and dispositioned before the test. Satisfactorily dispositioned action items will constitute verification that the testing agency and the test article are ready for test operations to begin.

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1.4 Test Procedures

Generic test procedures are presented in each of the performance test sections contained in this document. These procedures represent the general procedural steps required to evaluate the performance parameter of a given UAS or related system and should be used in conjunction with Appendices B and C to generate the final test procedure. The following discusses the importance of designing the flight test program around the operational mission of the test article, as well as the background for operational mission scenarios of UAS and ways to verify flight performance that will be derived from flight test data. The primary mission fulfilled by many UAS has been reconnaissance, surveillance, and target acquisition missions. Other mission objectives that may be employed by UAS include weapons drop, communications relay, air data relay, mission support, medical drops, and psychological operations (pamphlet delivery, radio delivery, etc.). Before flight testing, the test team must have a working knowledge of the UAS mission profile(s). Knowing this information enables the test engineer to plan a test program that combines UAS specific performance testing with mission profile testing, all in an effort to integrate the entire test effort into a single flight test program.

In general, flight test programs are planned to collect flight performance data over a given range of the operating envelop of the AV. With UAS, these operating envelops are largely determined by the payload requirements rather than the performance limitations of the AV. Each UAS must be understood as a whole before a detailed test plan can be generated for that particular UAS. If the UAS is equipped with multiple payloads, the system performance limits of those payloads should be well-defined and the flight test program should center on the data points of greatest interest.

It is recommended that actual range and endurance missions be included into the flight test program. The flight testing presented in Section 7 provides the theoretical basis for best case range and endurance values. However, mission profile testing will be required to verify these parameters in a real operational scenario. In addition, this type of testing lends itself to be useful in verifying many of the parameters found in the AV operator's manual, in particular the value for minimum fuel required for safe landing. In this scenario, a given number of mission sorties can be flown to observe the fuel flow and verify the reliability of the specifications found in the operator's manual.

UAS mission profile performance testing is used to verify the pertinent performance parameters that have been evaluated or extrapolated from other flight performance test events. These types of tests will verify that a given UAS can indeed perform the intended mission without incident. In addition, these tests typically involve operating the AV over an extended period of time, generally in a fully equipped configuration. As such, these tests present opportunities to the test engineer to evaluate several of the mission critical systems onboard the AV, such as the datalink range/margin, payload range and resolution, and the target location accuracy.

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Typically, mission profile testing will be accomplished towards the end of the flight test program. At this point during the test series, it is possible that the datalink limits of the system will not have been completely tested and/or verified. Therefore, the datalink signal strength should be carefully monitored for deteriorating conditions that would adversely affect the link margin of the system. Another important consideration pertaining to the datalink is the restriction associated with the line of sight (LOS). Terrain avoidance and terrain effects on LOS tools are available to the AVO and must be used during the planning stages of the test event to ensure that emergency and return home operational scenarios are achievable given the location of the UAS. UAS datalink testing has been accomplished by loitering the AV in a restricted airspace and then moving the GCS to a position that increases the range between the two systems. If moving the GCS is too problematic, the UAS can proceed outside the restricted area into the National Airspace System to evaluate the datalink. This will require that a Certificate of Authorization be obtained from the Federal Aviation Administration as well as see and avoid duties performed by either a manned chase plane or ground observers.

There are a number of safety considerations that must be addressed by the test engineer before conducting mission profile testing. Mission profile testing is designed to validate the derived points within the previously tested limits of the flight envelop. This type of testing will provide the required reliability for parameters such as maintenance cycles, fuel flow measurement systems, return home fuel status, avionics and sensors, and the datalink performance. Regarding the fuel status, the test engineer should have a working knowledge of the fuel system, how the fuel is monitored, the reliability of this monitoring system/scheme, and the maneuverability limitations of the UAS when operating near its minimum fuel state.

1.5 Data Required

The data required to evaluate the given performance parameter(s) for a particular UAS are summarized in each of the performance test sections in this document. The data requirements are provided for each of the engine types that fall within the scope of this TOP. In general, the data acquisition system used during the conduct of UAS performance testing must have the following capabilities:

1. **Data Transmitting and Recording**
2. **Analog/Digital Display**
3. **Plotting**

The data recording system must be capable of recording a complete set of test data. The data display system must permit "real-time" monitoring of the input levels to the test article and the corresponding flight test data. The testing agency must take still and video photography, using ground based or airborne assets, to document the overall test setup. This photographic coverage will provide an adequate visual record of the test. The data will be processed and delivered as part of the final test report.

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Data recorded during a stabilized test point are essential to a flight test, whether it be a manned or unmanned test. There is a greater degree of difficulty, however, in determining a stable test point during an unmanned test, as there is no test pilot to decide when the aircraft is stable. The data provided to the AVO in the GCS should be useful in determining a stable test point; however, it should be noted that each system will have its own set of tolerances for the data being relayed to the AVO. If these tolerances are such that the AVO data cannot be used to determine a stable test point, the test engineers may need to use real time telemetered test data in determining when the aircraft is stable. It should also be noted that data should, when applicable, be taken as soon as the aircraft can be deemed stable, due to the nature of most UAS. Most AV's are controlled by an autopilot or flight control system that may not be able to hold the AV stable as well as a pilot of a manned aircraft could.

1.6 Presentation of Data

A formal test report will be generated that pertains to the results of a given flight test event. This report will be generated by the testing agency, and it must faithfully represent the results of all test activities, including the as-run test procedure. The final report must explain any deviations from the approved test procedure, as well as all supplemental test data. Above all, the report must be written in such a way that all reasonable and relevant questions related to test objectives, test procedures, deviation from plan, data acquisition, and data analysis are anticipated and answered. The test report's various sections, tables, and graphs must form a legible, orderly, and cohesive whole from which sound technical assessments and engineering conclusions may be readily derived. It must be written to satisfy the informational needs of the authority (individual or organization) whose ultimate responsibility is to certify airworthiness, guarantee safety of flight for manned or unmanned vehicles, and assume legal liability for the consequences of normal flight operations.

References 1 and 2 and Appendices B and C contain detailed discussions in how to delineate and process/present the test data collected during a given performance flight test event. In addition, the aforementioned references and appendices contain brief discussions of the methods used to reduce and analyze the test data using various statistical techniques, charts, tables, and graphs. Properly presented, processed data will not require further elaboration to be understood by a reader who was not present at the actual flight test or post-flight processing. Following the data processing techniques as described in Appendices B and C should yield results that meet this criteria.

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SECTION 2

PITOT-STATIC PERFORMANCE

For most UAS, the pitot-static system performs a vital role in the overall performance of the AV. Most UAS flight profiles are generated using accurate measurements of pressure altitude, indicated airspeed, and vertical speed. These parameters are derived using the AV's pitot-static system. The following paragraphs contain detailed test procedures required to evaluate the performance of the pitot-static system. The various test methods discussed in this paragraph include the GPS Speed Course, GPS 3-Leg, Laser Tracking, Yaw/Angle of Attack/Pitot-Static (YAPS) Boom, Pace, Tower Flyby, Trailing Sources, and Radar Altimeter. These test procedures will be used to quantify the position error, and subsequent correction factors, for the AV. The various ground test activities required for calibration of the system are not covered under the scope of this document.

2.1 GPS Speed Course

2.1.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for pressure altitude, indicated airspeed, and vertical speed will be quantified and used in subsequent performance testing of the UAS.

2.1.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine the position accuracy of the AV. To accomplish this objective, the test approach will incorporate the GPS speed course technique. For complete details of the theory of this test technique, refer to Appendices B and C.

2.1.1.2 Test Article Configuration

The test article need not be in a flight configuration that is representative of the type of mission to be flown by the AV. For VTOL configurations, it is not imperative that parasitic or auxiliary devices requiring power from the engine be operational. The test article must be equipped with the necessary auxiliary instrumentation required to collect the test data needed to evaluate the pitot-static performance of the AV. The AV must also be equipped with instrumentation that allows for an accurate measurement of the AV's spatial position, preferably a differential GPS (DGPS) telemetry system. As required, the test instrumentation and related recording/telemetry systems should be powered using a separate source.

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2.1.2 Facilities and Instrumentation

2.1.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

2.1.2.2 Test Instrumentation

The test instrumentation required for the GPS speed course test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 2 contains a complete list of the instrumentation requirements needed to perform the GPS speed course test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

Table 2. Test Instrumentation for GPS Speed Course Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
GPS Ground Speed (V_G)	knots	All	Onboard GPS
Ground Track/Heading	degrees	All	Onboard GPS
Observed Airspeed (V_O)	knots	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data

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2.1.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 3 defines the test tolerances for the GPS speed course test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 3. GPS Speed Course Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft
Wind Speed	<3 knots
Heading	±2°
GPS Accuracy	4 Satellites (Minimum)

2.1.4 Test Procedures

The generic test procedures for the GPS speed course test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Takeoff AV using existing procedures, note configuration**
2. **Climb to desired test altitude, note and record altitude**
3. **Stabilize AV for 10 seconds before recording data**
4. **Begin test run holding heading and airspeed constant, and record required data for 30 seconds**
5. **Change heading by 180 degrees**
6. **Repeat Steps #2 and #4 for new heading**
7. **Repeat Steps #2 through #6 for subsequent airspeeds**
8. **Land AV using existing procedures**

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2.1.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

GPS Ground Speed (V_G)
Ground Track/Heading
Observed Airspeed (V_O)
Observed Pressure Altitude (H_{PO})
Outside Air Temperature (OAT)
Wind Speed and Direction
Ground Temperature and Pressure

2.1.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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2.2 GPS 3-Leg

2.2.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for pressure altitude, indicated airspeed, and vertical speed will be quantified and used in subsequent performance testing of the UAS.

2.2.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine the position accuracy of the AV. To accomplish this objective, the test approach will incorporate the GPS 3-Leg test technique. This test involves operating the UAS at a constant airspeed and altitude for three (3) or more legs of the flight test. Once the flight test has been completed and the data acquired, the correction factors can then be computed using simple geometric relationships. For complete details of the theory of this test technique, refer to Appendices B and C.

2.2.1.2 Test Article Configuration

Refer to Section 2.1.1.2 for the requirements for the test article configuration.

2.2.2 Facilities and Instrumentation

2.2.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

2.2.2.2 Test Instrumentation

The test instrumentation required for the GPS 3-Leg test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 4 contains a complete list of the instrumentation requirements needed to perform the GPS speed course test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 4. Test Instrumentation for GPS 3-Leg Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
GPS Ground Speed (V_G)	knots	All	Onboard GPS
Ground Track/Heading	degrees	All	Onboard GPS
Observed Airspeed (V_O)	knots	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data

2.2.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 5 defines the test tolerances for the GPS 3-Leg test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 5. GPS 3-Leg Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft
Wind Speed	<3 knots
Heading	±2°
GPS Accuracy	4 Satellites (Minimum)

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2.2.4 Test Procedures

The generic test procedures for the GPS 3-Leg test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Takeoff AV using existing procedures, note configuration**
2. **Climb to desired test altitude, note and record altitude**
3. **Stabilize AV for 10 seconds before recording data**
4. **Begin test run holding airspeed, altitude, and GPS ground track constant, record required data for 30 seconds**
5. **Change heading by at least 60 degrees, but not more than 120 degrees**
6. **Ensure the airspeed is the same as recorded in Step #4, repeat Steps #3 and #4**
7. **Change heading by at least 60 degrees, but not more than 120 degrees**
8. **Ensure the airspeed is the same as recorded in Steps #4 and #6, repeat Steps #3 and #4**
9. **A minimum of three (3) legs are required for data reduction; however more legs can be performed if more data points are required for accuracy and/or repeatability**
10. **Land AV using existing procedures**

2.2.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

GPS Ground Speed (V_G)
Ground Track/Heading
Observed Airspeed (V_O)
Observed Pressure Altitude (H_{PO})
OAT
Wind Speed and Direction
Ground Temperature and Pressure

2.2.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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2.3 Laser Tracking

2.3.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for pressure altitude, indicated airspeed, and vertical speed will be quantified and used in subsequent performance testing of the UAS.

2.3.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine the position accuracy of the AV. To accomplish this objective, the test approach will incorporate the laser tracking test technique. This test involves operating the UAS at a constant airspeed and altitude, much like the flight conditions discussed in the GPS speed course test method. Once the flight test has been completed and the data acquired, the correction factors can then be computed using simple geometric relationships. For complete details of the theory of this test technique, refer to References 1 and 2 and Appendices B and C.

2.3.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration. In addition, the AV must be equipped with a sufficient number of retro-reflectors consistent with the operational limitations of the ground-based laser tracking system. These reflectors must have three mutually perpendicular surfaces and a hypotenuse face. They must be positioned on the AV (e.g., wing tips, tip, tail, etc.) to meet the total internal reflection (TIR) limitations of the ground-based system.

2.3.2 Facilities and Instrumentation

2.3.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

2.3.2.2 Test Instrumentation

The test instrumentation required for the laser tracking test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 6 contains a complete list of the instrumentation requirements needed to perform the laser tracking test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 6. Test Instrumentation for Laser Tracking Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
GPS Ground Speed (V_G)	knots	All	Onboard GPS
Ground Track/Heading	degrees	All	Onboard GPS
Observed Airspeed (V_O)	knots	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data

2.3.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 7 defines the test tolerances for the laser tracking test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 7. Laser Tracking Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft
Wind Speed	<3 knots
Heading	±2°
GPS Accuracy	4 Satellites (Minimum)

2.3.4 Test Procedures

The generic test procedures for the laser tracking test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Takeoff AV using existing procedures, note configuration**
2. **Climb to desired test altitude, note and record altitude**
3. **Stabilize AV and ground-based laser tracking system for 10 seconds before recording data**
4. **Begin test run holding airspeed, altitude, and GPS ground track constant, and record required data for 30 seconds**
5. **Change heading by 180 degrees**
6. **Repeat Steps #2 and #4 for new heading**
7. **Repeat Steps #2 through #6 for subsequent airspeeds**
8. **Land AV using existing procedures**

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2.3.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

GPS Ground Speed (V_G)
Ground Track/Heading
Observed Airspeed (V_O)
Observed Pressure Altitude (H_{PO})
OAT
Wind Speed and Direction
Ground Temperature and Pressure

2.3.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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2.4 YAPS Boom - Fixed Wing Only

2.4.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for pressure altitude and indicated airspeed will be quantified and used in subsequent performance testing of the UAS.

2.4.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine altitude and airspeed accuracy of the AV. To accomplish this objective, the test approach will incorporate the YAPS Boom test technique. Note this test technique differs from the aforementioned GPS methods in that it cannot be used to evaluate the position accuracy of the AV. The YAPS Boom test is mostly suited for medium to large UAS that have the structural interfaces and weight capacity to accept the instrumentation boom.

This test involves operating the UAS at a constant airspeed and altitude, much like the flight conditions discussed in the previous GPS methods. Once the flight test has been completed and the data acquired, the correction factors can then be computed by comparing the data sets between the YAPS Boom and the AV pitot-static system. For complete details of the theory of this test technique, refer to Appendices B and C.

2.4.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration. In addition, this test technique generally involves mounting a calibrated boom to the airframe structure of the AV. The boom structure and related instrumentation should be calibrated before testing, and as such calibration activities for the YAPS Boom will not be addressed in this TOP. The YAPS Boom must be installed at a location forward of the fuselage, wing, or rotor system to eliminate potential interference with existing pitot-static systems or to minimize the effects of rotor downwash for some VTOL UAS. A thorough analysis of the weight and center of gravity impacts on the AV must be conducted to ensure the YAPS Boom will not adversely affect the performance of the AV.

2.4.2 Facilities and Instrumentation

2.4.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

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2.4.2.2 Test Instrumentation

The test instrumentation required for the YAPS Boom test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 8 contains a complete list of the instrumentation requirements needed to perform the YAPS Boom test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

Table 8. Test Instrumentation for YAPS Boom Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
YAPS Total Pressure (P_{TYAPS})	psf	All	YAPS Boom
YAPS Static Pressure (P_{SYAPS})	psf	All	YAPS Boom
YAPS Indicated Airspeed (V_{IYAPS})	knots	All	YAPS Boom
Angle of Attack (α_{YAPS})	degrees	All	YAPS Boom
Sideslip Angle (β_{YAPS})	degrees	All	YAPS Boom
Total Pressure (P_T)	psf	All	Existing pitot-static system on vehicle
Static Pressure (P_S)	psf	All	Existing pitot-static system on vehicle
Indicated Airspeed (V_I)	knots	All	Existing pitot-static system on vehicle
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature/Pressure	°C, in. Hg	All	Ground based met data

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2.4.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 9 defines the test tolerances for the YAPS Boom test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 9. YAPS Boom Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft
Wind Speed	<3 knots
Heading	±2°
GPS Accuracy	4 Satellites (Minimum)

2.4.4 Test Procedures

The generic test procedures for the YAPS Boom test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Takeoff AV using existing procedures, note configuration**
2. **Climb to desired test altitude, note and record altitude**
3. **Stabilize AV for 10 seconds before recording data**
4. **Begin test run holding airspeed and altitude constant and record required data for 30 seconds**
5. **If required, test at various combinations of airspeed and altitude, repeat Steps #2 through #4**
6. **Land AV using existing procedures**

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2.4.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

YAPS Total Pressure (P_{TYAPS})
YAPS Static Pressure (P_{SYAPS})
YAPS Indicated Airspeed (V_{IYAPS})
Angle of Attack (α_{YAPS})
Sideslip Angle (β_{YAPS})
Total Pressure (P_T)
Static Pressure (P_S)
Indicated Airspeed (V_I)
Observed Pressure Altitude (H_{PO})
OAT
Wind Speed and Direction
Ground Temperature and Pressure

2.4.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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2.5 Pace

2.5.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for pressure altitude, indicated airspeed, and vertical speed will be quantified and used in subsequent performance testing of the UAS.

2.5.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine the position accuracy of the AV. To accomplish this objective, the test approach will incorporate the Pace test technique. This test generally involves using a pace aircraft, complete with an independently calibrated pitot-static system, to observe the relative velocity and motion between the pace aircraft and the UAS. This test is valid over a wide range of altitudes and airspeeds, provided the pace aircraft and AV can operate within the same flight regimes. Once the flight test has been completed and the data acquired, the correction factors can then be computed using simple ratio methods. For complete details of the theory of this test technique, refer to Appendices B and C.

2.5.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration. As previously mentioned, the pace article must be configured with a calibrated pitot-static system. The pace aircraft must also be equipped with the necessary auxiliary instrumentation required to collect the test data needed to evaluate the pitot-static performance of the UAS. The data from this instrumentation package will have to be UTC time stamped to ensure accurate time correlation between the pace aircraft and the UAS.

2.5.2 Facilities and Instrumentation

2.5.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

2.5.2.2 Test Instrumentation

The test instrumentation required for the Pace test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data (for both the UAS and the pace aircraft) to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 10 contains a complete list of the instrumentation requirements needed to perform the Pace test for the various engine types covered under the scope of this document, namely gas turbine,

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reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

Table 10. Test Instrumentation for Pace Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
Ground Track/Heading	degrees	All	Onboard GPS
Observed Airspeed (V_O)	knots	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data

2.5.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 11 defines the test tolerances for the Pace test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 11. Pace Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Relative Airspeed Between Pace and UAS	±0.1 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft or limits of the system
Wind Speed	<3 knots
Heading	±2°
GPS Accuracy	4 Satellites (Minimum)

2.5.4 Test Procedures

The generic test procedures for the Pace test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Takeoff AV using existing procedures, note configuration**
2. **Climb AV and pace aircraft to test altitude, note observed altitude**
3. **Stabilize AV and pace aircraft for 30 seconds before recording data**
4. **Hold airspeed and altitude constant, record required data for 1 second**
5. **Repeat Steps #2 through #4 for all desired airspeed/altitude combinations**
6. **Land AV using existing procedures**

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2.5.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

- **Ground Track/Heading**
- **Observed Airspeed (V_O)**
- **Observed Pressure Altitude (H_{PO})**
- **Outside Air Temperature (OAT)**
- **Wind Speed and Direction**
- **Ground Temperature and Pressure**

2.5.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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2.6 Tower Flyby

2.6.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for altitude will be quantified and used in subsequent performance testing of the UAS.

2.6.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine the altitude accuracy of the AV. To accomplish this objective, the test approach will incorporate the Tower Flyby test technique. This test technique is suitable for all sizes of UAS, as defined in this TOP. This test involves operating the UAS on a ground track that is a known distance from the airfield tower, preferably the centerline of the runway. The airfield tower is equipped with a calibrated theodolite that is located at a known height. As the AV passes the tower, the aircraft will be sighted through the theodolite to record the incident angle between the tower and the AV. Once the flight test has been completed and the data acquired, the correction factors for altitude can be computed using trigonometric relationships. For complete details of the theory of this test technique, refer to Appendices B and C.

2.6.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

2.6.2 Facilities and Instrumentation

2.6.2.1 Test Facilities

Basic test facility requirements can be found in paragraph 1.2 of this document. As previously mentioned, the airfield tower must be equipped with a calibrated theodolite. This device must have a movable telescope capable of measuring in two mutually orthogonal axes, namely the horizontal axis and the sight axis. During calibration, the horizontal axis error and the collimation error must be determined and taken into account to eliminate their effect on the measurement results.

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2.6.2.2 Test Instrumentation

The test instrumentation required for the Tower Flyby test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. Table 12 contains a complete list of the instrumentation requirements needed to perform the Tower Flyby test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

Table 12. Test Instrumentation for Tower Flyby Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
Ground Track/Heading	degrees	All	Onboard GPS
Observed Airspeed (V_O)	knots	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data
Sight Axis Angle Measurement	arcsecond	All	Theodolite

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2.6.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 13 defines the test tolerances for the Tower Flyby test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 13. Tower Flyby Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft
Wind Speed	<3 knots
Heading	±2°
Sight Axis Angle	±10 arcsecond

2.6.4 Test Procedures

The generic test procedures for the Tower Flyby test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Takeoff AV using existing procedures, note configuration**
2. **Stabilize AV for 15 seconds before entering predetermined test pattern**
3. **Begin run, record required data onboard AV, hold airspeed and altitude constant during run. In tower, record AV distance from tower, angle to AV from observation point, and tower pressure altitude**
4. **Repeat Steps #2 and 3 as determined to reduce test error**
5. **Land AV using existing procedures.**

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2.6.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

- **Ground Track/Heading**
- **Observed Airspeed (V_O)**
- **Observed Pressure Altitude (H_{PO})**
- **Outside Air Temperature (OAT)**
- **Wind Speed and Direction**
- **Ground Temperature and Pressure**
- **Sight Axis Angle Measurement**

2.6.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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2.7 Trailing Source

2.7.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for indicated airspeed and altitude will be quantified and used in subsequent performance testing of the UAS.

2.7.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine the altitude and airspeed accuracy of the AV. To accomplish this objective, the test approach will incorporate the Trailing Source test technique. This test technique is appropriate for larger sized UAS that have the capability to carry the trailing source and its associated equipment. Generally, these trailing sources are large enough to preclude testing on small and medium-sized UAS.

For this test event, the trailing source is extended along a cable at a fixed distance below and/or behind the AV. For VTOL UAS, it is recommended that the trailing source be positioned at least one and a half rotor diameters below to fuselage. Regardless of the configuration of the UAS, the trailing source should be positioned a sufficient distance to eliminate aerodynamic interaction between the AV and the pitot-static instrumentation onboard the trailing source. The trailing source will have the necessary instrumentation to measure the airspeed and pressure altitude and transmit the data up the trailing source cable. Once the flight test has been completed and the data acquired, the correction factors for indicated airspeed and altitude can be computed using simple ratio techniques. For complete details of the theory of this test technique, refer to Appendices B and C.

2.7.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

2.7.2 Facilities and Instrumentation

2.7.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

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2.7.2.2 Test Instrumentation

The test instrumentation required for the Trailing Source test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. Table 14 contains a complete list of the instrumentation requirements needed to perform the Trailing Source test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

Table 14. Test Instrumentation for Trailing Source Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
Total Pressure ($P_{T,TS}$)	psf	All	Trailing Source
Static Pressure ($P_{S,TS}$)	psf	All	Trailing Source
Indicated Airspeed ($V_{I,TS}$)	knots	All	Trailing Source
Total Pressure (P_T)	psf	All	Existing pitot-static system on vehicle
Static Pressure (P_S)	psf	All	Existing pitot-static system on vehicle
Indicated Airspeed (V_I)	knots	All	Existing pitot-static system on vehicle
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data

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2.7.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 15 defines the test tolerances for the Trailing Source test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 15. Trailing Source Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft
Wind Speed	<3 knots
Heading	±2°
GPS Accuracy	4 Satellites (Minimum)

2.7.4 Test Procedures

The generic test procedures for the Trailing Source test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Configure AV with trailing source equipment**
2. **Takeoff AV using existing procedures, note configuration**
3. **Climb to test altitude, note observed altitude**
4. **Extend trailing source to required test distance from AV**
5. **Stabilize AV for 30 seconds before taking data**
6. **Hold airspeed and altitude constant, begin run and record required data for 60 seconds**
7. **Repeat Steps #3 through #5 for all desired airspeed/altitude combinations**
8. **Retract and stow trailing source**
9. **Land AV using existing procedures**

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2.7.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

Total Pressure (PT,TS)
Static Pressure (PS,TS)
Indicated Airspeed (VI,TS)
Total Pressure (PT)
Static Pressure (PS)
Indicated Airspeed (VI)
Observed Pressure Altitude (HPO)
Outside Air Temperature (OAT)
Wind Speed and Direction
Ground Temperature and Pressure

2.7.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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2.8 Radar Altimeter

2.8.1 Scope

The purpose of this test is to evaluate the performance of the AV's pitot-static system. The correction factors needed for altitude will be quantified and used in subsequent performance testing of the UAS.

2.8.1.1 Test Approach

One of the primary test objectives of this pitot-static performance test is to determine the altitude accuracy of the AV. To accomplish this objective, the test approach will incorporate the Radar Altimeter test technique. For this test technique, a calibrated radar altimeter is placed on a relatively level runway. The level runway is used to establish the reference altitude for the test. As the aircraft is flown within the vicinity of the radar, the calculated radar height is added to the reference altitude. This combined altitude is then compared to the data from the aircraft altimeter system. Once the flight test has been completed and the data acquired, the correction factors for altitude can be computed. For complete details of the theory of this test technique, refer to Appendices B and C.

2.8.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

2.8.2 Facilities and Instrumentation

2.8.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

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2.8.2.2 Test Instrumentation

The test instrumentation required for the Radar Altimeter test technique must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. Table 16 contains a complete list of the instrumentation requirements needed to perform the Radar Altimeter test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

Table 16. Test Instrumentation for Radar Altimeter Test Technique

Test Parameter	Units	Engine Type	Suggested Methods
Ground Track/Heading	degrees	All	Onboard GPS
Observed Airspeed (V_O)	knots	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Observed Pressure Altitude (H_{PO})	ft	All	Existing pitot-static system on vehicle. Auxiliary instrumentation for truth data
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data
Radar Height	ft	All	Radar Altimeter

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2.8.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 17 defines the test tolerances for the Tower Flyby test technique. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 17. Tower Flyby Test Tolerances

Test Parameter	Tolerance
Observed Airspeed (V_O)	±0.5 KIAS or limits of the system
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Altitude	±10 ft
Wind Speed	<3 knots
Heading	±2°
Radar Altimeter	±2 ft

2.8.4 Test Procedures

The generic test procedures for the Tower Flyby test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the pitot-static system performance of a given UAS.

1. **Takeoff AV using existing procedures, note configuration**
2. **Stabilize AV 15 seconds before entering pre-determined test pattern over level surface**
3. **Begin run, record required data over the length of the level surface, hold airspeed and altitude constant during run**
4. **Record pressure altitude at ground level of level surface**
5. **Repeat Steps #2 through #4 as required to reduce test error**
6. **Land AV using existing procedures**

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2.8.5 Data Required

The data required to evaluate the pitot-static performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

All Engine Types:

Ground Track/Heading
Observed Airspeed (V_O)
Observed Pressure Altitude (H_{PO})
Outside Air Temperature (OAT)
Wind Speed and Direction
Ground Temperature and Pressure
Radar Altimeter Measurement

2.8.6 Presentation of Data

The data collected during this test event will be used to analyze the pitot-static performance for a given UAS configuration. The data reduction and analysis techniques for the GPS speed course performance test are found in Section 2.5 and 2.6 of Appendix B for fixed wing UAS and Section 2.5 and 2.6 of Appendix C for VTOL UAS.

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SECTION 3

POWERPLANT ASSESSMENT – VTOL ONLY

3.1 Powerplant/Rotor Controls and Displays

3.1.1 Scope

The purpose of this test is to evaluate the autopilot inputs and feedback loop corrections for maintaining trim conditions of a VTOL AV. It also looks at the AVO controls of the AV and the displays associated with AV flight systems. The AV's flight critical data as well as visual and audio warnings are displayed to the AVO and commented on for effectiveness and ease of use.

3.1.1.1 Test Approach

One of the primary test objectives of this test is to analyze the data regarding the GCS by subjectively looking over the displays and AVO controls and noting layout efficiency and emergency situation handling. The observer is also looking at the effectiveness of the autopilot controls on the AV and the stability of flight as a result.

3.1.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

3.1.2 Facilities and Instrumentation

3.1.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

3.1.2.2 Test Instrumentation

There are no special instrumentation requirements for this test event. The test engineer must perform the evaluation of the control performance of the autopilot and the AVO control over the commands delivered to the flight computer.

3.1.3 Required Test Conditions

The required test conditions will be applicable from other flight tests. The controls/displays test will be performed during other flight tests and test condition guidelines for those tests should be followed.

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3.1.4 Test Procedures

The generic test procedures for the Powerplant/Rotor Controls and Displays test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the Powerplant/Rotor Controls and Displays of a given UAS.

1. **Record a description of the powerplant and rotor controls and displays in the GCS**
2. **Takeoff AV using existing procedures, note configuration**
3. **Run AV through the predetermined test routine to capture autopilot/control interactions during various phases of flight**
4. **Land AV using existing procedures**

3.1.5 Data Required

The data required to evaluate the Powerplant/Rotor Controls and Displays is generally a narrative of the test engineer's observation of the UAS controls and displays.

3.1.6 Presentation of Data

The data collected during this test event will be used to analyze the controls and display performance for a given UAS configuration. The data reduction and analysis techniques for the controls and display performance test are found in Section 3.5 and 3.6 of Appendix C.

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3.2 Powerplant/Rotor Operating Procedures

3.2.1 Scope

The purpose of this test is to evaluate the autopilot operations while maintaining proper control of a VTOL AV using hardware in the loop simulation HILSIM. The AVO is evaluated for the effectiveness of AV operation. This test also evaluates the flight termination system of the UAS and its effectiveness to operate the AV in an emergency situation.

3.2.1.1 Test Approach

One of the primary test objectives of this test is to analyze the operation of the GCS by subjectively monitoring AVO operations and noting efficiency and emergency situation handling. The observer is also monitoring the effectiveness of the autopilot through a simulated hardware test on the ground through the HILSIM.

3.2.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

3.2.2 Facilities and Instrumentation

3.2.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

3.2.2.2 Test Instrumentation

There are no special instrumentation requirements for this test event. The test engineer must perform the evaluation of the control performance of the autopilot and the AVO control over the commands delivered to the flight computer.

3.2.3 Required Test Conditions

The required test conditions will be applicable from other flight tests. This test will be performed during other flight tests and test condition guidelines for those tests should be followed.

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3.2.4 Test Procedures

The generic test procedures for the Powerplant/Rotor Operating Procedures test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the Powerplant/Rotor Operating Procedures of a given UAS.

Testing Using HILSIM:

1. **Install the HILSIM equipment to interact with the AV flight control system**
2. **Run the AV flight control system through the predetermined test routine to capture autopilot/control interactions during test**
3. **Uninstall HILSIM equipment**

Flight Testing:

1. **Takeoff AV using existing procedures, note configuration**
2. **Run AV through the predetermined test routine to capture autopilot/control interactions during various phases of flight**
3. **Land AV using existing procedures.**

3.2.5 Data Required

The data required to evaluate the Powerplant/Rotor Operating Procedures are summarized by the test engineer and their observed evaluation of the UAS through simulation and flight testing.

3.2.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant/rotor operating procedures for a given UAS configuration. The data reduction and analysis techniques for the controls and display performance test are found in Section 3.5 and 3.6 of Appendix C.

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3.3 Powerplant/Rotor Startup and Shutdown

3.3.1 Scope

The purpose of this test is to evaluate the startup and shutdown procedures of a VTOL AV. It addresses the issue of wind direction upon engine startup, the use of an auxiliary power unit, and temperature effects on the engine. Other areas of interest include engine startup functionality and power effects on other systems when in use.

3.3.1.1 Test Approach

This test analyzes the starting mechanics of the AV as well as any oscillations or power loss resulting from powerplant starting. It should be tested from the best starting case first to the worst-case scenario last. The AV can be tied down for testing these functions.

3.3.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

3.3.2 Facilities and Instrumentation

3.3.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

3.3.2.2 Test Instrumentation

The test instrumentation required for the Powerplant/Rotor startup and shutdown test is the use of data acquisition equipment that can record the sensor inputs and relay the engine information to the test engineer. The test engineer must also annotate ambient conditions, autopilot controls, and other engine data as required.

3.3.3 Required Test Conditions

The required test conditions will be applicable from other flight tests including ambient conditions, wind speeds, and precipitation limits. The startup and shutdown procedures will be performed while restrained on the ground and test condition guidelines from the manufacture for starting and shutdown should be followed.

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3.3.4 Test Procedures

The generic test procedures for the Powerplant/Rotor startup and shutdown test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the Powerplant/Rotor startup and shutdown of a given UAS.

1. **Start recording required data before powerplant startup, use auxiliary power if necessary**
2. **Start powerplant using manufacturer suggested methods in the most benign test conditions, record required data from startup to normal operating range, remove auxiliary power if used**
3. **Engage rotor drive system, record required data from rotor startup to normal operating range**
4. **Decelerate rotor system and powerplant, record required data**
5. **Repeat Steps #1 through 4 for all desired test conditions, usually progressing from most benign to most critical**
6. **Shutdown the AV powerplant and decelerate the rotor system**

3.3.5 Data Required

The data required to evaluate the Powerplant/Rotor startup and shutdown are summarized below for each engine type that fall within the scope of this operating procedure.

Gas Turbine:

- **Time history of engine/rotor parameters:**
- **Torque (Q)**
- **Gas generator speed (N_1)**
- **Free power turbine speed (N_2)**
- **Power turbine inlet temperature (T_5)**
- **Fuel flow (W_f)**
- **Rotor speed (N_R)**
- **Annotate engine/rotor control activation**
- **Annotate system operation**
- **OAT**
- **Pressure altitude (H_{Po})**
- **Wind speed and direction (relative to the nose of the AV)**
- **Wind gust spread**
- **Previous cooling period of the engines (if applicable)**

Reciprocating Engine:

- **Time history of engine/rotor parameters:**
- **Manifold Pressure (P_M)**
- **Crankshaft speed (N_1)**
- **Gearbox output shaft speed (N_2)**

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Cylinder Head Temperature (CHT)
Oil Temperature (T_{OIL})
W_f
N_R
Annotate engine/rotor control activation
Annotate system operation
OAT
H_{Po}
Wind speed and direction (relative to the nose of the AV)
Wind gust spread
Previous cooling period of the engines (if applicable)

Electric powerplants:

Time history of motor/rotor parameters:
Q
Motor output shaft speed (N_1)
Gearbox output shaft speed (N_2)
Motor case temperature (T_{case})
N_R
Source Voltage (V_S)
Source Current (A_S)
Source Temperature (T_S)
OAT
H_{Po}
Wind speed and direction (relative to the nose of the AV)
Wind gust spread
Previous cooling period of the motors (if applicable)

3.3.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant/rotor startup and shutdown performance for a given UAS configuration. The data reduction and analysis techniques for the controls and display performance test are found in Section 3.5 and 3.6 of Appendix C.

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3.4 Powerplant Acceleration

3.4.1 Scope

The purpose of this test is to evaluate the powerplant acceleration characteristics of a VTOL AV. It addresses the engine setting and time it takes the rotor to accelerate and compares the acceleration of the system. The rate of throttle application desired for normal operation is determined by this test.

3.4.1.1 Test Approach

This test analyzes the acceleration of the AV as well as any oscillations or yaw effects from hard powerplant acceleration. It should be tested from low acceleration rates to high rates. The AV can be tied down for testing these functions.

3.4.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

3.4.2 Facilities and Instrumentation

3.4.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

3.4.2.2 Test Instrumentation

The test instrumentation required for the powerplant acceleration test is the use of data acquisition equipment that can record the sensor inputs and relay the engine information to the test engineer. The test engineer must also annotate ambient conditions, autopilot controls, and other engine data as required. Table 18 defines the instrumentation test parameters required for Powerplant Acceleration testing.

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Table 18. Instrumentation for Powerplant Acceleration

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor

3.4.3 Required Test Conditions

The required test conditions will be applicable from other flight tests including ambient conditions, wind speeds, and precipitation limits. The powerplant acceleration procedures will be performed while restrained on the ground and test condition guidelines from the manufacture for engine acceleration should be followed.

3.4.4 Test Procedures

The generic test procedures for the powerplant acceleration test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the powerplant acceleration of a given UAS.

1. **Start the AV powerplant and set at flight or ground idle**
2. **Use the manufacturer recommended procedure for accelerating the powerplant and rotor system to takeoff condition, record required data during test**
3. **Repeat Step #2 for all desired test conditions, usually using an incremental buildup approach starting with the manufacturer recommended procedure as a base**

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4. **Shutdown the AV powerplant and decelerate the rotor system**

3.4.5 Data Required

The data required to evaluate the powerplant acceleration test are summarized below for each engine type that fall within the scope of this operating procedure.

Gas Turbine:

Time history of Q, N_1, N_2, T_5, W_f, N_R, and throttle position. Annotate system operation. Note ambient conditions: OAT, H_p, wind speed and direction (relative to the nose of the AV), wind gust spread, and previous cooling period of the engines (if applicable).

Reciprocating Engine:

Time history of P_M, N_1, N_2, T_5, T_{oil}, W_f, N_R, and throttle position. Annotate system operation. Note ambient conditions: OAT, H_p, wind speed and direction (relative to the nose of the AV), wind gust spread, and previous cooling period of the engines (if applicable).

Electric Powerplant:

Time history of Q, N_1, N_2, T_5, N_R, V_S, A_S, and T_S. Annotate system operation. Note ambient conditions: OAT, H_p, wind speed and direction (relative to the nose of the AV), wind gust spread, and previous cooling period of the motors (if applicable).

3.4.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant acceleration performance for a given UAS configuration. The data reduction and analysis techniques for the powerplant acceleration test are found in Section 3.5 and 3.6 of Appendix C.

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3.5 Powerplant/Rotor Stability – VTOL Only

3.5.1 Scope

The Powerplant/Rotor Stability tests evaluate the rotor and engine droop and overshoot while under load. These functions are performed by the autopilot and governor and preprogrammed by the manufacturer. This test will affirm proper control of the AV in all flight conditions including high loads and unloads of the rotor system.

3.5.1.1 Test Approach

This test analyzes the loading of the AV rotor head as well as any yaw effects from hard powerplant acceleration. This test should cover all droop ranges from high overshoot to maximum droop. Care needs to be taken to never reduce the rotor speed beyond a controllable region of flight.

3.5.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

3.5.2 Facilities and Instrumentation

3.5.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

3.5.2.2 Test Instrumentation

The test instrumentation required for the powerplant/rotor stability procedure is the use of data acquisition equipment that can record the sensor inputs and relay the engine information to the test engineer. Table 19 defines the instrumentation test parameters required for Powerplant Stability testing.

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Table 19. Instrumentation for Powerplant Stability

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor

3.5.3 Required Test Conditions

The required test conditions will be applicable from other flight tests including ambient conditions, wind speeds, and precipitation limits. The powerplant must have a fixed, trim throttle setting while collecting data, a single collective movement in one direction, and stable powerplant performance.

3.5.4 Test Procedures

The generic test procedures for the powerplant/rotor stability testing are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the powerplant acceleration of a given UAS.

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Hover Test:

1. **Takeoff AV using existing procedures, note configuration**
2. **Climb to test altitude, note observed altitude**
3. **Stabilize AV 30 seconds before beginning test point**
4. **Command predetermined climb rate starting with minimum rate of climb or, if applicable, increase collective to begin climb**
5. **Increase climb rate incrementally from minimum to maximum or, if applicable, increase collective in 5% increments from minimum to maximum. Record required data at each increment, once stabilized**
6. **Descend to Forward Flight Test altitude**

Forward Flight Test:

1. **Takeoff AV using existing procedures, note configuration**
2. **Command AV forward flight, note observed airspeed**
3. **Stabilize AV 30 seconds before beginning test point**
4. **Command predetermined climb rate starting with minimum rate of climb or, if applicable, increase collective to begin climb**
5. **Increase climb rate incrementally from minimum to maximum or, if applicable, increase collective in 5% increments from minimum to maximum. Record required data at each increment, once stabilized. Hold airspeed constant during run**
6. **Land AV using existing procedures**

3.5.5 Data Required

The data required to evaluate the Powerplant/Rotor Stability are summarized below for each engine type that fall within the scope of this operating procedure.

Gas Turbine:

Static Droop: Stabilized Q, N_R, and airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Transient Droop: Time history of Q, N_R, and collective servo position in order to determine blade pitch angle. Airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Reciprocating Engine:

Static Droop: Stabilized P_M, N_R, and airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Transient Droop: Time history of Q, N_R, and collective servo position in order to determine blade pitch angle. Airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Electric Powerplant:

Static Droop: Stabilized Q, N_R, and airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

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Transient Droop: Time history of Q, N_R, and collective servo position in order to determine blade pitch angle. Airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

3.5.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant/rotor stability performance for a given UAS configuration. The data reduction and analysis techniques for the controls and display performance test are found in Section 3.5 and 3.6 of Appendix C.

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3.6 Powerplant Limiting Characteristics

3.6.1 Scope

This test evaluates the operation of the powerplant overspeed protection typically consisting of a governor on the propeller or rotor head. This sensor acts as a feedback to the engine throttle setting and adjusts to maintain constant propeller/rotor rpm. Other limiting factors are the powerplant temperature and engine speed.

3.6.1.1 Test Approach

The powerplant limiting test is accomplished through testing the engine's rpm range and monitoring the rpm while testing the governor's reaction to rotor loading. The test should be incrementally run from the least loading to the highest loading while recording motor speed, temperature, and rotor rpm. The AV can be tied down for testing these functions.

3.6.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

3.6.2 Facilities and Instrumentation

3.6.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

3.6.2.2 Test Instrumentation

The test instrumentation required for the powerplant limiting characteristics procedure should have the capability to record the sensor inputs and relay the engine information to the test engineer. Table 20 defines the instrumentation test parameters required for Powerplant Limiting Characteristic testing.

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Table 20. Instrumentation for Powerplant Limiting Characteristic Testing

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor

3.6.3 Required Test Conditions

The required test conditions will be applicable from other flight tests including ambient conditions, wind speeds, and precipitation limits. The powerplant limiting characteristics will be analyzed while restrained on the ground and test condition guidelines from the manufacture for engine operation should be followed.

3.6.4 Test Procedures

The generic test procedures for the powerplant limiting characteristics test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the powerplant governor effectiveness of a given UAS.

1. **Start the AV powerplant and set at flight or ground idle**
2. **Use the manufacturer recommended procedure for accelerating the powerplant and rotor system to takeoff condition, record required data during test**
3. **Load the rotor by commanding collective pitch and record engine speed and rotor speed data.**

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4. **Repeat Step #3 for all desired test conditions, using an incremental buildup approach starting with the manufacturer recommended procedure as a base**
5. **Shutdown the AV powerplant and decelerate the rotor system**

3.6.5 Data Required

The data required to evaluate the Powerplant/Rotor governor and overspeed protection are summarized below for each engine type that fall within the scope of this operating procedure.

Gas Turbine:

Time history of Q, N_1, N_2, T_5, W_f, N_R as appropriate. Ambient conditions: OAT, H_P. Description of limiting feature.

Reciprocating Engine:

Time history of P_M, N_1, N_2, T_5, T_{oil}, W_f, N_R as appropriate. Ambient conditions: OAT, H_P. Description of limiting feature.

Electric Motor:

Time history of Q, N_1, N_2, T_5, N_R, V_S, A_S, and T_S, as appropriate. Ambient conditions: OAT, H_P. Description of limiting feature.

3.6.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant/rotor limiting characteristics for a given UAS configuration. The data reduction and analysis techniques for the powerplant/rotor limiting characteristics test are found in Section 3.5 and 3.6 of Appendix C.

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3.7 Powerplant Power Contribution – VTOL Only

3.7.1 Scope

This test analyzes the powerplant power contribution of a VTOL UAS during autorotative descent. It is performed only on aircraft that autorotate under autopilot control. In aircraft without autorotation capabilities, other recovery methods may be used for emergency situations.

3.7.1.1 Test Approach

The approach to this test is to start with the AV at an altitude high enough to allow for a safe recovery. The AV reduces the powerplant throttle to idle and enters a forward autorotating state. This is performed through the test altitude band, and then recovered to normal flying state.

3.7.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

3.7.2 Facilities and Instrumentation

3.7.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

3.7.2.2 Test Instrumentation

The test instrumentation required for the powerplant power contribution test procedure should have the capability to record the sensor inputs and relay the engine information to the test engineer. Table 21 defines the instrumentation test parameters required for Powerplant Contribution testing.

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Table 21. Instrumentation for Powerplant Contribution

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor

3.7.3 Required Test Conditions

The required test conditions will be applicable from other flight tests including ambient conditions, wind speeds, and precipitation limits. The powerplant power contribution procedures will be performed while descending in autorotative flight and test condition guidelines from the manufacture should be followed. The powerplant must have a fixed, idle throttle setting while collecting data, and be recoverable into a flying state.

3.7.4 Test Procedures

The generic test procedures for the powerplant power contribution are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the autorotation characteristics of a given UAS.

1. **Takeoff AV**
2. **Climb to a specified altitude that yields sufficient room to recover from an autorotative condition**
3. **Set the throttle position to idle and make sure the clutch is disengaged**

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4. **Record the descent rate while flying at an airspeed of level flight minimum**
5. **Return engine throttle position and climb to start position**
6. **Repeat descent test varying the airspeed in 10-kt increments for all ranges of descent requirements**
7. **Land AV**

3.7.5 Data Required

The data required to evaluate the powerplant power contribution include the following:

All Engine Types:

Altitude band
Time duration through band
AV airspeed
N_R
Gross weight (fuel used or remaining)
Ambient conditions

3.7.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant power contribution for a given UAS configuration. The data reduction and analysis techniques for the powerplant power contribution test are found in Section 3.5 and 3.6 of Appendix C.

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SECTION 4

POWERPLANT PERFORMANCE – VTOL ONLY

4.1 Inlet/Intake Performance

4.1.1 Scope

This test evaluates the inlet or intake performance of an AV powerplant. The inlet airflow and pressure changes with different flight states and must be quantified. For reciprocating engines, the intake performance is measured by the pressure changes during flight.

4.1.1.1 Test Approach

The powerplant Inlet/Intake Performance tests will be performed while performing standard flight maneuvers such as hover and forward flight. A sensor grid must be installed to measure pressure at various points across the powerplant inlet/intake.

4.1.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration. The test article in this test shall also be equipped with data acquisition equipment to measure the pressures around the intake region of the powerplant.

4.1.2 Facilities and Instrumentation

4.1.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

4.1.2.2 Test Instrumentation

The test instrumentation required for the inlet/intake performance procedure is the use of data acquisition equipment that can record the sensor inputs and relay any critical engine information to the test engineer. Table 22 defines the instrumentation test parameters required for Inlet/Intake Performance testing.

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Table 22. Instrumentation for Inlet/Intake Performance

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor
Compressor Face Static Pressure (P_{S2})	ft	Gas Turbine	Onoard pressure sensor
Compressor Face Total Pressure (P_{T2})	ft	Gas Turbine	Onboard pressure sensor
Compressor Face Static Temperature (T_{S2})	°C	Gas Turbine	Onboard temperature sensor
Engine Intake Manifold Pressure (P_M)	in. Hg	Reciprocating	Onboard manifold pressure sensor
Indicated Airspeed (Vo)	knots	All	Onboard pitot-static system

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4.1.3 Required Test Conditions

The required test conditions will be applicable from other flight tests including ambient conditions, wind speeds, and precipitation limits. The test condition guidelines from the manufacture should be followed. The powerplant must have the engine bleed air systems off and in normal flight mode. There must also be a fixed throttle setting (if possible with autopilot) with stabilized power demand while collecting data.

4.1.4 Test Procedures

The generic test procedures for the powerplant inlet/intake performance are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the inlet or intake characteristics of a given UAS.

Test #1 Turbine Inlet performance:

1. **Takeoff AV**
2. **Stabilize in a hover**
3. **Record data, including the pressure distribution across face of the inlet**
4. **Transition into forward flight and continue to record data at various airspeeds**
5. **Land AV**

Test #2 Reciprocating engine intake performance:

1. **Takeoff AV**
2. **Stabilize in a hover**
3. **Record the manifold pressure data**
4. **Transition into forward flight and continue to record data at various airspeeds**
5. **Land AV**

4.1.5 Data Required

The data required to evaluate the Powerplant Inlet/Intake performance are summarized below for each engine type that fall within the scope of this operating procedure.

Gas Turbine:

Static Pressure at the Engine Compressor Face (P_{S2}), Total Pressure at the Engine Compressor Face (P_{T2}), Static Temperature at the Engine Compressor Face (T_{S2}), Indicated Airspeed (Vo)

Reciprocating Engine:

Engine Intake Manifold Pressure (P_M), Engine Temperature (T_1), Indicated Airspeed (Vo)

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4.1.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant inlet/intake performance for a given UAS configuration. The data reduction and analysis techniques for the powerplant inlet/intake performance test are found in Section 4.5 and 4.6 of Appendix C.

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4.2 Powerplant Performance

4.2.1 Scope

This test evaluates the powerplant performance of an AV. The power required for different flight states must be quantified. For reciprocating engines, the powerplant performance is measured by the pressure changes during flight.

4.2.1.1 Test Approach

The Powerplant Performance test will be twofold. The gas turbine test is performed at extremes of powerplant demand in flight maneuvers such as high speed forward flight and high power climbs at high altitude. The power of the gas turbine is then measured in this flight state.

The alternative test is performed on electric and reciprocating engines. These motors do not have torque measurement installed in the engine and must be tested on a dynamometer.

4.2.1.2 Test Article Configuration

The gas turbine powerplant shall be in a flight configuration that is representative of the type of mission to be flown by the AV. The turbine shall be equipped with the standard equipment to perform a standard mission and data acquisition equipment to measure the torque output from the powerplant. For reciprocating engines, the powerplant must be set up to measure manifold pressure, rpm, and horsepower. Electric powerplants must be set up to measure power (in Watts) and rpm.

4.2.2 Facilities and Instrumentation

4.2.2.1 Test Facilities

Refer to section 1.2 for test facilities requirements for turbine engines. Reciprocating and electric dynamometer tests will be performed at separate facilities.

4.2.2.2 Test Instrumentation

The test instrumentation required for powerplant performance procedure is the use of data acquisition equipment that can record the engine sensor data. For dynamometer testing, the test setup will be instrumented to generate the power available graphs. Table 23 defines the instrumentation test parameters required for Powerplant Performance testing.

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Table 23. Test Instrumentation for Powerplant Performance

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Engine Case Temperature (T_{CASE})	°C	Electric	Onboard temperature sensor
Source Temperature (T_S)	°C	Electric	Onboard temperature sensor
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Engine Intake Manifold Pressure (P_M)	in. Hg	Reciprocating	Onboard manifold pressure sensor
Indicated Airspeed (Vo)	kt	All	Onboard Pitot-Static
Engine Temperature (CHT) or (EGT)	°C	Reciprocating	Onboard temperature sensor
Observed Pressure Altitude (H_{PO})	ft	All	On-board static port sensor

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4.2.3 Required Test Conditions

The required test conditions will be applicable from other flight tests including ambient conditions, wind speeds, and precipitation limits. The test condition guidelines from the manufacture should be followed. There must also be a fixed throttle setting with stabilized power demand while collecting data.

4.2.4 Test Procedures

The generic test procedures for the powerplant performance are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the power available of a given UAS.

Test #1 Turbine Engine Performance:

1. **Configure powerplant to run with installed intake and exhaust systems**
2. **Run engine with dynamometer test equipment to determine torque across the rpm range**
3. **Record required data during entire test**

Test #2 Reciprocating Engine Performance:

1. **Configure powerplant to run with installed intake and exhaust systems**
2. **Run engine with dynamometer test equipment to determine horsepower across the rpm range**
3. **Record required data during entire test measuring the manifold pressure associated with each rpm setting**
4. **Generate a power output curve showing the fuel flow/manifold pressure vs. horsepower at various rpm settings**

Test #3 Electric Motor Performance:

1. **Configure the electric power system to run with the motor and battery as installed**
2. **Run motor with dynamometer test equipment in a temperature-controlled environment**
3. **Record required data for the test runs**
4. **For electric motors, measure power available by recording the voltage and current used by the motor across the rpm range and temperature range**

4.2.5 Data Required

Gas Turbine:

Torque (Q)
Gas Generator Speed (N_1)
Power Turbine Inlet Temperature (T_5)
Fuel Flow (W_f)
Rotor Speed (N_R)

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(Vo)
Outside Air Temperature (OAT)
Pressure Altitude (H_{Po})

Reciprocating Engine:
Manifold Pressure (P_M)
Crankshaft Speed (N_1)
Cylinder Head Temperature (CHT) or
Exhaust Gas Temperature (EGT)
Oil Temperature (T_{oil})
(W_f)
(N_R)
(Vo)
OAT
Pressure Altitude (H_{Po})

Electric Powerplants:
Torque (Q)
Motor Output Shaft Speed (N_1)
Motor Case Temperature (T_{case})
(N_R)
(Vo)
Source Voltage Draw (V_S),
Source Current Draw (A_S)
Source Temperature (T_S)
OAT
Pressure Altitude (H_{Po})

4.2.6 Presentation of Data

The data collected during this test event will be used to analyze the powerplant performance for a given UAS configuration. The data output will produce power available curves for a given engine/motor. The data reduction and analysis techniques for the powerplant performance test are found in Section 4.5 and 4.6 of Appendix C.

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SECTION 5

HOVER – VTOL ONLY

5.1 Scope

The purpose of this test is to evaluate the hover performance of a VTOL AV. The AV's power required to hover will be determined and will be compared to the available power from the engine (as determined in Paragraph 4) to establish the AV hover performance characteristics. The power required to hover will be evaluated for both in-ground effects (IGE) and out-of-ground effects (OGE).

5.1.1 Test Approach

One of the primary test objectives of this hover performance test is to determine the power required to hover at various pressure altitudes given a constant AV gross weight. To accomplish this objective, the test approach will incorporate the free flight hover technique. For complete details of the theory of VTOL UAS hover performance testing, refer to Reference 2 and Appendix C.

5.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

5.2 Facilities and Instrumentation

5.2.1 Test Facilities

The general test facility requirements can be found in paragraph 1.2 of this document. However, some micro/mini class UAS might be susceptible to certain wind and/or weather conditions. For example, small winds can cause a rotor system to experience a gain in efficiency, thus creating the appearance that the tested hover performance is better than the actual hover performance. Therefore, under certain circumstances, indoor test facilities might be required to control the environmental and weather effects on this test event.

5.2.2 Test Instrumentation

The test instrumentation required for the free flight hover performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 24 contains a complete list of the instrumentation requirements needed to perform the hover performance test for the various engine types covered under the scope of this document, namely gas turbine,

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reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

Table 24. Test Instrumentation for Free-Flight Hover Performance

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Hover Height	ft	All	DGPS sensor (preferred)
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor
Ground Temperature and Pressure	°C, in. Hg	All	Ground based met data

5.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 25 defines the test tolerances for the free flight hover performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 25. Hover Performance Flight Test Tolerances

Test Parameter	Tolerance
Rotor Speed (N_R)	±2%
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Maximum Lateral and Axial Drift	±3 ft
IGE Hover Height	±1 ft
OGE Hover Height	±3 ft
Wind Speed	<3 knots
GPS Accuracy	4 Satellites (Minimum)

5.4 Test Procedures

The generic test procedures for the free flight hover performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the hover performance of a given UAS.

Test #1: Maximum Gross Takeoff Weight:

1. **Configure AV to maximum gross takeoff weight**
2. **Liftoff and stabilize in IGE hover**
3. **Stabilize in hover for 15 seconds, record data for 5 seconds (average data points)**
4. **Climb until the OGE transition altitude is achieved, noted by an increase in power to hover, record altitude**
5. **Climb an additional 2 rotor diameters to ensure OGE conditions, record altitude**
6. **Stabilize in hover for 15 seconds, record data for 5 seconds (average data points)**
7. **Land AV**

Test #2: Intermediate Gross Takeoff Weight:

1. **Reconfigure AV for intermediate gross takeoff weight**
2. **Liftoff and stabilize in IGE hover**
3. **Stabilize in hover for 15 seconds, record data for 5 seconds (average data points)**
4. **Climb to OGE altitude determined in Test #1, Step #5**
5. **Stabilize in hover for 15 seconds, record data for 5 seconds (average data points)**
6. **Land AV**

Test #3: Minimum Gross Takeoff Weight:

1. **Reconfigure AV for minimum gross takeoff weight**
2. **Liftoff and stabilize in IGE hover**

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3) **Stabilize in hover for 15 seconds, record data for 5 seconds (average data points)**
4) **Climb to OGE altitude determined in Test #1, Step #5**
5) **Stabilize in hover for 15 seconds, record data for 5 seconds (average data points)**
6) **Land AV**

5.5 Data Required

The data required to evaluate the hover performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

Gas Turbine Engine:
- **Torque (Q)**
- **N_1**
- **Rotor Speed (N_R)**
- **OAT**
- **Fuel Used (FU, FC) or Fuel Remaining**
- **Hover Height**
- **Wind Speed and Direction**
- **Observed Pressure Altitude (H_{Po})**
- **Ground Temperature & Pressure**

Reciprocating Engine:
- **Manifold Pressure (P_M)**
- **N_1**
- **N_R**
- **OAT**
- **Fuel Used (FU, FC) or Fuel Remaining**
- **Hover Height**
- **Wind Speed and Direction**
- **H_{Po}**
- **Ground Temperature & Pressure**

Electric Motor:
- **N_R**
- **OAT**
- **Source Voltage Draw (V_S)**
- **Source Current Draw (A_S)**
- **Hover Height**
- **H_{Po}**
- **Wind Speed and Direction.**

5.6 Presentation of Data

The data collected during this test event will be used to analyze the hover performance for a given UAS configuration. The data reduction and analysis techniques for the hover performance test are found in Section 5.5 and 5.6 of Appendix C.

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SECTION 6

VERTICAL CLIMB

6.1 Scope

The purpose of this test is to evaluate the vertical climb performance of the AV. The AV's power required (airframe power) for hover as determined in Section 7 can be combined with climb correction factor determined by the vertical climb test to predict vertical climb rates for any combination of weight, altitude, and rotor speed.

6.1.1 Test Approach

The objectives of vertical climb testing are to determine test day vertical climb performance and to compute a vertical climb correction factor. After obtaining this climb correction factor relationship from flight test, vertical climb rates can be predicted for any combination of aircraft gross weight, rotor speed, atmospheric parameters, or power margins. Vertical climb rate predictions are valid for the range of flight test conditions evaluated in building the climb correction factor model. Aircraft configuration (stores, weapons, fuel pods, etc.) is considered when grouping sets of data. For complete details of the theory for vertical climb performance testing, refer to Reference 2 and Appendix C.

6.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

6.2 Facilities and Instrumentation

6.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

6.2.2 Test Instrumentation

The test instrumentation required for the vertical climb performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 26 contains a complete list of the instrumentation requirements needed to perform the vertical climb performance test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 26. Test Instrumentation for Vertical Climb Performance

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor
Time History of Climb	seconds	All	UTC correlated GPS

6.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 27 defines the test tolerances for the vertical climb performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 27. Vertical Climb Performance Flight Test Tolerances

Test Parameter	Tolerance
Rotor Speed (N_R)	±1 rpm
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
Horizontal Velocity	±3 KIAS
Heading	±5°
Vertical Rate of Climb	Stable
Air Mass	Stable

6.4 Test Procedures

The generic test procedures for the vertical climb performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the vertical climb performance of a given UAS.

Test #1: Maximum Gross Takeoff Weight:

1. **Configure AV to maximum gross takeoff weight**
2. **Liftoff and stabilize in OGE hover, altitude determined from hover performance**
3. **Increase power to begin vertical climb**
4. **Stabilize in vertical rate of climb, as determined by accelerometers or vertical speed indicator, record climb history data through an altitude band or time limit**
5. **Once the data have been collected, the AVO increases forward speed and flies the UAS back to the initial OGE hover point for succeeding data points**
6. **Land and refuel/ballast AV to account for fuel burn**
7. **Repeat steps #1 through #6, incrementally increasing power until the maximum power available is obtained**
8. **Land AV**

Test #2: Intermediate Gross Takeoff Weight:

1. **Configure AV to maximum gross takeoff weight**
2. **Liftoff and stabilize in OGE hover, altitude determined from hover performance**
3. **Increase power to begin vertical climb**
4. **Stabilize in vertical rate of climb, as determined by accelerometers or vertical speed indicator, record climb history data through an altitude band or time limit**
5. **Once the data have been collected, the AVO increases forward speed and flies the UAS back to the initial OGE hover point for succeeding data points**
6. **Land and refuel/ballast AV to account for fuel burn**

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7. **Repeat steps #1 through #6, incrementally increasing power until the maximum power available is obtained**
8. **Land AV**

Test #3: Minimum Gross Takeoff Weight:

1. **Configure AV to maximum gross takeoff weight**
2. **Liftoff and stabilize in OGE hover, altitude determined from hover performance**
3. **Increase power to begin vertical climb**
4. **Stabilize in vertical rate of climb, as determined by accelerometers or vertical speed indicator, record climb history data through an altitude band or time limit**
5. **Once the data has been collected, the AVO increases forward speed and flies the UAS back to the initial OGE hover point for succeeding data points**
6. **Land and refuel/ballast AV to account for fuel burn**
7. **Repeat steps #1 through #6, incrementally increasing power until the maximum power available is obtained**
8. **Land AV**

6.5 Data Required

The data required to evaluate the vertical climb performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

Gas Turbine Engine:

- **OAT**
- **N_1**
- **N_R**
- **Q**
- **FC**
- **H_{Po}**
- **AV configuration**
- **Time history of climb**

Reciprocating Engine:

- **OAT**
- **N_R**
- **P_M**
- **N_1**
- **FC**
- **H_{Po}**
- **AV configuration**
- **Time history of climb**

Electric Motor:

- **OAT**
- **H_{Po}**
- **N_1**

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N_R
V_S
A_S
AV configuration
Time history of climb

6.6 Presentation of Data

The data collected during this test event will be used to analyze the vertical climb performance for a given UAS configuration. The data reduction and analysis techniques for the vertical climb performance test are found in Section 6.5 and 6.6 of Appendix C.

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SECTION 7

LEVEL FLIGHT

7.1 Scope

The purpose of this test is to evaluate the level flight performance of the AV. The AV's power required (airframe power) for level flight will be determined from the prescribed tests to establish airspeeds for maximum endurance and maximum airspeed; the airframe power for level flight will be compared to the available power from the engine (as determined in Section 4) to establish the AV maximum level flight airspeed.

7.1.1 Test Approach

One of the primary test objectives of this level flight performance test is to determine the power required to maintain level flight at various combinations of pressure altitude, AV gross weight, and airspeed. To accomplish this objective, the test approach will incorporate the constant W/σ or constant W/δ technique. For complete details of the theory for level flight performance testing, refer to References 1 and 2 and Appendices B and C.

7.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

7.2 Facilities and Instrumentation

7.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

7.2.2 Test Instrumentation

The test instrumentation required for the level flight performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 28 contains a complete list of the instrumentation requirements needed to perform the level flight performance test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 28. Test Instrumentation for Level Flight

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q) - (VTOL Only)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Indicated/Observed Airspeed (V_o)	knots	All	Existing pitot-static instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R) - (VTOL Only)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Propeller Speed (N_P) - (Fixed Wing Only)	rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor
Ground Temperature/Pressure	°C, in. Hg	All	Ground based met data

7.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 29 defines the test tolerances for the level flight performance test. The parameters listed in

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this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 29. Level Flight Performance Flight Test Tolerances

Test Parameter	Tolerance
Rotor Speed (N_R) - (VTOL Only)	±0.2%
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
V_o	±1 KIAS
H_{Po}	±20 ft
Power Required	Stabilized, Constant

7.4 Test Procedures

The generic test procedures for the level flight performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the level flight performance of a given UAS.

Test #1: Maximum Gross Takeoff Weight:

1. **Configure AV to maximum gross takeoff weight**
2. **Liftoff and stabilize in level flight attitude at the first chosen altitude and airspeed.**
3. **Stabilize in level flight for 10 seconds, record data for 5 seconds (average data points)**
4. **If sufficient fuel (as determined by test team) has burned, increase altitude to maintain constant W/σ; if negligible amount of fuel has been burned, maintain altitude.**
5. **Increase airspeed to next target airspeed**
6. **Repeat steps #3 through #5 until a minimum of eight airspeeds across the operational envelope have been tested for a constant W/σ**
7. **Land AV**

Test #2: Intermediate Gross Takeoff Weight:

1. **Reconfigure AV for intermediate gross takeoff weight**
2. **Liftoff and stabilize in level flight attitude at the first chosen altitude and airspeed**
3. **Stabilize in level flight for 10 seconds, record data for 5 seconds (average data points)**

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4. **If sufficient fuel (as determined by test team) has burned, increase altitude to maintain constant W/σ; if negligible amount of fuel has been burned, maintain altitude.**
5. **Increase airspeed to next target airspeed**
6. **Repeat steps #3 through #5 until a minimum of eight airspeeds across the operational envelope have been tested for a constant W/σ**
7. **Land AV**

Test #3: Minimum Gross Takeoff Weight:

1. **Reconfigure AV for minimum gross takeoff weight**
2. **Liftoff and stabilize in level flight attitude at the first chosen altitude and airspeed.**
3. **Stabilize in level flight for 10 seconds, record data for 5 seconds (average data points)**
4. **If sufficient fuel (as determined by test team) has burned, increase altitude to maintain constant W/σ; if negligible amount of fuel has been burned, maintain altitude.**
5. **Increase airspeed to next target airspeed**
6. **Repeat steps #3 through #5 until a minimum of eight airspeeds across the operational envelope have been tested for a constant W/σ**
7. **Land AV**

7.5 Data Required

The data required to evaluate the level flight performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

Gas Turbine Engine:

- **Indicated/Observed Airspeed (V_o)**
- **Gross Weight**
- **Configuration**
- **Torque (Q) - (VTOL Only)**
- **Rotor Speed (N_R) - (VTOL Only)**
- **Propeller Speed (N_P) - (Fixed Wing Only)**
- **Gas Generator Speed (N_1)**
- **Indicated/Observed Altitude (H_{Po})**
- **OAT**
- **Fuel Used (FU, FC) or Fuel Remaining**

Reciprocating Engine:

- **Indicated/Observed Airspeed (V_o)**
- **Gross Weight**
- **Configuration**
- **Manifold Pressure (P_M)**
- **N_R - (VTOL Only)**
- **N_P - (Fixed Wing Only)**

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Crankshaft speed (N_1)
H_{Po}
OAT
Fuel Used (FU, FC) or Fuel Remaining

Electric Motor:

Indicated/Observed Airspeed (V_o)
Gross Weight
Configuration
Motor Output Shaft Speed (N_1)
N_R - (VTOL Only)
N_P - (Fixed Wing Only)
H_{Po}
OAT
Source Voltage Draw (V_S)
Source Current Draw (A_S)

7.6 Presentation of Data

The data collected during this test event will be used to analyze the level flight performance for a given UAS configuration, namely the airspeeds for maximum endurance and maximum range and the maximum level flight airspeed. The data reduction and analysis techniques for the level flight performance test are found in Section 4.5 and 4.6 of Appendix B for fixed wing UAS and Section 7.5 and 7.6 of Appendix C for VTOL UAS.

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SECTION 8

CLIMB AND DESCENT

8.1 Sawtooth Climb and Descent

8.1.1 Scope

The purpose of this test is to evaluate the climb and descent performance of the AV. The intent is to characterize the effects of weight, altitude, and power setting on the airspeed for maximum rate of climb and the minimum rate of descent. UAS climb rates are important to determine time to climb and performance characteristics which help for mission planning and suitability.

8.1.1.1 Test Approach

One of the primary test objectives of the saw-tooth climb and descent test is to determine the airspeed for the best rate of climb and the airspeed for minimum rate of descent at various combinations of pressure altitude and AV gross weight. For complete details of the theory for level flight performance testing, refer to References 1 and 2 and Appendices B and C.

8.1.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

8.1.2 Facilities and Instrumentation

8.1.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

8.1.2.2 Test Instrumentation

The test instrumentation required for the saw-tooth climb and descent test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 30 contains a complete list of the instrumentation requirements needed to perform the saw-tooth climb and descent test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 30. Test Instrumentation for Climb and Descent

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q) - (VTOL Only)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R) - (VTOL Only)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Time to climb	seconds	All	UTC correlated GPS
Wind Speed and Direction	knots, degrees	All	Onboard met data sensor
Sideslip Angle (β_o)	degrees	All	YAPS boom
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor
Observed Airspeed (V_o)	KIAS	All	Onboard pitot-static system

8.1.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 31 defines the test tolerances for the saw-tooth climb and descent test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 31. Sawtooth Climb and Descent Performance Flight Test Tolerances

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Test Parameter	Tolerance
Rotor Speed (N_R) - (VTOL Only)	±0.5 rpm
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
V_o	±2 KIAS
Heading	±5°
Air Mass	Stable
Climb Rate	Stabilized

8.1.4 Test Procedures

The generic test procedures for the saw-tooth climb and descent performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the climb and descent performance of a given UAS.

1. **Takeoff AV**
2. **Stabilize in level, forward flight**
3. **Increase power setting to maximum**
4. **Initiate climb and stabilize at test airspeed for 10 seconds**
5. **Record data for the test altitude band**
6. **Reduce power and descend to starting altitude**
7. **Increase airspeed by 10 kts and repeat steps #3 through 5**
8. **Repeat tests for all necessary airspeeds with 5-kt increments near edge of flight envelope**
9. **Land AV**

8.1.5 Data Required

The data required to evaluate the saw-tooth climb and descent performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

Gas Turbine Engine:

- **Gross Weight**
- **Configuration**
- **OAT**
- **Indicated Altitude (H_{Po})**
- **Gas Generator Speed (N_1)**
- **Torque (Q)**

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Fuel Remaining (FC)
Indicated Airspeed (V_O)
Sideslip Angle (β_O) [Preferably measured with YAPS boom]
AV configuration
Time history of climb

Reciprocating Engine:
Gross Weight
Configuration
OAT
H_{Po}
Manifold Pressure (P_M)
Engine Crankshaft Speed (N_1)
FC
V_O
β_O
AV configuration
Time history of climb

Electric Motor:
Gross Weight
Configuration
OAT
H_{Po}
Motor Output Shaft Speed (N_1)
Source Voltage Draw (V_S)
Source Current Draw (A_S)
V_O
β_O
AV configuration
Time history of climb

8.1.6 Presentation of Data

The data collected during this test event will be used to analyze the climb and descent performance for a given UAS configuration, namely the airspeeds for maximum rate of climb and minimum rate of descent. The data reduction and analysis techniques for the climb and descent performance test are found in Section 7.5, 7.6, 8.5, and 8.6 of Appendix B for fixed wing UAS and Section 6.5 and 6.6 of Appendix C for VTOL UAS.

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8.2 Check Climb Test

8.2.1 Scope

The purpose of this test is to evaluate the service ceiling of the AV. Depending on the specifications or requirements of the UAS under test, the conventional service ceiling climb rate limit of 100 fpm may not apply. If this is the case, the service ceiling climb test should be conducted until the climb rate limitation specified for the particular UAS under test is reached.

8.2.1.1 Test Approach

One of the primary test objectives of the climb performance test is to determine the AV service ceiling, conventionally noted by the altitude where the maximum rate of climb is 100 fpm (although the specific UAS may have a different specified limit). To accomplish this objective, the test approach will incorporate the continuous climb technique from the surface until the service ceiling is attained. For complete details of the theory for climb performance testing, refer to References 1 and 2 and Appendices B and C.

8.2.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

8.2.2 Facilities and Instrumentation

8.2.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

8.2.2.2 Test Instrumentation

The test instrumentation required for the continuous climb test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 32 contains a complete list of the instrumentation requirements needed to perform the continuous climb test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 32. Test Instrumentation for Continuous Climb Test

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q) - (VTOL Only)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R) - (VTOL Only)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Sideslip Angle (β_o)	degrees	All	YAPS boom
Observed Pressure Altitude (H_{PO})	ft	All	Onboard static port sensor
Observed Airspeed (V_o)	KIAS	All	Onboard pitot-static system
Rate of Climb (ROC)	fpm	All	Onboard vertical speed indicator

8.2.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 33 defines the test tolerances for the continuous climb test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 33. Continuous Climb Flight Test Tolerances

Test Parameter	Tolerance
Rotor Speed (N_R) - (VTOL Only)	±0.5 rpm
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
V_o	±2 KIAS
Heading	±5°
Climb rate	Stable
Air Mass	Stable

8.2.4 Test Procedures

The generic test procedures for the continuous climb performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the level flight performance of a given UAS.

1. **Takeoff AV**
2. **Climb at minimum power required airspeed determined from previous climb tests**
3. **Terminate climb when AV climb rate falls below 100 ft/min or other specified rate**
4. **Record this altitude as service ceiling**

8.2.5 Data Required

The data required to evaluate the level flight performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

Gas Turbine Engine:

- **Gross Weight**
- **OAT**
- **Rotor Speed (N_R)**
- **Gas Generator Speed (N_1)**
- **Torque (Q)**
- **Indicated Airspeed (V_O)**
- **Fuel Remaining (FC)**
- **Sideslip Angle (β_O)**
- **Aircraft configuration**
- **Rate of Climb (ROC)**

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Note: These data elements are generally continuously recorded on a data system and specific data elements reduced at altitude increments.

Reciprocating Engine:

Gross Weight
OAT
N_R
Engine Crankshaft Speed (N_1)
Manifold Pressure (P_M)
V_O
FC
β_O
Aircraft Configuration
ROC
Note: These data elements are generally continuously recorded on a data system and specific data elements reduced at altitude increments.

Electric Motor:

Gross Weight
OAT
N_R
Motor Output Shaft Speed (N_1)
V_O
Source Voltage Draw (V_S)
Source Current Draw (A_S)
β_O
Aircraft Configuration
ROC
Note: These data elements are generally continuously recorded on a data system and specific data elements reduced at altitude increments.

8.2.6 Presentation of Data

The data collected during this test event will be used to analyze the continuous climb performance for a given UAS configuration, namely the altitude at which the service ceiling threshold climb rate is attained. The data reduction and analysis techniques for the climb and descent performance test are found in Section 7.5, 7.6, 8.5, and 8.6 of Appendix B for fixed wing UAS and Section 6.5 and 6.6 of Appendix C for VTOL UAS.

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8.3 Autorotation Descent – VTOL Only

8.3.1 Scope

The purpose of this test is to evaluate the autorotation descent (VTOL, or engine out for fixed wing) performance of the AV. The intent is to characterize the effects of weight and altitude on the best airspeed for minimum descent rate and maximum glide distance. The main rotor speed may also be varied to record the effects of rotor speed on glide capability. Other parameters for autorotation testing include stability in a descent, cyclic control of the UAS, and turn rates of the AV while autorotating. The requirements of the specification are examined where applicable; however, the primary emphasis of this test is based on service suitability and mission suitability.

8.3.1.1 Test Approach

One of the primary test objectives of the autorotation descent performance test is to determine the airspeed for minimum rate of descent and the airspeed for maximum glide range must be determined. To accomplish this objective, the test approach will incorporate the minimum power glide for autorotative (engine out) performance. For complete details of the theory for level flight performance testing, refer to References 1 and 2 and Appendices B and C.

8.3.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

8.3.2 Facilities and Instrumentation

8.3.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

8.3.2.2 Test Instrumentation

The test instrumentation required for the autorotation descent performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 34 contains a complete list of the instrumentation requirements needed to perform the autorotation descent performance test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 34. Test Instrumentation for Autorotative Descent Performance

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Observed Airspeed (V_o)	KIAS	All	Onboard pitot tube
Time history of descent	seconds	All	UTC correlated GPS

8.3.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 35 defines the test tolerances for the autorotation descent performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 35. Autorotation Descent Performance Flight Test Tolerances

Test Parameter	Tolerance
Rotor Speed (N_R)	±1 rpm
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
V_o	±2 KIAS
Heading	±5°
Air Mass	Stable
Rate of Descent	Stable

8.3.4 Test Procedures

The generic test procedures for the autorotation descent performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the autorotation descent performance of a given UAS.

1. **Takeoff AV**
2. **Climb to a specified altitude that yields sufficient room to recover from an autorotative condition**
3. **Set the throttle position to idle and make sure the clutch is disengaged**
4. **Record the descent rate while flying at an airspeed of level flight minimum**
5. **Return engine throttle position and climb to start position**
6. **Repeat descent test varying the airspeed in 10-kt increments for all ranges of descent requirements**
7. **Land AV**

8.3.5 Data Required

The data required to evaluate the autorotation descent performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

Gas Turbine:

Gross Weight
OAT
Rotor Speed (N_R)
Gas Generator Speed (N_1)
Torque (Q)
Indicated Airspeed (V_O)

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Fuel Remaining (FC)
Aircraft Configuration
Time History of Descent

Reciprocating Engine:
Gross Weight
OAT
N_R
Engine Crankshaft Speed (N_1)
Manifold Pressure (P_M)
V_O
FC
Aircraft Configuration
Time History of Descent

Electric Motor:
Gross Weight
OAT
N_R
Output Shaft Speed (N_1)
Source Voltage Draw (V_S)
Source Current Draw (A_S)
V_O
Aircraft Configuration
Time History of Descent

8.3.6 Presentation of Data

The data collected during this test event will be used to analyze the autorotation descent performance for a given UAS configuration, namely the airspeeds for maximum glide range and minimum rate of descent. The data reduction and analysis techniques for the climb and descent performance test are found in Section 6.5 and 6.6 of Appendix C.

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8.4 Autorotation Turn – VTOL Only

8.4.1 Scope

The purpose of this test is to evaluate the autorotation turn performance of a VTOL AV. The intent is to characterize the effects of weight and altitude on the quickest rate of turn and best rate of turn while minimizing the rate of descent of the AV. The main rotor speed may also be varied to record the effects of rotor speed on glide capability. Other parameters for autorotation testing include stability in a descent, cyclic control of the UAS, and turn rates of the AV while autorotating. The requirements of the specification are examined where applicable; however, the primary emphasis of this test is based on service suitability and mission suitability.

8.4.1.1 Test Approach

One of the primary test objectives of the autorotation turn performance test is to determine the maximum turn rate and best rate of turn while minimizing the descent rate of the AV in autorotation. For complete details of the theory for level flight performance testing, refer to References 1 and 2 and Appendices B and C.

8.4.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

8.4.2 Facilities and Instrumentation

8.4.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

8.4.2.2 Test Instrumentation

The test instrumentation required for the autorotation turn performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 36 contains a complete list of the instrumentation requirements needed to perform the autorotation turn performance test for the various engine types covered under the scope of this document, namely gas turbine, reciprocating, and electric. Suggested instrumentation methods and techniques are also provided in this table.

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Table 36. Test Instrumentation for Autorotative Turn Performance

Test Parameter	Units	Engine Type	Suggested Methods
Torque (Q)	% psi	Gas Turbine	Existing engine instrumentation on vehicle
Gas Generator Speed (N_1)	% rpm	Gas Turbine	Existing engine instrumentation on vehicle
Crank Shaft Speed (N_1)	% rpm	Reciprocating	Existing engine instrumentation on vehicle
Motor Shaft Speed (N_1)	% rpm	Electric	Existing engine instrumentation on vehicle
Manifold Pressure (P_M)	in. Hg	Reciprocating	Existing engine instrumentation or added pressure transducer
Rotor Speed (N_R)	% rpm	All	Existing engine instrumentation or added Hall Effect sensor
Outside Air Temperature (OAT)	°C	All	Onboard AV thermocouple
Fuel Used/Remaining	lb	Gas Turbine, Reciprocating	Onboard fuel flow meter or algorithm
Battery Voltage (V_S)	volts	Electric	Existing battery instrumentation
Battery Current (A_S)	amperes	Electric	Existing battery instrumentation or added meter/Coulomb counter
Bank Angle (ϕ)	degrees	All	Onboard accelerometer
Observed Airspeed (V_o)	KIAS	All	Onboard pitot tube
Time history of turn rate and descent	seconds	All	UTC correlated GPS

8.4.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 37 defines the test tolerances for the autorotation turn performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 37. Autorotation Turn Performance Flight Test Tolerances

Test Parameter	Tolerance
Rotor Speed (N_R)	±3 rpm
Ambient Air Temperature	25°C ± 10°C
Temperature Inversion	None
Atmospheric Pressure	Site Pressure
Time Duration (Events > 8 Hours)	±5 minutes
Time Duration (Events < 8 Hours)	±1%
V_o	±3 KIAS
ϕ	±2°
Air Mass	Stable
Rate of Descent	Stable

8.4.4 Test Procedures

The generic test procedures for the autorotation turn performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the level flight performance of a given UAS.

1. **Takeoff AV**
2. **Climb to a specified altitude that yields sufficient room to recover from an autorotative condition**
3. **Set the throttle position to idle and make sure the clutch is disengaged**
4. **Descend according to the recommended guidelines and best performance speed**
5. **Test turn capability and controllability**
6. **Land AV**

8.4.5 Data Required

The data required to evaluate the autorotation turn performance of a given UAS are summarized below for each of the engine types that fall within the scope of this TOP.

Gas Turbine:

OAT
Rotor Speed (N_R)
Gas Generator Speed (N_1)
Torque (Q)
Indicated Airspeed (V_O)
Fuel Remaining (FC)
Bank Angle (φ)

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Aircraft Configuration
Time History of Turn Rate and Descent

Reciprocating Engine:

OAT
N_R
Engine Crankshaft Speed (N_1)
Manifold Pressure (P_M)
V_O
FC
Bank Angle (φ)
Aircraft Configuration
Time History of Turn Rate and Descent

Electric Motor:

OAT
N_R
Output Shaft Speed (N_1)
Source Voltage Draw (V_S)
Source Current Draw (A_S)
V_O
Bank Angle (φ)
Aircraft Configuration
Time History of Turn Rate and Descent

8.4.6 Presentation of Data

The data collected during this test event will be used to analyze the autorotative turn performance for a given UAS configuration, namely its ability to navigate during powerplant failure. The data reduction and analysis techniques for the autorotative turn performance test are found in Section 6.5 and 6.6 of Appendix C.

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SECTION 9

TAKEOFF AND LANDING – FIXED WING ONLY

9.1 Conventional Takeoff

9.1.1 Scope

The purpose of this test is to evaluate the conventional takeoff performance of a fixed wing UAS. The test will determine the complete takeoff performance of the AV up to 50 ft obstacle clearance or reaching climb airspeed, whichever comes last.

9.1.1.1 Test Approach

One of the primary test objectives of a conventional takeoff performance test is to determine flight performance parameters from ground roll initiation to 50 ft obstacle clearance or climb airspeed. To accomplish this objective, the test approach will be to run the AV through a typical takeoff profile while recording all required data. For complete details of the theory of fixed wing UAS conventional takeoff performance testing, refer to Reference 1 and Appendix B.

9.1.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

9.1.2 Facilities and Instrumentation

9.1.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

9.1.2.2 Test Instrumentation

The test instrumentation required for a conventional takeoff performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 38 contains a complete list of the instrumentation requirements needed to perform a conventional takeoff performance test. Suggested instrumentation methods and techniques are also provided in this table.

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Table 38. Test Instrumentation for Conventional Takeoff Performance

Test Parameter	Units	Suggested Methods
Takeoff Airspeed (V_{TO})	knots	Existing pitot-static instrumentation on vehicle
Distance to Liftoff	ft	Theodolite, Runway Camera, Paint Gun, Observers, Laser Tracker, DGPS (with time correlated weight on wheels sensor or accelermometers)
Pitch Attitude, Angle of Attack on Rotation/Initial Climb	degrees	Existing instrumentation on vehicle, YAPS boom
Distance to 50 ft Altitude, Distance to Reach Climb Airspeed	ft	Existing instrumentation on vehicle, DGPS
Time to Liftoff, 50 ft Altitude, Climb Airspeed	seconds	Existing instrumentation on vehicle, DGPS
Runway Wind Conditions and Temperature	degrees, knots, ^{o}C	Existing test facility meteorological station or portable meteorological station
Elevator/Horizontal Stabilizer Position	degrees	Existing instrumentation on vehicle
Fuel Used/Remaining	lb	Onboard fuel flow meter or algorithm
Powerplant Parameters (rpm, EGT, etc.)	NA	Existing powerplant instrumentation, may need to add thermocouple or pressure transducer, depending on powerplant used.

9.1.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 39 defines the test tolerances for a conventional takeoff performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 39. Conventional Takeoff Performance Flight Test Tolerances

Test Parameter	Tolerance
Runway Wind Conditions	**Within operational limits of UAS**
Ambient Air Temperature	**25°C ± 10°C**

9.1.4 Test Procedures

The generic test procedures for the conventional takeoff performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the conventional takeoff performance of a given UAS. If there is more than one flap setting for takeoff, all possible settings should be tested; if only clean configuration (No Flaps) will be used, only tests 1, 3, and 5 will be necessary.

Test #1: Maximum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to maximum gross takeoff weight, 0° flap setting**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #2: Maximum Gross Takeoff Weight, Flaps Deployed:

1. **Configure AV to maximum gross takeoff weight, takeoff flap setting (15°, 30°, 60°, etc.)**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #3: Intermediate Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to maximum gross takeoff weight, 0° flap setting**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

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Test #4: Intermediated Gross Takeoff Weight, Flaps Deployed:

1. **Configure AV to maximum gross takeoff weight, takeoff flap setting (15°, 30°, 60°, etc.)**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #5: Minimum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to maximum gross takeoff weight, 0° flap setting**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #6: Minimum Gross Takeoff Weight, Flap Setting:

1. **Configure AV to maximum gross takeoff weight, takeoff flap setting (15°, 30°, 60°, etc.)**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

9.1.5 Data Required

The data required to evaluate the conventional takeoff performance of a given UAS are summarized below.

All Engine Types:

- **Takeoff airspeed, V_{TO}**
- **Distance to lift off obtained by:**
  - **Theodolite**
  - **Runway camera**
  - **Paint gun**
  - **Observers**
  - **Laser**
  - **DGPS (with time-correlated weight on wheels sensors or accelerometers)**
- **Pitch attitude on rotation/initial climb**
- **Distance to 50 ft/distance to climb airspeed**
- **Time to lift off/50 ft/climb airspeed**

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Angle of attack at rotation/climb out
Fuel used, brake release to climb airspeed
Runway wind conditions
Runway temperature
Runway composition/runway condition reading (RCR)
Field elevation
Altimeter
Runway gradient
Aircraft gross weight/center of gravity
Power parameters: rpm, EGT, etc
Elevator/horizontal tail position

9.1.6 Presentation of Data

The data collected during this test event will be used to analyze the conventional takeoff performance for a given UAS configuration. The data reduction and analysis techniques for the conventional takeoff performance test are found in Section 9.5 and 9.6 of Appendix B.

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9.2 Conventional Landing

9.2.1 Scope

The purpose of this test is to evaluate the conventional landing performance of a fixed wing UAS. The test will determine the complete landing performance of the AV from 50 ft obstacle clearance to full stop.

9.2.1.1 Test Approach

One of the primary test objectives of a conventional landing performance test is to determine flight performance parameters from 50 ft obstacle clearance to full stop. To accomplish this objective, the test approach will be to run the AV through a typical landing profile while recording all required data. For complete details of the theory of fixed wing UAS conventional landing performance testing, refer to Reference 1 and Appendix B.

9.2.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

9.2.2 Facilities and Instrumentation

9.2.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

9.2.2.2 Test Instrumentation

The test instrumentation required for a conventional landing performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 40 contains a complete list of the instrumentation requirements needed to perform a conventional landing performance test. Suggested instrumentation methods and techniques are also provided in this table.

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Table 40. Test Instrumentation for Conventional Landing Performance

Test Parameter	Units	Suggested Methods
Distance from 50 ft above the ground through the flare to touch down, distance from touchdown to full stop	ft	Existing instrumentation on vehicle, DGPS (with time-correlated weight on wheels sensor or accelermometers)
Glide Slope Angle	degrees	Existing instrumentation on vehicle, YAPS boom
Time from 50 ft altitude to touchdown, touchdown to full stop	seconds	Existing instrumentation on vehicle, DGPS
Runway Wind Conditions and Temperature	degrees, knots, ^{o}C	Existing test facility meteorological station or portable meteorological station
Powerplant Parameters (rpm, EGT, etc.)	NA	Existing powerplant instrumentation, may need to add thermocouple or pressure transducer, depending on powerplant used

9.2.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 41 defines the test tolerances for a conventional landing performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 41. Conventional Landing Performance Flight Test Tolerances

Test Parameter	Tolerance
Runway Wind Conditions	Within operational limits of UAS
Ambient Air Temperature	25°C ± 10°C

9.2.4 Test Procedures

The generic test procedures for the conventional landing performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the conventional landing performance of a given UAS. If there is

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more than one flap setting for landing, all possible settings should be tested; if only clean configuration (No Flaps) will be used, only tests 1, 3, and 5 will be necessary.

Test #1: Maximum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to minimum gross takeoff weight, no flaps**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #2: Maximum Gross Takeoff Weight, Flaps Deployed:

1. **Configure AV to minimum gross takeoff weight, landing flap settings (15°, 30°, 60°, etc.)**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #3: Intermediate Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to minimum gross takeoff weight, no flaps**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #4: Intermediated Gross Takeoff Weight, Flap Setting:

1. **Configure AV to minimum gross takeoff weight, landing flap settings (15°, 30°, 60°, etc.)**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #5: Minimum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to minimum gross takeoff weight, no flaps**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #6: Minimum Gross Takeoff Weight, Flap Setting:

1. **Configure AV to minimum gross takeoff weight, landing flap settings (15°, 30°, 60°, etc.)**
2. **Launch AV using existing procedures**

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3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

9.2.5 Data Required

The data required to evaluate the conventional landing performance of a given UAS are summarized below.

Distance from 50 ft above the ground through the flare to touch down
Glideslope angle
Distance from touch down to full stop
Runway wind conditions
Runway temperature
Runway composition/runway condition reading
Field elevation
Altimeter
Runway gradient
Aircraft gross weight/center of gravity
Configuration
Power parameters during landing and roll out: rpm, EGT

9.2.6 Presentation of Data

The data collected during this test event will be used to analyze the conventional landing performance for a given UAS configuration. The data reduction and analysis techniques for conventional takeoff performance test are found in Section 9.5 and 9.6 of Appendix B.

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9.3 Augmented Takeoff

9.3.1 Scope

The purpose of this test is to evaluate the augmented takeoff performance of a fixed wing UAS. Augmented takeoff testing requires determining the energy imparted to AV from the augmenting system. For launchers or boosters, accelerometers must be placed on the AV so that the energy supplied can be calculated from the change in velocity. The methods discussed for conventional takeoffs for tracking the climb distance should apply for augmented takeoffs as well.

9.3.1.1 Test Approach

One of the primary test objectives of an augmented takeoff performance test is to determine flight performance parameters from launch to 50 ft obstacle clearance or climb airspeed. To accomplish this objective, the test approach will be to run the AV through a typical augmented takeoff profile while recording all required data. For complete details of the theory of fixed wing UAS augmented takeoff performance testing, refer to Reference 1 and Appendix B.

9.3.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

9.3.2 Facilities and Instrumentation

9.3.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

9.3.2.2 Test Instrumentation

The test instrumentation required for an augmented takeoff performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 42 contains a complete list of the instrumentation requirements needed to perform an augmented takeoff performance test. Suggested instrumentation methods and techniques are also provided in this table.

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Table 42. Test Instrumentation for Augmented Takeoff Performance

Test Parameter	Units	Suggested Methods
Takeoff Airspeed (V_{TO})	knots	Existing pitot-static instrumentation on vehicle
Acceleration Upon Launch	ft/s^2	Installed accelerometers or flight test instrumentation package equipped with accelerometers
Pitch Attitude, Angle of Attack on Rotation/Initial Climb	degrees	Existing instrumentation on vehicle, YAPS boom
Distance to 50 ft Altitude, Distance to Reach Climb Airspeed	ft	Existing instrumentation on vehicle, DGPS
Time to Liftoff, 50 ft Altitude, Climb Airspeed	seconds	Existing instrumentation on vehicle, DGPS
Runway Wind Conditions and Temperature	degrees, knots, ^oC	Existing test facility meteorological station or portable meteorological station
Elevator/Horizontal Stabilizer Position	degrees	Existing instrumentation on vehicle
Fuel Used/Remaining	lb	Onboard fuel flow meter or algorithm
Powerplant Parameters (rpm, EGT, etc.)	NA	Existing powerplant instrumentation, may need to add thermocouple or pressure transducer, depending on powerplant used.

9.3.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 43 defines the test tolerances for an augmented takeoff performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

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Table 43. Augmented Takeoff Performance Flight Test Tolerances

Test Parameter	Tolerance
Runway Wind Conditions	**Within operational limits of UAS**
Ambient Air Temperature	**25°C ± 10°C**

9.3.4 Test Procedures

The generic test procedures for the augmented takeoff performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the augmented takeoff performance of a given UAS. If there is more than one flap setting for takeoff, all possible settings should be tested; if only clean configuration (No Flaps) will be used, only tests 1, 3, and 5 will be necessary.

Test #1: Maximum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to maximum gross takeoff weight, 0° flap setting**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #2: Maximum Gross Takeoff Weight, Flaps Deployed:

1. **Configure AV to maximum gross takeoff weight, takeoff flap setting (15°, 30°, 60°, etc.)**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #3: Intermediate Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to maximum gross takeoff weight, 0° flap setting**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

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Test #4: Intermediated Gross Takeoff Weight, Flaps Deployed:

1. **Configure AV to maximum gross takeoff weight, takeoff flap setting (15°, 30°, 60°, etc.)**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #5: Minimum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to maximum gross takeoff weight, 0° flap setting**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

Test #6: Minimum Gross Takeoff Weight, Flap Setting:

1. **Configure AV to maximum gross takeoff weight, takeoff flap setting (15°, 30°, 60°, etc.)**
2. **Increase throttle for maximum power**
3. **Takeoff or launch AV using existing procedures**
4. **Lift off and climb out to 50 ft AGL, recording data from first movement to 50 ft AGL**
5. **Land AV (can be combined with landing test)**

9.3.5 Data Required

The data required to evaluate the conventional takeoff performance of a given UAS are summarized below.

Takeoff airspeed, V_{TO}
Acceleration upon launch
Pitch attitude on rotation/initial climb
Distance to 50 ft / distance to climb airspeed
Time to lift off / 50 ft / climb airspeed
Angle of attack at rotation/climb out
Fuel used, brake release to climb airspeed
Runway wind conditions
Runway temperature
Runway composition/runway condition reading (RCR)
Field elevation
Altimeter
Runway gradient

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Aircraft gross weight/center of gravity
Power parameters: rpm, EGT, etc.
Elevator/horizontal tail position

9.3.6 Presentation of Data

The data collected during this test event will be used to analyze the conventional takeoff performance for a given UAS configuration. The data reduction and analysis techniques for conventional takeoff performance test are found in Section 9.5 and 9.6 of Appendix B.

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9.4 Augmented Landing

9.4.1 Scope

The purpose of this test is to evaluate the augmented landing performance of a fixed wing UAS. Similar to augmented takeoff, augmented landing tests require determining the energy imparted to AV from the augmenting system. Accelerometers must be placed on the AV so that the energy supplied can be calculated from the change in velocity of the AV. The test will determine the complete landing performance of the AV from 50 ft obstacle clearance to full stop.

9.4.1.1 Test Approach

One of the primary test objectives of an augmented landing performance test is to determine flight performance parameters from 50 ft obstacle clearance to full stop. To accomplish this objective, the test approach will be to run the AV through a typical landing profile while recording all required data. For complete details of the theory of fixed wing UAS augmented landing performance testing, refer to Reference 1 and Appendix B.

9.4.1.2 Test Article Configuration

Refer to Section 1.3 for the requirements for the test article configuration.

9.4.2 Facilities and Instrumentation

9.4.2.1 Test Facilities

Refer to Section 1.2 for the test facility requirements.

9.4.2.2 Test Instrumentation

The test instrumentation required for an augmented landing performance test must be capable of sensing, processing, storing, and/or transmitting a complete set of test data to an auxiliary ground-based recording station. It is preferred that all test data be time correlated to the UTC using GPS techniques. Table 44 contains a complete list of the instrumentation requirements needed to perform an augmented landing performance test. Suggested instrumentation methods and techniques are also provided in this table.

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Table 44. Test Instrumentation for Augmented Landing Performance

Test Parameter	Units	Suggested Methods
Distance from 50 ft above the ground through the flare to touch down, distance from touchdown to full stop	ft	Existing instrumentation on vehicle, DGPS (with time-correlated weight on wheels sensor or accelerometers)
Acceleration Upon Engaging Augmented Landing System	ft/s^2	Installed accelerometers or flight test instrumentation package equipped with accelerometers
Glide Slope Angle	degrees	Existing instrumentation on vehicle, YAPS boom
Time from 50 ft altitude to touchdown, touchdown to full stop	seconds	Existing instrumentation on vehicle, DGPS
Runway Wind Conditions and Temperature	degrees, knots, ^oC	Existing test facility meteorological station or portable meteorological station
Powerplant Parameters (rpm, EGT, etc.)	NA	Existing powerplant instrumentation, may need to add thermocouple or pressure transducer, depending on powerplant used.

9.4.3 Required Test Conditions

The test tolerances defined in this TOP will be applied to the nominal test values. Table 45 defines the test tolerances for an augmented landing performance test. The parameters listed in this table are suggested values, noting that many UAS will have limitations in meeting these requirements.

Table 45. Augmented Landing Performance Flight Test Tolerances

Test Parameter	Tolerance
Runway Wind Conditions	Within operational limits of UAS
Ambient Air Temperature	25°C ± 10°C

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9.4.4 Test Procedures

The generic test procedures for the augmented landing performance test are summarized below. The procedural steps listed below represent a summary of the primary tasks required to evaluate the augmented landing performance of a given UAS. If there is more than one flap setting for landing, all possible settings should be tested; if only clean configuration (No Flaps) will be used, only tests 1, 3, and 5 will be necessary.

Test #1: Maximum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to minimum gross takeoff weight, no flaps**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #2: Maximum Gross Takeoff Weight, Flaps Deployed:

1. **Configure AV to minimum gross takeoff weight, landing flap settings (15°, 30°, 60°, etc.)**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #3: Intermediate Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to minimum gross takeoff weight, no flaps**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #4: Intermediated Gross Takeoff Weight, Flap Setting:

1. **Configure AV to minimum gross takeoff weight, landing flap settings (15°, 30°, 60°, etc.)**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

Test #5: Minimum Gross Takeoff Weight, Clean Configuration:

1. **Configure AV to minimum gross takeoff weight, no flaps**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

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Test #6: Minimum Gross Takeoff Weight, Flap Setting:

1. **Configure AV to minimum gross takeoff weight, landing flap settings (15°, 30°, 60°, etc.)**
2. **Launch AV using existing procedures**
3. **Lift off and climb out to local pattern**
4. **Configure appropriate flap setting**
5. **Land AV using existing procedures (start at 50 ft AGL)**

9.4.5 Data Required

The data required to evaluate the augmented landing performance of a given UAS are summarized below.

Distance from 50 ft above the ground through the flare to touch down.
Acceleration upon engaging augmented landing system
Glideslope angle.
Distance from touch down to full stop.
Runway wind conditions.
Runway temperature.
Runway composition / runway condition reading.
Field elevation.
Altimeter.
Runway gradient.
Aircraft gross weight / center of gravity.
Configuration.
Power parameters during landing and roll out: rpm, EGT.

9.4.6 Presentation of Data

The data collected during this test event will be used to analyze the augmented landing performance for a given UAS configuration. The data reduction and analysis techniques for conventional takeoff performance test are found in Section 9.5 and 9.6 of Appendix B.

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APPENDIX A

REFERENCES

1. **United States Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108), Fixed Wing Performance, Naval Test Pilot School, Patuxent River, Maryland, September 30, 1992.**

2. **United States Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), Rotary Wing Performance, Naval Test Pilot School, Patuxent River, Maryland, December 31, 1996.**

3. **DTC Regulation No. 25-30, Department of the Army Headquarters, U.S. Army Developmental Test Command, Aberdeen Proving Ground, Maryland, August 4, 2005.**

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Appendix B

UNMANNED AERIAL VEHICLE (UAV)

FLIGHT TEST METHODOLOGY

"FIXED WING PERFORMANCE"

3 August 2005

Prepared For:

U.S. Army Aviation Technical Test Center
Fort Rucker, Alabama 36362

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This manual provides supplemental information for use in testing the performance of fixed wing unmanned aerial vehicles (UAVs). It modifies and adds to the fixed wing performance theory and test techniques given in the U.S. Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108), "Fixed Wing Performance." It has been prepared to address the unique test requirements for UAVs as opposed to manned aircraft. Both manuals should be used in conjunction in developing a performance flight test program for fixed wing UAV systems. This manual follows the same numbering scheme and format as the USNTPS-FTM-No. 108 to facilitate easy reference between the two manuals. Additionally, each chapter is stand alone and incorporates an independent Table of Contents.

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Appendix I: Chapter 4 Propeller Efficiency Charts

Appendix II: Chapter 4 Propeller Efficiency Spreadsheet

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CHAPTER 1

INTRODUCTION TO UAV FLIGHT TESTING

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1. Introduction to UAV Flight Testing

1.1 Approach to Testing

A vital concept that should be stressed before any flight testing discussion begins is that flight testing should be balanced with system simulation. The Department of Defense (DoD) policy (AR 5-11) stresses a synergistic approach to the development of military systems. The concept is to use the benefits of both simulation and flight testing to compliment one another. Simulation in conjunction with flight testing supports a build-up test methodology by effectively supporting unmanned aerial vehicle (UAV) operator training, while at the same time providing a model of the expected behavior of the air vehicle (AV) under controlled input conditions. This combination aids the flight testing process since the simulation facilitates improved air vehicle operator (AVO) mission planning and maneuver execution while allowing test engineers to better predict actual AV performance. Spiral testing and development allow for refined aircraft models and generation of better AV simulations which feedback to benefit the development process of the UAV and the ultimate customer, the U.S. Warfighter.

Before flight test, considerable ground testing of key systems should occur. Whenever possible, the propulsion system should be tested in an altitude chamber to model the power performance and limiting parameters with independent variation of pressure and temperature of the atmosphere. The data are crucial to proper analysis of the flight test data, as well as establishment of the baseline performance model for the system. The wing and overall AV should be tested in a wind tunnel, or extensively modeled by Computational Fluid Dynamics (CFD), to determine the drag polar and aerodynamic characteristics of the UAV. CFD methods can be less expensive than wind tunnel testing; however, with the low Reynolds numbers that could be observed when flying smaller UAVs and with the complex and frequently blunt fuselage configurations typical of UAVs, CFD methods can be much less robust. The flight test will further supplement and substantiate AV performance; however, good theoretical knowledge of the AV performance is critical to effective test planning, execution, and evaluation.

Another important consideration is the planning and use of test instrumentation to map the parameters of the flight deck. These vital pieces of information can aid in deterring mishaps and critical component failures. Whenever possible, AVs under test should be extensively instrumented (air data booms, control surface potentiometers, current measurements, temperatures, pressures, etc.). Experience with testing UAVs has shown that most engines integrated into the AVs will develop cooling problems at some point in their flight envelope expansion. To obtain a better understanding of the airflow and cooling properties for the engine, extensive instrumentation should be used in the engine intake so that these problems can be addressed in developmental/operational testing rather than by the Warfighter in the field. If feasible, the essential parameters should be instrumented and fed through the downlink to the ground control station (GCS) so that the data can be monitored in real time by the AVOs and test engineers. It is preferred practice to partition the data into the smaller flow of flight critical parameters

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directed to the AVO and Mission Commander and the larger flow of detailed information presented to a test engineering team. However, an alternate datalink may be required to provide sufficient bandwidth for transferring required engineering data.

The U.S. Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108) “Fixed Wing Performance” should be used as a baseline for developing a customized fixed wing UAV flight performance test plan supplemented/modified by the information in this manual.

Flight testing for development aircraft can often lead to changes and improvements to the UAV system. The AV can be altered such that the standard gross take-off weights and configurations of the test aircraft may be different from the production aircraft, leading to difference in the performance from testing. Also, the GCS flight control system software may have numerous upgrades throughout the development and initial fielding effort. The software version should always be monitored as software changes may dramatically change performance characteristics and change the test techniques used for specific test points.

1.2 UAV vs. Manned Aircraft

When starting a flight test program for UAVs, it is important to remember that there are many differences in performance and operation from manned aircraft. Certain flight testing procedures cannot be accomplished with some UAVs due to control and/or performance limitations. Also, the AVO has the difficult task of performing flight test maneuvers without the same level of situational awareness available to a pilot of a manned vehicle.

One difference is the flight control system. A good understanding of the particular UAV flight control system is essential. Most autonomous UAVs have a flight control system that severely limits the types of maneuvers that may be performed by the aircraft. An example would be flying with intentional sideslip on the aircraft to evaluate pitot-static performance. This test technique could pose a problem because a number of UAV control systems integrate roll and yaw together when making a turn, and there may not be a provision in the flight control logic to allow a pure yaw input. In this case, the maneuver could not be performed. Another example of the flight control system potentially limiting test capabilities is for climbing flight. Many autopilot systems use a single airspeed when a climb is commanded. This precludes the testing of multiple airspeed test points necessary for validating the airspeeds for maximum rate of climb, maximum climb angle, minimum time to climb, and minimum fuel burn. If this is the case for the test aircraft, the flight control logic will need to have supplementary code so that various airspeed test points can be flown. The alteration of the flight control software from the production configuration to the test configuration will require flight release from the airworthiness authority.

Some UAV flight control systems use pitch control to command airspeed while losing altitude; others use application of throttle. If the flight control system of the UAV

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uses pitch for airspeed, a test that requires a tight altitude hold tolerance would not be appropriate. For UAVs that perform high-g turns, the trailing bomb test might not be appropriate due to the stress on the tether. However, as of this time, UAVs capable of performing the trailing bomb test usually make turns with large radii, resulting in low-g turns.

A solid understanding of each AV's control logic sequencing and lost link procedures, including failure modes, are central to the planning process for the flight test engineering team. As most UAVs operate without redundant data links, quick response to failure modes must be possible with limited situational awareness and interface with the AV. On these lines, a well understood lost link procedure is essential to risk management.

Most modern UAV flight control designs do not require the AVO to "close the loop" on altitude or airspeed. Modern UAVs are designed to be "hands off" requiring the AVO to "fly" the aircraft using only a map display and a mouse to input a desired waypoint. This interface frequently affects the fidelity of the downlink data; therefore, the test team and AVO need to be aware that some of the flight critical data such as airspeed and altitude being transmitted through the downlink may not be of the same resolution as that being supplied to the onboard autopilot.

Some UAVs also use a tactic known as "airfoil flexing" to reduce AV drag. This procedure involves the use of flaps at negative angle settings. This will cause the induced drag on the wing to drop drastically, as well as reducing the aerodynamic moment induced on the wing allowing the UAV to be flown with less elevator trim, further reducing the total induced drag. This tactic is usually employed during cruise, when a greater airspeed is desirable to get the UAV on station as quickly as possible using as little fuel as possible.

Consideration should be given to the UAVs sensitivity to weather and terrain. Turbulent weather can have a dynamic impact on the flight control system's ability to maintain condition for flight test points. This is particularly the case for the smaller UAVs, but the flight test conditions can be difficult to sustain with larger UAVs in a highly unstable weather environment. This may lead to repeating selected test procedures to ensure test accuracy.

1.3 UAV Classifications

The following UAV classification system can be used to better apply test techniques to different sized UAVs. The system is not an attempt to change future combat system (FCS) classification standards. These classes are meant to help the test team plan for flight days and set standards for weather conditions and repetition of test procedures. Some UAVs may bridge two categories and techniques may apply from one or both classes. Most UAVs are large enough to cause serious injury in the event of a mishap, and test personnel should maintain safe distances during takeoff and landing despite the small size of certain classes of UAV systems.

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1.3.1 Small UAV

The Small UAV class is defined as aircraft that are capable of carrying up to a 10 lb payload for up to 5 hours or up to 30 km at typical altitudes of up to 15,000 ft mean sea level (MSL); however, their payloads typically mandate operations at or below 1,000 ft above ground level (AGL). This class would include FCS Class I and II UAVs. Examples of UAVs in this class are Raven, Pointer, and DragonEye. The small size of this class makes these aircraft very sensitive to weather and especially wind conditions. For this reason, these UAVs should be tested on days with calm winds (winds <3 kt). High winds could exceed the UAV's ability to maintain course and in extreme conditions, could result in loss of AV control and possible injury to test personnel.

1.3.2 Medium UAV

The Medium UAV class is defined as aircraft that are capable of carrying up to a 70 lb payload for up to 10 hours or up to 40 km at typical altitudes up to 20,000 ft MSL. Based on payload considerations, operating altitudes range from 500 to 8,000 ft AGL. This class would include FCS Class III UAVs. Examples of UAVs in this class are Shadow 200 and ScanEagle. These UAVs are less sensitive to weather conditions but care should still be taken to test on calm days (winds <3 kt) to increase the accuracy of test data.

1.3.3 Large UAV

The Large UAV class is defined as aircraft that are capable of carrying more than a 70 lb payload for longer than 10 hours or greater than 40 km at typical altitudes over 20,000 ft MSL. This class would include FCS Class IV and above UAVs. Examples of UAVs falling into this class are Hunter, Fire Scout, Shadow 600, and ER/MP. These UAVs are the least sensitive to weather and wind conditions due to their size, but calm days (winds <3 kt) for testing are still desirable.

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CHAPTER 2

PITOT-STATIC SYSTEM PERFORMANCE

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2. Pitot-Static System Testing

2.1 Introduction

Performing tests of the unmanned aerial vehicle (UAV) pitot-static system is a vital part of the overall flight testing since most UAV flights profiles are managed by pressure altitude, indicated airspeed (IAS), and vertical speed, each derived from the pitot-static system. The instruments must transmit precise readings of both altitude and IAS to the ground control station (GCS) since the testing will often call for the air vehicle operator (AVO) to hold a specific airspeed or altitude or both. Additionally, errors in the pitot-static system have critical implications for flying at speeds near V_{ne}, maneuvering, loitering with sensitive payloads, or instrument flight rules (IFR) missions. Ground testing in preparation for this type test is described in Section 2.3 and flight test methods are detailed in Section 2.4. The U.S. Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108) Chapter 2, "Pitot-static System Performance," is referenced in a number of the methods and should be used in conjunction with this chapter.

2.2 Purpose of the Test

The intent of the test is to fully characterize the instrument error and the position error of the air vehicle (AV). This will determine the airspeed and altimeter correction for the remainder of the flight testing. The instrument calibration is accomplished thru the ground test and the position error will be determined by one of the flight test methods. Additionally, the temperature recovery factor is mentioned in USNTPS-FTM-No. 108 Section 2.5.4 and those procedures can be followed, which will determine how closely the total temperature sensor observes the actual total temperature.

2.3 Ground Test

Ground testing of the AV's pitot-static system involves connecting a calibrated test set directly to the aircraft's pitot tube and static ports to simulate various altitudes and indicated airspeed combinations. The intent is to fully characterize the instrument error portion of the total pitot-static error budget. The airframe technician will control the test set and the AVO will observe the response of the aircraft pitot-static system across the datalink in the GCS. The GCS altimeter is set to 29.92 in. Hg (1013.25 millibars) to correlate to the replicated static pressure for the static port from the test set. This provides a check of the system before takeoff so that the pitot-static system can be calibrated for its instrument error. The lag in the pitot-static system can also be determined by comparing the input-to-response time difference between the test set and the AV pitot-static system. Additionally, any lag in the UAV GCS or telemetry datalinks must be assessed and can be found by observing the total lag of the system, and then subtracting the pitot-static system lag. When calibrating the pitot-static system from flight test data, the high resolution data logged by the onboard telemetry package should be used. The intent of the lower resolution data being sent to the GCS is for the AVO to maintain situational awareness.

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2.4 Flight Test Methods

Stall speed is an important parameter for all flight testing programs. Most UAVs, however, lack a control system that allows them to be flown at or near stall; therefore, actual stall speed is rarely determined for a UAV. The theoretical stall speed may be given by the manufacturer, but if it is not, the following equation can be used to compute the theoretical stall speed of the aircraft. The factors that must be known are the weight of the aircraft (W), density of the air (ρ), the wing plan form area (S), and the maximum lift coefficient ($C_{L_{MAX}}$). The effect of sideslip on high aspect ratio, high L/D wings such as those found on most UAVs is a considerable increase in stall speed. Zero-sideslip during maneuvers performed in the lower airspeed region is paramount. Refer to Chapter 3 for the full discussion of stall speed determination.

$$V_{Stall} = \sqrt{\frac{2W}{\rho S C_{L_{MAX}}}} \tag{2.1}$$

For manned aircraft, the traditional approaches to the flight component of pitot-static system testing can be accomplished by one of six methods that are fully detailed in Section 2.4 of the USNTPS FTM 108. These include the measured course method, trailing source, tower Flyby, space positioning, radar altimeter, and pace aircraft. Additional means of testing the pitot-static system for UAVs comprise the YAPS (Yaw, Angle of Attack, Pitot-static) and two global positioning system (GPS) methods.

The test program should be a matrix of airspeed, weight, altitude, flaps, and, if possible, yaw. The tolerances are set under each test method. However, the flight control system of each UAV dictates the spacing between the indicated airspeed data points, since the “airspeed hold” tolerance of the flight control system may be greater than the airspeed increments. The test points should be concentrated in the lower airspeed regime. Variation in airspeed alters the angle of attack of the aircraft, more so at low airspeeds, and this change in attitude position of the AV relative to the free stream air could drastically alter pitot tube and static port performance. All flap settings should be tested, if discrete. Refer to Figure 2.4.1 for an illustration of the concentration of test points at the lower end of the airspeed envelope. For this example, the AV theoretical stall airspeed is 50 knots indicated airspeed (KIAS) and the max airspeed is 74 KIAS. Figure 2.4.2 diagrams the impact of airspeed on the rate of change in angle of attack (α). UAV control systems are usually built to obtain near zero sideslip (beta), but if possible, the general trend in static pressure defect should be mapped at least $\pm$5 degrees, preferably 10, depending on the safety of the alpha/beta combinations to be flown.

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	Indicated Airspeed (KIAS)																		
Flaps	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74
-15°					X	X	X	X		X		X		X		X				X
0°		X	X	X	X	X	X	X		X		X		X		X				X
15°	X	X	X	X	X	X	X	X		X		X		X		X				
30°	X	X	X	X	X	X	X	X		X		X		X						
60°	X	X	X	X	X	X	X	X		X		X								

Figure 2.4.1 Data Point Concentration Illustration

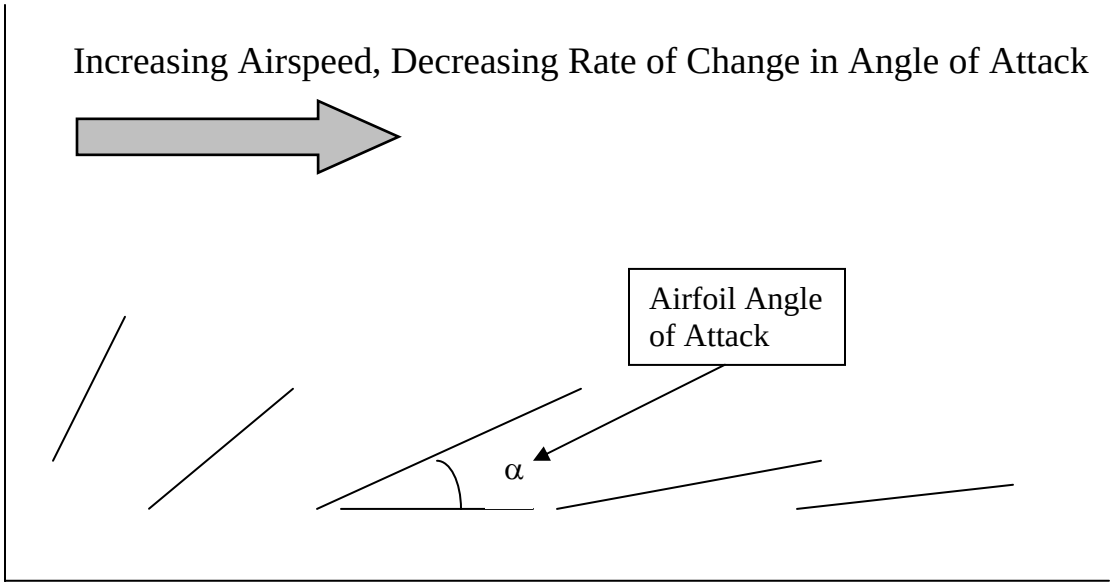


Figure 2.4.2 Diagram of Impact of Airspeed Variation on Angle of Attack Rate of Change

In order to fly the aircraft at a desired angle of attack (α, alpha), the airspeed for the given weight (W) must be calculated to produce the desired α. This can be accomplished as long as the change in lift coefficient with α (C_{L_α}), weight (W), wing planform area (S), and air density (ρ) are known. First, the desired α is selected and the lift coefficient (C_L) at that α is calculated. C_{L_α} should be provided by the UAV manufacturer. If this is not the case, this information can be found by using experimental

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C_L vs. α charts. To find the true airspeed (V_T) at which the UAV must be flown to attain the desired α, the following equation can be used.

$$V_T = \sqrt{\frac{2W}{\rho S C_L}} \tag{2.2}$$

2.4.1 Medium and Large UAV Systems - Onboard Differential Pressure Measurement - YAPS Preferred

Experience with UAV systems over the last 15 years suggests that for medium and large UAVs, there are certain pitot-static calibration methods that are most useful. A preferred testing method uses an onboard differential pressure transducer to compare the aircraft static port reading with a test static source (provided that the AVs have the space, weight, and power capability for additional instrumentation). The transducers must have an accuracy of 0.001 psi or less with a range of 0.03 psi. An onboard telemetry data logging system should be integrated into the UAV. This will provide high-resolution, time-correlated data to be used for data reduction and analysis, with the capability of sending low-resolution data during the flight tests through the datalink to the test control center for situational awareness. The test techniques for large UAVs are YAPS boom, GPS measured course, GPS 3-Leg, Space Positioning, Pace Aircraft, Tower Flyby, Trailing Cone, and Trailing Bomb.

2.4.2 Small UAV Systems - Tower Flyby Preferred

For UAVs that are sensitive (do not have the space, weight and power (SWAP) capability) to additional onboard systems and can fly at low AGL altitudes, the tower flyby is the preferred option. Other options include GPS measured course, GPS 3-Leg, and Space Positioning.

2.4.3 YAPS Boom

2.4.3.1 Overview

The YAPS boom is suitable for medium and large UAVs. This method involves mounting a calibrated boom assembly at a distance of at least two wing-chord lengths in front of the wing or at least two fuselage diameters in front of the nose of the aircraft. Desirably, the boom installation should extend significantly beyond the AV's nose or wing to mitigate any interference effects. Care must be taken to mount the boom as rigidly as possible. This can usually be accomplished by securing the boom mounting bracket to the airframe structure. Refer to Figure 2.4.3 for the YAPS boom installation on a RQ-7A Shadow 200.

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Figure 2.4.3 YAPS installation on RQ-7A Shadow 200

The change in center of gravity (CG) resulting from boom installation can be counteracted in a number of ways. The payload of the UAV should already be removed and replaced with an aerodynamically similar "dummy" payload for flight testing due to the risks involved with losing that asset. This will move the CG of the UAV rearward, assuming a forward mounted unit, requiring ballast toward the nose of the aircraft to move the CG into the acceptable range. With the YAPS boom installed, the CG would move forward, and to counteract this, ballast from the nose could be removed. The boom is usually calibrated by the manufacturer, but if there is a question as to the boom accuracy, a pace aircraft can be used to verify a set of speed/altitude data points. The manufacturer calibration method should be obtained for use in explaining the error correction.

The data from the UAV pitot-static system can be compared to the data from the boom, and position error may be calculated. The pressure transducers and the sense points should be mounted as close as is practical for the particular UAV to be tested, but, if the boom is mounted away from the pressure transducers, the tubing and connectors between the pitot-static ports and the transducers must be leak free. The YAPS boom also provides time and location correlated Angle of Attack (α) and Sideslip (β) measurement, and thus provides both the direct measurement of independent and dependent parameters.

The test procedure is to fly the UAV along a predetermined course to allow the boom and UAV pitot-static system to record data. The test can then be repeated at various combinations of altitude, airspeed, weight, configuration and yaw (if the Flight Control System is capable of performing a pure yaw maneuver). This will determine correction values for the UAV pitot-static system.

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2.4.3.2 Data Required

$P_{T_{YAPS}}$, $P_{S_{YAPS}}$, $V_{i_{YAPS}}$, α_{YAPS}, β_{YAPS}, P_T, P_S, V_i, T_i, GW, Configuration

2.4.3.3 Test Criteria

1. Check AV and YAPS pitot-static systems for leaks
2. Conduct test in as little wind as possible
3. Constant altitude
4. Constant airspeed
5. Constant heading (except pure yaw maneuver, if possible)

2.4.3.4 Data Requirements

1. Stabilize AV for 5 seconds before recording
2. Record Data for 30 seconds
3. $V_i \pm 0.5$ KIAS or limits of flight control system
4. Altitude ± 10 ft or limits of flight control system

2.4.3.5 Safety Considerations

There are no safety issues directly relating to this technique differing from UAV standard operating procedures.

2.4.4 GPS Measured Course

2.4.4.1 Overview

The two-leg GPS method is similar to the standard measured course method detailed in USNTPS-FTM-No. 108, Section 2.4.1, and has been developed by Dr. Ralph Kimberlin at University of Tennessee Space Institute (UTSI). This method has been used recently on a Piper Navajo with Dr. Kimberlin during a flight test course at UTSI.

The procedure is to fly reciprocal headings at the same indicated airspeed and altitude. The indicated airspeed, altitude, heading, and outside air temperature (OAT) are recorded in the avionics suite. The ground track and start and end points are recorded by the GPS so that the course and distance can be determined. Once the UAV is stable for 5 to 10 seconds, the ground track should be recorded for the desired run length. It should be at least 2 minutes long so that any variation in airspeed can be averaged. The AVO must ensure a constant heading during the run. The ground velocity (*GV*) is the vector distance flown divided by the time of the run. The calculated ground speed (*GS*) in the direction of the heading is the magnitude of ground velocity times the cosine of the difference between the Heading and the Ground Track angles (θ). The same reference, magnetic or true, should be used for heading and ground. The calculated ground speed

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for the reciprocal runs can be averaged to determine the true airspeed (V_T) for that test point. Refer to Figure 2.4.4 for an illustration of the GPS method. Use altitude and OAT to convert to Calibrated Airspeed. The calibrated airspeed from the GPS can then be compared with the instrument readings from the pitot-static system. The accuracy of the data depends upon the wind vector remaining constant for both legs.

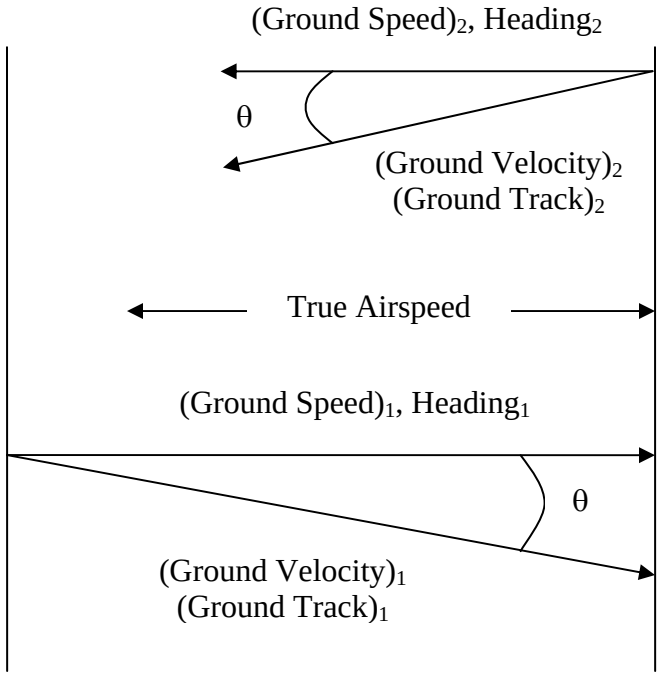


Figure 2.4.4 GPS Method Graphical Representation

The equations are as follows:

$$GV = \frac{(GroundVector)}{\Delta t} \tag{2.3}$$

$$GS = GV * \cos(\theta) \tag{2.4}$$

$$V_T = \frac{GS_1 + GS_2}{2} \tag{2.5}$$

An efficient way to fly the conditions is to fly the first speed leg, then turn around and fly the second leg and the first leg of the next speed before turning around. Continue to fly two legs (different speeds) before each turn.

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2.4.4.2 Data Required

Ground Vector, Ground Track, V_i , H_{P_i} , T_i , GW, Configuration

2.4.4.3 Test Criteria

1. Steady level flight
2. Constant altitude
3. Constant airspeed
4. Constant heading
5. Constant wind speed and direction

2.4.4.4 Data Requirements

1. Stabilize AV for 30 seconds before recording
2. Record data for 2 minutes (each leg)
3. $V_i \pm 0.5$ KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system

2.4.4.5 Safety Considerations

There are no safety issues directly relating to this technique differing from UAV standard operating procedures.

2.4.5 GPS 3-Leg Method

2.4.5.1 Overview

The GPS method of calibrating the pitot-static system involves flying the UAV at a constant airspeed and altitude over three or more legs. Stabilize the UAV and record indicated airspeed, pressure altitude, and OAT. Once the UAV is stable for 5 to 10 seconds, the ground speed and ground track should be recorded for the desired run length. It should be at least 5 minutes long so that any variation in airspeed can be averaged. Change the heading by at least 60° but not more than 120° and repeat the previous step at the same altitude and airspeed for the second leg. Airspeed is the more important parameter to maintain. Following the second run, change the heading by at least 60° but not more than 120° for the third leg, flying at the same airspeed and altitude. Once at least three legs are completed, the desired information can be calculated. In lieu of completing these three legs, a weather balloon may be used to acquire winds aloft (speed and direction), allowing for a single test run along one leg.

The AVO must ensure a constant ground track during the run. The onboard GPS, or an installed GPS, is used to determine ground speed and ground track. It is assumed that the UAV true airspeed along with wind speed and direction are constant during all three or more runs. Wind speed, wind direction, and true airspeed can then be

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determined by solving three equations to find the three unknowns. The *GV* (speed and direction) is the vector sum of the true airspeed and wind. This is illustrated in Figure 2.4.5.

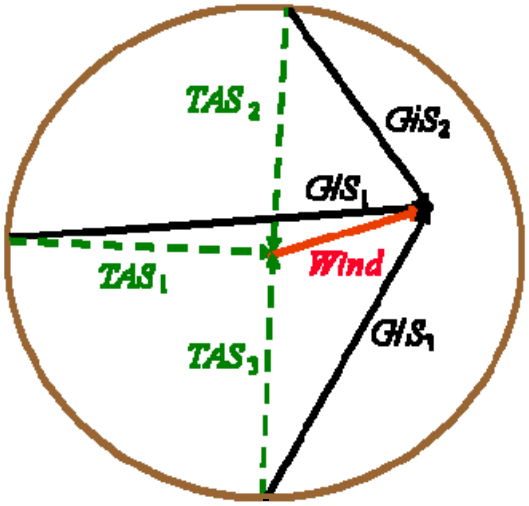


Figure 2.4.5 GPS Method Graphical Representation

A downloadable Excel solution, "GPS_PEC.xls." is available on the National Test Pilot School's website: www.ntps.edu/downloads.htm. The general solution shown by Doug Gray in "Using GPS to Accurately Determine True Airspeed" is given below in Figure 2.4.6. The solution is given in a form to expedite entry into a spreadsheet program with example speeds and tracks shown in cells B1 to B6.

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	A	B	C
1	Speed1	140	140
2	Track1	192	192
3	Speed2	112	112
4	Track2	283	283
5	Speed3	120	120
6	Track3	20	20
7	X1	=B1*SIN(PI()*(360-B2)/180)	
8	Y1	=B1*COS(PI()*(360-B2)/180)	
9	X2	=B3*SIN(PI()*(360-B4)/180)	
10	Y2	=B3*COS(PI()*(360-B4)/180)	
11	X3	=B5*SIN(PI()*(360-B6)/180)	
12	Y3	=B5*COS(PI()*(360-B6)/180)	
13	M1	=-1*(B9-B7)/(B10-B8)	
14	B1	=(B8+B10)/2-B13*(B7+B9)/2	
15	M2	=-1*(B11-B7)/(B12-B8)	
16	B2	=(B8+B12)/2-B15*(B7+B11)/2	
17	WX	=(B14-B16)/(B15-B13)	
18	WY	=B13*B17+B14	
19	Wind Speed	=SQRT(B17^2+B18^2)	20.6
20	Wind Direction	=MOD(540-(180/PI()*ATAN2(B18,B17)),360)	314.8
21	TAS	=SQRT((B7-B17)^2+(B8-B18)^2)	130
22	Heading 1	=MOD(540-(180/PI()*ATAN2(B18-B8,B17-B7)),360)	200
23	Heading 2	=MOD(540-(180/PI()*ATAN2(B18-B10,B17-B9)),360)	287.8
24	Heading 3	=MOD(540-(180/PI()*ATAN2(B18-B12,B17-B11)),360)	11.7

Figure 2.4.6 Excel Calculator for GPS Method

There is no technical requirement to use a specially certified or differential GPS. The indication of accuracy of the derived true airspeed can result from the repeatability of the derived wind vector. If several points are flown at different speeds (each "point" being at least three legs at the same speed) and result in the same derived wind speed and direction, then a high degree of confidence can be placed in those results. Regardless of the sophistication of the GPS equipment, the data can be unreliable from one of several factors: wind or temperature variation, failure to hold airspeed and/or heading constant for each leg, altering airspeed between legs of a data point, or poor GPS satellite coverage. Figure 2.4.7 shows examples of calculations of GPS-derived true airspeeds on a certification program and the calculated wind remained very nearly constant over the approximate 30 minute data gathering period. The average wind speed varied less than one knot (18.1 to 18.8 kt) and the direction changed a maximum of 3 degrees. The pitot-static errors derived from those calculations are shown in Figure 2.4.8 and compare favorably to other data gathered from a traditional ground course in the same aircraft. Lastly, Figure 2.4.9 shows data from a different certification project comparing GPS results to those obtained with a wing mounted boom. Both Figures 2.4.8 and 2.4.9

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illustrate the type of accuracy possible using the GPS flight test technique. The data shown in both of these figures were taken using a handheld Garmin 92 GPS unit.

	Point 1		Point 2		Point 3		Point 4	
	Vg	Track	Vg	Track	Vg	Track	Vg	Track
	116	236	75	221	43	204	58	204
	131	135	92	117	64	109	77	115
	153	46	111	41	79	40	95	42
	134	316	90	315	61	318	75	316
Vtrue	**134.0**		**91.5**		**60.7**		**75.6**	

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
1	116	236	131	135	153	46	18.9	219	134.2	
	131	135	153	46	134	316	19.4	221	133.7	
	153	46	134	316	116	236	18.7	223	134.3	
	134	316	116	236	131	135	18.1	221	133.6	
							18.8	**221**	**134.0**	Average
									0.4	Std Dev

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
2	75	221	92	117	111	41	18.4	209	93.1	
	92	117	111	41	90	315	20.9	219	90.1	
	111	41	90	315	75	221	18.2	229	93.0	
	90	315	75	221	92	117	14.9	220	89.9	
							18.1	**219**	**91.5**	Average
									1.8	Std Dev

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
3	43	204	64	109	79	40	18.2	217	60.8	
	64	109	79	40	61	318	18.4	218	60.6	
	79	40	61	318	43	204	18.2	219	60.8	
	61	318	43	204	64	109	17.9	218	60.6	
							18.2	**218**	**60.7**	Average
									0.2	Std Dev

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
4	58	204	77	115	95	42	18.7	213	76.5	
	77	115	95	42	75	316	20.1	218	75.0	
	95	42	75	316	58	204	18.9	222	76.1	
	75	316	58	204	77	115	17.2	219	74.8	
							18.7	**218**	**75.6**	Average
									0.9	Std Dev

Figure 2.4.7 GPS Example Calculation

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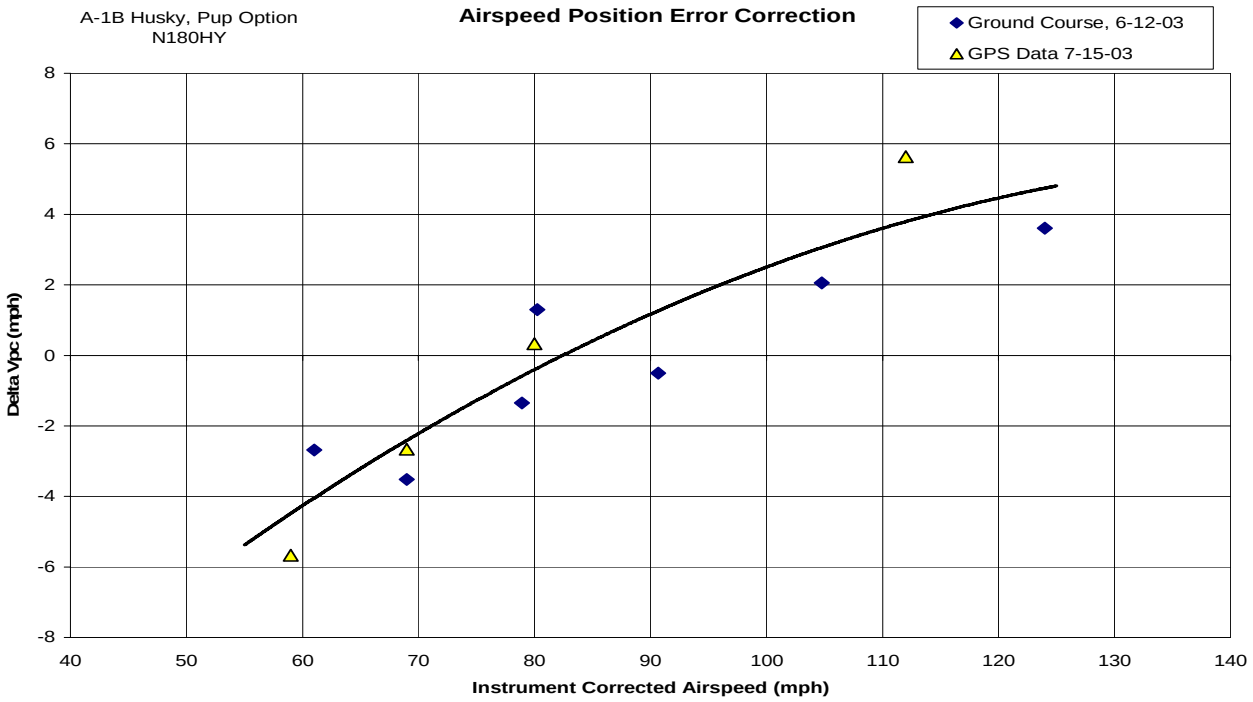


Figure 2.7.8 Position Error Correction Data Compared with Ground Course

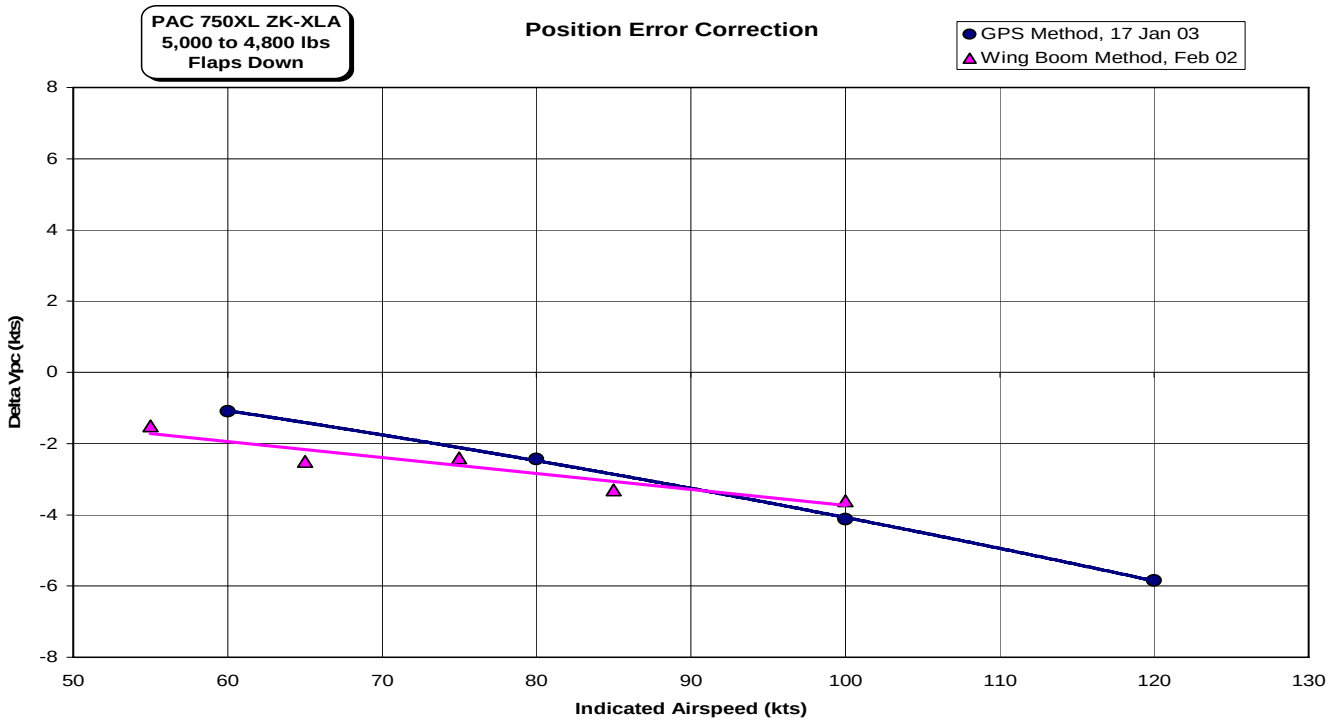


Figure 2.4.9 Position Error Correction Data Compared with Wing Boom Method

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2.4.5.2 Data Required

V_G , Ground Track, V_i , H_{P_i} , T_i , GW, Configuration

2.4.5.3 Test Criteria

1. Steady level flight
2. Constant altitude
3. Constant airspeed
4. Constant heading
5. Constant wind speed and direction

2.4.5.4 Data Requirements

1. Stabilize AV for 30 seconds before recording
2. Record Data for 5 minutes (each leg)
3. Vi $\pm$ 0.5 KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system

2.4.5.5 Safety Considerations

There are no safety issues directly relating to this technique differing from UAV standard operating procedures.

2.4.6 Space Positioning

2.4.6.1 Overview

This method of pitot-static testing involves the use of three ground-based tracking stations, or one self-ranging station, which acquire and track test aircraft either by radar lock or visual processes. The tracking station computer will record angular and linear displacements and determine a solution of the tapeline altitude and ground speed for the test aircraft.

A pace aircraft should be used for a pressure survey of the area. If one is not available, the position error of the test aircraft must be known for at least one value of airspeed at the test altitude. The pace aircraft flies at constant airspeed and altitude so that, since the pace altimeter position error and radar measured tapeline altitude are known, the actual pressure altitude will be determined. For an accurate calculation of the test AV's true airspeed, the wind speed and direction must be known. This can be accomplished through use of a weather balloon to acquire the atmospheric conditions at the test altitude.

Soon after the completion of the pressure survey, the test UAV should follow along the same ground track with an onboard tracking beacon, both for accurate radar

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ranging and so that ground controllers can provide course corrections. The pressure altitude of the test aircraft is the tapeline difference between the test and pace aircraft corrected for nonstandard temperature. Computer algorithms use the raw radar data to triangulate the position, velocity, and acceleration data of the test aircraft. Use of an onboard transponder will improve the radar tracking. Additionally, compensating for the tracking errors due to random drift from the prearranged target point on the AV can augment the accuracy of the data.

This test is particularly useful when higher airspeeds render the tower flyby or pace tests unsuitable. Since a pace aircraft should be used, this test is probably unsuitable for the small and mid-sized UAV. It requires detailed planning and coordination as well as an exact time correlation between onboard and ground recorders. The benefits are the large volume of data generated in a relatively short period of time and the large variety of flight conditions (climbing or descending, accelerating or decelerating) time correlated as long as systems are synchronized. The primary drawbacks are the complexity and cost. It is similar to the GPS three-leg method except that one run is sufficient to capture the necessary data.

2.4.6.2 Data Required

V_i, H_{P_i}, T_i, GW, Configuration

2.4.6.3 Test Criteria

1. Steady level flight
2. Constant heading
3. Constant airspeed
4. Constant altitude
5. Constant wind vector

2.4.6.4 Data Requirements

1. Stabilize AV for 30 seconds before recording
2. Record Data for 15 seconds
3. $V_i \pm 0.5$ KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system

2.4.6.5 Safety Considerations

The safety of flight is paramount and flight testing should not risk the minimum safe altitudes or airspeeds at any point during the flight. Also, the test and pace aircraft pilots should coordinate the responsibility for providing safe separation distance between the aircraft and coordinate each AV emergency procedure and break-off maneuver during lost link emergencies and test termination.

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2.4.7 Pace Aircraft

2.4.7.1 Overview

This method of pitot-static system testing can be useful as a way of attaining the test AV position error information accurately in a short period of time. It can cover any altitude and airspeed that the pace aircraft and test aircraft are compatible. However, since the pace aircraft requires precise observation of relative velocity between the two AVs, this test may not be suitable for small and medium UAVs due to the inability to follow the test AV at excessively low airspeeds. The pace will have independently calibrated pitot-static systems that will require periodic checks on speed courses or with space positioning systems.

The UAV will lead the formation. The start point is set by the lead with the trailing AV stabilized in position, following closely enough to observe minor changes in relative velocity without interfering with its own pitot-static system (at least one wing span separation). The pace aircraft provides calibrated pressure altitude and airspeed so that the position error associated with pressure and airspeed can be evaluated.

2.4.7.2 Data Required

V_i, H_{Pi}, T_i, $H_{P_{i,ref}}$, $V_{i_{ref}}$, $T_{i_{ref}}$, GW, Configuration

2.4.7.3 Test Criteria

1. Steady level flight
2. Constant heading
3. Constant airspeed
4. Constant altitude (or stable rate of change during climbs or descents)
5. Zero relative velocity between two aircraft

2.4.7.4 Data Requirements

1. Stabilize AV for 30 seconds before recording
2. Record data for 60 seconds
3. $V_i \pm 0.5$ KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system

2.4.7.5 Safety Considerations

Many of the risks associated with manned flight testing can be mitigated since the UAV flight computer will transmit the data via the downlink to be recorded in the GCS so the AVO may concentrate on flying the pattern. The pace aircraft should have a second crewmember to alleviate the pilot from the task of accurately reading the airspeed, altimeter, and other instruments. As always, safety of flight is paramount and flight

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testing should not risk the minimum safe altitudes or airspeeds at any point during the flight. Also, the test and pace aircraft pilots should coordinate the responsibility for providing safe separation distance between the aircraft as well as each AV's emergency procedures and break-off maneuver at test termination. Lost link procedures must also be fully understood by the pace aircraft pilot for each stage of UAV flight. Knowledge of the direction of turn and behavior of the UAV under lost link can be used to plan the pace aircraft's position while chasing and response when a lost link event occurs. Two way communications between the pace aircraft and the AVO are critical.

2.4.8 Tower Flyby

2.4.8.1 Overview

This method is suitable for all sizes of UAVs and involves flying the AV on a track that is a known distance from the tower, which is at a known height and has a calibrated altimeter to determine the static free-stream pressure (Figure 2.4.10). For UAVs that use GPS only for altitude, this method would not be useful due to the UAV lack of static pressure measurement for altitude. For each pass, the aircraft will be sighted through a theodolite to record the angle from the tower to the aircraft as well as the tower altimeter setting. The angle and the distance to the aircraft can be used to calculate the height above the tower through trigonometric relationships. The aircraft instrument and the tower reference instrument should be placed side by side to check the output from the altimeters. If any discrepancy exists, it should be taken into account in the data reduction.

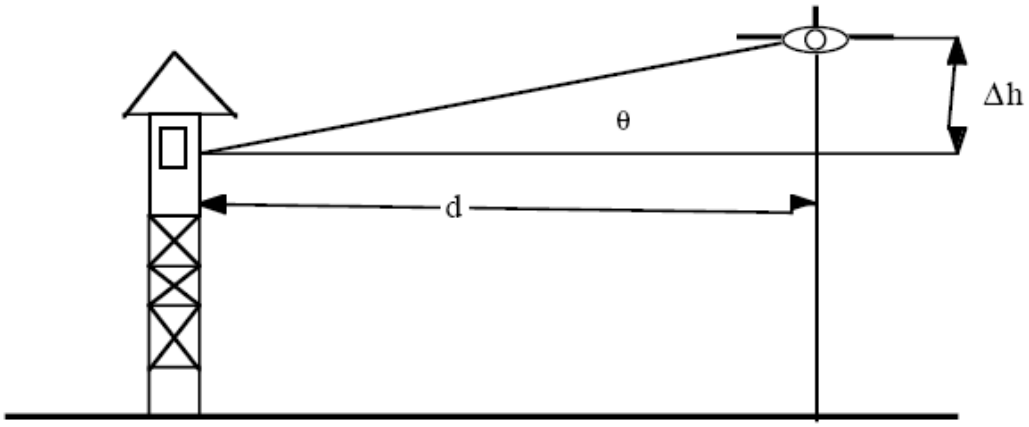


Figure 2.4.10 Tower Flyby

Another way to use the tower Flyby is to take a grid photograph of the aircraft as it passes the tower. The scaled AV length and scaled height above the tower can be used with the actual length to determine the actual height above the tower. This alteration relieves the AVO of flying over the predetermined path. Only the camera would need to be aligned so the picture is not taken at an angle. This test will generate an accurately calibrated pressure altitude, but not a calibrated airspeed. Since the tower is at an

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extremely low altitude, the altitude variation should not exceed 15 feet above or below the target altitude.

2.4.8.2 Data Required

V_i, H_{Pi}, $H_{P_{c,twr}}$, T_i, $T_{a_{ref}}$, GW, Configuration

2.4.8.3 Test Criteria

1. Steady level flight
2. Constant airspeed
3. Constant altitude
4. Constant heading and track over predetermined path

2.4.8.4 Data Requirements

1. Stabilize AV for 15 seconds before tower
2. Ground Track $\pm$ 1% of Stand-off Distance
3. V_i $\pm$ 0.5 KIAS or limits of flight control system
4. Altitude $\pm$10 ft, or limits of flight control system

2.4.8.5 Safety Considerations

This technique requires the AVO to pay considerable attention in such proximity to the tower and be fully aware of the normal and emergency procedures of the UAV. Lost link procedures must also be known and when adding a lost link waypoint to the flight profile, it should be placed to turn the aircraft away from the tower and any test personnel.

2.4.9 Trailing Cone

2.4.9.1 Overview

This technique is appropriate for medium and large UAVs and entails suspending a cone from the trailing pressure tube to stabilize it and keep it taut. This helps to dampen the cable oscillation. The trailing source should be constantly monitored for accurate data. The static source again determines the pressure altitude and thus the altimeter position error. The calibration of the cone installation can be accomplished during a tower flyby or by a pace aircraft. A properly installed cone can be used to check the aircrafts instruments or to initially validate an aircraft of the same model. Figure 2.4.11 illustrates the trailing cone test arrangement.

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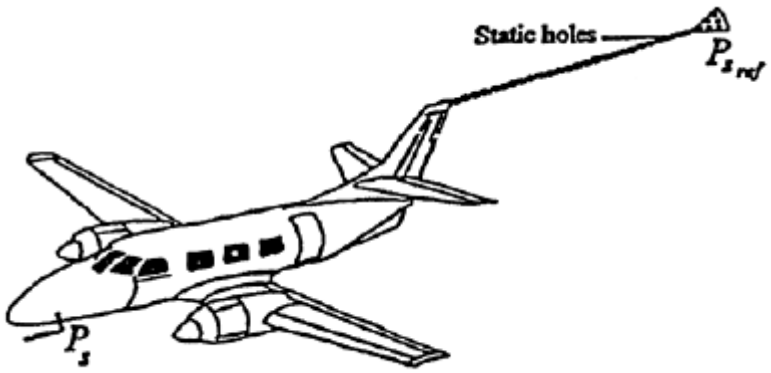


Figure 2.4.11 Trailing Cone Test

2.4.9.2 Data Required

V_i, H_{P_i}, $H_{P_{i,ref}}$, T_i, GW, Configuration

2.4.9.3 Test Criteria

1. Steady level flight
2. Constant airspeed
3. Constant heading
4. Constant altitude
5. Stabilized instrument readings

2.4.9.4 Data Requirements

1. Stabilize AV for 30 seconds before recording
2. Record Data for 60 seconds
3. $V_i \pm 0.5$ KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system

2.4.9.4 Safety Considerations

The AVO should be able to jettison the tube and probe if they become unstable to the point that recovery cannot be made. This requirement may result in software modification to the AV/GCS. Observers should always maintain the minimum safe distance from the ground track of the UAV. Lost link procedures and the turn rates of the UAV must be understood before testing; a tight turn could cause the tether to break or interfere with the control surfaces or propulsion system.

2.4.10 Trailing Bomb

2.4.10.1 Overview

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This approach is appropriate for large UAVs and requires having a bomb-shaped static source extended on a cable at least two wingspans in length (Figure 2.4.12). The bomb size is large enough to preclude testing small and medium UAVs by this method. The static pressure will be measured and transmitted up the cable to the aircraft where the value will be converted into a pressure altitude. The altimeter position error can then be determined since the difference in altitude between the trailing source and aircraft source will be known. The trailing bomb must be calibrated in a wind tunnel to determine static source error. The source can acquire the free-stream pressure altimeter setting below the aircraft if connected via the cable to an altimeter. One potential problem is instability of the cable and trailing source, which may occur at both low and high airspeeds. This leaves midrange airspeeds that can be flown by the aircraft. Additionally, there should be a means to constantly monitor the trailing source so that data is recorded when both the airspeed and the trailing source are stable. The source can acquire the free-stream pressure altimeter setting below the aircraft if connected via the cable to an altimeter. The method of cable extension and retrieval, such as a winch, might also cause SWAP requirement issues which need to be addressed before testing using the trailing bomb method.

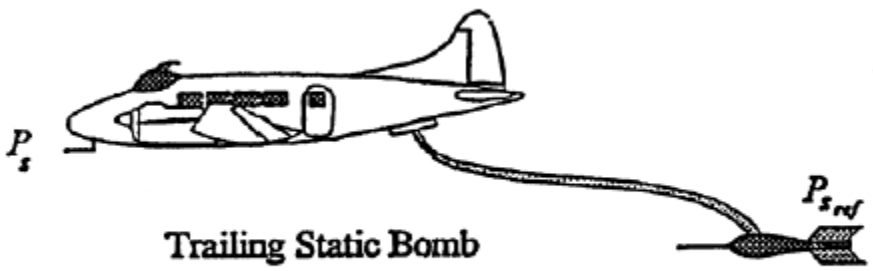


Figure 2.4.12 Trailing Bomb Test

2.4.10.2 Data Required

V_i, H_{P_i}, $H_{P_{i,ref}}$, T_i, GW, Configuration

2.4.10.3 Test Criteria

1. Steady level flight
2. Constant heading
3. Constant airspeed
4. Constant altitude
5. Steady instrument readings

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2.4.10.4 Data Requirements

1. Stabilize AV for 30 seconds before recording
2. Record Data for 60 seconds
3. $V_i \pm 0.5$ KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system

2.4.10.5 Safety Considerations

The AVO should have control of the trailing source system to jettison the tube and probe if these are unstable beyond a point where recovery action can be taken. This requirement may result in software modification to the AV/GCS. Observers should always maintain a safe distance from the ground track of the aircraft to prevent exposure to danger. Lost link procedures and the turn rates of the UAV must be understood before testing; a tight turn could cause the tether to break or interfere with the control surfaces or propulsion system.

2.4.11 Advantages and Disadvantages for each Method

2.4.11.1 YAPS

The advantages of the YAPS boom are relatively inexpensive cost, highly accurate data for pitot and static calibration, and less restrictive tolerances for holding altitude, airspeed, and heading. Additionally, it gives both angle of attack and sideslip angle so that flight at various attitudes may be explored. The disadvantages are AV integration of the YAPS boom to include the SWAP requirements, potential mounting issues, and displacement of the aircraft center of gravity.

2.4.11.2 GPS Speed Course Method

The GPS speed course method is relatively inexpensive, may use existing hardware on the aircraft, and offers highly accurate data for the true airspeed determination. Negative contributions include stringent airspeed and heading tolerances and little data for determining altitude position error.

2.4.11.3 GPS 3-Leg Method

The GPS 3-Leg method has similar advantages and disadvantages to the GPS speed course method. The advantages are the relatively inexpensive cost, utilization of existing hardware, and highly accurate data for the true airspeed determination. The disadvantages include stringent airspeed and ground track tolerances and little data for determining altitude position error.

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2.4.11.4 Space Positioning

The space positioning technique is highly accurate, results in a large amount of useful airspeed data, and requires little AV hardware integration. However, it is quite expensive and complex to plan and execute, has rigid tolerances for airspeed and ground track, and gives less precise altitude position error.

2.4.11.5 Pace Aircraft

The pace aircraft method is an accurate way to determine the airspeed and altitude position error. It involves minimum system integration for the AV and is an existing capability of most test pilot schools and aviation test centers. The UAV test aircraft and pace aircraft must have similar flight performance capabilities and two pace aircraft crewmembers are typically required to maintain a high level of safety.

2.4.11.6 Tower Flyby

The tower flyby method is an accurate, low-cost means to attain the altitude position error without adding hardware to the UAV; however, this method will not provide data for the airspeed position error information and requires vigilance from the AVO and tower personnel to maintain situational awareness and safety of flight.

2.4.11.7 Trailing Source

The trailing source method, in a trailing cone or trailing bomb configuration, is a reliable means to attain accurate static pressure data. However, it can only be used for a mid-range of airspeeds; requires integration of the tether, pressure transducer, extension and retrieval device; could cause stability issues for the aircraft; can only be used for large UAVs (bomb); and necessitates careful observation for flight safety and software modification. Also, pitot-static testing is usually performed early in the aircraft life cycle and this method may not be best suited to development aircraft.

2.5 Data Reduction

The reduction of the acquired data from the flight test should follow the procedures of Section 2.5 of the USNTPS-FTM-No. 108. That manual should be referenced for the data reduction of each of the tests discussed here; the exception is the GPS 3-leg method, which has the data performance evaluation explained in Section 2.4.5.

2.6 Data Analysis

Section 2.6 of the USNTPS-FTM-No. 108 provides full detail of the analysis and evaluation of the test data. These steps should be followed to validate the altimeter position error and the airspeed position error and analyze qualitatively the pitot-static system performance from the quantitative test data.

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2.7 Mission Suitability

USNTPS-FTM-No. 108 Section 2.7 is applicable to UAVs of all sizes. This is particularly true when discussing the seriousness of the airspeed or position errors affecting automated systems, since flight control systems direct the AV. Additionally, the future UAVs are moving toward autonomous takeoff and landing systems, and pitot-static errors could cause severe mishaps that could endanger test personnel.

2.8 Specification Compliance

Section 2.8 of the USNTPS-FTM-No. 108 references military standards (MIL-I-5072-1 and MIL-I-6115A) as well as Federal Aviation Regulations (Parts 27.1323 and 27.1335). These establish the standard for instrument installation and system performance, in addition to tolerances and maneuvers (pull-up, pushover, yawing, and rough air). These documents should be referenced as the benchmarks for pitot-static system compliance.

2.9 Glossary

2.9.1 Notations

a	Speed of sound	kt
a	Temperature lapse rate	°/ft
ARDC	Arnold Research and Development Center	
a_{ssl}	Standard sea level speed of sound	661.483 kt
a_{ssl}	Standard sea level temperature lapse rate	.001981 °K/ft
C_L	Lift coefficient	
D	Course length	nmi
d	Horizontal distance (Tower to aircraft)	ft
Δh	Aircraft height above tower	ft
ΔH_{pic}	Altimeter instrument correction	ft
$\Delta H_{pic\ ref}$	Reference altimeter instrument correction	ft
ΔH_{pos}	Altimeter position error	ft
ΔM_{pos}	Mach position error	
ΔP	Static pressure error	psf
$\frac{\Delta P}{qc}$	Static pressure error coefficient	
$\frac{\Delta P}{qc_i}$	Indicated static pressure error coefficient	
Δt	Elapsed time	s
ΔT_{ic}	Temperature instrument correction	°C
ΔV_c	Compressibility correction	kt
ΔV_{ic}	Airspeed instrument correction	kt
ΔV_{pos}	Airspeed position error	kt

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e	Base of natural logarithm	
FAR	Federal Aviation Regulations	
g	Gravitational acceleration	ft/s2
gc	Conversion constant	32.17 lbm/slug
gssl	Standard sea level gravitational acceleration	32.174049 ft/s2
GW	Gross weight	lb
H	Geopotential	ft
h	Tapeline altitude	ft
HP	Pressure altitude	ft
HP ref	Reference pressure altitude	ft
HPc	Calibrated pressure altitude	ft
HPc twr	Tower calibrated pressure altitude	ft
HPi	Indicated pressure altitude	ft
HPi ref	Reference indicated pressure altitude	ft
HPo	Observed pressure altitude	ft
HPo ref	Reference observed pressure altitude	ft
IFR	Instrument flight rules	
KT	Temperature recovery factor	
La/c	Length of aircraft	ft
M	Mach number	
Mi	Indicated Mach number	
MTest	Test Mach number	
NACA	National Advisory Committee on Aeronautics	
nz	Normal acceleration	g
P	Pressure	psf
Pa	Ambient pressure	psf
Ps	Static pressure	psf
$P_{S_{YAPS}}$	Static pressure measured by the YAPS boom	psf
Pssl	Standard sea level pressure	2116.217 psf 29.9219 inHg
PT	Total pressure	psf
$P_{T_{YAPS}}$	Total pressure measured by the YAPS boom	psf
PT'	Total pressure at total pressure source	psf
q	Dynamic pressure	psf
qc	Impact pressure	psf
qci	Indicated impact pressure	psf
R	Engineering gas constant for air	96.93 ft-lbf/lbm-°K
Re	Reynold's number	
SAS	Stability augmentation system	
T	Temperature	°C or °K
Ta	Ambient temperature	°C or °K
Ta ref	Reference ambient temperature	°C or °K

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T_i	Indicated temperature	°C or °K
T_o	Observed temperature	°C
T_{ssl}	Standard sea level temperature	15°C or 288.15°K
T_{Std}	Standard temperature (At tower)	°K
T_T	Total temperature	°K
T_{Test}	Test temperature (At tower)	°K
USNTPS	U.S. Naval Test Pilot School	
V	Velocity	kt
V_c	Calibrated airspeed	kt
V_{cStd}	Standard calibrated airspeed	kt
V_{cTest}	Test calibrated airspeed	kt
V_{cW}	Calibrated airspeed corrected to standard weight	kt
V_e	Equivalent airspeed	kt
V_{eStd}	Standard equivalent airspeed	kt
V_{eTest}	Test equivalent airspeed	kt
V_G	Ground speed	kt
V_i	Indicated airspeed	kt
$V_{i_{YAPS}}$	Indicated airspeed measured by the YAPS boom	kt
V_{iTest}	Test indicated airspeed	kt
V_{iW}	Indicated airspeed corrected to standard weight	kt
V_{ne}	Never Exceed Airspeed	kt
V_o	Observed airspeed	kt
$V_{o\ ref}$	Reference observed airspeed	kt
V_T	True airspeed	kt
V_w	Wind velocity	kt
W	Weight	lb
W_{Std}	Standard weight	lb
W_{Test}	Test weight	lb
x	Scaled length of aircraft	
y	Scaled height of aircraft above tower	

2.9.2 Greek Symbols

α (alpha)	Angle of attack	deg
α_{YAPS}	Angle of attack measured by the YAPS boom	deg
β (beta)	Sideslip angle	deg
β_{YAPS}	Sideslip angle measured by the YAPS boom	deg
δ (delta)	Pressure ratio	
γ (gamma)	Ratio of specific heats	
λ (lambda)	Lag error constant	
λ_s	Static pressure lag error constant	
λ_T	Total pressure lag error constant	
θ (theta)	Angle, Temperature ratio	deg

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ρ (rho)	Air density	slug/ ft3
ρ_a	Ambient air density	slug/ ft3
ρ_{ssl}	Standard sea level air density	0.0023769 slug/ ft3
σ (sigma)	Density ratio	

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CHAPTER 3

STALL SPEED DETERMINATION

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3. Stall Speed Determination

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CHAPTER 4

LEVEL FLIGHT PERFORMANCE

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4. *Level Flight Performance Testing*

4.1 Introduction

This chapter will focus on the performance of the air vehicle (AV) in unaccelerated, level flight. This includes the maximum range, maximum endurance, and full throttle airspeed. Also important is the airspeeds for maximum range and endurance so that the air vehicle operator (AVO) can tailor the flight to the specific mission requirements. The first section of the chapter deals with fixed pitch propeller theory, which is the powerplant configuration used by many unmanned aerial vehicles (UAVs). Specific fuel consumption, maximum range, maximum endurance, the airspeeds for maximum range and endurance, and the full throttle airspeed should be determined theoretically before flight testing so data analysts will have a benchmark for these parameters. Additionally, the flight control system of the UAV may limit the types of maneuvers that can be performed. For example, the maximum throttle setting allowed is determined from the calculations of the flight control system based on data from its sensor package. Refer to Chapter 1 for a more in-depth discussion of the differences between UAVs and manned aircraft.

There are particular challenges to be dealt with when performing level flight testing on a battery-electric powered vehicle. These stem from the nature of the battery discharge under the motor load. The net result is that the output power of the propulsion system varies significantly, particularly during high power operations. It is akin to testing a conventional air vehicle with a degrading engine, and thus requires some special provisions. Some detailed information on the challenges and options for conducting performance flight test with electric vehicles is outlined in Section 4.4.4.

If a variable pitch propeller is employed on a UAV, there is a separate control system for varying the pitch angle and propeller speed. Also, the propeller efficiency will incorporate the variance in propeller pitch for both the theory and data reduction. Additionally, different wood propellers of the same design have shown variability in load handling performance. The testing should have periodic checks of a couple of different propellers for the same conditions as the initial propeller. This should give the load variability that reflects the operational environment. During testing, all timing devices should remain synchronized to the standard, preferably GPS, with a deviation less than 1 second over any individual test day.

4.2 Purpose of the Test

The intent of testing the level flight performance of the UAV is to determine maximum range (and associated airspeed), maximum endurance (associated airspeed), and full throttle airspeed. These parameters contribute to the overall flight envelope of the AV which will be defined over the course of the flight testing. The flight test airspeeds for maximum range and endurance can be compared to the theoretical values (refer to equations 4.13 for max range and 4.22 for max endurance). The theoretical

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values for maximum range and endurance are the maximum attainable and will probably be greater than the flight test's performance model range and endurance. Since the true airspeed is in the numerator in the range equation, 4.11, but in the denominator in the endurance equation, 4.21, the test airspeeds for maximum range and endurance will converge if the flight performance model range and endurance are less than the theoretical.

4.3 Fixed Pitch Propeller Theory

4.3.1 Specific Fuel Consumption Determination

In order to determine the maximum range and endurance for a UAV, the specific fuel consumption (SFC) must first be determined. Additionally, the fuel flow, fuel density, true airspeed, engine horsepower, and the propeller efficiency must be known or calculated. This propeller efficiency (η_p) can be found in chart form in Appendix I of this document and in spreadsheet form in Appendix II of this document using the advance ratio.

$$J = \frac{V_T}{nD_p} \tag{4.1}$$

the Power Coefficient,

$$C_P = \frac{550(bhp)}{\rho n^3 D_p^5} \tag{4.2a}$$

and the Activity Factor

$$AF = \frac{10^5 c_{prop,root}}{16 D_p}[0.25 - (1-\lambda_p)0.2] \tag{4.2b}$$

The engine horsepower will change with altitude and a horsepower vs. altitude chart should be provided by the engine manufacturer. If such a chart is not available, the propulsion system should be tested in an altitude chamber to provide the test team with an accurate determination of horsepower versus altitude. True airspeed can be determined from meteorological data and the total pressure given by the pitot-static system. The definition for true airspeed is given below.

$$V_T = \sqrt{\frac{2\gamma}{\gamma-1}\frac{P_a}{\rho_a}\left[\left(\frac{P_T - P_a}{P_a}+1\right)^{\frac{\gamma-1}{\gamma}} - 1\right]} \tag{4.3}$$

Once η_p is determined, the propeller efficiency corrected for tip Mach number can be found using the following equations:

$$a = \sqrt{\gamma RT} \tag{4.4}$$

$$M_{tip} = \frac{\sqrt{V_{tip}{}^2 + (n\pi D)^2}}{a} \tag{4.5}$$

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$$\eta_{p,corrected} = \eta_p - (M_{tip} - 0.89)\left(\frac{0.16}{0.48 - 3\frac{t}{c}}\right);\ M_{tip} > 0.89 \tag{4.6}$$

The derivation of SFC for propeller aircraft is as follows:

First, find the Propeller Specific Fuel Consumption (C_{bhp})

$$\frac{gal}{hr} \times \frac{lb}{gal} \times \frac{1}{bhp_{Engine}} = C_{bhp}\left(\frac{lb}{\frac{hr}{bhp}}\right) \tag{4.7}$$

Then,

$$SFC = \frac{C_{bhp} V_T}{550\eta_{p,corrected}} \tag{4.8}$$

4.3.2 Range Determination

Now that SFC is known, the lift-to-drag ratio (L/D) of the UAV must be determined. If the manufacturer has provided a chart of lift coefficient versus drag coefficient (C_L vs. C_D) for the aircraft, L/D can be easily found. If the aircraft is in level flight at a constant velocity, lift (L) is equal to the weight of the aircraft, assuming negligible horizontal stabilizer down forces as should be the case for a well designed aircraft in steady, level flight. This value can then be used to find C_L using equation 4.9.

$$L = 0.5\rho V_T^2 S C_L \tag{4.9}$$

C_D can be found using the manufacturer's provided chart. L/D is then the same value as C_L/C_D. Another way to find L/D in unaccelerated, level flight is to use the weight of the aircraft for L and set the thrust produced equal to the drag (D). (If the aircraft is at a constant velocity, all thrust being produced is being used to counteract drag, and is therefore the same value as the drag on the aircraft.) Thrust in forward flight can be found using the following equation.

$$T = \frac{550 * bhp * \eta_{p,corrected}}{V_T} \tag{4.10}$$

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It is important to note that the equations, listed below, for maximum range and endurance assume that *L/D*, V_T, and C_{bhp} are held constant. As the weight decreases from fuel burn, the lift would exceed the weight and the aircraft would climb to an altitude where the lift would decrease to equal the weight. This results in a cruise-climb, which will maximize the range and endurance of the aircraft. Since steady, level flight requires altitude (ρ) and the true airspeed to be held constant, the decreasing lift requires that the drag and therefore power must decrease as well. The power decrease necessitates the throttle be pulled back periodically over the mission to attain the maximum range (or endurance). This is true for both manned and unmanned aircraft; but in a UAV, the airspeed command controls the throttle position via the flight control system. The entire mission can be broken into smaller segments and calculated using the corresponding V_T, *L/D*, and C_{bhp} for each of the smaller segments. Refer to Figure 4.1 for an illustration of how the throttle position will need to change to burn the least amount of fuel for the mission. This method will produce a reasonable approximation depending on the amount of smaller segments the entire mission is broken into. The more pieces the entire mission segment is broken into, the greater the accuracy of the approximation.

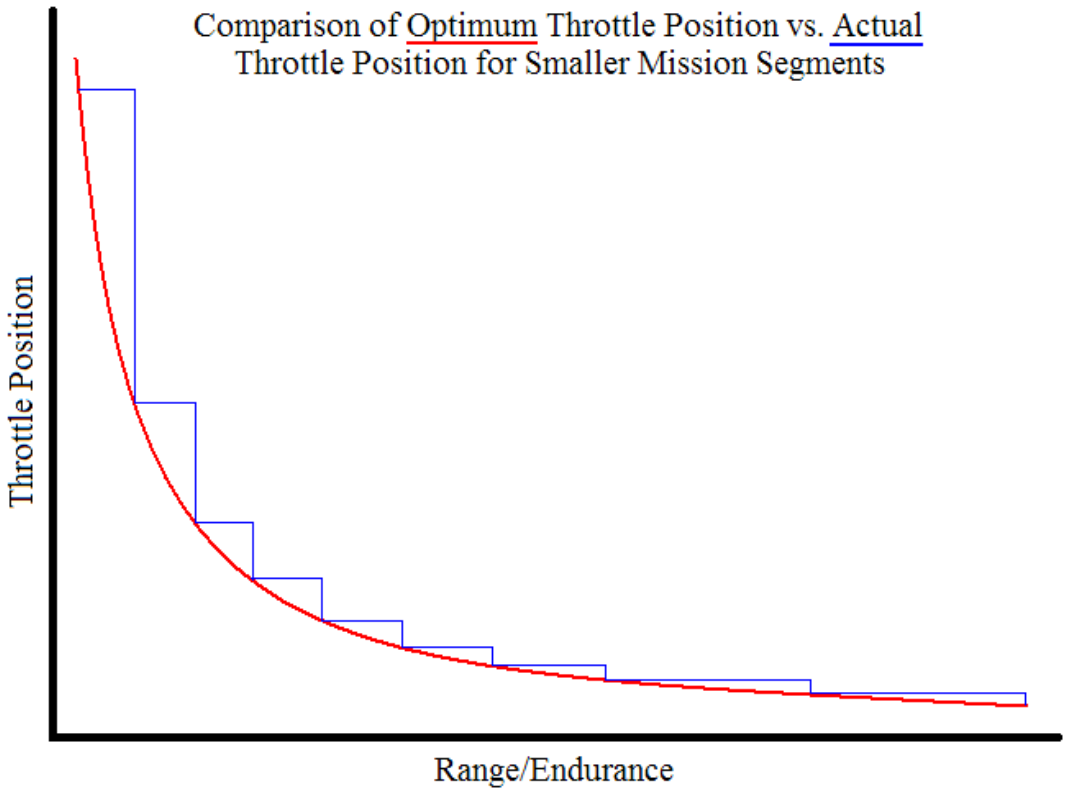


Figure 4.1 Illustration of Throttle Position Change with Time

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The following equation is true for determining the range of an aircraft powered by any type of engine.

$$R = \frac{V_T}{SFC}\frac{L}{D}\ln\left(\frac{W_i}{W_f}\right) \tag{4.11}$$

For propeller aircraft, the specific parameters are as follows:

$$R = \frac{550\eta_{p,corrected} * 3600}{C_{bhp}}\frac{L}{D}\ln\left(\frac{W_i}{W_f}\right) \tag{4.12}$$

In order to maximize the range of a propeller aircraft, the aircraft should fly at the velocity and lift coefficient for minimum drag. These can be found using the following equations:

$$V_{MinDrag} = \sqrt{\frac{2W}{\rho S}\sqrt{\frac{K}{C_{D_0}}}} \tag{4.13}$$

$$K = \frac{1}{e\pi AR} \tag{4.14}$$

Good estimations of Oswald efficiency factor (e) can be found using one of the following equations:

Straight Wing Aircraft:

$$e = 1.78(1 - 0.045AR^{0.68}) - 0.64 \tag{4.15}$$

Swept Wing Aircraft ($\Lambda_{LE} > 30$ deg):

$$e = 4.61(1 - 0.045AR^{0.68})(\cos\Lambda_{LE})^{0.15} - 3.1 \tag{4.16}$$

The following equations will determine the Lift and Drag at minimum drag:

$$C_{L_{MinDrag}} = \sqrt{\frac{C_{D_0}}{K}} \tag{4.17}$$

$$L_{MinDrag} = \frac{1}{2}\rho V_{MinDrag}^2 S C_{L_{MinDrag}} \tag{4.18}$$

$$D_{MinDrag} = \frac{1}{2}\rho V_{MinDrag}^2 S(2C_{D_0}) \tag{4.19}$$

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4.3.3 Endurance Determination

The following equation is true for determining the range of an aircraft powered by any type of engine.

$$E = \frac{L}{D}\frac{1}{SFC}\ln\left(\frac{W_i}{W_f}\right) \tag{4.20}$$

For propeller aircraft, the specific parameters are as follows:

$$E = \frac{L}{D}\frac{550\eta_{p,corrected}}{C_{bhp}V_T}\ln\left(\frac{W_i}{W_f}\right) \tag{4.21}$$

The true airspeed for maximum endurance can be found using the following equations, stated in Section 4.3.2:

$$K = \frac{1}{e\pi AR} \tag{4.14}$$

Again, e can be estimated using one of the following equations:

Straight Wing Aircraft:

$$e = 1.78(1 - 0.045AR^{0.68}) - 0.64 \tag{4.15}$$

Swept Wing Aircraft ($\Lambda_{LE} > 30$ deg):

$$e = 4.61(1 - 0.045AR^{0.68})(\cos\Lambda_{LE})^{0.15} - 3.1 \tag{4.16}$$

$$V_T = \sqrt{\frac{2W}{\rho S}\sqrt{\frac{K}{3C_{D_0}}}} \tag{4.22}$$

4.4 Test Methods

UAVs in the past have typically been tested for an entire range of airspeeds at a particular altitude without burning a significant amount of fuel. The constant weight to pressure ratio method has not typically been necessary since the fuel weight does not change significantly between test points.

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4.4.1 Piston Propeller

An onboard data logger (used to prevent any corruptions to the data from datalink latency) should capture the fuel flow, the target indicated airspeed, the actual indicate airspeed, indicated altitude, and engine horsepower. This can be used in conjunction with the meteorological data from weather balloons as well as the preflight and postflight weight to determine the necessary range and endurance information, which can then be compared to the theoretical values for the UAV. Refer to Figure 4.2 below for an illustration of a data card at a specific altitude.

Altitude	XX feet MSL		
Target Indicated Airspeed	Actual Indicated Airspeed	Fuel Flow	Engine Horsepower
Vmin	XX	XX	XX
Vmin + ΔV	XX	XX	XX
Vmin + 2*ΔV	XX	XX	XX
Vmin + 3*ΔV	XX	XX	XX
Vmin + 4*ΔV	XX	XX	XX
Vmin + 5*ΔV	XX	XX	XX
Vmin + 6*ΔV	XX	XX	XX
Vmin + 7*ΔV	XX	XX	XX
Vmin + 8*ΔV	XX	XX	XX
Vmax	XX	XX	XX

Figure 4.2 Illustration of Data Card for Particular Altitude

4.4.1.1 Maximum Airspeed Determination

4.4.1.1.1 Overview

Determining the maximum airspeed of a piston propeller aircraft will set the upper bound for the range and endurance determination. The AVO should place the throttle position sensor in its most open position, allowing for the greatest fuel flow possible. The AV should be stabilized in this position before the test run begins. The tests should be performed at five different altitudes, originating at the minimum safe operational altitude and increasing by equal increments up to the surface ceiling. All possible configurations should be tested although modifications that affect only one configuration may allow for a reduced set of test points.

4.4.1.1.2 Data Required

V_i, H_P, Fuel Flow, GW, Configuration, Meteorological Data, Total Fuel Consumed in Flight.

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4.4.1.1.3 Test Criteria

1. Stable flight for 5 minutes
2. Wings level
3. Full throttle
4. Winds less than 20% of stall airspeed

4.4.1.1.4 Data Requirements

1. Stabilize AV 15 seconds before start of run
2. Record data for at least 5 minutes
3. Throttle position sensor (TPS) set at maximum (limits of the AV)
4. Altitude $\pm$10 ft or limits of the AV flight control system

4.4.1.1.5 Safety Considerations

There are no special safety considerations other than those required for standard flight operations of UAVs. The manufacturer should be consulted for the minimum safe operational altitude or minimum parachute opening altitude, if equipped. Refer to Chapter 1 for a discussion of the standard operating procedures for UAV.

4.4.1.2 Range and Endurance Determination

4.4.1.2.1 Overview

For the range and endurance flight testing, the AV will perform runs of at least 5 minutes in length at constant airspeed and constant altitude. These will be repeated for the envelope of airspeeds. At least 10 different airspeeds should be used for each of the altitudes from the maximum airspeed test, although more airspeed data points improve reliability of the resulting performance curves. The experimental stall speed (refer to USNTPS FTM 108 Chapter 3) or the theoretical stall speed with the margin of safety (usually 10% above) will set the lower limit for airspeed, and the dash speed (refer to Section 4.3.1.1) will set the upper bound on the airspeed envelope. The difference between the maximum and minimum airspeeds should be evenly divided for a minimum of eight airspeeds in addition to the minimum and maximum. If the test AV has an airspeed envelope which causes the increments to be less than 1 knot indicated airspeed (KIAS), then a 1 KIAS increment should be used over the span of airspeeds. Through fuel mapping, data for the range and endurance for any airspeed can be determined. Flights should also be performed at all configurations necessary for the test program as well as all gross weights. This will allow the AVO or mission planner to easily determine the range and endurance for any combination of altitude, airspeed, gross weight, and configuration of the UAV. All timing devices should be synced to a standard, preferably time from a GPS satellite; the data recorded onboard the AV can be easily identified with the 5 minute runs called by the test conductor.

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4.4.1.2.2 Data Required

V_i, H_P, Fuel Flow, GW, Configuration, Meteorological Data, Total Fuel Consumed in Flight.

4.4.1.2.3 Test Criteria

1. Stable flight for 5 minutes
2. Wings level
3. 10 airspeeds at each altitude
4. Winds less than 20% stall airspeed

4.4.1.2.4 Data Requirements

1. Stabilize AV 15 seconds before start of run
2. Record data for at least 5 minutes
3. $V_i \pm 1$ KIAS or limits of the AV
4. Altitude ± 50 ft or limits of the AV

4.4.1.2.5 Safety Considerations

There are no special safety considerations other than those required for standard flight operations of UAVs. The lost link procedures should be well known by all test team members to prevent a mishap. The manufacturer should be consulted for the minimum safe operational altitude or minimum parachute opening altitude, if equipped. Refer to Chapter 1 for a discussion of the standard operating procedures for UAV.

4.4.2 Jet Aircraft

This section is a placeholder for future UAV capabilities.

4.4.3 Turboprop

This section is a placeholder for future UAV capabilities.

4.4.4 Battery Electric Air Vehicles

The primary complications involved in testing vehicles with electric propulsion come from two sources. These are the need for unique instrumentation and the means to test in the presence of varying battery output.

The lesser challenge is instrumentation. Instead of fuel flow, manifold pressure, and RPM instrumentation, the primary need is for measurement of the current and voltage level at two locations – leaving the battery and entering the motor. The challenge is compounded by two competing effects – typically electric power UAVs are extremely

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weight critical and the current values used in electric propulsion are frequently quite large (i.e., 30 to 70 ampere) in climb. Thus, the instrumentation design must provide a lightweight compact way to examine the electrical system without onerous burden on the airframe. There may also be a strong need for temperature data on the motor, speed control, and the battery itself. They are all prone to significant heating and heat-related performance offsets.

The larger challenge is the fact that all batteries drop output voltage during discharge. An example of the characteristic of a representative battery technology is shown in Figure 4.3. Some battery types experience more pronounced effects during different segments of their discharge curves. Propulsion motors are typically direct current machines, either brushed or brushless designs. They generally have fixed values of the rpm to voltage constant, and thus will run at different rpm and current draw and output power as the battery supply voltage drops. There is also a variation in the discharge behavior with the current being pulled from the battery. This will result in a climb performance that rolls off as the battery is discharged. Thus, performing repeated climb assessments, as in the process for sawtooth climb evaluation of best climb speed, will not result in the variation in climb performance solely as a function of airspeed. It will also contain a component of variation from the battery output. Also bear in mind that the smaller battery electric UAVs operate their props in a very high propeller tip Mach number and low Reynolds number regime where predictability is low and variability in efficiency with varying rpm can be quite large.

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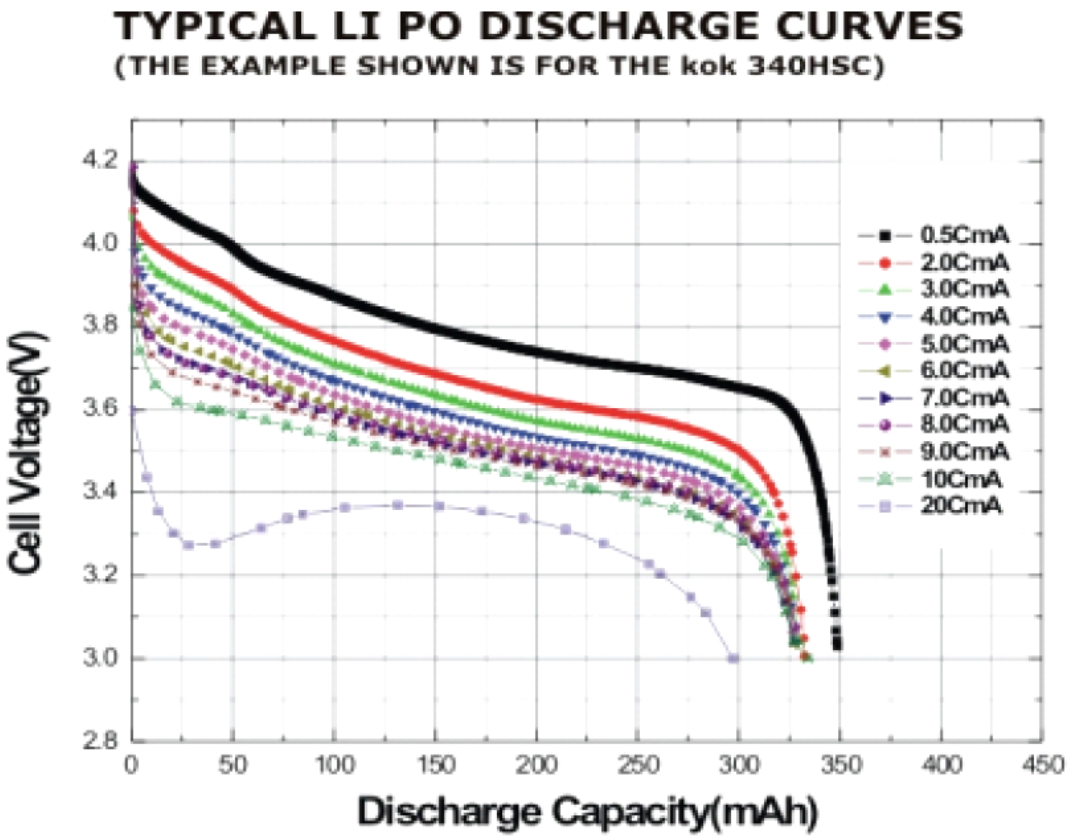


Figure 4.3 Battery Voltage in Discharge

For any of the testing options, gaining an understanding of the article under test will be critical. The tester must have a reasonable idea of the discharge curve behavior in terms of thrust output versus time at throttle settings in order to correctly plan the test mission. If the discharge behavior is not understood, then test points that should be at the electric equivalent of full throttle will be taken at much lower power levels and the results skewed. The option space for planning this type of testing includes:

a. Perform only one or a few test points per charge/flight to keep the battery in its high voltage segment of the discharge curve. This will provide the data on the absolute best performance of the air vehicle. As an example, to explore climb performance, repeated missions could be performed, each at different climb airspeeds. Repeating the maneuvers during each flight would gain insight into the climb variation with battery state and allow for development of a climb performance matrix of vertical speed as a function of airspeed and battery discharge state.

b. Perform the maneuvers on both sides of the “average” power point of discharge and standardize the results as being mid-point values.

c. Perform testing with a larger capacity battery less time sensitive, and adjust the results for the weight deltas. This may be made possible with smaller weight

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deltas by using a primary or nonrechargeable battery. Given the problems with adjusting for weight in the context of low Reynolds aerodynamics, this approach is not recommended if it requires significant weight growth to conduct.

d. Augment flight testing with full scale, flight vehicle wind tunnel testing, which can be powered by fixed power supplies, and then extrapolate to the power supply available in flight from the battery. This is made possible generally by the small scale of current battery UAVs.

e. Higher altitude test points may need to be taken by relocating the test site, as the extended climb to reach high altitudes may preclude getting a good examination of full system performance at those upper altitudes.

f. While these comments have been made concerning climb performance, they will also apply to the upper end of the airspeed envelope, and dash testing will have to be performed in consideration of the battery variation with time/discharge to effectively perform those trials.

4.5 Data Reduction

4.5.1 Piston Propeller

4.5.1.1 Dash Speed Determination

Since the ambient pressure and density will be known at each test altitude from the meteorological data and the total pressure will be known from the pitot-static system, true airspeed for full throttle can easily be determined from equation 4.3 in Section 4.3.1.

4.5.1.2 Range and Endurance Determination

Each 5 minute performance segment can be used to calculate the range and endurance of the AV for that particular altitude, configuration, and gross weight. The propeller specific fuel consumption can be calculated from the fuel flow. The true airspeed can be determined from the meteorological data and the pitot-static system (equation 4.3). The propeller tip velocity and rotational speed can be used to calculate a corrected propeller efficiency for tip Mach effects, if necessary. The fuel flow will also be used to determine the AV weight and thus the lift throughout the run. The drag can be backed out of the thrust produced from the powerplant, knowing the propeller efficiency, true airspeed, and engine brake horsepower. The gross take-off weight and AV empty weight are the remaining parameters necessary to determine the range and endurance for a particular airspeed, configuration, gross weight, and altitude from equations 4.12 and 4.21. Additionally, the effect of the change in AV weight due to fuel burn on the range and endurance calculations is negligible compared to the change in airspeed for every run, so the gross weight can be assumed reasonably constant for weight changes of up to 10%.

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4.5.2 Jet Aircraft

This section is a placeholder for future UAV capabilities.

4.5.3 Turboprop

This section is a placeholder for future UAV capabilities.

4.6 Data Analysis

The range and endurance will be calculated for the spectrum of possible airspeeds at each gross weight, configuration, and altitude. Both can be plotted against airspeed to determine the maximum value and associated airspeed for range and endurance at each altitude, gross weight, and configuration. These graphs can be used as reference performance curves for the AVO to operate the aircraft.

4.7 Mission Suitability

In addition to the items in USNTPS-FTM-No. 108 Section 4.7, consideration should be given to the primary role of UAVs, maximum endurance for reconnaissance, surveillance, and target acquisition.

4.8 Specification Compliance

The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAV system will provide qualifications for which the UAV was designed to meet. Expected performance of the UAV will also be rooted in the modeling and simulation of the UAV. The detailed technical manuals for the UAV system can then be written from the developmental testing as a reference source for all future system testing and military operations.

4.9 Glossary

4.9.1 Notations

a	Speed of Sound	ft/s
AR	Wing Aspect Ratio	
bhp	Engine Horsepower	bhp
C_{bhp}	Propeller Specific Fuel Consumption	lb/hr/bhp

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C_{D_0}	Drag Coefficient at $C_L = 0$	
$C_{L_{MinDrag}}$	Minimum Drag Lift Coefficient	
C_p	Power Coefficient	
$c_{prop,root}$	Propeller Root Chord	ft
D_p	Diameter of Propeller	ft
D	Drag	lb
$D_{MinDrag}$	Drag at Minimum Drag	lb
e	Oswald Efficiency Factor	
E	Endurance	hr
J	Advance Ratio	1/rev
L	Lift	lb
$L_{MinDrag}$	Lift at Minimum Drag	lb
M_{Tip}	Propeller Tip Mach Number	
n	Propeller Angular Velocity	rev/s
P_a	Ambient Pressure	psf
P_T	Total Pressure	psf
R	Gas Constant	1716 $\frac{ft \cdot lb}{slug \cdot {}^\circ R}$
R	Range	ft
S	Wing Plan Form Area	ft^2
SFC	Specific Fuel Consumption	1/hr
T	Temperature	Rankine
T	Thrust	lb
$\frac{t}{c}$	Propeller Thickness to Chord Ratio	
V_T	True Airspeed	ft/s
$V_{MinDrag}$	Minimum Drag Velocity	ft/s
W	AV Weight	lb
W_f	AV Weight Empty Weight	lb
W_i	AV Weight Gross Take-off Weight	lb

4.9.2 Greek Symbols

γ	Ratio of Specific Heats	1.4 for Air
λ_p	Propeller Taper Ratio	
ρ	Air Density	$slug / ft^3$
ρ_a	Ambient Air Density	$slug / ft^3$
η_p	Propeller Efficiency	
$\eta_{p,corrected}$	Corrected Propeller Efficiency	

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CHAPTER 5

EXCESS POWER

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5. ***Excess Power***

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CHAPTER 6

TURN PERFORMANCE

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6. Turn Performance

This section is not applicable to Unmanned Aerial Vehicles.

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CHAPTER 7

CLIMB PERFORMANCE

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7. Climb Performance Testing

7.1 Introduction

Performance of the air vehicle (AV) in unaccelerated, climbing flight is the focus of this chapter. Section 7.3 pertains to piston engine driven fixed pitch propeller theory for steady state climbing flight. The test procedures are outlined in Section 7.4 of the chapter. The primary reference for the test methods is Chapter 7 of the U.S. Naval Test Pilot School Flight Test Manual (USNTPS-FTM-No. 108). An understanding of the unmanned aerial vehicle (UAV) autopilot control logic is necessary for understanding which parameters the flight control system will alter in order to attain the desired output. Refer to Chapter 1 for a more in depth discussion of the differences between UAVs and manned aircraft.

Additionally, experience with UAVs has shown that most engines integrated into the AVs will develop cooling problems at some point during testing. To obtain a better understanding of the airflow and cooling properties for the engine, extensive instrumentation should be utilized in the engine intake, including mass flow sensors, cowl flap position sensors, and temperature gauges. This will allow for these problems, which will likely arise, to be addressed in developmental/operational testing rather than by the warfighter in the field.

If a variable pitch propeller is employed on a UAV, there is a separate control system for varying the pitch angle and propeller speed. Also, the propeller efficiency will incorporate the variance in propeller pitch for both the theory and data reduction. Additionally, different wood propellers of the same design have shown variability in load handling performance in the past. The testing should have periodic checks of a couple of different propellers for the same conditions as the first propeller. This should give the load variability that could be expected operationally. During testing, all timing devices should remain synchronized to the standard, preferably global positioning system (GPS), with a deviation less than 1 second over any individual test day.

7.2 Purpose of the Test

The intent of testing the climb performance of the UAV is to identify performance parameters such as maximum rate of climb and associated airspeed, the maximum climb angle (associated airspeed), the minimum time to climb to altitude (airspeed), and the airspeed that will result in the minimum amount of fuel burned that can be obtained. The parameters should be determined theoretically before flight testing to provide a benchmark for data analysis.

7.3 Fixed Pitch Propeller Theory

To find the theoretical steady-state, maximum rate of climb (ROC) for a fixed-pitch propeller, piston engine airplane, the following equations can be used:

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$$ROC_{Max} = \frac{550 \cdot bhp \cdot \eta_{p,corrected}}{W} - \left[\frac{2}{\rho_a}\sqrt{\frac{K}{3 \cdot C_{D_0}}}\left(\frac{W}{S}\right)\right]^{\frac{1}{2}} \frac{1.155}{\left(\frac{L}{D}\right)_{Max}} \tag{7.1}$$

(Where K is given by equation 4.14)
or,

$$ROC_{Max} = \frac{550 \cdot bhp \cdot \eta_{p,corrected}}{W} - \left(\frac{D \cdot V_{MinPower}}{W}\right) \tag{7.2}$$

(Where $V_{MinPower}$ is the velocity for best endurance from Chapter 4 given by equation 4.22)

The maximum climb angle is also easily found using the following equations, but it should be noted that the theoretical optimum velocity for best climb angle tend to be too low due to the separation drag at high angles of attack. The graphical method is the preferred method if good thrust and drag data are available at low speeds, but as an approximation, most propeller driven airplanes have an airspeed for best climb angle about 10-15% less than the airspeed for best rate of climb.

$$V_{\theta,Max} \approx \frac{4\left(\frac{W}{S}\right)K}{\rho_a\left(\frac{550 \cdot bhp}{W}\right)\eta_{p,corrected}} \tag{7.3}$$

(Where K is given by equation 4.14)

$$\theta_{Max} = \sin^{-1}\left(\frac{\eta_{p,corrected} \cdot 550 \cdot bhp}{W \cdot V_{\theta,Max}} - 0.5 \cdot \rho_a \cdot C_{D_0} \cdot V_{\theta,Max}^2\left(\frac{W}{S}\right)^{-1} - \left(\frac{W}{S}\right)\frac{2K}{\rho_a \cdot V_{\theta,Max}^2}\right) \tag{7.4}$$

or,

$$\theta_{Max} = \sin^{-1}\left[\frac{550 \cdot bhp \cdot \eta_{p,corrected}}{W \cdot V_{\theta,Max}} - \frac{D}{W}\right] \tag{7.5}$$

7.4 Test Methods

7.4.1 Sawtooth Climb Test

7.4.1.1 Overview

The sawtooth climb test method detailed in USNTPS-FTM-No. 108 Section 7.4.1 will be applicable for testing climb performance of UAVs of all sizes. A difference for some UAVs is that the autopilot may set the climbing airspeed automatically. If this is the case, the flight control logic may need to have supplementary code so that various airspeeds can be explored, provided the airworthiness authority can accommodate a flight

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release. This will allow airspeeds to be commanded from the ground control station (GCS) rather than from the climb schedules embedded in the autopilot. The altitude band should be 1,000 ft below the target altitude to 1,000 ft above for each airspeed test point. The time should be recorded for a minimum of 2 minutes, but longer recordings help attain better data for performance modeling, particularly the nonlinearity in climb performance near the surface ceiling. The time limit should be set at 5 minutes to give an upper bound for each test point. The altitude band should be centered about the target altitude if the time limit is used. If range constraints allow, the climbs should be performed perpendicular to the wind vector so that the wind will be a second order effect on the climb performance of the AV.

7.4.1.2 Data Required

1. Time: Record elapsed time from the beginning of the altitude band to the end or the time limit (2 minute minimum – 5 minute maximum), whichever comes first.
2. Altitude: Record the observed pressure altitude H_{P_o} band for each point.
3. Velocity: Record observed airspeed, V_o.
4. OAT or T_A: Record ambient air temperature from onboard instrumentation at target altitude (may be obtained from direct observation).
5. Fuel weight: Record the fuel remaining to determine aircraft gross weight.
6. Miscellaneous: Record other information desired such as rpm, angle of attack, or torque.

7.4.1.3 Test Criteria

The Sawtooth Climb Test Scenario should begin 2,000 ft below the target altitude so that it can reach steady state conditions for the engine, airspeed, and wind effects before entering the test altitude band. Smoothness is just as important as in acceleration runs and for the same reasons. If a small airspeed error is made while setting the airplane up, it is better to maintain the incorrect speed as accurately as possible, rather than try to correct it and risk aborting the entire run.

7.4.1.4 Data Requirements

The following are the data requirements from USNTPS-FTM-No. 108. If the time limit is used rather than the altitude band, the theoretical average rate of climb should be used to evenly split the climbing time below and above the target altitude.

1. Test altitude band ± 1,000 ft about a target altitude
2. V_o ± 1 kt with smooth corrections
3. Normal acceleration ± 0.1 g

7.4.1.5 Safety Considerations

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As stated in Section 7.3, the theoretical values for maximum climb angle airspeed may be less than the stall margin of the flight control system. In this case, the approximation of 10% to 15% less than the airspeed for maximum rate of climb should be used rather than altering the flight control logic as described in Section 7.4.1.1, negating any possibility of inadvertent stall. Supplementary code to the flight control logic, if used, should have a flight release issued by the airworthiness authority. For specific AVs, the extensive instrumentation for the engine may be deemed a safety of flight issue by the airworthiness authority. In this case, the instrumentation should be removed before flight test.

7.4.2 Check Climb Test

7.4.2.1 Overview

The check climb test method detailed in USNTPS-FTM-No. 108 Section 7.4.2 will be applicable for testing climb performance of UAVs of all sizes. The test should be conducted as described in USNTPS-FTM-No. 108 except that the data should be recorded with an onboard data logger. This will prevent any corruptions to the data from datalink latency. The predetermined climb schedule will be set by the autopilot. If a different airspeed is needed, supplemental code may be necessary for the flight control logic. If range constraints allow, the climbs should be performed perpendicular to the wind vector so that the wind will be a second order effect on the climb performance of the AV.

7.4.2.2 Data Required

Record the following data at each climb increment which should be as often as possible but no greater than each 5,000 ft or 2 minutes for handheld data:

1. Time (t)
2. Observed airspeed (V_o)
3. Observed pressure altitude (H_{Po})
4. Temperature (OAT or T_a)
5. Fuel remaining (W_f), or start and end fuel weight
6. Distance (d)
7. Miscellaneous as desired

7.4.2.3 Test Criteria

1. Maintain coordinated wings level flight.
2. Keep turns to a minimum and use less than 10 degrees bank to keep normal acceleration near 1.0.
3. No more than 30 degrees heading change.

7.4.2.4 Data Requirements

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1. Airspeed ± 5 kt or 0.01 M
2. Normal acceleration ± 0.1 g

7.4.2.5 Safety Considerations

As stated in Section 7.3, the theoretical values for maximum climb angle airspeed may be less than the stall margin of the flight control system. In this case, the approximation of 10% to 15% less than the airspeed for maximum rate of climb should be used rather than altering the flight control logic as described in Section 7.4.2, negating any possibility of inadvertent stall. Supplementary code to the flight control logic, if used, should have a flight release issued by the airworthiness. For specific AVs, the extensive instrumentation for the engine may be deemed a safety of flight issue by the airworthiness authority. In this case, the instrumentation should be removed before flight test.

7.5 Data Reduction

Section 7.5.1 gives an example of the data reduction for fixed pitch propeller UAV. The correction for standard day conditions is detailed in USNTPS-FTM-No. 108 Section 7.5

7.5.1 Data Reduction Sample Calculation for Fixed Pitch Propeller

7.5.1.1 Maximum Rate of Climb (ROC_{Max}) Calculation using Eq. 7.2

$$ROC_{Max} = \frac{550 \cdot bhp \cdot \eta_{p,corrected}}{W} - \left(\frac{D \cdot V_{MinPower}}{W} \right)$$

Known Values:

$bhp = 100$ hp	$AR = 20$
$W = 2250$ lb	$S = 162 \quad ft^2$
$C_{D_0} = 0.0186$	$n = 105$ rev/s
$\rho = 0.002043 \quad slug / ft^3$ (@ 5000 ft MSL)	
$D_P = 3$ ft	$a = 1097$ ft/s
$\theta = 6^\circ$	$\left(\frac{t}{c} \right)_{prop} = 0.14$

Calculations:

Find the Oswald efficiency factor (e) (Eq. 4.15),

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$$e = 1.78(1 - 0.045AR^{0.68}) - 0.64$$
$$e = 1.78(1 - 0.045(20)^{0.68}) - 0.64$$
$$\underline{e = 0.5258}$$

Use e to find K (Eq. 4.14),

$$K = \frac{1}{e\pi AR}$$
$$K = \frac{1}{(0.5258)\pi(20)}$$
$$\underline{K = 0.03027}$$

Find the minimum power required true airspeed (Eq. 4.22),

$$V_{MinPower} = \sqrt{\frac{2W}{\rho S}\sqrt{\frac{K}{3C_{D_0}}}}$$
$$V_{MinPower} = \sqrt{\frac{2(2250)}{(0.002043)(162)}\sqrt{\frac{0.03027}{3(0.0185)}}}$$
$$\underline{V_{MinPower} = 100 \text{ (ft/s)}}$$

Now that $V_{MinPower}$ is known, we use J (Eq. 4.1) and C_P (Eq. 4.2) to find the propeller efficiency.

$$J = \frac{V_{MinPower}}{nD_P} \qquad C_P = \frac{550 \cdot bhp}{\rho n^3 D^5}$$
$$J = \frac{100}{(105)(3)} \qquad C_P = \frac{550(100)}{(0.002043)(105)^3(3)^5}$$
$$\underline{J = 0.317}$$
$$\underline{C_P = 0.096}$$

Go to the J vs. C_P chart for the particular propeller in use to find η_p; in this case, we will assume $\underline{\eta_p = 0.795}$.

The effect on the tips of the propeller due to Mach number then needs to be addressed (Eq. 4.5, 4.6).

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$$M_{tip} = \frac{\sqrt{(V_{MinPower})^2 + (n\pi D)^2}}{a}$$

$$M_{tip} = \frac{\sqrt{(100)^2 + ((104)\pi(3))^2}}{1097}$$

$$\underline{M_{tip} = 0.907}$$

$$\eta_{p,corrected} = \eta_p - (M_{tip} - 0.89)\left(\frac{0.16}{0.48 - 3\frac{t}{c}\Big)_{prop}}\right)$$

$$\eta_{p,corrected} = (0.795) - (0.907 - 0.89)\left(\frac{0.16}{0.48 - 3(0.14)}\right)$$

$$\underline{\eta_{p,corrected} = 0.75}$$

Drag can be determined by assuming that thrust is equal to drag plus the weight component of the aircraft in the opposite direction of thrust (Eq. 4.10).

$$T = \frac{550 \cdot bhp \cdot \eta_{p,corrected}}{V_{MinPower}}$$

$$T = \frac{550(100)(0.75)}{100}$$

$$\underline{T = 412.5 \text{ (lb)}}$$

$$T = D + W\sin\theta$$

$$412.5 = D + (2250)\sin(6^\circ)$$

$$\underline{D = 177 \text{ (lb)}}$$

The values for D, W, $V_{MinPower}$, bhp, and $\eta_{p,corrected}$ can now be substituted into equation 7.2.

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$$ROC_{Max} = \frac{550 \cdot bhp \cdot \eta_{p,corrected}}{W} - \frac{D \cdot V_{MinPower}}{W}$$

$$ROC_{Max} = \frac{550(100)(0.75)}{2250} - \frac{(177)(100)}{2250}$$

$$\underline{ROC_{Max} = 10.5} \text{ (ft/s) (= 630 (ft/min))}$$

7.6 Data Analysis

The USNTPS-FTM-No. 108 Section 7.6 for data analysis applies for all sizes of UAVs.

7.7 Mission Suitability

In addition to the items listed in Section 7.7 of USNTPS-FTM-No 108, consideration should be given to the restriction of the airspace on the UAV. This might be the case in a test program or in tactical wartime environment where the AV might need to terminate a climb to remain in the designated area. Because UAVs are typically more of a gliding aircraft, their primary climbing airspeed is the one that gives the maximum rate of climb.

7.8 Specification Compliance

The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAV system will provide qualifications for which the UAV was designed to meet. Expected performance of the UAV will also be rooted in the modeling and simulation of the UAV. The detailed technical manuals for the UAV system can then be written from the developmental testing as a reference source for all future system testing and military operations.

7.9 Glossary

7.9.1 Notations

bhp	Engine Horsepower	bhp
C_{D_0}	Drag Coefficient at $C_L = 0$	
D	Drag	lb
L	Lift	lb

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ROC_{Max}	Maximum Rate of Climb	ft/s
S	Wing Plan Form Area	ft^2
$V_{MinPower}$	Velocity for Minimum Power	ft/s
$V_{\theta,Max}$	Velocity for Maximum Climb Angle	ft/s
W	AV Weight	lb

7.9.2 Greek Symbols

ρ_a	Ambient Air Density	$slug / ft^3$
η_p	Propeller Efficiency	
$\eta_{p,corrected}$	Corrected Propeller Efficiency	
θ_{Max}	Maximum Climb Angle	rad, deg

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CHAPTER 8

DESCENT PERFORMANCE

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8. Descent Performance Testing

8.1 Purpose of the Test

This chapter will focus on the performance of the air vehicle (AV) in steady, engine idle, descending flight. The theory will not change from U.S. Naval Test Pilot School Flight Test Manual (USNTPS-FTM-No. 108) for unmanned aerial vehicles (UAVs) since the descent performance is not dependent upon engine and propeller type. The test procedures are outlined in Section 8.4 of the chapter. The primary reference for the test methods is Chapter 8 of USNTPS-FTM-No. 108. An understanding of the UAV autopilot control logic is necessary for knowing which parameters the flight control system will alter to attain the desired output. Refer to Chapter 1 for a more in depth discussion of the differences between UAVs and manned aircraft.

As stated in Section 7.1, the engine intake duct should be instrumented to obtain a better understanding of the potential over-cooling problems associated with descent performance testing, primarily caused by large descent rates. The precautionary engine out approach test scenario could be difficult for UAVs since the engine might not restart if shutdown. The command for shutdown and restart are accomplished through the data link so an actual lost link situation with the engine shutdown is dangerous. Therefore, an intentional engine shutdown is not recommended. During testing, all timing devices should remain synchronized to the standard, preferably global positioning system (GPS), with a deviation less than 1 second over any individual test day.

8.2 Purpose of the Test

The purpose of this test is to determine the UAV's time/fuel required and distance to descend, minimum glide angle for maximum gliding range and the associated airspeed, the minimum sink rate (associated airspeed), as well as determination of parameters for precautionary and engine-out approach patterns. The minimum glide angle and minimum sink rate should be determined theoretically before flight testing to provide a benchmark for data analysis.

8.3 Theory

The USNTPS-FTM-No. 108 Section 8.3 for theory applies for all sizes of UAVs.

8.4 Test Methods

8.4.1 Sawtooth Descent

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8.4.1.1 Overview

The sawtooth descent test method fully detailed in USNTPS-FTM-No. 108 Section 8.4.1 will be applicable for UAVs of all sizes. A difference for some UAVs is that the autopilot may set the descent airspeed automatically. If this is the case, the flight control logic may require augmentation with supplementary code so that various airspeeds can be explored, provided the airworthiness authority issues a flight release for the changes. This will allow airspeeds to be commanded from the ground control station (GCS) rather than from the descent schedules embedded in the autopilot. The altitude band should be 1,000 ft above the target altitude to 1,000 ft below for each airspeed test point. The time limit should be set at 5 minutes to give an upper bound for each test point; longer recordings help attain better data for performance modeling. The altitude band should be centered about the target altitude if the time limit is used. If range constraints allow, the descents should be performed perpendicular to the wind vector so the wind will be a second order effect on the descent performance of the AV.

8.4.1.2 Data Required

1. Time: Record elapsed time from the beginning of the altitude band to the end or the 5 minute time limit.
2. Altitude: Record the observed pressure altitude (H_{Po}) band for each point.
3. Velocity: Record observed airspeed, V_o.
4. OAT or T_A: Record ambient air temperature from onboard instrumentation at target altitude (may be obtained from direct observation).
5. Fuel weight: Record the fuel remaining to determine aircraft gross weight.
6. Miscellaneous: Record other information desired such as rpm, angle of attack, or torque.

Additional data and information associated with the engine management criteria are important for the descent evaluation. For example:

1. Jet exhaust nozzle position at idle thrust and the minimum thrust setting to keep the nozzle closed.
2. Heating/air conditioning, pressurization, and anti-ice systems operation at low power.
3. Engine and other aircraft systems operation limits at low power settings.

8.4.1.3 Test Criteria

The Sawtooth Descent Test Scenario should begin above the test altitude band so steady state conditions for the engine, airspeed, and wind effects may be achieved. The AV should begin descending so that at least 30 seconds of stabilization will occur before test data is recorded. Smoothness is just as important as in acceleration runs and for the same reasons. If a small airspeed error is made while establishing the test conditions, it is better to maintain the incorrect speed as accurately as possible, rather than try to correct it and risk aborting the entire run.

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1. Balanced (ball centered) wings level, unaccelerated flight
2. Engine bleed air system OFF or operated in a normal flight mode
3. Stabilized engine thrust/power setting
4. Stabilize before the altitude band
5. Altimeter: 29.92, standby
6. Conduct test 90° to wind direction
7. Bank angle ≤ 10°
8. Heading change ≤ 30°
9. Normal acceleration ± 0.1 g

8.4.1.4 Data Requirements

The data requirements are the same as USNTPS-FTM-No. 108 unless the time limit is used rather than the altitude band. The theoretical average rate of descent would be assumed to evenly split the descent time above and below the target altitude.

1. Test altitude band 1,000 ft above and below the target altitude.
2. Vo ±1 knot with smooth corrections.

8.4.1.5 Safety Considerations

If the UAV powerplant uses heavy fuel (diesel, JP8, etc.), the cylinder head temperatures should be monitored carefully since overcooling could limit the ability of the engine to reacquire full power or greatly extend the time to do so. This is due to the temperature and the compression required to develop the power of the engine will be decreased dramatically from the air cooling. As stated in Section 7.4.1.5, for specific AVs, the extensive instrumentation for the engine may be deemed a safety of flight issue by the airworthiness authority. In this case, the instrumentation should be removed before flight test. Supplementary code to the flight control logic, if used, should have flight release issued by the airworthiness authority.

8.4.2 Check Descent

8.4.2.1 Overview

The check descent test method detailed in USNTPS-FTM-No. 108 Section 8.4.2 will be applicable for testing descent performance of UAVs of all sizes. The test should be conducted as described in USNTPS-FTM-No. 108 except that the data should be recorded with an onboard data logger. This will prevent any corruptions to the data from datalink latency. The predetermined descent schedule will be set by the autopilot. If a different airspeed is needed, supplemental code may be necessary for the flight control logic. If range constraints allow, the climbs should be performed perpendicular to the wind vector so that the wind will be a second order effect on the climb performance of the AV.

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8.4.2.2 Data Required

Record the following data with the onboard data logger as often as possible:

1. Time (t)
2. Observed airspeed (Vo)
3. Observed pressure altitude (HPo)
4. Temperature (OAT or Ta)
5. Fuel remaining (Wf) or start and end fuel weight
6. Distance (d)

8.4.2.3 Test Criteria

1. Balanced (ball centered) wings level, unaccelerated flight
2. Engine operated in a normal flight mode
3. Stabilized engine thrust/power setting
4. Stabilize before the altitude band
5. Altimeter: 29.92, standby
6. Conduct test 90° to wind direction
7. Bank angle £ 10°
8. Heading change £ 30°
9. Normal acceleration ± 0.1 g

8.4.2.4 Data Requirements

1. Airspeed ±5 kt or 0.01 M

8.4.2.5 Safety Considerations

If the UAV powerplant uses heavy fuel (diesel, JP8, etc.), the cylinder head temperatures should be monitored carefully since overcooling could limit the ability of the engine to reacquire full power or greatly extend the time to do so. This is due to the fact that the temperature and compression required to develop the power of the engine will be decreased dramatically from the air cooling. As stated in Section 7.4.1.5, for specific AVs, the extensive instrumentation for the engine may be deemed a safety of flight issue by the airworthiness authority. In this case, the instrumentation should be removed prior to flight test. Supplementary code to the flight control logic, if used, should have flight release issued by the airworthiness authority.

8.4.3 Precautionary/Flameout Approach

The precautionary engine out approach test could cause a serious mishap to occur if the engine, controlled by the GCS through the datalink, will not restart once shutdown due to a loss of link with the GCS. The engine should not be intentionally shutdown.

8.5 Data Reduction

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The USNTPS-FTM-No. 108 Section 8.5 for data reduction applies for all sizes of UAVs.

8.6 Data Analysis

The USNTPS-FTM-No. 108 Section 8.6 for data analysis applies for all sizes of UAVs.

8.7 Mission Suitability

In addition to the items listed in USNTPS-FTM-No. 108 Section 8.7, consideration should be given to the restriction of the airspace on the UAV. This might be the case in a test program or in tactical wartime environment where the AV might need to break off its descent in order to remain in the restricted area. As stated in Section 8.4.1.5 above, the rate of descent should never bring the heavy fuel engine cylinder head temperatures below the levels specified in the operator's handbook.

8.8 Specification Compliance

The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAV system will provide qualifications for which the UAV was designed to meet. Expected performance of the UAV will also be rooted in the modeling and simulation of the UAV. The detailed technical manuals for the UAV system can then be written from the developmental testing as a reference source for all future system testing and military operations.

8.9 Glossary

The Glossary in USNTPS-FTM-No. 108 will apply since the theory, data reduction, and analysis sections of USNTPS-FTM-No. 108 apply to UAVs of all sizes.

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CHAPTER 9

TAKEOFF AND LANDING PERFORMANCE

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9. *Takeoff and Landing Performance Testing*

9.1 Introduction

This chapter will focus on the performance of the air vehicle (AV) at the origin and termination of the flight. This includes the total takeoff distance (ground roll and climbing to clear a 50-ft obstacle), the rotation airspeed, the lift-off airspeed, the approach airspeed, the touchdown airspeed, and the total landing distance (approach and ground roll). These parameters will define the operational footprint of the AV required for takeoff and landing. All parameters should be determined theoretically before flight testing so the data analysts will have a benchmark for these parameters. Throughout the flight tests, data should be recorded for all takeoffs and landings. This may provide sufficient statistical basis for evaluating takeoff and landing performance; however, additional sorties that are devoted completely to takeoff and landing may be needed to fully characterize the takeoff and landing performance. The U.S. Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108) Chapter 9 should be used as the basis for developing a test plan unless a particular theory or test method from this section is more applicable to UAV specific testing.

Section 9.3 of the chapter deals with fixed pitch propeller theory, which is the powerplant configuration used by many unmanned aerial vehicles (UAVs). This chapter is divided into conventional takeoff and landing performance and augmented takeoff and landing performance. Conventional takeoffs are those whereby the AV uses its own powerplant for attaining the rotation and lift-off airspeeds; conventional landings are those in which the AV touches down and applies brakes to come to a complete stop. Augmented takeoffs or landings are those in which the AV requires a system outside of the normal flight operations to supplement launch (takeoff) and/or recovery (landing).

The stall margins used for calculating takeoff and landing performance vary for different UAV systems. Some UAVs use angle of attack to maintain safe stall margins while others use airspeed. In some cases, the convention of using 10% stall airspeed margin for liftoff or touchdown and 20% stall airspeed margin for climb or approach is not applicable to UAVs. If a UAV uses angle of attack measurements to maintain safe operational limits, the stall margin could be on the order of 5%. If a particular UAV has dangerous stall characteristics, the stall margins might rise to 20% for liftoff or touchdown and 30% for climb and approach to insure operations well within the flight envelope.

For UAVs with autonomous launch and recovery capability, the control laws should be thoroughly understood before testing. Some autonomous systems restrict operations for a particular mode of flight. For takeoff and landing, the most aggressive takeoff (landing) or the most benign may not be allowed by the autopilot. In this case, supplementary code is needed so that the full envelope of takeoff and landing performance can be tested. Additionally, simulation and flight testing need to be partnered together since the software is a major component of the UAV system.

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Hardware in the loop or software in the loop testing should occur in a lab environment to build confidence in the system prior to flight test. Hardware and software failure mode analysis should be obtained from the manufacturer.

In the future, the technology may exist that allows the air vehicle operator (AVO) to simply set the start point and the end point of the mission with the UAV autonomously planning the best route to accomplish the mission as well as autonomously operating the payload. This would transition the role of the AVO to the mission commander rather then the pilot. If that technology is developed in the future, testing must focus even more on the software of the flight control system.

9.2 Purpose of the Test

The intent of testing the takeoff and landing performance of the UAV is so that important performance parameters can be calculated. These include total takeoff distance, rotational airspeed, liftoff airspeed, approach airspeed, touchdown airspeed, landing ground roll distance, and maximum airspeed at which the brakes may be applied. These parameters contribute to the overall flight performance of the AV and should be determined theoretically for comparison to the flight test data.

9.3 Takeoff and Landing Theory (Fixed Pitch Propeller)

9.3.1 Conventional Takeoff and Landing

Conventional takeoff and landing requires the AV to use only its own power for takeoff and involves a ground roll, transition/flare, and climb/approach. The primary interest in conventional takeoff and landing performance is the total takeoff and landing distance as well as the airspeeds at each stage of the takeoff and landing. These distances can be found by summing the ground roll and rotation distance, transition period/flare, and climb/descent.

A first order effect on the conventional landing performance is the scatter of the touchdown point from the expected point. This will have a dramatic impact on determining the total landing distance. Since the primary impact of testing for the landing distance is to determine the safe operational limits of the UAV, a sufficient number of test data points should be taken to ascertain the statistical nature of the scatter and potential causes of the scatter. Table 9.1 demonstrates the approach distances for varying uphill or downhill slopes for a Shadow B model UAV. Also important are the design decisions imbedded in the UAV flight control system for the time between rotation and liftoff on takeoff as well as the time between touchdown and brakes applied (free roll) on the landing.

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	Level Runway (0 degree slope)	For runways sloping upward from approach edge toward TDP (values shown in degrees)									
	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Minimum Runway Approach Distance to TDP (feet)	100	96	93	89	86	83	80	78	75	73	71
		For runways sloping downward from approach edge toward TDP (values shown in degrees)									
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Minimum Runway Approach Distance to TDP (feet)		105	110	115	121	127	134	142	151	161	172

Table 9.1 Approach Distance as a Function of Runway Slope

The equations in Sections 9.3.1.1 and 9.3.1.2 include assumptions such as the rotation time and free roll time. The rotation time can be approximated by 3 seconds for a large aircraft to 1 second for a small aircraft; however, the flight control system, in an automated takeoff role, may dictate the time from rotation to liftoff. The equations are still in general form and have the flexibility to apply to any UAV with a fixed pitch propeller powerplant. When values are given for n or V_{TO}, these are provided for approximations; data for a particular UAV should be verified. It should also be noted that these equations calculate minimum takeoff and landing distances. For nonstandard takeoff and landing operations (different flap deflections, high drag devices, lift reducing flap settings, etc.), the equations given may need to be revised or new coefficients developed to account for the new variables being introduced.

9.3.1.1 Takeoff Performance

The total takeoff distance approximation is made by summing all segments (ground roll, rotation, transition, and climb) given by the calculations below. The stall velocity that should be used is for max gross weight with the AV in takeoff configuration.

$$\text{Total Takeoff Distance} = S_{1_G} + S_{1_R} + S_{2_T} + S_{2_C} \tag{9.1}$$

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9.3.1.1.1 Ground Roll Approximation

The following equation will find the ground roll distance for any initial velocity to any final velocity:

$$S_{1_G} = \left(\frac{1}{2gK_{A,TO}}\right)\ln\left(\frac{K_{T,TO} + K_{A,TO} * V_f^2}{K_{T,TO} + K_{A,TO} * V_i^2}\right) \tag{9.2}$$

For takeoff, the initial velocity will be 0, and the final velocity will be V_{TO}; usually V_{TO} is $1.1V_{Stall}$. This will change the preceding equation to:

$$S_{1_G} = \left(\frac{1}{2gK_{A,TO}}\right)\ln\left(\frac{K_{T,TO} + K_{A,TO} * V_{TO}^2}{K_{T,TO}}\right) \tag{9.3}$$

The coefficients $K_{A,TO}$ and $K_{T,TO}$ can be found using the following equations. The coefficients K and μ are given by equation 4.14 and USNTPS-FTM No. 108, Table 9.2, respectively.

$$K_{A,TO} = \frac{\rho}{2(W/S)}(\mu C_L - C_{D_o} - KC_L^2) \tag{9.4}$$

$$K_{T,TO} = \left(\frac{550 \cdot bhp \cdot \eta_p}{\left(\frac{1}{\sqrt{2}}(V_f - V_i) + V_i\right)W}\right) - \mu \tag{9.5}$$

The preceding equation will give K_T for any velocity rise from an initial (V_i) to a final velocity (V_f). The approximation for ground roll can be made more accurate by breaking it into smaller segments of velocity rise and using the corresponding K_T for each of the smaller segments. If the entire ground roll is being taken as one segment, and the UAV is accelerating from 0 to V_{TO}, the following equation can be used:

$$K_{T,TO} = \left(\frac{550 \cdot bhp \cdot \eta_p}{\frac{1}{\sqrt{2}}V_{TO}W}\right) \tag{9.6}$$

As stated in Section 9.3.1, the rotation time can be approximated by 3 seconds for a large aircraft to 1 second for a small aircraft; however, the flight control system, in an

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automated takeoff role, may dictate the time from rotation to liftoff. The rotation distance (S_{1_R}), then, can be approximated by this time multiplied by the takeoff velocity.

9.3.1.1.2 Transition

During the transition, the aircraft will accelerate from $1.1V_{Stall}$ to $1.2V_{Stall}$. The average velocity during this acceleration (V_{TR}) can then be assumed to be $1.15V_{Stall}$. The average load factor (n) can be assumed to be 1.2. However, a different load factor can be used if different for a particular UAV. This allows calculation of the transition radius using the following equation:

$$R_{TR} = \frac{V_{TR}^2}{g(n-1)} \tag{9.7}$$

The climb angle, the angle at which the transition stops, can be found using the following equation:

$$\sin(\gamma_{climb}) = \frac{550 \cdot bhp \cdot \eta_p}{V_{TR} W} - \frac{D}{W} \tag{9.8}$$

If drag is not known, the following equation will offer a good approximation of the result of equation 9.7:

$$\sin(\gamma_{climb}) \cong \frac{550 \cdot bhp \cdot \eta_p}{V_{TR} W} - \frac{1}{(L/D)} \tag{9.9}$$

The distance traveled during the transition and height attained at the end of transition can then be found using the following equations:

$$S_{2_T} = R_{TR} \sin(\gamma_{climb}) \tag{9.10}$$

$$h_{TR} = R_{TR}(1 - \cos(\gamma_{climb})) \tag{9.11}$$

If equation 9.10 yields a number greater that 50 ft or the chosen obstacle height, then the following equation can be used to determine transition distance:

$$S_{2_T} = \sqrt{R_{TR}^2 - (R_{TR} - h_{TR})^2} \tag{9.12}$$

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9.3.1.1.3 Climb

The equation to find the horizontal distance traveled during the climb to clear an obstacle, usually 50 ft, is given below:

$$S_{2_2} = \frac{h_{obstacle} - h_{TR}}{\tan(\gamma_{climb})} \tag{9.13}$$

9.3.1.2 Landing Performance

The total landing distance is found by summing the landing distance segments (approach, flare, free roll, ground roll with brakes applied) given by the calculations below. The stall velocity used in the calculations should be when the AV is in the landing configuration at max gross weight.

$$\text{Total Landing Distance} = S_{3_A} + S_{3_f} + S_{4_r} + S_{4_B} \tag{9.14}$$

9.3.1.2.1 Approach

The approach starts with clearance over an obstacle usually 50ft. A good approximation of approach speed is $1.2V_{Stall}$. The steepest approach angle can be calculated using the following equation with idle thrust and drag with flaps deflected:

$$\sin(\gamma_{approach}) = \frac{T - D}{W} \cong \frac{T}{W} - \frac{1}{(L/D)} \tag{9.15}$$

The approach distance is then found using the flare height (h_f) given by equation 9.16.

$$h_f = R_f(1 - \cos(\gamma_{approach})) \tag{9.16}$$

$$S_{3_A} = \frac{h_{obstacle} - h_f}{\tan(\gamma_{approach})} \tag{9.17}$$

9.3.1.2.2 Flare

Some UAVs do not flare before touchdown so this component of landing distance will be zero. If the approach velocity is $1.2V_{Stall}$ and the touchdown velocity is $1.1V_{Stall}$, then the average flare velocity (V_f) is $1.15V_{Stall}$. The flare radius, flare height, and horizontal distance covered during flare can then be found using the following equations (9.16 is a repeat from Section 9.3.1.2.1):

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$$R_f = \frac{V_f^2}{g(n-1)} \tag{9.18}$$

$$h_f = R_f(1-\cos(\gamma_{approach})) \tag{9.16}$$

$$S_{3_f} = R_f \sin(\gamma_{approach}) \tag{9.19}$$

9.3.1.2.3 Ground Roll Approximation

After touchdown, the aircraft rolls for 1 to 3 seconds before the brakes are applied, or the time delay programmed into the autopilot. The distance covered during this time (S_{4_r}) can be found by multiplying the touchdown velocity (V_{TD}) by the delay. The ground roll can be approximated by the following equations assuming idle thrust without thrust reversing:

$$S_{4_B} = \left(\frac{1}{2gK_{A,Land}}\right)\ln\left(\frac{K_{T,Land}}{K_{T,Land}+K_{A,land}V_{TD}^2}\right) \tag{9.20}$$

$$K_{A,Land} = \frac{\rho}{2(W/S)}(\mu_2 C_L - C_{D_O} - KC_L^2) \tag{9.21}$$

$$K_{T,Land} = \left(\frac{T}{W}\right) - \mu_2 \tag{9.22}$$

9.3.2 Augmented Takeoff and Landing Performance

Different means of augmenting takeoff and landing have been developed since the inception of the UAV. Augmented takeoff and landing systems fall into the ground system category and can be tested along with the rest of the ground systems. However, the data acquisition process for augmented takeoff and landing systems lends itself to testing during the flight test phase of the test program, where they can be used for their designated purpose and tested simultaneously.

9.3.2.1 Augmented Takeoff

Augmented takeoff systems include catapult, booster, hand launch, and air drop. Most of these systems have been developed to allow extremely short takeoff distances from unimproved airstrips, when no airfield is available, or when the AV cannot takeoff by conventional means. Refer to Figures 9.1, 9.2, and 9.3 for illustrations of catapult launcher, booster launch, and hand launch, respectively.

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Figure 9.1 Shadow UAV Catapult Launch



Figure 9.2 Hunter UAV with Booster Launch

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Figure 9.3 Raven UAV Hand Launch

The catapult and booster are closely related; the difference between the two is that the catapult uses mechanical energy to augment takeoff and the booster uses chemical energy. Tube-launching, as in the case of the BatCam from an AH-64 Apache, is also closely related to the booster launch, except that it combines the booster launch and the air drop in the same augmented takeoff system. These systems should have manufacturer provided data concerning maximum gross takeoff weight that can be used with the system and the AV exit airspeed. If the data are not provided or if the test team wishes to validate the manufacturer data, an accelerometer can be installed on the AV and a series of launches can be performed. This will provide acceleration vs. time data that can be integrated to find the total velocity change during augmented takeoff. The data can then be used to determine the total energy imparted by the takeoff system. Using the data, the test team can develop augmented takeoff performance sheets and calculate the AV maximum gross takeoff weight where climb airspeed ($1.2V_{Stall}$, or airspeed set by the autopilot control logic) can be attained.

Whenever using an accelerometer, it is important the instrumentation be designed to both have sufficient measurement bandwidth and sampling rate to fully resolve the launch event. The acceleration data provided by either downlink or onboard logging of navigation relevant accelerations is generally not of sufficient rate to fully resolve the launch dynamics. The energy method described above would be especially useful for tests of a new augmented takeoff system. The test team would be able to determine the new max gross takeoff weight and also the min gross takeoff weight, the latter being determined from the point at which the augmented takeoff system would cause the AV to exceed the structural limits of the airframe. Another important factor to understand is the change in acceleration over a change in time $\left(\frac{da}{dt}\right)$, referred to as "jerk." This can be found by differentiating the acceleration vs. time data. Acceleration data are normally acquired by onboard accelerometers. Electronics and avionics can be sensitive to jerk

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and during high-g takeoff operations, this factor must be addressed as well as the acceleration during takeoff. Two sample plots of launcher accelerometer data in terms of g's vs. time are shown in Figure 9.4.

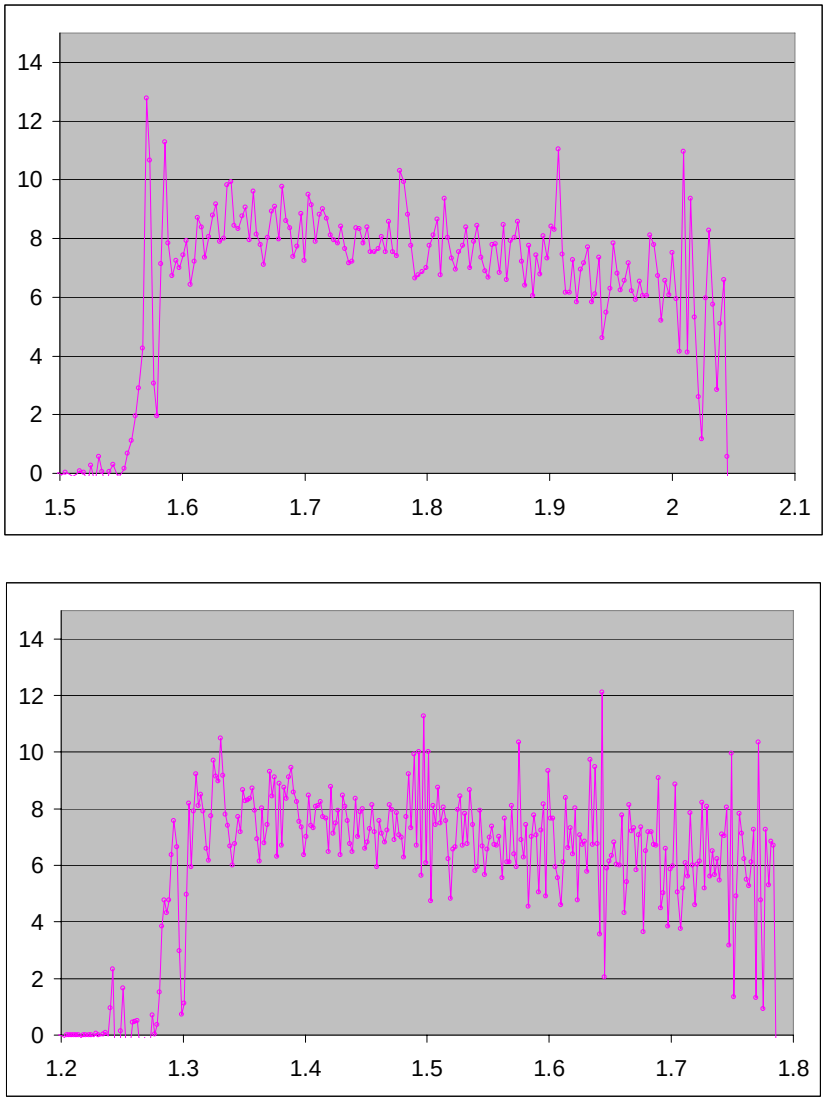


Figure 9.4 Launcher Sample Data at 500Hz

Hand launching is usually employed by extremely small UAVs. This method is very difficult to test and even standardize due to the variability in the launching system, the human operator. In this case, the technique for hand launching should be well understood by the test team so as to minimize the variation in launches from one takeoff to the next. Another testing method would be to design a launch system capable of reasonably simulating the launch from a human hand. The launcher could then be used to test all possible launch situations and the data could be used to determine the operational limits of the UAV. This would develop the proper throwing technique for the UAV.

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Ground vehicle launching can be employed for some UAVs with suitable stall margins for such launching. In this case, the UAV is mounted on the bed of a High Mobility Multipurpose Wheeled Vehicle (HMMWV) or held in the hand of a passenger. As the HMMWV reaches takeoff speed, the AV is released and continues its planned mission. Accelerometer data could again be used to determine the total necessary energy that must be transferred to the AV, and the liftoff speed of the AV and its configuration should be noted.

Air dropping is when a UAV is dropped from a larger carrier aircraft. This aircraft can be another UAV or a manned aircraft. The main factors that need to be understood in this technique are altitude and airspeed. The UAV being carried might have to lose altitude to attain the velocity necessary for sustained flight. Some air-dropped UAVs use a parafoil as the lifting device, and, in this case, the g-loading imposed on the aircraft from the opening of the parafoil must be quantified to remain within the structural limits of the AV. The Snow Goose UAV can takeoff from the back of a HMMWV or be dropped from another aircraft with the parafoil as shown in Figure 9.5.



Figure 9.5 Snow Goose UAV

9.3.2.2 Augmented Landing

Augmented landing systems include nets, ropes, and arresting cables. Most of these systems were designed to allow extremely short landing distances on unimproved airstrips when no airfield is available or for AVs without conventional braking systems.

Nets and ropes are very similar in the method for capturing the incoming UAV. The net is designed so that the AVO will be able to fly the AV directly into the net, and the net will collapse around the UAV, catching it. The rope method, as is used by the

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Scan Eagle, involves clips on the end of the wings of the UAV that will clip onto a rope if the UAV is flown into it. The main concerns with these two methods are to keep the acceleration of the AV below the threshold that will overstress the airframe and to ensure that the augmented landing system is not overstressed. Manufacturer data for the augmented landing system should provide information as to the limits of the energy which can be absorbed by the augmented landing system, including limitations on AV gross weight and airspeed. If the augmented landing system is to be tested by the test team, accelerometers can be installed on the AV and then used to measure the acceleration that the AV encounters when striking the augmented landing system. This method will allow the test team to set the limits of max gross weight and airspeed to be used with the augmented landing system. The data can also be used to determine the energy and force being applied to the augmented landing system to ensure that the structure of the system is not being compromised. As with the takeoff, "jerk" must be considered in assessing potential issues with onboard equipment. The required field size to support the relatively flat approach profile of this landing method should also be considered and tested.

Arresting cables are used on many UAVs such as the Shadow 200 and Hunter. This method involves the landing of the AV conventionally with a cable strung across the airstrip. The AV then has an arrestor hook mounted underneath that snags the cable and allows the AV to come to a stop much sooner than with a conventional landing. The main concerns with this system are overstressing the hook and its mounting structure and overstressing the cable. The arrestor hook can also present the potential issue of oscillation when the hook comes into contact with the ground, which can lead to the hook missing the cable. The energy being absorbed by the cable and the force with which the hook pulls on the mounting structure can be found by installing an accelerometer on the AV and then landing using the arrestor cable. The test team will be able to determine the forces acting on the hook, its mounting structure, and the cable by finding the total energy being transferred and absorbed by the hook and cable. This will allow the test team to determine max gross weight and landing speeds that can be used with the system.

9.4 Test Methods

Specific weights and configurations (external stores, flaps, etc.) should be tested for any takeoff or landing, conventional or augmented. The takeoff and landing performance will be affected by meteorological conditions, and the parameters that effect density altitude should be recorded so that test performance can be correlated to the expected performance in standard weather conditions.

Some UAVs are designed for short takeoff and landing (STOL). This requires fundamental understanding of the control logic governing the takeoff and landing. Additionally, the instrument errors, datalink latency, relative time difference among instrumentation should be checked and understood before testing. As stated in Section 9.1, the flight control system may need supplementary code for testing the full takeoff and landing operational envelope. This might include STOL operations for short

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or tactical fields or operating with lower power settings that will result in prolonging the life of the engine. All settings for takeoff and landing should be tested.

Systems with autolaunch and autorecovery frequently have automatic sequencing through those events. Typical concepts might be for runway steering up from V1 for a rolling takeoff with a mode switch to rotation, followed by a mode switch to climb-away, followed by an entry to point navigation. Similarly, augmented takeoffs may incorporate sequencing through a hold pitch/roll angle segment up to a certain altitude or airspeed, followed by airspeed loop closure, and then waypoint guidance closure laid on top of that. The test program, instrumentation, and approach should verify that the system will reliably perform these sequences, on time and reliably. If there is any determination in the safety analysis that a possibility might exist for "ill-sequencing," then consideration should be given to live flight or Hardware-In-The-Loop Simulation (HILSIM) testing of that situation. This might, for example, include an airspeed or altitude glitch due to acceleration on an augment launch leading to premature transition out of hold pitch mode into airspeed loop closure. The key to a safe and effective test program will be garnering the understanding of the control system functionality and taking all opportunities to explore the what-ifs of anomalies in the data and sequencing and human interface of that control system. Refer to Figure 9.6 for an illustration of a UAV auto-launch sequence.

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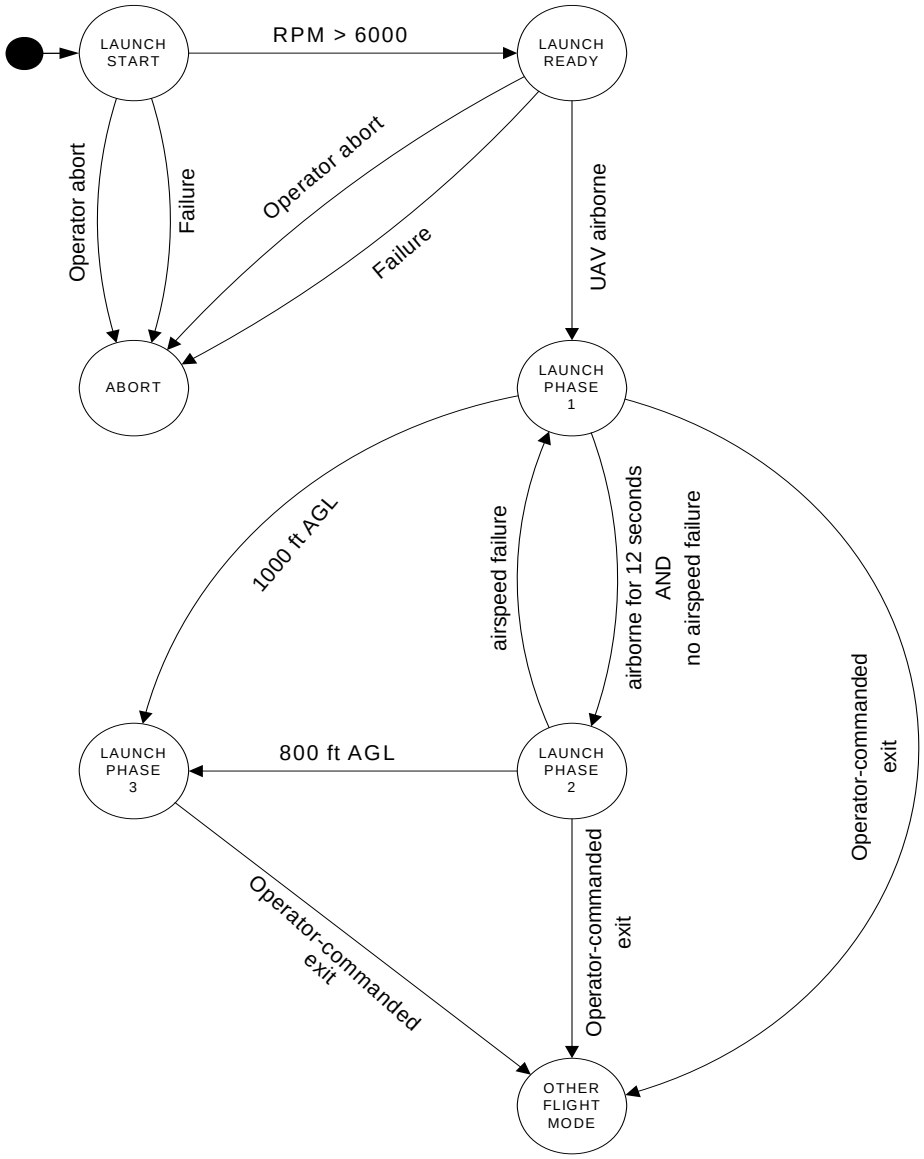


Figure 9.6 Example Sequence in Auto-Launch

Although the future of UAVs is full autonomy, some legacy systems still use an external pilot (EP) to takeoff or land the AV. In this case, the UAV takeoff and landing will more closely approximate manned aircraft, since a person will be deciding on time from rotation to liftoff and from touchdown to brakes applied. Refer to Figures 9.7 and 9.8 for an illustration of a Hunter UAV controlled by an EP at takeoff and the tactical automated landing system (TALS) controlling Shadow 200 UAV during approach and arrested landing, respectively.

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Figure 9.7 EP Controlling Hunter UAV during Liftoff
[Note arresting gear pendant just to right of right-hand EP and arresting gear drum located at end of pendant, just off runway]



Figure 9.8 TALS controlling Shadow 200 UAV during approach and arrested landing [Note Flare point and small bounce after touchdown]

Since the requirements for takeoff and landing distances differ among the surface material (paved, dirt, grass) and shape (slope, crown, etc.), the test team should consider the varying surfaces in which the UAV will takeoff and land. If possible, testing should

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incorporate as many variations as possible in the testing without overstressing the test budget or timetable. Takeoff performance can be very significantly affected by soft surfaces and their rolling friction effect. Control of the AV via nose wheel steering can be significantly degraded on loose surfaces or across the crown of the runway or road. Excess crown can also impact the effectiveness of arresting gear equipment. It is crucial to obtain sufficient tension in the pendant for it to spring back into position to catch the hook after the landing gear have crossed the pendant. If the pendant is too high relative to the runway due to excess crown or surface irregularities, then it may have a much lower likelihood of catching the hook to complete the arrestment.

9.4.1 Conventional Takeoff

9.4.1.1 Overview

This technique is detailed in USNTPS-FTM-No. 108 Section 9.4.1 and the overview described in USNTPS-FTM-No. 108 Section 9.4.1.1 would apply to UAVs capable of conventional takeoffs. Two additional methods for measuring the ground roll distance and climb distance are an onboard differential global positioning system (DGPS) and high resolution cameras. The DGPS can only be used when the takeoff distance is sufficiently long enough to negate any position error from the DGPS. For UAVs without DGPS, cameras may be placed on the runway centerline with reference lines for determining lift-off point. Additionally, a trustworthy weight-on-wheels sensor(s) on the landing gear may be employed to determine when liftoff occurs. With either method, exact time correlation among all onboard and ground systems is vital for data accuracy. Any variables associated with density altitude effects should be recorded.

For measurement of climb distance during takeoff, several methods may be considered. A ground-based laser tracking system with onboard reflectors could track the AV during takeoff to attain the total takeoff distance. For accurate measurements, ground placement of the laser tracking system for these tests must be considered. Another technique in obtaining distances may be a marked or measured course using ground based cameras or observers. Small UAV testing may employ observers much easier than for large aircraft.

In an automated takeoff, correct sequencing of the control logic is vital. If the autopilot does not sequence properly, the takeoff should be aborted and expected outcome should be tested in the simulator.

9.4.1.2 Data Required

1. Takeoff airspeed, V_{TO}
2. Distance to lift off obtained by:
   a. Theodolite
   b. Runway camera
   c. Paint gun
   d. Observers

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e. Laser
f. “Eyes right” - check runway marker
g. DGPS (if conditions in 9.4.1.1 are met)

3. Pitch attitude on rotation/initial climb
4. Distance to 50 ft/distance to climb airspeed
5. Time to lift off/50 ft/climb airspeed
6. Angle of attack at rotation/climb out
7. Fuel used, brake release to climb airspeed
8. Runway wind conditions
9. Runway temperature
10. Runway composition/Runway Condition Reading (RCR)
11. Field elevation
12. Altimeter
13. Runway gradient
14. Aircraft gross weight/center of gravity
15. Power parameters: rpm, EGT
16. Elevator/horizontal tail position

9.4.1.3 Test Criteria

1 Establish takeoff trim setting.
2. No brakes (limit use of rudder/nose wheel steering.).
3. If turbine engine, bleed air system OFF or in normal mode.

9.4.1.4 Data Requirements

1. Engine stabilized before roll (if feasible).
2. Surface winds (head, tail, and cross) should within the operational limits of the UAV.
3. Specified weight (Note: weights may differ from test platform to production AV, refer to Chapter 1).
4. Rate of climb from liftoff to intercepting climb schedule < one half the maximum rate of climb.
5. Time correlation between all data acquisition systems.

9.4.1.5 Safety Considerations

Calculation and analysis of the “safety fan” at takeoff must be performed. All safety considerations such as the propeller cone, structural failure, and foreign object damage (FOD) prop-strike and their corresponding safety distances should be determined before the test to determine the limits of the proximity of test personnel. The AVO should not have the AV deviate from the runway centerline until well past test personnel and air traffic control facilities. Also, the logic tree for lost link, engine failure, and navigation system loss should be known so that the risk of serious injury to test personnel can be mitigated. Additionally, if the autopilot control logic does not sequence properly,

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the takeoff should be aborted; for the aborted takeoff, the engine shutdown should be correctly sequenced to prevent a mishap from occurring.

9.4.2 Conventional Landing

9.4.2.1 Overview

This technique is detailed in USNTPS-FTM-No. 108 Section 9.4.2 and the overview described in USNTPS-FTM-No. 108 Section 9.4.2.1 would apply to UAVs capable of conventional takeoffs. The additional methods such as those described in Section 9.4.1.1 above may be used to determine approach and landing distances.

For measurement of approach distance during landing, several methods may be considered. A ground-based laser tracking system with onboard reflectors could track the AV during landing to attain the total landing distance. For accurate measurements, ground placement of the laser tracking system for these tests must be considered. Another technique in obtaining distances may be a marked or measured course using ground based cameras or observers. Small UAV testing may employ observers much easier than for large aircraft.

In an automated landing, correct sequencing of the control logic is vital. If the autopilot does not sequence properly, the landing should be aborted and expected outcome should be tested in the simulator.

9.4.2.2 Data Required

1. Distance from 50 ft above the ground through the flare to touch down
2. Glideslope angle
3. Distance from touch down to full stop
4. Runway wind conditions
5. Runway temperature
6. Runway composition/RCR
7. Field elevation
8. Altimeter
9. Runway gradient
10. Aircraft gross weight/center of gravity
11. Configuration
12. Power parameters during landing and roll out: rpm, EGT

9.4.2.3 Test Criteria

If turbine engine, bleed air system OFF or in normal mode.

9.4.2.4 Data Requirements

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1. Surface winds (head, tail, and cross) should within the operational limits of the UAV.
2. Specified weight (Note: weights may differ from test platform to production AV, refer to Chapter 1).
3. Time correlation among all onboard and ground systems.

9.4.2.5 Safety Considerations

Calculation and analysis of the "safety fan" at landing must be performed. All safety considerations such as the propeller cone, structural failure, and FOD prop-strike and their corresponding safety distances should be determined before the test to determine the limits of the proximity of test personnel. The AVO should not have the AV deviate from the runway centerline until well past test personnel and ATC facilities. Also, the logic tree for lost link, engine failure, and navigation system loss should be known so that the risk of serious injury to test personnel can be mitigated. Additionally, if the autopilot control logic does not sequence properly during landing, the landing should be waved off to prevent a mishap.

9.4.3 Augmented Takeoff

9.4.3.1 Overview

Augmented takeoff testing requires determining the energy imparted to AV from the augmenting system. For launchers or boosters, accelerometers must be placed on the AV so that the energy supplied can be calculated from the change in velocity. The methods discussed for conventional takeoffs for tracking the climb distance should apply for augmented takeoffs as well.

Hand launches pose a particular difficulty since the potential for system performance variation is relatively high compared to other augmented takeoff systems. Consideration should be given to varying techniques and physical abilities of the operator. One user may be able to impart a great deal of energy in the launch for sustained flight while others may be incapable.

For measurement of climb distance during takeoff, several methods may be considered. A ground-based laser tracking system with onboard reflectors could track the AV during takeoff to attain the total takeoff distance. For accurate measurements, ground placement of the laser tracking system for these tests must be considered. Another technique in obtaining distances may be a marked or measured course using ground based cameras or observers. Small UAV testing may employ observers much easier than for large aircraft.

In an automated takeoff, correct sequencing of the control logic is vital. If the autopilot does not sequence properly, the takeoff should be aborted and expected outcome should be tested in the simulator.

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9.4.3.2 Data Required

1. Takeoff airspeed, V_{TO}
2. Distance to lift off obtained by:
   a. Theodolite
   b. Runway camera
   c. Paint gun
   d. Observers
   e. Laser
   f. "Eyes right" - check runway marker
   g. DGPS (if conditions in 9.4.1.1 are met)
3. Pitch attitude on rotation/initial climb
4. Distance to 50 ft/distance to climb airspeed
5. Time to liftoff/50 ft/climb airspeed
6. Angle of attack at rotation/climb out
7. Fuel used, brake release to climb airspeed
8. Runway wind conditions
9. Runway temperature
10. Runway composition/RCR
11. Field elevation
12. Altimeter
13. Runway gradient
14. Aircraft gross weight/center of gravity
15. Power parameters: rpm, EGT
16. Elevator/horizontal tail position

Accelerometer data, time correlated with all onboard and ground data recording devices, should provide the necessary information for determining the energy exchange between the AV and the augmenting system. It can also quantify the acceleration change with respect to time. The other data required matches with USNTPS-FTM-No. 108 Section 9.4.1.2.

9.4.3.3 Test Criteria

1. Ensure the pitot-static system is free from obstructions.
2. For hand launch UAVs, consider the human factor elements which may skew launch data.
3. Assure consistent energy supplied during mechanically assisted launches from the pneumatic pressure or spring tension.

9.4.3.4 Data Requirements

1. Surface winds (head, tail, and cross) should within the operational limits of the UAV.
2. Specified weight (Note: weights may differ from test platform to production AV, refer to Chapter 1).

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3. Rate of climb from lift off to intercepting climb schedule < one half the maximum rate of climb.
4. Time correlation between all data acquisition systems.

9.4.3.5 Safety Considerations

Calculation and analysis of the "safety fan" of a particular augmented takeoff system also must be performed. All safety considerations such as the propeller cone, structural failure, and FOD prop-strike and their corresponding safety distances should be established before the test to determine the limits of the proximity of test personnel. The AVO should not have the AV deviate from the runway centerline until well past test personnel and ATC facilities. Also, the logic tree for lost link, engine failure, and navigation system loss should be known so that the risk of serious injury to test personnel can be mitigated. Additionally, if the autopilot control logic does not sequence properly, the takeoff should be aborted; for the aborted takeoff, the engine shutdown should be correctly sequenced to prevent a mishap from occurring.

9.4.4 Augmented Landing

9.4.4.1 Overview

Similar to augmented takeoff, augmented landing tests require determining the energy imparted to AV from the augmenting system. Accelerometers must be placed on the AV so that the energy supplied can be calculated from the change in velocity.

For measurement of approach distance during landing, several methods may be considered. A ground-based laser tracking system with on-board reflectors could track the AV during landing to attain the total landing distance. For accurate measurements, ground placement of the laser tracking system for these tests must be considered. Another technique in obtaining distances may be a marked or measured course using ground based cameras or observers. Small UAV testing may employ observers much easier than for large aircraft.

In an automated landing, correct sequencing of the control logic is vital. If the autopilot does not sequence properly, the landing should be aborted and expected outcome should be tested in the simulator.

9.4.4.2 Data Required

1. Distance from 50 ft above the ground through the flare to touch down
2. Glideslope angle
3. Distance from touch down to full stop
4. Runway wind conditions
5. Runway temperature
6. Runway composition/RCR
7. Field elevation

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8. Altimeter
9. Runway gradient
10. Aircraft gross weight/center of gravity
11. Configuration
12. Power parameters during landing and roll out: rpm, EGT

Accelerometer data time correlated with all onboard and ground systems should provide the necessary information for determining the energy exchange between the AV and the augmenting system. It can also quantify the acceleration change with respect to time. The other data required matches with USNTPS-FTM-No. 108 Section 9.4.2.2.

9.4.4.3 Test Criteria

1. Ensure the pitot-static system is free from obstructions.
2. Assure consistent energy absorbed during mechanically assisted landings (cable tension, net tension, or rope tension).

9.4.4.4 Data Requirements

1. Surface winds (head, tail, and cross) should be within the operational limits of the UAV.
2. Specified weight (Note: weights may differ from test platform to production AV, refer to Chapter 1).
3. Time correlation among all onboard and ground systems.

9.4.4.5 Safety Considerations

Calculation and analysis of the “safety fan” of a particular augmented landing system also must be performed. All safety considerations such as the propeller cone, structural failure, and FOD prop-strike and their corresponding safety distances should be determined before the test to determine the limits of the proximity of test personnel. The AVO should not have the AV deviate from the runway centerline until well past test personnel and ATC facilities. Also, the logic tree for lost link, engine failure, and navigation system loss should be known so that the risk of serious injury to test personnel can be mitigated. Additionally, if the autopilot control logic does not sequence properly during landing, the landing should be waved off to prevent a mishap from occurring. Consideration should also be given to the possibility of the engine ill-sequencing during shutdown and how the engine continuing to operate might affect structural integrity of the augmented landing system or the AV. Attention should be given to the arrestor hook dynamics, as stated in Section 9.3.2.2. A photographic example is given in Figures 9.9 and 9.10 showing how a hook developed an oscillation that precluded pendant capture due to interface with the soil.

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Figure 9.9 UAV Tail Hook with Shoe

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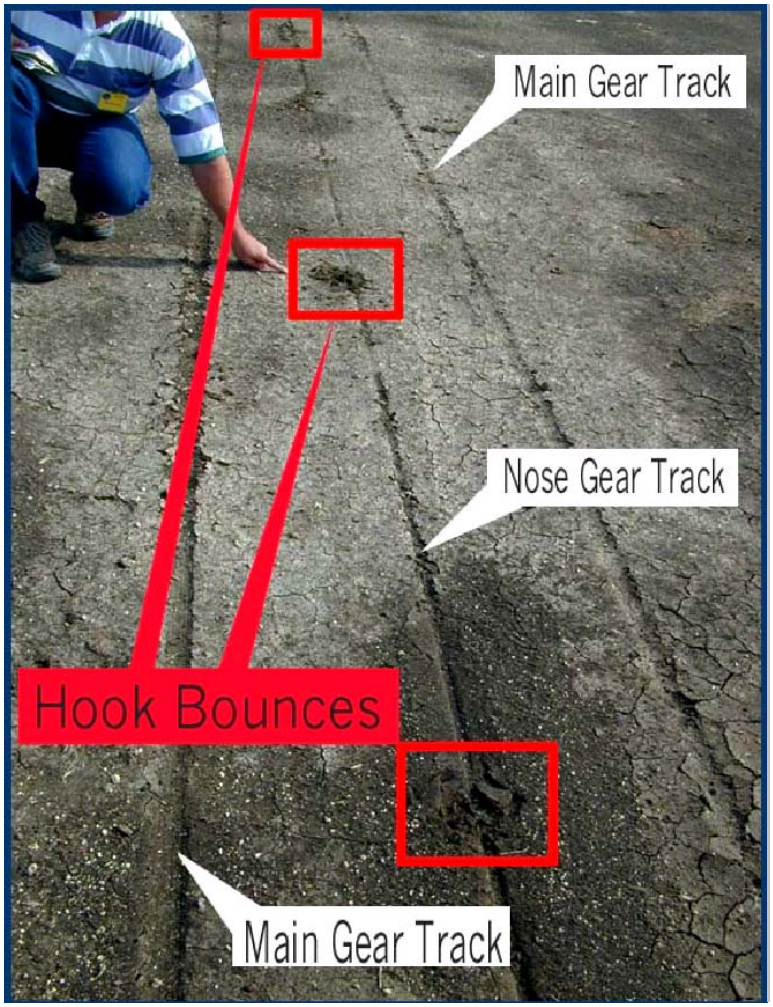


Figure 9.10 Oscillating Tail Hook Behavior.

9.5 Data Reduction

9.5.1 Takeoff

The conventional takeoff will have the same data reduction as USNTPS-FTM-No. 108 Section 9.5.1.

For augmented launches the following analysis will be used. The flight crew will perform “maximum safe and reliable” launcher pressure/performance on all launches. Following launch, the AV should be flown as to safely climb to a local pattern at not less than 500 ft above ground level. The end of stroke launch speed will be read from either a radar gun or integrated from accelerometer data. This will be used, along with the end of stroke change in height (determined from the test launcher configuration), to calculate the total work done by the launcher:

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$$W = mg\Delta h + 0.5 * m * \Delta V^2$$

Where:

W	is the work done by launcher
m	is the mass of the AV plus the launch sled
g	is acceleration due to gravity
Δh	is the change in height during launch
ΔV	is the change in velocity during launch

The target end of stroke velocity at maximum specification density altitude will be found from:

- Calculate the specification/tech manual maximum gross takeoff weight (MGTOW) using the empty weight, adjustments to the minimum payload weight, and the fuel capacity
- Convert the greater of the stall (MGTOW) or minimum controllable airspeed, for the launch configuration, from indicated to true airspeed at the maximum specification density altitude, and multiply by 1.1 to get the target velocity, V_{target} (includes 10% stall/min. control margin)
- Adjust V_{target} for a 20 kt crosswind
- Determine the standard day (9,000 ft) change in height, Δh_{std}, from the standard day launcher configuration.

Calculate the velocity at the standard weight and launcher angle using:

$$\Delta V_{std} = \sqrt{\frac{2 * (W - m_{MGTOW} g \Delta h_{std})}{m_{MGTOW}}}$$

ΔV_{std} must be equal to or greater than V_{target} for the system to be graded capable of the launch requirement.

(9.21)

9.5.2 Landing

The conventional landing will have the same data reduction as USNTPS-FTM-No. 108 Section 9.5.2. The accelerometer data can be integrated to provide the velocity decrease during landing. The total energy absorbed by the augmenting system can be determined from equation 9.21.

The landing scatter will be measured across the wind and weight and altitude range of the system. Statistical limits on the most upwind and downwind landing positions will be developed and the total field setup will make provision for the worst-case short landing (at the statistical level set by the specification or the

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airworthiness authority), the longest landing, any necessary bounce and settle down space, and the distance to stop the AV using either the brakes or the augmentation system. Weather and surface friction effects will also be applied to the braking and or augmentation system.

9.6 Data Analysis

The data analysis in USNTPS-FTM-No. 108 Section 9.6 will apply to UAVs.

9.7 Mission Suitability

USNTPS-FTM-No. 108 Section 9.7 applies to UAVs of all sizes. The mission suitability of the UAV is dependent upon its ability to satisfy the takeoff and landing requirements for each particular UAV.

9.8 Specification Compliance

The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAV system will provide qualifications for which the UAV was designed to meet. Expected performance of the UAV will also be rooted in the modeling and simulation of the UAV. The detailed technical manuals for the UAV system can then be written from the developmental testing as a reference source for all future system testing and military operations.

9.9 Glossary

9.9.1 Notations

a	Speed of Sound	ft/s
AR	Wing Aspect Ratio	
bhp	Engine Brake Horsepower	bhp
C_{bhp}	Propeller Specific Fuel Consumption	lb/hr/bhp
C_{D_0}	Drag Coefficient at $C_L = 0$	
$C_{L_{MinDrag}}$	Minimum Drag Lift Coefficient	
C_p	Power Coefficient	
D	Diameter of Propeller	ft
D	Drag	lb

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$D_{MinDrag}$	Drag at Minimum Drag	lb
e	Oswald Efficiency Factor	
E	Endurance	hr
J	Advance Ratio	1/rev
$K_{A,Land}$	Landing Coefficient	
$K_{T,Land}$	Landing Coefficient	
$K_{A,TO}$	Takeoff Coefficient	
$K_{T,TO}$	Takeoff Coefficient	
L	Lift	lb
$L_{MinDrag}$	Lift at Minimum Drag	lb
M_{Tip}	Propeller Tip Mach Number	
n	Propeller Angular Velocity	rev/s
P_a	Ambient Pressure	psf
P_T	Total Pressure	psf
R	Gas Constant	1716 $\frac{ft \cdot lb}{slug \cdot {}^{\circ}R}$
R	Range	ft
S	Wing Plan Form Area	ft^2
S_{1_G}	Ground Roll Distance	ft
S_{1_R}	Rotation Distance	ft
S_{2_T}	Transition Distance	ft
S_{2_C}	Climb Distance	ft
S_{3_A}	Approach Distance	ft
S_{3_f}	Flare Distance	ft
S_{4_r}	Free Roll	ft
S_{4_B}	Ground Roll Distance with Brakes	ft
SFC	Specific Fuel Consumption	1/hr
T	Temperature	Rankine
T	Thrust	lb
$\frac{t}{c}$	Propeller Thickness to Chord Ratio	
V_T	True Airspeed	ft/s
$V_{MinDrag}$	Minimum Drag Velocity	ft/s
W	AV Weight	lb
W_f	AV Weight Empty Weight	lb
W_i	AV Weight Gross Take-off Weight	lb

9.9.2 Greek Symbols

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γ	Ratio of Specific Heats	1.4 for Air
ρ	Air Density	$slug / ft^3$
ρ_a	Ambient Air Density	$slug / ft^3$
η_p	Propeller Efficiency	
$\eta_{p,corrected}$	Corrected Propeller Efficiency	

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CHAPTER 10

STANDARD MISSION PROFILES

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10. Standard Mission Profiles

10.1 Introduction

Typical mission profiles flown by unmanned aerial vehicles (UAVs) will be the focus of this chapter. Section 10.4 of the chapter discusses the UAV specific mission profiles. The primary role fulfilled by many UAVs has been reconnaissance, surveillance, and target acquisition (RSTA) missions. With larger UAVs, the attack mission has been added to the arsenal of potential mission profiles. Even with the advent of munitions on UAVs, the aircraft will still perform a RSTA mission profile in conjunction with the attack profile. Other mission objectives that may be employed by UAVs include communications relay and air data relay. The U.S. Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108) Chapter 10 should be used for reference unless a particular section in this chapter offers UAV specific information differing from USNTPS-FTM-No. 108.

10.2 Purpose of the Test

UAV mission profile performance testing is used to verify the projected endurance and aerodynamic range data extrapolated from the level flight performance testing. This will be achieved in full mission execution. Also, these tests typically involve long duration flights, normally in a full equipped configuration, which may be suitable for performing a variety of other system related tests. These include tests of the datalink range and robustness, payload range and resolution, and target location accuracy.

10.3 Theory

USNTPS-FTM-No. 108 Section 10.3 discusses the relationship of the operational requirements of the aircraft to the technical requirements that drive the test criteria. This will apply for UAVs as well. The flight test data will be used to generate electronic flight performance models of the specific UAV to be given to the air vehicle operators (AVO) so that they will know the operational envelope of the UAV. The flight test data will also facilitate the development of mission planning software for the UAV.

Flight test programs are usually conducted so as to gather the most performance information near the data points at which the airplane will most likely be flying. In the case of UAVs, these data points are mostly determined by the payload requirements, not the performance limitations or even the specific mission. What this amounts to is that each UAV system must be understood as a whole before the detailed test plan can be written for that particular UAV. If the UAV is equipped with multiple payloads, the limits of those payloads must be known and the flight test program should center on the data points of most interest. As mentioned in Chapter 4, performance parameters at various altitudes are important. Though the sensors operate at a standard above ground level (AGL) altitude, different altitudes still need to be tested to simulate various terrain conditions, such as high desert or mountainous areas.

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It is important that some testing be done on the system as a whole. The way that the flight control system interacts with the AV during flight testing needs to be taken into account for performance testing due to the possibility of altered performance during full system operation. One example is called camera guide; the flight control system has a tendency to "pull" the AV toward whatever the payload is locked on to. Navigation accuracy can also affect flight performance. Navigation accuracy has become more important in the UAV world over recent years due to increasing mission profiles that require the AV to follow a road. The result of the need to test the system as a whole is that it is necessary to test the system as it would be used in real combat situations. All the systems that would be operating during combat need to be operating during flight test, so that the test team can be reasonably sure that the behavior of the AV will not differ significantly from flight test to actual combat.

It is also a good idea to include in the flight test program actual range and endurance missions. The analysis from Chapter 4 will provide theoretical best case range and endurance values, but if possible, these values should be validated by actually performing these missions at some point during the flight test program. Additionally, the test team can prove the manufacturer's value for the minimum fuel allowed for safe landing with a 30 minute reserve.

These flight test missions would also provide a good opportunity for payload "piggybacking." This allows good use of the time required for the range and endurance validation missions by giving the test team ample time to test payload parameters. Performance testing also gives the test team an outstanding opportunity to test the datalink range. In the past, this has been accomplished by allowing the AV to remain loitering about a point in restricted airspace and moving the ground control station (GCS) position to increase the range.

10.4 Safety

There are a number of safety considerations for the test engineer and test director to address in mission profile testing. The mission profile testing will be accomplished near the end of the flight developmental testing. The test points will have developed the flight envelope of the UAV, but the mission profile test is to validate the interpolated flight envelope. There will be much more confidence about the accuracy of the maintenance cycles, the fuel flow measurement system, the return home fuel status (bingo fuel), the avionics and sensors, and the datalink performance. The test team should discuss the minimum fuel quantity with the manufacturer and verify how the value is determined for reliability. The datalink will not have been tested at the limits of the ground antenna range up to this point, and the datalink signal strength should be carefully monitored when testing the range limits.

10.5 Mission Profiles

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UAV standard profiles are usually comprised of the following:

- a launch from a set altitude (which may range up to 9,000' density altitude)
- a climb segment (with or without transit downrange)
- an ingress/cruise segment
- a loiter or mission payload operations segment
- an egress, descent, and landing

Profiles also address:

- required mission payloads and standard equipment package, along with guidance on minimum and maximum speeds for operation
- rules for the reserve fuel

A goal of mission profile testing is that the minimum quantity of fuel upon landing will be tested with the range and endurance validation. A number of sorties with the fuel brought down to the minimum will offer reliability to the value for the operator's manual.

Frequently the mission and loiter segment involves transiting between named areas of interest (NAI) and/or target areas of interest (TAI); however, these are usually much more local than the distance from the GCS. In executing mission profiles, AVOs must remain cognizant of:

- their fuel state
- the winds expected on the return
- the bingo or minimum fuel/battery guidance for that vehicle type
- airspace restrictions and the overall air defense environment

Another major consideration will be datalink operations which are usually strictly line of sight (LOS) and may also require a minimum (typically 1 degree) elevation angle above the strict LOS value. Terrain avoidance and terrain effect on LOS tools are provided to AVOs, and must be used to plan both the basic and the emergency and return home scenarios.

A significant consideration is that UAV altitudes for payload operations are typically quoted as AGL values. With the need to operate the systems with terrain altitudes from sea level to approximately 9,000', the mission profiles should be tested and analyzed with that variation in terms of MSL altitudes as well.

10.5.1 Small UAV (SUAV) Example Requirement

Army SUAVs support company and platoon-sized units, and as such, have small ranges (typically 10 to 15km maximum) in keeping with the tactical area of interest of

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those units. The normal profile for these systems involves a rapid climb to 500 to 700'AGL in proximity to the launch site, transit to the mission reconnaissance waypoints/NAIs/TAIs, performing surveillance between 300' to 700' AGL. Small UAV missions involve only short periods in climb and cruise and normally operate in surveillance and data gathering mode. Airspace deconfliction and management is pre-eminent in SUAV mission planning and monitoring. Typical mission profile data for SUAV type operations is provided in Figures 10.1 and 10.2. Mission times of up to 90 minutes can be expected in this class.

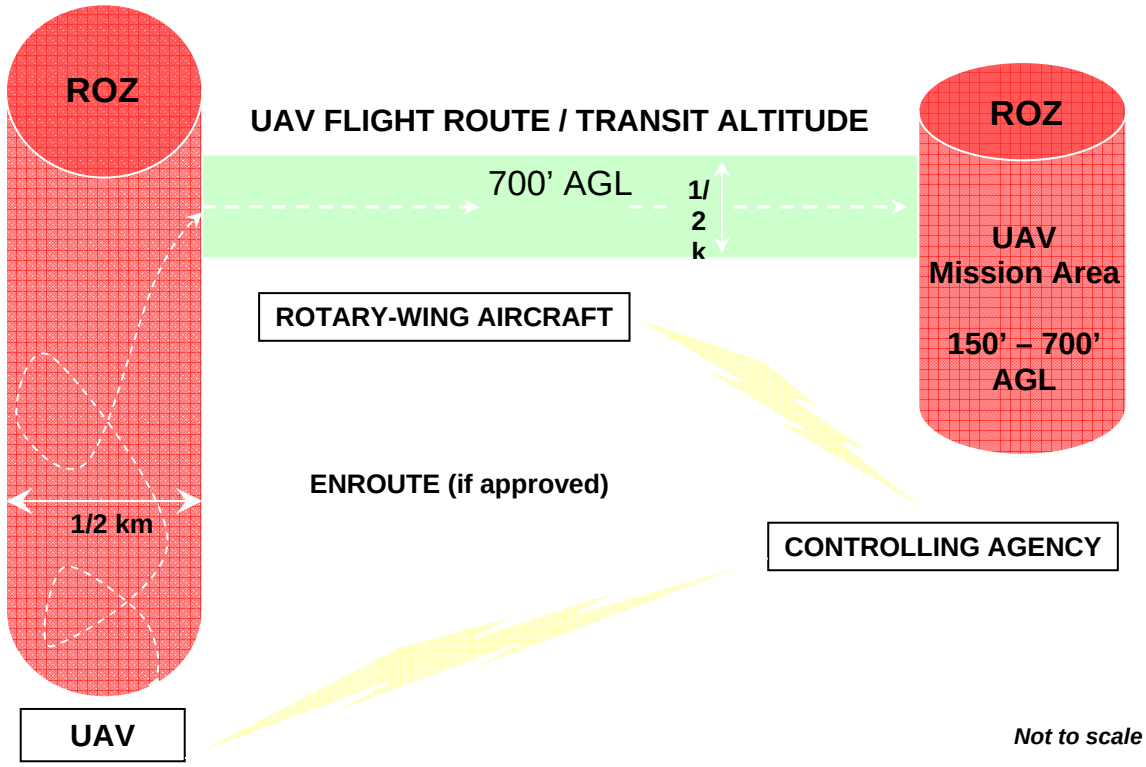


Figure 10.1 SUAV Typical Altitude Profile

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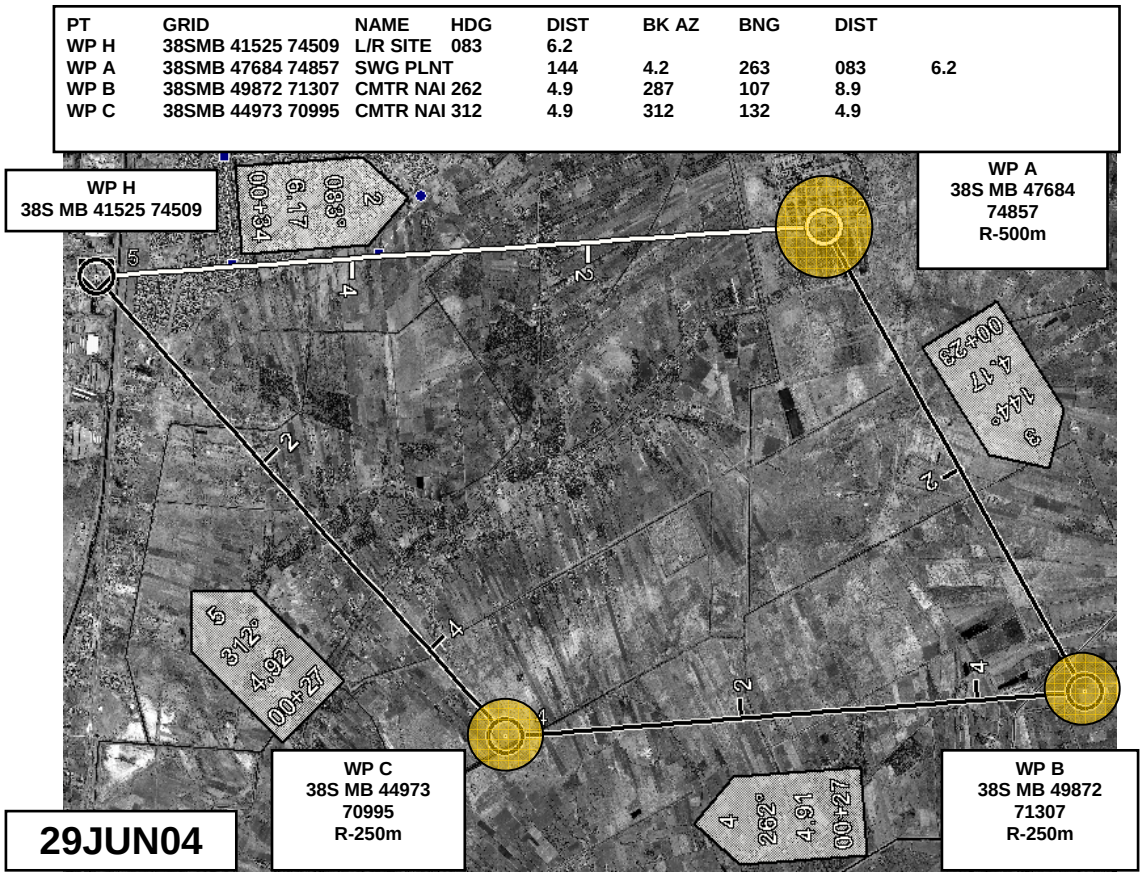


Figure 10.2 SUAV Typical Ground Track Profile

Some UAVs (usually small) have also performed mission segments which require the AV to be as acoustically covert as possible. If Raven were used for this mission segment, the AV would climb to about 1,000' AGL before reducing engine power to the lowest speed control setting, then gliding down to 300' AGL as the AV starts its surveillance run over the target, and gliding to about 100' AGL at the engine power-up point. Some AVOs will actually plan the power-up point to be behind a ridgeline so as to reduce further the chance of detection on climb-out. The AV will then climb back up to 1,000' AGL and perform the run again until all necessary information is acquired.

10.5.2 Tactical UAV (TUAV) Example Requirement and Performance

Tactical class UAVs support battalion, brigade, and occasionally division sized Army elements. As such, they have significantly larger areas of regard. The Shadow UAV system was procured to address a need for a 50 km range. The delivered system has a 125 km range, which is very useful to supporting certain missions at the brigade and up level. Most missions involve at least one handoff of control and is typically between the launch/recovery site and a forward GCS located at a tactical operations center (TOC). It is not uncommon for a mission to support multiple TOCs and require numerous handoffs. Although this is not a first order impact on mission profile

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performance, the need to place the AV in an advantageous location for datalink reception by sending and receiving control stations can be an issue, as well as any "hold" time required to coordinate the handoff while keeping the AV at a precoordinated point. Mission times of up to 8 hrs can be expected in this class.

A general example of a TUAV standard mission profile is:

The AV shall have an airborne on-station time of 4 hours at a 50 km range carrying the standard EO/IR payload as defined by the technical requirements and operational requirements documents. This flight profile assumes a standard day, cruise at 90 KTAS @ 11,000 ft MSL altitude, loiter at less than or equal to 70 KIAS @ 8,000 ft MSL altitude, with a 30-minute fuel reserve.

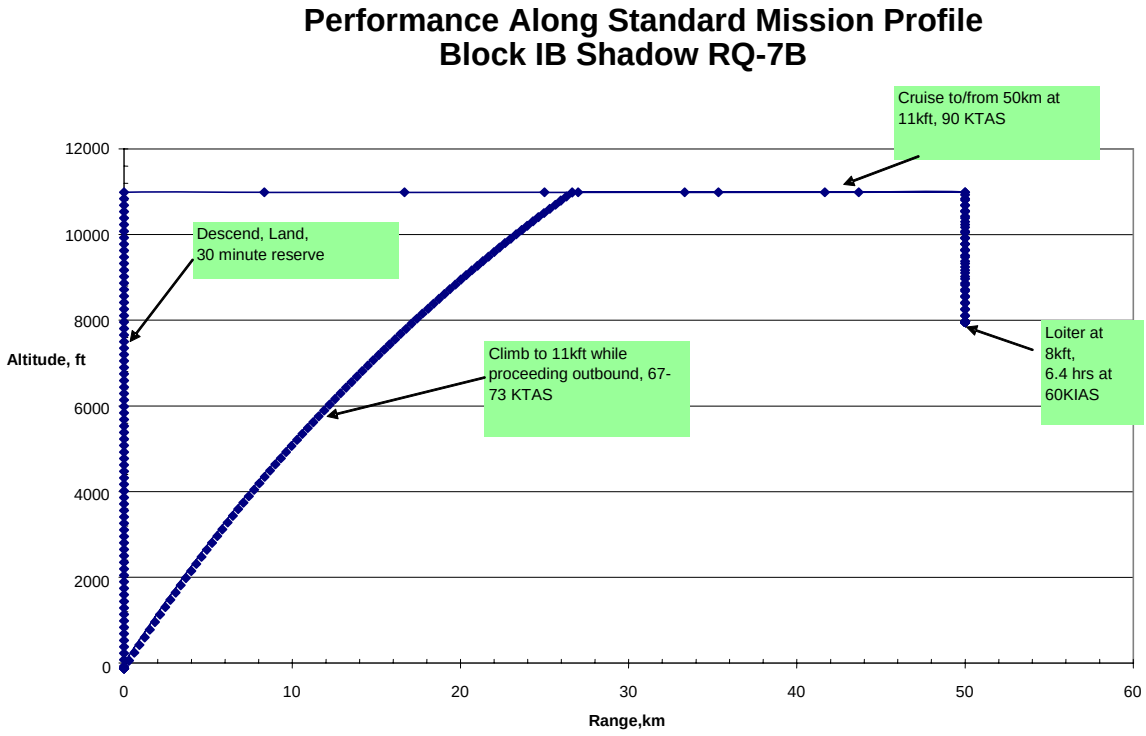


Figure 10.3 Standard Mission Performance- RQ-7B Shadow

More recently, TUAV systems have taken on counter-mine and counter-improvised explosive device (IED) missions, the discriminator there being a need to obtain high resolution of much smaller objects. TUAV systems were originally envisaged as performing RSTA against vehicle-sized targets. With IEDs down to 81mm mortars and smaller, it has become necessary to move along and closer to the road of interest to perform this mission. Typical altitudes for this mission type are 2,500' AGL,

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but altitudes as low as 700' have been employed at times. Since these missions have at times used fixed or nearly fixed cameras staring straight down, it is more critical that the AV navigate precisely along the road or area undergoing imaging.

10.5.3 Large (ER/MP class) UAV Example Requirement

Large Army UAVs, supporting division and corps level formations differ from their smaller counterparts with larger altitude envelopes for both the AV and their payloads. Additionally, they are much more likely to be weaponized. As such, the number of mission profiles to be considered is larger.

The following mission profiles are provided as a basis for calculating range and endurance performance of the extended range/multipurpose (ER/MP) AV. Zero winds are assumed for each segment of the Standard Mission Profiles, and the airspeed shall, at all times, provide no less than 10% margin above the stall and minimum controllable airspeeds. If applicable, the EO/IR shall be extended throughout the flight except for takeoff and landing operations. Refer to Figures 10.4, 10.5, 10.6 for illustrations of standard mission profiles A, B, and C, respectively.

10.5.3.1 Standard Mission Profile A

1 Warm-up, preflight checks, taxi, and automatic takeoff from sea level.
2. Climb to 25,000 ft DA within 25 km of the takeoff site.
3. Dash at 25,000 ft DA at maximum airspeed to a range of 300 km from the takeoff site (T), 500 km from the takeoff site (O).
4. Loiter on-station at a calibrated airspeed between 60 and 75 kt, at 25,000 ft DA, for 12 hours (T), 24 hours (O).
5. Dash at 25,000 ft DA at maximum airspeed to within 25 km of the takeoff site.
6. Descend to sea level within the remaining 25 km of the takeoff site.
7. Perform automatic landing and taxi, maintaining a usable fuel reserve for a 30-minute loiter at 5,000 ft DA following completion of the mission.

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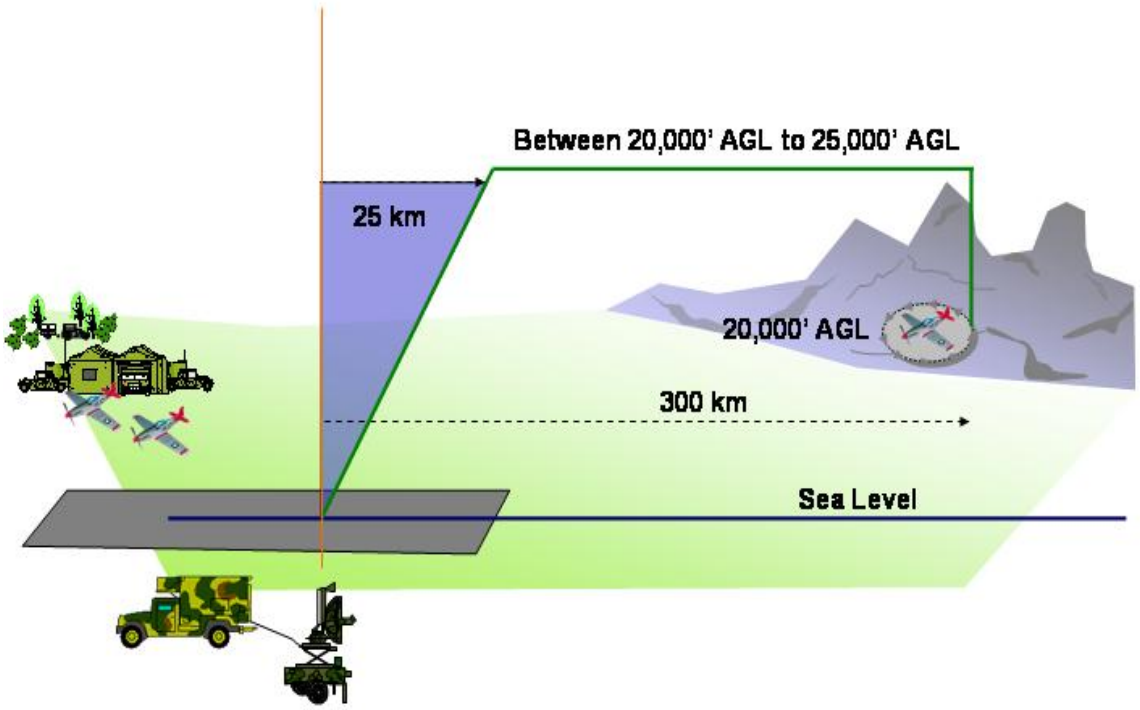


Figure 10.4 Mission Profile A – RSTA Hi-Hi

10.5.3.2 Standard Mission Profile B

1. Warm-up, preflight checks, taxi, and automatic takeoff from sea level.
2. Climb to 8,000 ft DA within 10 km of the takeoff site.
3. Dash at 8,000 ft DA at 130 kt true airspeed to a range of 300 km from the takeoff site (T), 500 km from the takeoff site (O).
4. Loiter on-station at a calibrated airspeed between 60 and 75 kt, at 8,000 ft DA, for 12 hours (T), 24 hours (O).
5. Dash at 8,000 ft DA at 130 kt true airspeed to within 10 km of the takeoff site.
6. Descend to sea level within the remaining 10 km of the takeoff site.
7. Perform automatic landing and taxi, maintaining a usable fuel reserve for a 30-minute loiter at 5,000 ft DA following completion of the mission.

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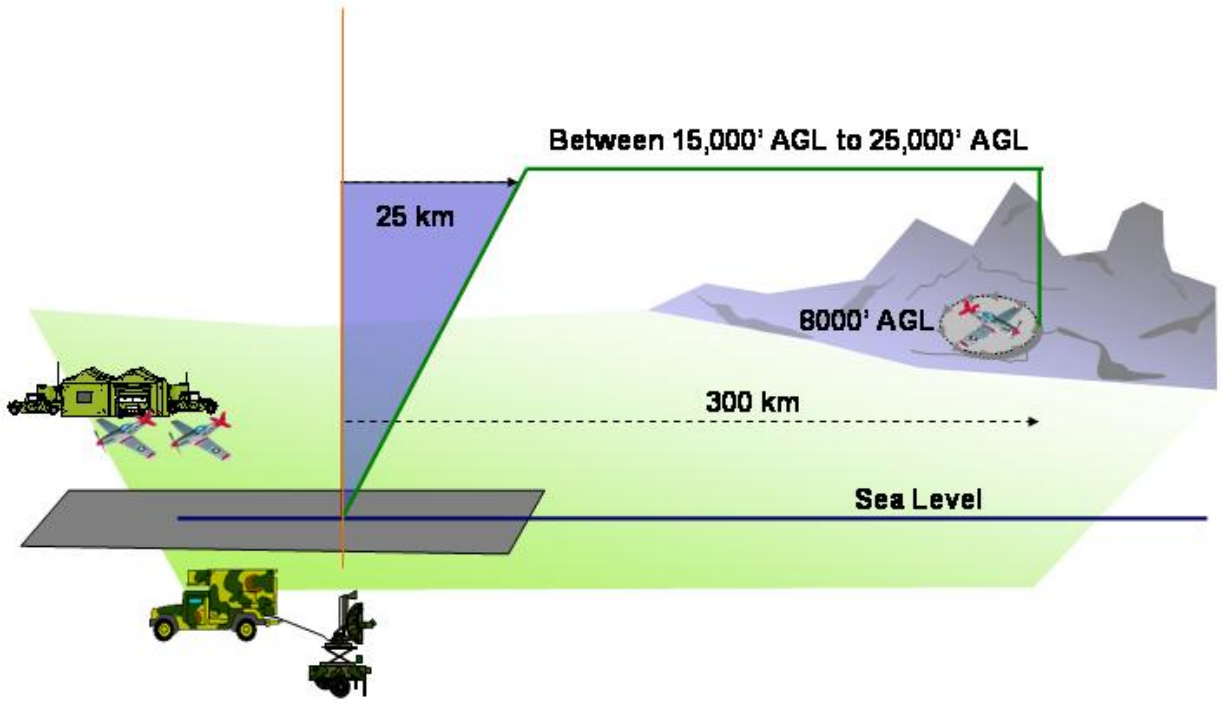


Figure 10.5 Mission Profile B - RSTA Hi-Lo

10.5.3.3 Standard Mission Profile C

A mix of payload and fuel can be offloaded to support the weapons mission. At a minimum, 150 lb of EO/IR/LD/LRF shall be installed and operational while carrying 400 lb of weapon stores (or simulated ordnance).

1. Warm-up, preflight checks, taxi, and automatic takeoff from sea level. Climb to 15,000 ft DA within 15 km of the takeoff site.
2. Dash at 15,000 ft DA at maximum airspeed to a range of 300 km from the takeoff site (T), 500 km from the takeoff site (O).
3. Descend to 8,000 ft DA within range of 295 to 300 km from the takeoff site (T), 495 to 500 km (O).
4. Loiter on-station at a calibrated airspeed between 60 and 75 kt, at 8,000 ft DA, for 6 hours.
5. Climb to 15,000 ft DA within a range of 295 to 300 km (T) from takeoff site, 495 to 500 km (O) from the takeoff site.
6. Dash at 15,000 ft DA at maximum airspeed to within 15 km of the takeoff site.

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7. Descend to sea level within the remaining 15 km of the takeoff site. Perform automatic landing and taxi, maintaining a usable fuel reserve for 30-minute loiter at 5,000 ft DA following completion of the mission.

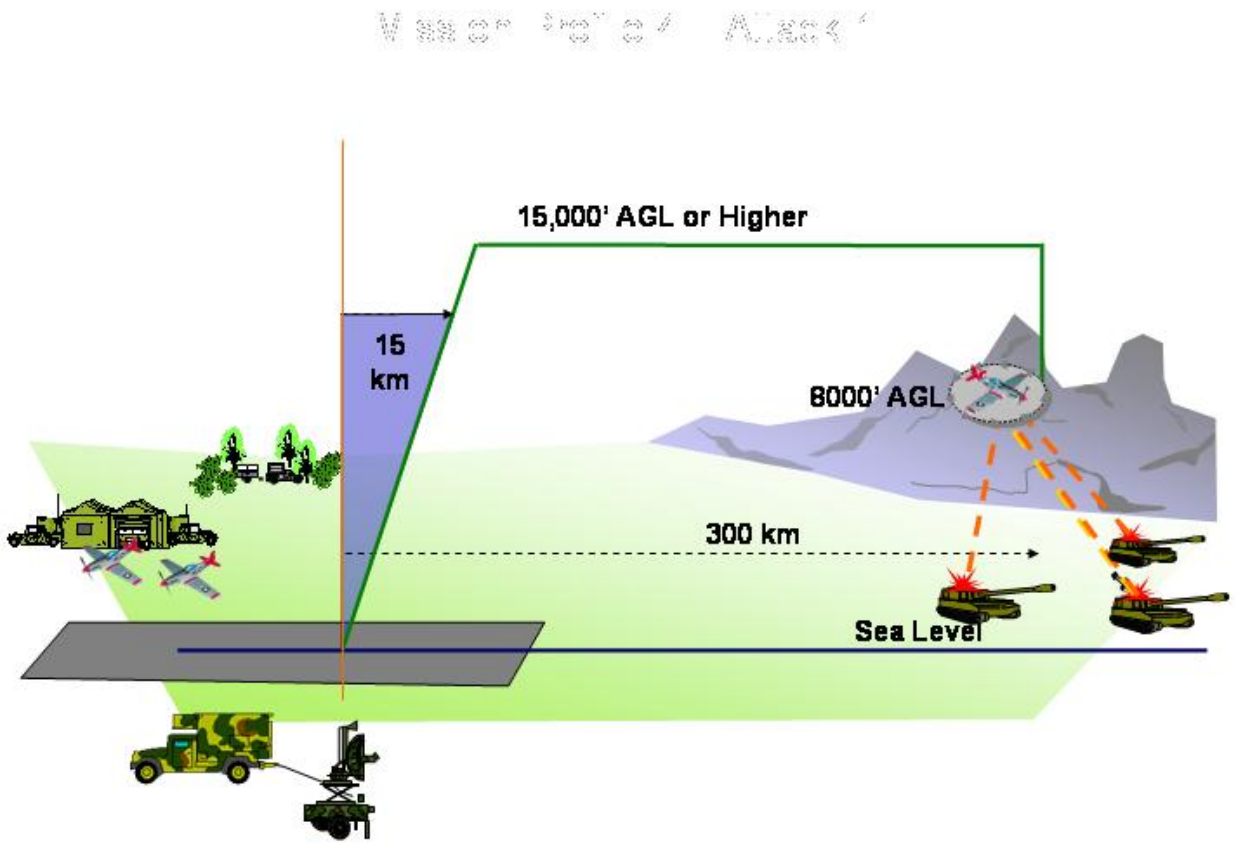


Figure 10.6 Mission Profile C - Attack 1

10.6 Conclusions

Technology development may lead to more mission objectives that are possible or desired for UAVs. The mission profiles discussed in this chapter represent the missions that UAVs have and are currently carrying out.

10.7 Glossary

10.7.1 Notations

a	Speed of Sound	ft/s
AR	Wing Aspect Ratio	
bhp	Engine Brake Horsepower	bhp

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C_{bhp}	Propeller Specific Fuel Consumption	lb/hr/bhp
C_{D_0}	Drag Coefficient at $C_L = 0$	
$C_{L_{MinDrag}}$	Minimum Drag Lift Coefficient	
C_p	Power Coefficient	
D	Diameter of Propeller	ft
D	Drag	lb
$D_{MinDrag}$	Drag at Minimum Drag	lb
e	Oswald Efficiency Factor	
E	Endurance	hr
J	Advance Ratio	1/rev
L	Lift	lb
$L_{MinDrag}$	Lift at Minimum Drag	lb
M_{Tip}	Propeller Tip Mach Number	
n	Propeller Angular Velocity	rev/s
P_a	Ambient Pressure	psf
P_T	Total Pressure	psf
R	Gas Constant	1716 $\frac{ft \cdot lb}{slug \cdot {}^\circ R}$
R	Range	ft
S	Wing Plan Form Area	ft^2
SFC	Specific Fuel Consumption	1/hr
T	Temperature	Rankine
T	Thrust	lb
$\frac{t}{c}$	Propeller Thickness to Chord Ratio	
V_T	True Airspeed	ft/s
$V_{MinDrag}$	Minimum Drag Velocity	ft/s
W	AV Weight	lb
W_f	AV Weight Empty Weight	lb
W_i	AV Weight Gross Take-off Weight	lb

10.7.2 Greek Symbols

γ	Ratio of Specific Heats	1.4 for Air
ρ	Air Density	$slug / ft^3$
ρ_a	Ambient Air Density	$slug / ft^3$
η_p	Propeller Efficiency	
$\eta_{p,corrected}$	Corrected Propeller Efficiency	

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Appendix C

UNMANNED AIRCRAFT SYSTEM (UAS)

FLIGHT TEST METHODOLOGY

"VERTICAL TAKEOFF & LANDING AIRCRAFT PERFORMANCE"

19 May 2006

Prepared For:

U.S. Army Aviation Technical Test Center
Fort Rucker, Alabama 36362

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This manual provides supplemental information for use in testing the performance of vertical takeoff and landing (VTOL) unmanned aircraft systems (UAS). It modifies and adds to the rotary wing performance theory and test techniques given in the U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), "Rotary Wing Performance." It has been prepared to address the unique test requirements for UAS as opposed to manned aircraft. Both manuals should be used in conjunction in developing a performance flight test program for fixed wing UAS systems. This manual follows the same numbering scheme and format as the USNTPS-FTM-No. 106 to facilitate easy reference between the two manuals. Additionally, each chapter is stand alone and incorporates an independent Table of Contents.

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Chapter 8: Climb and Descent Performance

Chapter 9: Transition Flight

Chapter 10: Standard Mission Profiles

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CHAPTER 1
INTRODUCTION TO UAS FLIGHT TESTING

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1. Introduction to UAS Flight Testing

1.1 Objective

This manual provides supplemental information for use in testing the performance of vertical takeoff and landing (VTOL) unmanned aircraft systems (UAS). It modifies and supplements the rotary wing performance theory and test techniques given in the U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM No.-106), "Rotary Wing Performance." It has been prepared to address the unique test requirements for UAS as opposed to manned aircraft. This manual is not intended to replace the USNTPS-FTM-No. 106 and will only reiterate relevant material that will facilitate a more stand alone document. The goal is to aid test engineers in developing a performance flight test program for VTOL UAS.

1.2 Organization

1.2.1 Document Organization

This manual is organized into individual chapters. While an attempt is made to treat each subject chapter completely and independently, that is not always possible due to the very nature of the interdependence of performance parameters. For example, engine performance data (power available) will influence hover performance. Likewise, power available data and level flight power required data will influence climb performance. Where applicable, reference to a preceding chapter will be highlighted to allow for correlation.

Chapter 1 provides an overview of the manual. Described with that chapter are unique differences between UAS and manned aircraft, UAS size classification, VTOL classification, and particular safety, instrumentation, and facility considerations for UAS flight testing.

Chapter 2 is concerned with pitot/static system performance. The source of system errors is discussed along with some methods of reducing those errors. Test methods of global positioning system (GPS) Speed Course, GPS 3-Leg, Laser Tracking, Pace, Tower Flyby, Trailing Sources, and Radar Altimeter are described.

Chapter 3 covers the area of powerplant assessment for turbine, reciprocating, and electric power plants. Discussion includes engine/rotor controls and displays, engine/rotor operating procedures, engine/rotor start, runup, and shutdown characteristics, engine acceleration characteristics, engine trim response, engine/rotor stability, torque matching, engine limiting characteristics, and power contribution in descent.

Chapter 4 describes the area of powerplant performance characteristics for turbine, reciprocating, and electric power plants. This chapter includes the relationship of

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corrected engine parameters: corrected engine shaft horsepower, corrected engine gas generator speed, corrected fuel flow, corrected power turbine inlet temperature, and specific fuel consumption. Inlet recovery is discussed along with specification compliance.

Chapter 5 details the hover performance and establishes the relationship between engine performance (power available) and airframe performance (power required). Free-flight test methods are detailed in this chapter. Test methods for the free-flight hover performance include ground reference using the hover height measuring device and air reference methods.

Chapter 6 deals with vertical climb, with both test-day vertical climb performance and generalized vertical climb performance. The relationship of power available and hover power required are brought together in the vertical climb. Ground referenced and air referenced methods are discussed.

Chapter 7 discusses level flight performance. This chapter deals with both test-day data and generalized level flight performance. Again, this chapter draws together engine performance (power available) with airframe performance (power required) and develops important aircraft performance parameters (range, endurance, and maximum level flight speed). The theory will differentiate for non-augmented rotor/lift systems and augmented rotor/lift systems. The test method using the constant weight to pressure ratio (W/δ) and rotor speed to temperature ratio ($N_R/\sqrt{\theta}$) is compared to the constant weight to density ratio (W/σ) method. This method will work for the turbine and reciprocating power plants; however, electric power plants require some different test techniques to prevent battery discharge which has profound effects on the test data.

Chapter 8 discusses climb and descent performance. This chapter covers test-day data as well as generalized climb and descent performance. The chapter brings together engine performance (power available) and airframe performance (power required in level flight) to predict aircraft performance parameters (rate of climb or descent). The theory will differentiate for non-augmented rotor/lift systems and augmented rotor/lift systems. Test methods of power correction factor (K_P), weight correction factor (K_W), generalized climb performance, and test day performance are discussed.

Chapter 9 discusses the standard mission profiles pertinent to VTOL UAS. This chapter culminates the performance testing by bringing together the necessary information for testing the UAS as a whole. It also mentions system and payload testing that, even though outside the scope of this document, can be accomplished simultaneously while capturing certain performance parameters.

1.2.2 Chapter Organization

For continuity, each chapter is organized similar to others, where possible. Following the chapter introduction, Section 2 gives the purpose of the test. Section 3 is a review of the applicable theory. Section 4 discusses the test methods and techniques

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along with the data requirements and safety precautions applicable to those methods. Section 5 discusses data reduction, both manual and computer aided. Section 6 is the data analysis. Section 7 covers the relevant mission suitability aspects of the performance parameter. Section 8 discusses the relevant specification compliance issues. Section 9 is a glossary of terms introduced in the chapter. Finally, Section 10 is a listing of applicable references.

1.3 UAS vs. Manned Aircraft

Important differences between manned and unmanned vehicle testing can be found in Section 3 of each chapter of this document. Differences or necessary additions for UAS testing theory will also be explained in this section of each chapter.

When starting a flight test program for UAS, it is important to remember that there are many differences in performance and operation from manned aircraft. Certain flight testing procedures cannot be accomplished with some UAS due to control and/or performance limitations. Also, the air vehicle operator (AVO) has the difficult task of performing flight test maneuvers without the same level of situational awareness available to a pilot of a manned vehicle.

One difference is the flight control system. A good understanding of the particular UAS flight control system is essential. Most autonomous UAS employ a flight control system that severely limits the types of maneuvers performed by the aircraft. An example of where the flight control system could potentially limit test capabilities is climbing flight. Many autopilot systems use a single airspeed when a climb is commanded, which precludes the testing of multiple airspeed test points when validating the associated airspeeds for maximum rate of climb, maximum climb angle, minimum time to climb, and minimum fuel burn. If this is the case for the test aircraft, the flight control logic will need to have supplementary code so that various airspeed test points can be flown. The alteration of the flight control software from the production configuration to the test configuration will require flight release from the airworthiness authority.

A solid understanding of each AVs control logic sequencing and lost link procedures, including failure modes, are central to the planning process for the flight test engineering team. As many UAS operate without redundant data links, quick response to failure modes must be possible with limited situational awareness and interface with the AV. With this, a well understood lost link procedure is essential to risk management.

Most modern UAS flight control designs do not require the AVO to “close the loop” on altitude or airspeed; many are designed to be “hands off” requiring the AVO to “fly” the aircraft using only a map display and a mouse-like control for input of desired waypoints. This interface frequently affects the fidelity of the downlink data; therefore, the test team and AVO need to be aware that some of the flight critical data such as airspeed and altitude being transmitted through the downlink may not be of the same resolution as that being provided to the onboard autopilot. An onboard data logger is the

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preferred method of recording high-fidelity data for use in most flight test operations. After the test flight is completed, the data are retrieved from the data logger for use by the flight test engineers.

Consideration should be given to the UAS sensitivity regarding weather and terrain. Turbulent weather can have a dynamic impact on the flight control system's ability to maintain condition for flight test points. This is particularly a concern for the smaller UAS, but acceptable flight test conditions can also be difficult to sustain with larger UAS in a highly unstable weather environment. This may lead to repeating selected test procedures to ensure test accuracy. The weather tolerances described in Section 4 of all chapters of this document offer desirable test condition guidelines for reliable test data.

1.4 UAS Size Classifications

The following UAS classification system can be used to better apply test techniques to different sized UAS and is based on the classifications given in Aviation Engineering Directorate Standard Operating Procedure AE70-62-1-UAS. The system is meant to help the test team plan for flight days and set standards for weather conditions and repetition of test procedures. Some UAS may bridge two categories and techniques may apply from one or both classes. Most UAS are large enough to cause serious injury in the event of a mishap and test procedures should be designed to ensure personnel maintain safe distances during takeoff and landing for all classes.

1.4.1 Small UAS

The Small UAS class is defined as aircraft weighing up to 55 lb with maximum airspeed of 120 knots indicated airspeed (KIAS). Examples of UAS in this class are Honeywell Micro AV, Naval Research Labs Spider, and TAG-M80. The small physical size of this class creates aircraft sensitive to even slight weather phenomenon to include thermal activity and wind For this reason, it is suggested these UAS be tested on days with calm winds (winds <3 kt). Any winds, especially at the surface with gusts, would most likely produce poor test results, could exceed the UAS ability to maintain course, and in extreme conditions, result in loss of AV control and possible injury to ground personnel. For very small systems, it may become necessary to perform certain maneuvers in large enclosed buildings.

1.4.2 Light UAS

The Light UAS class is defined as aircraft weighing from 55 to 1,320 lb and with maximum airspeeds of 200 KIAS. Examples of UAS in this class are Yamaha R-Max, Sikorsky DragonWarrior (Cypher II), and Schiebel Camcopter. These UAS are less sensitive to high wind conditions and turbulence; nevertheless, it is always more prudent to test on calm wind days.

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1.4.3 Medium/Large UAS

The Large UAS class is defined as aircraft weighing greater than 1,320 lb with no limit on airspeed. Examples of this class are Fire Scout, Hummingbird, SAIC Vigilante, Bell Eagle Eye, Boeing Heliwing, and Boeing Rotary Wing Canard.

1.5 VTOL Classifications

A VTOL UAS trade study was conducted to look out in the 5- to 7-year timeframe for types of current and future VTOL UAS aircraft which are likely to be employed. The study resulted in 14 classes of systems:

1. Standard helicopter
2. Off-axis tandem
3. Shrouded tandem
4. Coaxial
5. Tilt-rotor
6. Vectored-thrust
7. Tail sitter
8. Rotary wing canard
9. Rotor-gyro
10. Augmented ducted fan
11. Tilt-prop
12. Tilt-wing
13. Tilt-duct
14. Tilt-shaft

The first 10 classes have been developed or are being developed as of 2006 while the last 4 (tilt-prop, tilt-wing, tilt-duct, tilt-shaft) are manned systems which may be converted to an unmanned system in the 5- to 7-year timeframe.

All the classes fall into one of two categories: (1) nonaugmented rotor/lift system and (2) augmented rotor/lift system. The former uses the same propulsion system for hover and forward flight while the latter employs an augmented system for hover. Figure 1.1 and Table 1.1 illustrate the results of the study and summarize the differences among classes of VTOL UAS for how major modes of flight are handled, respectively.

The two categories will be used throughout this manual to differentiate when prescribing techniques which may be pertinent only to that group. Additionally, some chapters may refer to the specific class of VTOL systems, addressing unique test techniques for that specific configuration.

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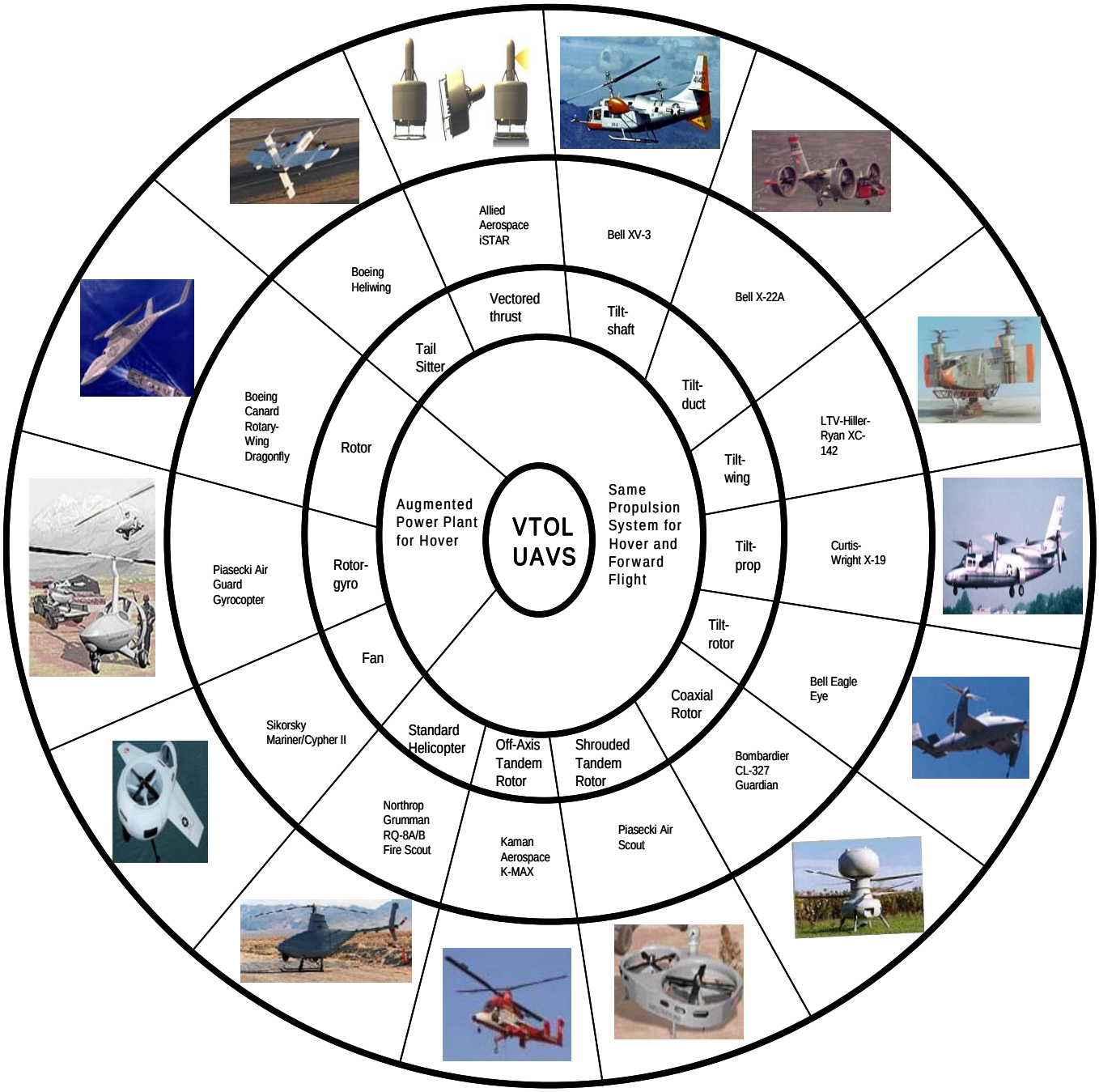


Figure 1.1 VTOL Wheel

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VTOL UAV	Anti-Torque	Transition (VTOL - FWD FLT)	Turns	Engine Failure	Landing	Example	
STD Helicopter	Tail Rotor	Main Rotor Pitch Change	Main Rotor Pitch/Roll + Tail Rotor	Autorotation	Vertical	RQ-8A/B Firescout	Non-Augmented Rotor/Lift System
Off Axis Tandem	Counter-rotating blades	Main Rotor Pitch	Main Rotor Pitch/Roll + Differential Roll + Speed Differential	Autorotation	Vertical	K-max	
Shrouded Tandem Rotor	Counter-rotating blades	Main Rotor Pitch Change	Main Rotor Pitch + Differential Thrust + Differential Speed	Autorotation	Vertical	Piasecki Air Scout	
Coaxial	Counter-rotating blades	Main Rotor Pitch + Speed Differential	Main Rotor Pitch/Roll + Speed Differential	Autorotation	Vertical	CL-327 Guardian	
Tilt Rotor	Counter-rotating blades + Vertical Stabilizer	Rotate Rotors gradually to FWD	VTOL Main Rotor Pitch + Differential Thrust + Vertical Stabilzer; FWD Aileron + Rudder	Autorotation VTOL, Glide FWD	Vertical or Rolling Conventional	Eagle Eye	
Tilt Shaft	Counter-rotating blades + Vertical Stabilizer	Rotate Shaft gradually to FWD	VTOL Main Rotor Pitch + Differential Thrust + Vertical Stabilzer; FWD Aileron + Rudder	Autorotation VTOL, Glide FWD	Vertical or Rolling Conventional	Bell XV-3	
Tilt Propeller	Counter-rotating blades + Vertical Stabilizer	Rotate Propeller gradually to FWD	VTOL Main Rotor Pitch + Differential Thrust + Vertical-Stabilzer; FWD Aileron + Rudder	Autorotation VTOL, Glide FWD	Vertical or Rolling Conventional	Curtis-Wright X-19	
Tilt Wing	Counter-rotating blades + Vertical Stabilizer + Ailerudders (VTOL)	Rotate Wings gradually to FWD	VTOL Main Rotor Pitch + Differential Thrust + Vertical Stabilzer; FWD Aileron + Rudder	Autorotation VTOL, Glide FWD	Vertical or Rolling Conventional	LTV-Hiller-Ryan XC-142	
Tilt Duct	Counter-rotating blades + Vertical Stabilizer	Rotate Rotors gradually to FWD	VTOL Main Rotor Pitch + Differential Thrust + Vertical Stabilzer; FWD Aileron + Rudder	Autorotation VTOL, Glide FWD	Vertical or Rolling Conventional	Bell X-22A	
Vectored Thrust Ducted-Fan	Vectored-thrust	Vectored-thrust	Vectored-thrust	Glide	Vertical	iSTAR	
Tail Sitter	Counter-rotating blades + Vertical Stabilizer	Gradual Pitch over	VTOL Main Rotor Pitch + Speed Differential; FWD Aileron + Rudder	Autorotation VTOL, Glide FWD	Vertical or Rolling Conventional	Heliwing	
Augmented Rotor	Air blown out rotor tips	Engine Flow bypass Power Turbine to exhaust nozzle for pure turbofan, rotor becomes fixed	VTOL Main Rotor Pitch + Vertical Stabilzer; FWD Aileron + Rudder	Autorotation VTOL, Glide FWD	Vertical or Rolling Conventional	Dragonfly	Augmented Rotor/Lift System
Augmented Rotor-gyro	N/A	Main Unloaded, Ducted Fan FWD Thrust	Vectored Thrust/Rudder	Autorotation	Vertical w/ small roll	Gyrocopter	
Augmented Fan	Counter-rotating blades + Vectored-thrust	Horizontal Ducted Fan Uses more of available power	VTOL Main Rotor Pitch/Roll + Speed Differential; FWD Aileron + Vectored-Thrust	VTOL Autorotation; FWD Glide	Vertical or Rolling Conventional	Cypher II	

Table 1.2 VTOL Classification

1.6 Reasons for Performance Flight Testing

1.6.1 Performance Testing and Simulation

A vital concept to be emphasized before any initial flight testing discussion is that flight testing should be balanced with system simulation. The Department of Defense policy (AR 5-11) stresses a synergistic approach to the development of military systems. The concept is to use the benefits of both simulation and flight testing to compliment one another. Physics based modeling and simulation, though expensive, can be used to support flight testing as well as failure and mishap investigation. Nonphysics based or lower fidelity simulation, in conjunction with flight testing, supports a buildup test

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methodology by effectively supporting UAS operator training, while at the same time providing a less expensive model of the expected behavior of the AV under controlled input conditions. This combination aids the flight testing process since the simulation facilitates improved AVO mission planning and maneuver execution while allowing test engineers to better predict actual AV performance. Spiral testing and development allow for refined aircraft models and generation of better AV simulations which feedback to benefit the development process of the UAS and the ultimate customer, the U.S. Warfighter.

Performance flight testing for development aircraft can often lead to changes and improvements to the UAS system. The AV can be altered such that the standard gross takeoff weights and configurations of the test aircraft may be different from the production aircraft, leading to inaccurate performance data. Additionally, the ground control station (GCS) flight control system software may have numerous upgrades throughout the development and initial fielding effort. The software configuration should always be monitored as software changes may dramatically change performance characteristics and affect the test techniques needed to acquire specific test points.

1.6.2 Mission Suitability

Each aircraft or weapon system is developed to meet or fulfill a mission requirement. This mission is comprised of many sub-missions and tasks and is performed in many areas of the world under different conditions. Mission suitability is the ability of the VTOL aircraft to fulfill its mission. To evaluate mission suitability, the test engineer must have a thorough understanding of the entire mission and the environment in which it is to be performed. Inherent to mission suitability determination is the conclusion whether the aircraft can perform the intended mission safely, whether capability can be substantially improved, or whether a particular characteristic should be avoided in the future. Based on the conclusions of the flight test engineers, a decision can be made on whether the platform can be modified to carry weapons, and if so, which weapons can be carried and launched/dropped safely from the UAS. Additionally, during performance evaluations, specific characteristics may be identified that, if appropriately applied during certain phases of flight, may enhance mission capability and should be recommended as standard operating procedure.

It is important to remember that while UAS have primarily been used for reconnaissance, surveillance, and target acquisition (RSTA) missions in the past, their role is ever expanding to include attack missions and possible future intercept missions. Mission suitability will likely dramatically change over time as the role of UAS expands.

1.6.3 Specification Compliance

The VTOL UAS may be developed by a manufacturer to meet certain specifications. If applicable and to insure contract compliance, the government must check the performance parameters of the system against the contractual requirements and/or specifications. This consideration demands accurate quantitative data derived as the result of flight testing.

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1.6.4 Operating Manual Performance

On occasion, the performance flight testing of the VTOL UAS will contribute to the development or update of a handbook for the operator. Additionally, the performance data may integrate into development of a flight performance model for simulation activities. If vendor-supplied performance data are available, it may become necessary to verify such data and augment as appropriate per the test objectives.

1.7 Approach to Performance Flight Testing

1.7.1 Safety/Risk Management

The paramount concern during any flight, and especially during flight testing, is the safety of those involved. Although risk is inescapable in any test activity, it can be minimized through proper risk management techniques. Thus, risk management is a tool whereby safety can be optimized. All flight operations and all flight testing must be conducted with the principle of safety in mind. Safety is promoted by detailed knowledge of the system, thorough planning, adherence to the plan, complete crew coordination (both flight crew and support crew), a rigorous lookout policy, and disciplined flight conduct. The key to risk management is knowledge, planning, and execution. Remember, the first priority is to fly the aircraft safely; the evaluation process becomes secondary.

Medium and large UAS where a pilot could occupy the cockpit, a man-in-the-loop could dramatically aid in risk reduction for a developmental aircraft. This might require the UAS to undergo alterations so that the pilot could control the AV from the cockpit with the type of controls in a manned aircraft (i.e., standard helicopter cyclic, collective, rudder). This feeds into the concept of incremental buildup described in Section 1.7.3 where the testing would move from a place of known response (manned aircraft) to unknown response (unmanned). If man-in-the-loop testing is determined to be feasible for a particular UAS, the concepts and test approaches described in USNTPS-FTM-No. 106 would apply, with the exception of reciprocating engine assessment and performance, if applicable, described in chapters 3 and 4 of this document. Man-in-the-loop testing may also require air worthiness certification or an airworthiness release (AWR) before flight testing can begin.

1.7.2 Concept

The foundation of flight testing in general, and performance testing in particular, is dependent upon identifying the applicable variables for a given condition based on sound theory with appropriately applied assumptions and simplifications. Ideally, flight test is conducted by changing one parameter, observing the result, while maintaining other parameters consistent and stable. This, however, does not preclude the combination of tests from different chapters or sections into one flight test plan. Educated and efficient test planning and careful observation yield test results that address specific

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aspects of performance which then can be analyzed in the context of the intended mission.

1.7.3 Incremental Buildup

The concept of incremental buildup is one of the most important practical and philosophical concepts in the flight test community. Practically, buildup is the process of proceeding from the known to the unknown in an incremental, methodical pattern. Philosophically, buildup is the manner in which flight testing should be structured; beginning with the best documented, least hazardous data points and proceeding toward the desired end points, always conscious of the aircraft, AVO, and test limits. Buildup is applicable to both the overall structure of the evaluation and to each of the subtasks. Although there should be no surprises in flight testing, unexpected results are encountered. A major theme contained in the buildup concept is that in the event a data point yields an unexpected result or a series of data points suggests an unexpected trend, the flight test ceases until the results are analyzed and explained. Repeating previously completed tests may be necessary to re-establish the buildup concept once unexpected results are analyzed.

1.8 Test Facilities

The facilities should be considered before starting a flight test program. In particular, the typical weather at a test facility and the impact that weather might have on the UAS under test must be considered. For flight test operations leaving restricted airspace, a Certificate of Authorization from the Federal Aviation Administration may be required. An AWR may also be required before a flight test program can begin. Each test center will have a definition of what they consider a stable air mass. It will be a combination of wind speed and direction variance, temperature variance, and pressure variance. The temperature and pressure variation often lead to changes in the wind speed and direction. The specific test center should be consulted for these tolerances so that the test criteria which require stable air mass can be quantified. Throughout this document where stable air mass is used it will refer to this section.

1.9 Generalized Performance Data vs. Test-Day Data

The performance of the UAS can be evaluated under the conditions that exist at the time of evaluation. Data analyzed and conclusions made relative to the performance of the UAS for that given set of test day conditions are usually termed test-day data. Usually, data of this type are termed test-day data. When the test-day data are analyzed using a generalized method of data reduction, the performance of the UAS can be determined at conditions of interest other than those at which the data were gathered. The distinction of test-day data versus generalized data reduction methods will be made throughout this manual as appropriate.

1.10 Instrumentation

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Another important consideration is the planning and use of test instrumentation to map parameters that would normally be associated with a manned aircraft flight deck. These vital pieces of information can aid in deterring mishaps and critical component failures. Whenever possible, UAS should be extensively instrumented during testing (air data booms, control surface potentiometers, current measurements, temperatures, pressures, etc.). For example, experience with testing UAS has shown that most engines integrated into the AV will develop cooling problems at some point in their flight envelope expansion. To obtain a better understanding of the airflow and cooling properties of the powerplant, extensive instrumentation should be used in the cooling air inlet as well as instrumentation to monitor critical powerplant temperatures so that these problems can be addressed in developmental/operational testing rather than by the Warfighter in the field. When dealing with UAS rotorcraft and rotorcraft in general, it is good practice to instrument for vibration and consider possible vibration impact on the rest of the AV. External cameras or a rotor monitoring system may also be mounted to aid testers in determining AV behavior during flight test.

If feasible, the essential parameters should be instrumented and fed through the downlink to the GCS so that the data can be monitored in real time by the AVO and test engineers. It is preferred practice to partition the data into the smaller flow of flight critical parameters directed to the AVO and Mission Commander and the larger flow of detailed information presented to a test engineering team. However, an alternate data link may be required to provide sufficient bandwidth for transferring required engineering data. The data sampling rates are test dependent; some tests require more data points for reliable data reduction and analysis.

1.11 Glossary

1.11.1 Notations

AV	Air Vehicle	
AVO	Air Vehicle Operator	
FTM	Flight Test Manual	
GCS	Ground Control Station	
HHMD	Hover height measuring device	
KP	Power correction factor	
KW	Weight correction factor	
RSTA	Reconnaissance, Surveillance, & Target Acquisition	
UAS	Unmanned Aircraft System	
$N_R/\sqrt{\theta}$	Rotor speed to root of temperature ratio	%, RPM
$\frac{W}{\delta}$	Weight to pressure ratio	lb
$\frac{W}{\sigma}$	Weight to density ratio	lb

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1.12 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 2
PITOT/STATIC SYSTEM PERFORMANCE

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2. Pitot/Static System Performance

2.1 Introduction

Performing tests of the unmanned aircraft system (UAS) pitot/static system is a vital part of the overall flight testing since most UAS flight profiles are managed by pressure altitude, indicated airspeed (IAS), and vertical speed, each derived from the pitot/static system. The instruments must transmit precise readings of both altitude and IAS to the ground control station (GCS) since the testing will often call for the air vehicle operator (AVO) to hold a specific airspeed, altitude, or both. Additionally, errors in the pitot/static system have critical implications for flying at speeds near never exceed airspeed (V_{NE}), maneuvering, loitering with sensitive payloads which are dependent on precise airspeeds, or instrument flight rules (IFR) missions. Flights in the National Airspace System (NAS) require obtaining a Certificate of Authorization (COA) with the Federal Aviation Administration (FAA). For detailed requirements refer to AFS-400 UAS Policy 05-01. Ground testing in preparation for this type test is described in Section 2.3.1 and flight test methods are detailed in Section 2.4. The U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106) Chapter 2, "Pitot/Static System Performance," is referenced in a number of the methods and should be used in conjunction with this chapter.

2.2 Purpose of the Test

The intent of the test is to fully characterize the instrument error and the position error of the air vehicle (AV). This will determine the airspeed and altimeter correction for the remainder of the flight testing. The instrument calibration is accomplished through ground test and the position error is determined by one of the flight test methods discussed in the following sections.

2.3 UAS vs. Manned Aircraft

2.3.1 Differences between UAS and Manned Aircraft

For rotary wing UAS with standard pitot/static systems, the theory set forth in USNTPS-FTM-No. 106 "Rotary Wing Performance" will be applicable. For UAS that do not use a standard pitot/static system or any pitot/static system, the theory provided below will need to be understood.

2.3.2 UAS Specific Theory

2.3.2.1 UAS with Pitot/Static System

Ground testing of the AV pitot/static system involves connecting a calibrated test set directly to the aircraft pitot tube and static ports to simulate various altitudes and

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indicated airspeed combinations. The intent is to fully characterize the instrument error portion of the total pitot-static error budget. The airframe technician will control the test set and the AVO will observe the response of the aircraft pitot-static system across the data link in the GCS. The GCS altimeter is set to 29.92 in. Hg (1013.25 millibars) to correlate to the replicated static pressure for the static port from the test set. This provides a check of the system before flight test so that the pitot-static system can be calibrated for its instrument error. The lag in the pitot/static system can also be determined by comparing the input-to-response time difference between the test set and the AV pitot/static system. Additionally, any lag in the UAS GCS data link must be assessed and can be found by observing the total lag of the system and then subtracting the pitot/static system lag. When calibrating the pitot/static system from flight test data, the high resolution data logged by the onboard instrumentation package should be used. The intent of the lower resolution data being sent to the GCS is for the AVO to maintain situational awareness.

2.3.2.2 UAS without Pitot/Static System

UAS without a pitot/static system may require alternative testing to calibrate the speed instrumentation. This would apply to UAS systems which employ global positioning system (GPS), inertial navigation systems (INS), radar altimeters, or a combination in providing speed and altitude information. It may be necessary, however, to use a modified method in this chapter to provide redundant or truth information if necessary to evaluate those non-standard systems. In obtaining a GPS altitude solution, it is not uncommon to realize elevation errors of 20 ft or more, unless differentially corrected. However, ground speed error is usually less than 1 kt. Differential GPS (DGPS), while being used on some UAS for virtually eliminating altitude errors, will still not be used for landing where the altitude error becomes important. The altitude error from GPS will become insignificant at appreciable altitudes.

2.4 Test Methods and Techniques

For manned aircraft, the traditional approaches to the flight component of pitot/static system testing can be accomplished by one of six methods that are fully detailed in Section 2.4 of the USNTPS-FTM-No. 106. These include the measured course, trailing source, tower flyby, space positioning, radar altimeter, and pace aircraft methods. Additional means of testing the pitot/static system for UAS are ground-based laser tracking and two GPS methods, one of which (GPS measured course) is simply a new twist on the USNTPS measured course method.

The test program should be a matrix of airspeed, weight, altitude, and yaw. The tolerances are set under each test method. However, the flight control system of each UAS dictates the spacing between the indicated airspeed data points, since the “airspeed hold” tolerance of the flight control system may be greater than the airspeed increments. The test points should be concentrated in the regimes where both the effects of rotor downwash and the rate of change of rotor blade angle of attack with respect to airspeed are greatest. For a standard helicopter, the effects of rotor downwash will be the greatest

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at slower forward airspeeds and the rate of change of rotor blade angle of attack with respect to airspeed will be the greatest at high airspeeds. These conditions have the most drastic impact on pitot tube and static port performance. The rotor downwash could change the boundary layer on the airframe leading to an incorrect static pressure reading, as well as leading to incorrect total pressure indication. The angle of attack rate of change influences the inflow angle to pitot tube and can lead to lower total pressure measurement at higher angles of attack.

2.4.1 GPS Speed Course (2-Leg Method)

2.4.1.1 Overview

The two-leg GPS method is similar to the standard measured course method detailed in USNTPS-FTM-No. 108, Section 2.4.1, and has been developed by Dr. Ralph Kimberlin at University of Tennessee Space Institute (UTSI). This method has been used recently on a Piper Navajo with Dr. Kimberlin during a flight test course at UTSI.

The procedure is to fly reciprocal headings at the same indicated airspeed and altitude. The indicated airspeed, altitude, heading, and outside air temperature (OAT) are recorded in the onboard data logger. The ground track and start and end points are recorded by the GPS so that the course and distance can be determined. Once the ground speed and ground track are stable for 5 to 10 seconds, these parameters should be recorded for the desired run length. The AVO must ensure a constant heading during the run ($\pm$ 2°). The GPS ground speed (*Ggps*) is the speed given by the onboard GPS. The calculated ground speed (*GS*) in the direction of the heading is the GPS ground speed times the cosine of the difference between the heading and the ground track angles (θ). The same reference, magnetic or true, should be used for heading and ground. The calculated ground speed for the reciprocal runs can be averaged to determine the true airspeed (V_T) for that test point. Refer to Figure 2.1A for an illustration of the GPS method. Use altitude and OAT to convert to calibrated airspeed. The calibrated airspeed from the GPS can then be compared with the instrument readings from the pitot/static system. The accuracy of the data depends upon the wind vector remaining constant for both legs. Wind data should be taken at the location of the speed course from special instrumentation such as portable weather station that measures wind speed and direction, surface temperature, and surface pressure. The changes in temperature and pressure typically forecast changes in wind speed and/or direction.

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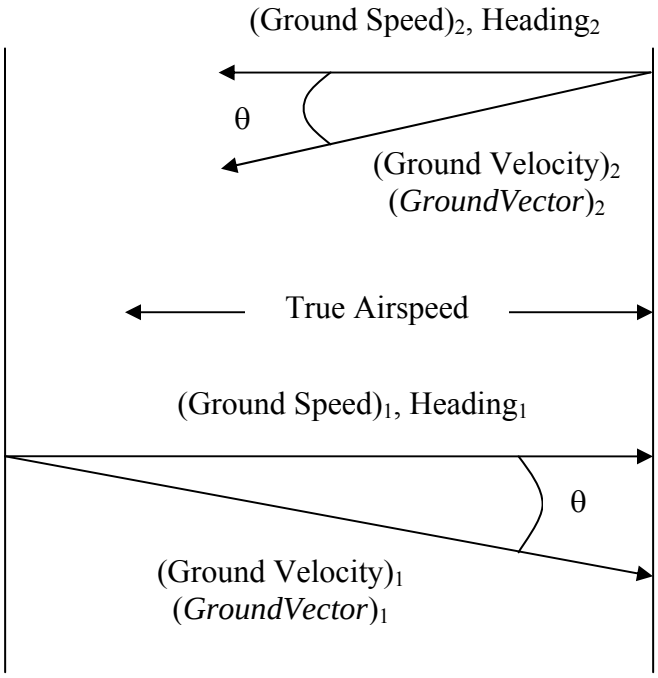


Figure 2.1A GPS Method Graphical Representation

The equations are as follows:

$$GS = Ggps * \cos(\theta) \tag{2.3}$$

$$V_T = \frac{GS_1 + GS_2}{2} \tag{2.4}$$

An efficient way to fly the conditions is to fly the first speed leg, then turn around and fly the second leg and the first leg of the next speed before turning around. Continue to fly two legs (different speeds) before each turn.

2.4.1.2 Data Required

GPS Ground Speed, Ground Track, Heading, V_o, H_{Po}, OAT, N_R, Q, CG, GW, α_o, β_o , Aircraft Configuration (weapons, external fuel pods, other aerodynamic effects, clean configuration, etc.), wind speed and direction

2.4.1.3 Test Criteria

1. Steady level flight
2. Constant altitude
3. Constant airspeed
4. Constant heading

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5. Stable air mass as defined in Section 1.8

2.4.1.4 Data Requirements/Tolerances

1. Stabilize AV for 30 seconds before recording
2. Stabilize ground speed and ground track
3. $V_o \pm 0.5$ KIAS or limits of flight control system
4. Altitude $\pm$ 10 ft or limits of flight control system
5. $N_R \pm 2\%$
6. Heading $\pm 2^o$

2.4.1.5 Safety Considerations/Risk Management

There are no safety issues directly relating to this technique differing from UAS standard operating procedures.

2.4.2 GPS 3-Leg Method

2.4.2.1 Overview

The 3-Leg GPS method of calibrating the pitot/static system involves flying the UAS at a constant airspeed and altitude over three or more legs. Stabilize the UAS and record indicated airspeed, pressure altitude, and OAT. Once the ground speed and ground track are stabilized, these parameters should be recorded for the desired run length. Change the heading by at least 60°, but not more than 120°, and repeat the previous step at the same altitude and airspeed for the second leg. Airspeed is the more important parameter to maintain. Following the second run, change the heading by at least 60°, but not more than 120° for the third leg, flying at the same airspeed and altitude. Once at least three legs are completed, the desired information can be calculated.

The AVO must ensure a constant ground track during the run. The onboard GPS, or an installed GPS, is used to determine ground speed and ground track. It is assumed that the UAS true airspeed along with wind speed and direction are constant during all three or more runs. Wind speed, wind direction, and true airspeed can then be determined by solving three equations to find the three unknowns. The ground velocity (speed and direction) is the vector sum of the true airspeed and wind. This is illustrated in Figure 2.1.

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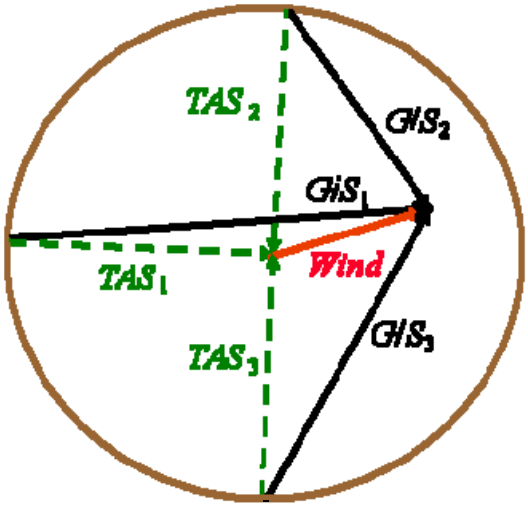


Figure 2.1 GPS Method Graphical Representation

A downloadable Excel solution, "GPS_PEC.xls" is available on the National Test Pilot School's website: www.ntps.edu/downloads.htm. The general solution shown by Doug Gray in "Using GPS to Accurately Determine True Airspeed" is given in Figure 2.2. The units for airspeed are knots and the units for ground track are degrees. The solution is given in a form to expedite entry into a spreadsheet program with example speeds and tracks shown in cells B1 to B6.

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	A	B	C
1	Speed1	140	140
2	Track1	192	192
3	Speed2	112	112
4	Track2	283	283
5	Speed3	120	120
6	Track3	20	20
7	X1	=B1*SIN(PI()*(360-B2)/180)	
8	Y1	=B1*COS(PI()*(360-B2)/180)	
9	X2	=B3*SIN(PI()*(360-B4)/180)	
10	Y2	=B3*COS(PI()*(360-B4)/180)	
11	X3	=B5*SIN(PI()*(360-B6)/180)	
12	Y3	=B5*COS(PI()*(360-B6)/180)	
13	M1	=-1*(B9-B7)/(B10-B8)	
14	B1	=(B8+B10)/2-B13*(B7+B9)/2	
15	M2	=-1*(B11-B7)/(B12-B8)	
16	B2	=(B8+B12)/2-B15*(B7+B11)/2	
17	WX	=(B14-B16)/(B15-B13)	
18	WY	=B13*B17+B14	
19	Wind Speed	=SQRT(B17^2+B18^2)	20.6
20	Wind Direction	=MOD(540-(180/PI()*ATAN2(B18,B17)),360)	314.8
21	TAS	=SQRT((B7-B17)^2+(B8-B18)^2)	130
22	Heading 1	=MOD(540-(180/PI()*ATAN2(B18-B8,B17-B7)),360)	200
23	Heading 2	=MOD(540-(180/PI()*ATAN2(B18-B10,B17-B9)),360)	287.8
24	Heading 3	=MOD(540-(180/PI()*ATAN2(B18-B12,B17-B11)),360)	11.7

Figure 2.2 Excel Calculator for GPS Method

There is no technical requirement to use a specially certified or differential GPS. The indication of accuracy of the derived true airspeed can result from the repeatability of the derived wind vector. If several points are flown at different speeds (each "point" being at least three legs at the same speed) and result in the same derived wind speed and direction, then a high degree of confidence can be placed in those results. Regardless of the sophistication of the GPS equipment, the data can be unreliable from one of several factors: wind or temperature variation, failure to hold airspeed and/or heading constant for each leg, altering airspeed between legs of a data point, or poor GPS satellite coverage. Figure 2.3 shows examples of calculations of GPS-derived true airspeeds on a certification program and the calculated wind remained very nearly constant over the approximate 30 minute data gathering period. The average wind speed varied less than 1 kt (18.1 to 18.8 kt) and the direction changed a maximum of 3 degrees. The pitot/static errors derived from those calculations are shown in Figure 2.4 and compare favorably to other data gathered from a traditional ground course in the same aircraft. Lastly, Figure 2.5 shows data from a different certification project comparing GPS results to those obtained with a wing mounted boom. Both Figures 2.4 and 2.5 illustrate the type of

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accuracy possible using the GPS flight test technique. The data shown in both of these figures were taken using a handheld Garmin 92 GPS unit.

	Point 1		Point 2		Point 3		Point 4	
	Vg	Track	Vg	Track	Vg	Track	Vg	Track
	116	236	75	221	43	204	58	204
	131	135	92	117	64	109	77	115
	153	46	111	41	79	40	95	42
	134	316	90	315	61	318	75	316
Vtrue	**134.0**		**91.5**		**60.7**		**75.6**	

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
1	116	236	131	135	153	46	18.9	219	134.2	
	131	135	153	46	134	316	19.4	221	133.7	
	153	46	134	316	116	236	18.7	223	134.3	
	134	316	116	236	131	135	18.1	221	133.6	
							18.8	**221**	**134.0**	Average
									0.4	Std Dev

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
2	75	221	92	117	111	41	18.4	209	93.1	
	92	117	111	41	90	315	20.9	219	90.1	
	111	41	90	315	75	221	18.2	229	93.0	
	90	315	75	221	92	117	14.9	220	89.9	
							18.1	**219**	**91.5**	Average
									1.8	Std Dev

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
3	43	204	64	109	79	40	18.2	217	60.8	
	64	109	79	40	61	318	18.4	218	60.6	
	79	40	61	318	43	204	18.2	219	60.8	
	61	318	43	204	64	109	17.9	218	60.6	
							18.2	**218**	**60.7**	Average
									0.2	Std Dev

	Vg 1		Vg 2		Vg 3		Vwind		Vtrue	
	(kts)	(deg)	(kts)	(deg)	(kts)	(deg)	(mph)	(deg)	(mph)	
4	58	204	77	115	95	42	18.7	213	76.5	
	77	115	95	42	75	316	20.1	218	75.0	
	95	42	75	316	58	204	18.9	222	76.1	
	75	316	58	204	77	115	17.2	219	74.8	
							18.7	**218**	**75.6**	Average
									0.9	Std Dev

Figure 2.3 GPS Example Calculation

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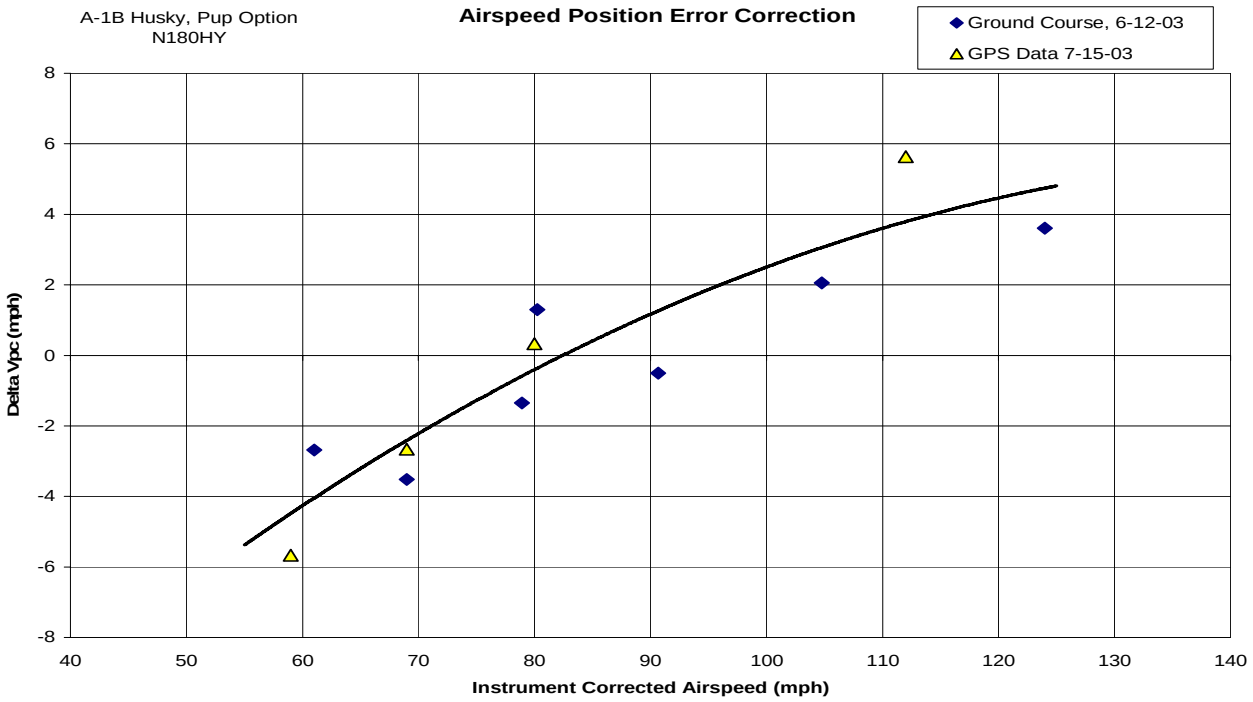


Figure 2.4 Position Error Correction Data Compared with Ground Course

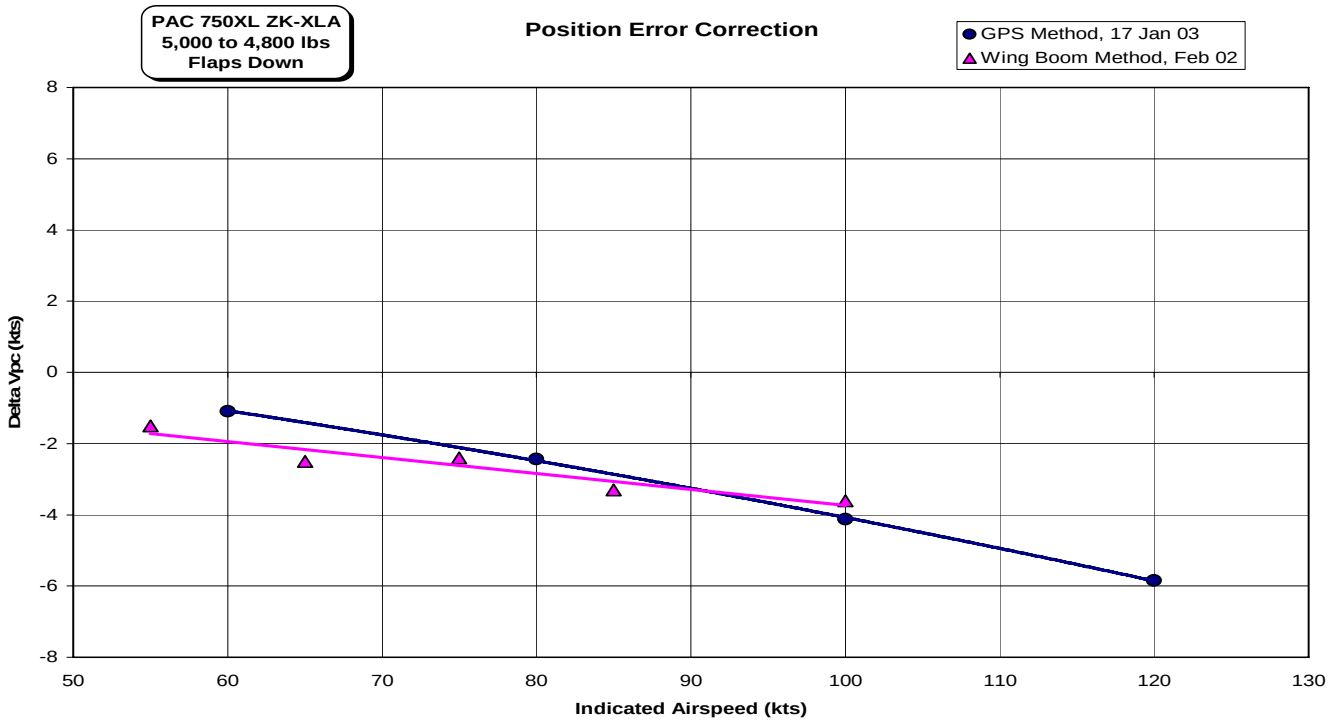


Figure 2.5 Position Error Correction Data Compared with Wing Boom Method

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2.4.2.2 Data Required

V_G, Ground Track, V_o, H_{Po}, OAT, N_R, Q, CG, GW, α_o, β_o, Aircraft Configuration (clean, weapons, etc.)

2.4.2.3 Test Criteria

1. Steady level flight
2. Constant altitude
3. Constant airspeed
4. Constant haeading
5. Stable air mass (Section1.8)

2.4.2.4 Data Requirements/Tolerances

1. Stabilize AV for 30 seconds before recording
2. Stabilize ground track and ground speed
3. V_o $\pm$0.5 KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system
5. N_R $\pm$2%
6. Heading $\pm 2^o$

2.4.2.5 Safety Considerations/Risk Management

There are no safety issues directly relating to this technique differing from UAS standard operating procedures.

2.4.3 Ground-Based Laser Tracking

2.4.3.1 Overview

Use of a ground-based laser tracking station, along with reflectors on board the aircraft, allows the flight test engineers to calibrate the pitot/static system for airspeed, as well as altitude, during the same test. This method is similar to the GPS speed course method described above in that the aircraft is tracked in a particular heading and is then required to turn 180 degrees to complete the second run. The data taken from the laser tracking station will give the ground speed, altitude, and ground track of the aircraft, similar to the data taken during the GPS speed course method. The equations given in Section 2.4.1.1 can be used for data reduction for this method.

2.4.3.2 Data Required

GPS Ground Speed, Ground Track, Heading, V_o, H_{Po}, OAT, N_R, Q, CG, GW, α_o, β_o, Aircraft Configuration

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2.4.3.3 Test Criteria

1. Steady level flight
2. Constant altitude
3. Constant airspeed
4. Constant heading
5. Stable air aass (Section 1.8)

2.4.3.4 Data Requirements/Tolerances

1. Stabilize AV for 30 seconds before recording
2. Stabilize ground speed and ground track
3. V_o $\pm$0.5 KIAS or limits of flight control system
4. Altitude $\pm$10 ft or limits of flight control system
5. N_R $\pm$2%
6. Heading $\pm 2^o$

2.4.3.5 Safety Considerations/Risk Management

Eye hazards are typically associated with laser tracking systems. A safety procedure based on the system manufacturer's guidelines must be implemented to mitigate eye-safe issues. Such processes will include establishing laser azimuth and elevation restrictions, laser operating indicators such as warning lights, announcements when activated, and the establishment of personnel restriction areas. Safety issues related to the payload onboard the UAS should be considered; however, since the payload should be removed before flight performance testing and many auto-landing systems use laser based tracking for precision landings, it will probably not have a detrimental effect on the UAS payload.

2.4.4 Pace Aircraft/Vehicle

2.4.4.1 Overview

This method of pitot/static system testing can be useful as a way of attaining the test AV position error information accurately in a short period of time. It can cover any altitude and airspeed where the pace aircraft and test aircraft are compatible. However, since the pace aircraft requires precise observation of relative velocity between the two AVs, this test may not be suitable for small and medium UAS due to the inability to follow the test AV at excessively low airspeeds. The pace aircraft will have independently calibrated pitot/static systems that will require periodic checks by accepted methods.

The UAS will lead the formation. The start point is set by the lead with the trailing pace aircraft stabilized in position, following closely enough to observe minor changes in relative velocity without interfering with its own pitot/static system. The pace aircraft provides calibrated pressure altitude and airspeed so that the position error

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associated with pressure and airspeed can be evaluated. One problem associated with using the pace aircraft method of calibrating UAS pitot/static system is synching the time that the data are taken from the UAS and pace aircraft pitot/static systems. The UAS and pace aircraft must have an established system of determining when to take data and how to mark the time that the data were taken in order for the data to be useful to the test engineers.

2.4.4.2 Data Required

V_o, H_{Po}, OAT, N_R, Q, GW, CG, α_o, β_o, Aircraft Configuration.

Note: Velocity and altitude data must be recorded for each system to be calibrated as well as pace aircraft pitot/static data.

2.4.4.3 Test Criteria

1. Steady level flight
2. Constant heading
3. Constant airspeed
4. Constant altitude (or stable rate of change during climbs or descents)
5. Zero relative velocity between two aircraft
6. Stable air mass (Section 1.8)

2.4.4.4 Data Requirements/Tolerances

1. Stabilize AV for 30 seconds before recording
2. Record instantaneous data upon stabilization
3. V_o $\pm$0.5 KIAS or limits of flight control system
4. H_{Po} $\pm$10 ft or limits of flight control system
5. N_R $\pm$2%
6. Heading $\pm 2^o$

2.4.4.5 Safety Considerations/Risk Management

Many of the risks associated with manned flight testing can be mitigated since control of the AV is accomplished by the autopilot and the AVO can concentrate on situational awareness (SA) and communication with the chase plane pilot or copilot. The pace aircraft must have a second crewmember to alleviate the pilot from the task of accurately reading the airspeed, altimeter, and other instruments. As always, safety of flight is paramount and flight testing should not risk the minimum safe altitudes or airspeeds at any point during the flight. Also, the UAS operator and pace aircraft pilot should coordinate the responsibility for providing safe separation distance between the aircraft as well as each AV's emergency procedures break-off maneuver at test termination and return to base procedures. Lost link procedures must also be fully understood by the pace aircraft pilot for each stage of UAS flight. Knowledge of the direction of turn and behavior of the UAS under lost link can be used to plan the pace

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aircraft's position while chasing and response when a lost link event occurs. Two way communications between the pace aircraft and the AVO are critical. Procedures for loss of communication between the AVO and pace aircraft should also be fully understood before proceeding to flight test.

2.4.5 Tower Flyby

2.4.5.1 Overview

This method is suitable for all sizes of UAS and involves flying the AV on a track that is a known distance from the flyby tower, which is at a known height and has a calibrated altimeter to determine the static free-stream pressure (see Figure 2.8). For UAS that use GPS only for altitude, this method would not be useful due to the lack of static pressure measurement for altitude. For each pass, the AV will be sighted through a theodolite to record the angle from the tower to the aircraft as well as the tower altimeter setting. The angle and the distance to the aircraft can be used to calculate the height above the tower through trigonometric relationships. The aircraft instrument and the tower reference instrument should be placed side by side to check the output from the altimeters. If any discrepancy exists it should be taken into account in the data reduction.

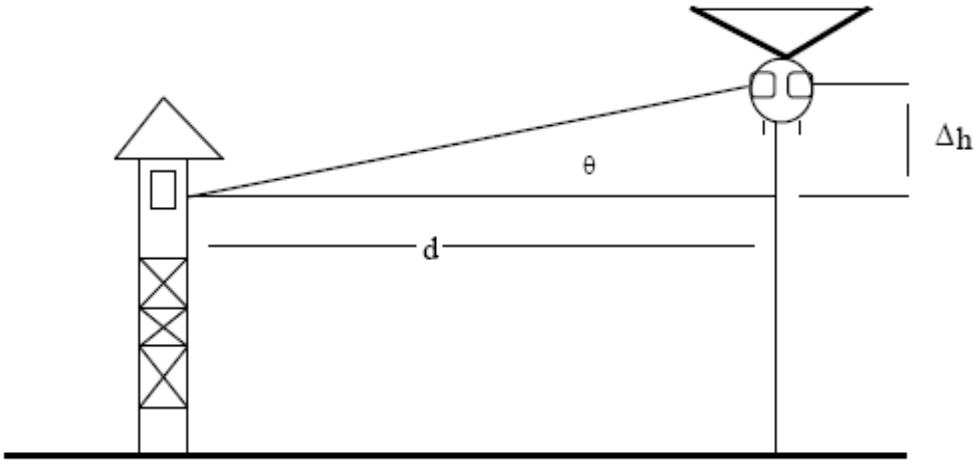


Figure 2.8 Tower Flyby Geometry

Another way to use the tower flyby is to take a grid photograph of the aircraft as it passes the tower. The scaled AV length and scaled height above the tower can be used with the actual length to determine the actual height above the tower. This alteration relieves the AVO of flying over the predetermined path. Only the camera would need to be aligned so the picture is not taken at an angle. This test will generate an accurately calibrated pressure altitude but not a calibrated airspeed. Since the tower is at an extremely low altitude, the altitude variation should not exceed 15 ft above or below the target altitude.

2.4.5.2 Data Required

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V_o, $V_{o,ref}$, $H_{Po,twr}$, OAT, N_R, Q, GW, CG, α_o, β_o, Aircraft Configuration.

Note: Velocity and altitude data must be recorded for each system to be calibrated as well as tower reference pressure altitude, temperature, and aircraft geometric height data (d, θ).

2.4.5.3 Test Criteria

1. Steady level flight
2. Constant airspeed
3. Constant altitude
4. Constant ground track over predetermined path
5. Stable air mass (Section 1.8)

2.4.5.4 Data Requirements/Tolerances

1. Stabilize AV for 15 seconds before tower
2. Ground track $\pm$1% of stand-off distance
3. V_o $\pm$0.5 KIAS or limits of flight control system
4. H_{Po} $\pm$10 ft or limits of flight control system
5. N_R $\pm$2%

2.4.5.5 Safety Considerations/Risk Management

This technique requires the AVO to pay considerable attention in such close proximity to the tower and be fully aware of the normal and emergency procedures of the UAS. Lost link procedures must also be known and, when adding a lost link waypoint to the flight profile, it should be placed to turn the aircraft away from the tower and any test personnel.

2.4.6 Trailing Source

2.4.6.1 Overview

The trailing bomb technique is appropriate for large UAS and requires having a bomb-shaped static source extended on a cable at least one and a half rotor diameters below the fuselage. The bomb size is large enough to preclude testing small and medium UAS by this method. The static pressure will be measured and transmitted up the cable to the AV where the value will be converted into a pressure altitude. The altimeter position error can then be determined since the difference in altitude between the trailing source and AV source will be known. If the bomb uses an electrical method to send static pressure data to the AV, the position of the bomb relative to the AV must be known. This can be avoided if the bomb is equipped with a pressure tube to transmit the static data to the test aircraft, thus eliminating position error. Some trailing bombs are also equipped with a pitot system, allowing the calibration of the entire pitot/static system

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of the AV. The trailing bomb must be calibrated in a wind tunnel to determine static source error. The source can acquire the free-stream pressure altimeter setting below the AV if connected via the cable to an altimeter. One potential problem is instability of the cable and trailing source, which may occur at both low and high airspeeds. This leaves midrange airspeeds that can be flown by the AV up to 150 kt. Additionally, there should be a means to constantly monitor the trailing source so that data are recorded when both the airspeed and the trailing source are stable. The method of cable extension and retrieval, such as a winch, might also cause size, weight, and power (SWAP) requirement issues which need to be addressed before testing using the trailing bomb method. A chase aircraft or ground observers could be useful for determining the onset of instability for the trailing bomb as well as providing SA.

2.4.6.2 Data Required

V_o, $V_{o,ref}$, H_{Po}, $H_{Po,ref}$, OAT, N_R, Q, GW, CG, α_o, β_o, Aircraft Configuration.

Note: Velocity and altitude data must be recorded for each system to be calibrated as well as the trailing source system (reference data).

2.4.6.3 Test Criteria

1. Steady level flight
2. Constant airspeed
3. Constant heading
4. Constant altitude
5. Stabilized instrument readings
6. Stable air mass (Section 1.8)

2.4.6.4 Data Requirements/Tolerances

1. Stabilize AV for 30 seconds before recording
2. Record data for 60 seconds
3. V_o $\pm$0.5 KIAS or limits of flight control system
4. H_{Po} $\pm$10 ft or limits of flight control system
5. N_R $\pm$2%
6. Heading $\pm 2^o$

2.4.6.4 Safety Considerations/Risk Management

The AVO should be able to jettison the trailing bomb if they become unstable to the point that recovery cannot be made. This requirement may result in software modification to the AV/GCS. Observers should always maintain the minimum safe distance from the ground track of the UAS. Lost link procedures and the turn rates of the UAS must be understood before testing; a tight turn could cause the tether to break or interfere with the control surfaces, propulsion system, or tail rotor.

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2.4.7 Radar Altimeter

A calibrated radar altimeter can be used to determine altimeter position errors present in level flight at low altitude. A level runway is required to establish a reference altitude for the test. A sensitive pressure altimeter should be placed at the runway elevation during test runs. The radar height is added to the runway pressure altitude and compared with the aircraft altimeter system indication to determine position errors. The aircraft altimeter may be used as the reference altimeter if the AV is landed on the runway reference point and the rotor disengaged. With the rotor not engaged, the service system altimeter reading before and after the test runs can be used to establish a base pressure altitude to which the radar height can be added. This method is recommended only for rough approximation of gross altimeter position errors. Flight above the level surface should be conducted low enough to provide accurate height information with the radar altimeter but not so low that the pressure field around the AV will be affected by ground proximity. A good guideline is generally two main rotor diameters of altitude.

2.4.7.1 Data Required

Surface pressure altitude, V_o , H_{Po}, T_o, N_R , Q, GW, CG, α_o, β_o, radar altitude. Aircraft Configuration.

2.4.7.2 Test Criteria

1. Balanced, level flight
2. Constant aircraft heading
3. Constant airspeed
4. Constant radar altimeter altitude
5. Stable air mass (Section 1.8)
6. Level runway (< 0.5% grade, or 5 ft elevation change for every 1000 feet of runway)

2.4.7.3 Data Requirements/Tolerances

1. Stabilized flight for 15 s
2. V_o ±0.5 KIAS
3. H_{Po} ±10 ft
4. Radar altimeter altitude ±3 ft
5. N_R $\pm$2%
6. Heading $\pm 2^o$

2.4.7.4 Safety Considerations/Risk Management

There are no special safety considerations for this test. The test team should understand UAS normal and emergency procedures.

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2.5 Data Reduction

The reduction of the acquired data from the performance flight test should follow the procedures of Section 2.5 of the USNTPS-FTM-No. 108. That manual should be referenced for the data reduction of each of the tests discussed here; the exceptions are the GPS speed course method, the GPS 3-leg method, and the ground-based laser tracking method, which explain the data reduction in their respective overviews.

2.6 Data Analysis

Section 2.6 of the USNTPS-FTM-No. 108 provides full detail of the analysis and evaluation of the test data. These steps should be followed to validate the altimeter and airspeed position errors and analyze qualitatively the pitot/static system performance from the quantitative test data.

2.7 Mission Suitability

USNTPS-FTM-No. 108 Section 2.7 is applicable to UAS of all sizes. This is particularly true when discussing the seriousness of the airspeed or position errors affecting automated systems, since flight control systems direct the AV. Additionally, the future UAS are moving toward autonomous takeoff and landing systems, and pitot/static errors could cause severe mishaps that could endanger test personnel.

2.8 Specification Compliance

Section 2.8 of the USNTPS-FTM-No. 108 references military standards (MIL-I-5072-1 and MIL-I-6115A) as well as Federal Aviation Regulations (Parts 27.1323 and 27.1335). These establish the standard for instrument installation and system performance, in addition to tolerances and maneuvers (pull-up, pushover, yawing, and rough air). These documents should be referenced as the benchmarks for pitot/static system compliance.

2.9 Glossary

2.9.1 Notations

D	Course length	nmi
d	Horizontal distance (Tower to aircraft)	ft
FAR	Federal Aviation Regulations	
g	Gravitational acceleration	ft/s^2
g_c	Conversion constant	32.17 lb$_m$/slug
g_{ssl}	Standard sea level gravitational acceleration	32.17405 ft/s^2

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GW	Gross weight	lb
H_{Po}	Observed pressure altitude	ft
$H_{Po\ ref}$	Reference observed pressure altitude	ft
IFR	Instrument flight rules	
NACA	National Advisory Committee on Aeronautics	
nz	Normal acceleration	g
OAT	Outside Air Temperature	°C or °K
P	Pressure	psf
P_a	Ambient pressure	psf
P_s	Static pressure	psf
$P_{S_{YAPS}}$	Static pressure measured by the YAPS boom	psf
P_{ssl}	Standard sea level pressure	2116.217 psf 29.9219 inHg
P_T	Total pressure	psf
P_T'	Total pressure at total pressure source	psf
q	Dynamic pressure	psf
R	Engineering gas constant for air	96.93 ft-lbf/lbm-°K
Re	Reynold's number	
SAS	Stability augmentation system	
T	Temperature	°C or °K
T_a	Ambient temperature	°C or °K
$T_{a\ ref}$	Reference ambient temperature	°C or °K
T_i	Indicated temperature	°C or °K
T_o	Observed temperature	°C
T_{ssl}	Standard sea level temperature	15°C or 288.15°K
USNTPS	U.S. Naval Test Pilot School	
V	Velocity	kt
V_c	Calibrated airspeed	kt
V_e	Equivalent airspeed	kt
V_G	Ground speed	kt
V_{ne}	Never Exceed Airspeed	kt
V_o	Observed airspeed	kt
$V_{o\ ref}$	Reference observed airspeed	kt
V_T	True airspeed	kt
V_w	Wind velocity	kt
W	Weight	lb
x	Scaled length of aircraft	
y	Scaled height of aircraft above tower	

2.9.2 Greek Symbols

α (alpha)	Angle of attack	deg
α_o	Angle of attack measured by the YAPS boom	deg

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β (beta)	Sideslip angle	deg
β_o	Sideslip angle measured by the YAPS boom	deg
δ (delta)	Pressure ratio	
γ (gamma)	Ratio of specific heats	
λ (lambda)	Lag error constant	
λ_s	Static pressure lag error constant	
λ_T	Total pressure lag error constant	
θ (theta)	Angle, Temperature ratio	deg
ρ (rho)	Air density	slug/ ft3
ρ_a	Ambient air density	slug/ ft3
ρ_{ssl}	Standard sea level air density	0.0023769 slug/ ft3
σ (sigma)	Density ratio	

2.10 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 3
POWERPLANT ASSESSMENT

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3. Powerplant Assessment

3.1 Introduction

3.1.1 General

To adequately evaluate the overall performance of the unmanned aircraft system (UAS) as an aircraft system, the individual components of the air vehicle (AV) must be evaluated. The two major contributors to the overall AV performance are the powerplant performance (power available) and the airframe performance (power required). We must, therefore, evaluate both components of AV performance: powerplant and airframe. To evaluate powerplant performance, we must first evaluate the air vehicle operator's (AVO) interface with the powerplant and the autopilot or vehicle mission computer control of that powerplant. This chapter will examine these issues using powerplant assessment techniques. Data recorded, whether it be onboard or downlinked to the ground control station (GCS), should be time-tagged with UTC time, usually from a global positioning system (GPS) receiver. Time-tagging the data allows test engineers to determine latencies in the system that may, if not understood and quantified, complicate data reduction and analysis.

3.1.2 Types of Powerplants

In general, there are two main categories of vertical takeoff and landing (VTOL) propulsion systems, tip driven rotors, where the powerplant or part of it is blade mounted; and shaft driven rotors, where the powerplant is located in the fuselage and connected to the rotor system by a shaft. By far, the most common installation today is the shaft driven rotor. Of the shaft driven rotors, the powerplant may be a turbine engine, a reciprocating engine, or an electric motor. Of these three types of powerplants, most UAS will have turbine power in the largest AV, reciprocating in the medium size AV's, and electric power in the smallest AV. Most electric UAS drive low inertia rotor systems due to the small AV size; however, large inertia rotor systems can be accommodated by electric motors.

3.1.2.1 Gas Turbine

Within the turbine engine design, there are two types: the free power turbine and the fixed shaft turbine engine, with the most common being the free power turbine. A few of the advantages of the free power turbine are less power required to start the engine, the engine can be started with the rotor stopped, engine rotor matching is easier to achieve, and simple load sharing is possible in a multi-engine installation. Since the gas turbine engines most commonly used in UAS are free power turbine engines, this chapter will only discuss this type of gas turbine engine.

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Different engine manufacturers use varying shorthand notations to refer to the internal parts of a turbine engine. Engine gas generator speed may be referred to as N_1 or N_G. Engine power turbine speed may be referred to as N_2 or N_P. The engine temperature which the manufacturer chooses to measure and display on the GCS may be power turbine inlet temperature commonly referred to as interstage turbine temperature (ITT), power turbine outlet temperature (TOT), exhaust gas temperature (EGT), or turbine gas temperature (TGT). Within this manual, N_1 will be used for engine gas generator speed, N_2 will be used for engine power turbine speed, and T_5 will be used to represent the measured and displayed engine temperature and is further defined as power turbine inlet temperature.

3.1.2.2 Reciprocating

A reciprocating engine can consist of internal combustion chambers driving pistons into and rotating a central crankshaft or a rotary combustion chamber driving a reciprocating rotor which rotates a crankshaft. The crankshaft is coupled to the rotor head by a belt drive or clutch mechanism which engages and drives the rotor shaft as throttle is increased.

The engine rotating speed (N_1) is measured at the output shaft of a reciprocating engine. The output shaft rotating speed of any gearboxes attached or internal gearing to reduce the engine's rotations per minute (RPM) would be used as (N_2). Cylinder head temperature and exhaust gas temperature will be referred to as CHT and EGT respectively for all reciprocating engines. Oil temperature will be referred to as T_{oil}.

3.1.2.3 Electric

Electric power systems are particularly suited for small UAS and the technology is rapidly evolving due to increased military and commercial demand for small, light, high energy density power systems. This new technology brings quiet (low decibel) power and fast startup times for UAS, as well as consistent torque at all RPM. Unlike turbine and reciprocating engines which require engine adjustments for even small changes in ambient conditions, electric motors require very limited maintenance and adjustment when operating in different weather conditions. Electric power output however is still highly dependent on ambient temperature.

The primary concern of an electric power system is the dependency on the battery and the available power from the battery cells. The second power concern is the power drain or voltage decrease of the battery installed and how it degrades over time. This phenomenon must be tested and quantified by testing the entire powerplant as a system, rather than testing individual components.

The motor rotating speed (N_1) is measured at the output shaft of the motor. The output shaft rotating speed of any gearboxes used to reduce RPM will be used as N_2. The

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outside case temperature of the motor will be referred to as T_{case} and the temperature of the battery or source will be referred to as $T_{S,}$

3.2 Purpose of the Test

The purpose of these tests is to evaluate the powerplant/rotor compatibility and suitability for the mission of the host AV. Specific tests that are conducted include an evaluation of the powerplant controls and displays, powerplant operating procedures, both normal and emergency, powerplant start and shutdown characteristics, powerplant acceleration characteristics, powerplant trim response, powerplant/rotor stability, both static droop and transient droop, powerplant torque matching, powerplant limiting characteristics, and powerplant power contribution during minimum power descents.

Due to the fact that most UAS are controlled by the flight control systems rather than a pilot (as in the case of a manned aircraft), the autopilot input and control data needs to be recorded either onboard the AV with a self-contained power source or downlinked and recorded by the GCS. Flight test data recorded, whether recorded onboard or downlinked, should be time-tagged with UTC time usually from a GPS receiver. Data sampling rates are test specific and should be appropriate for the particular system under test. If all data are recorded in the GCS, the data will need to be corrected for the latencies in the data link system. Though the data being supplied to the AVO are of a lower fidelity than data being collected for reduction by the test engineers, the data may be sufficient to determine when the AV is within data tolerances on a stable data point.

3.3 UAS vs. Manned Aircraft Theory

3.3.1 General

The guidelines set forth in USNTPS-FTM-No. 106 Sections 3.1 and 3.2 in the sections concerning powerplant selection and rotor system explanation apply to VTOL UAS. The powerplant guidelines are practical and applicable to all AVs, regardless of configuration or whether being manned or unmanned. All vehicles will benefit from selecting a powerplant with the characteristics given by FTM-106. The rotor system explanation section is also applicable to UAS with standard or close to standard rotor configurations.

3.3.2 Governor

The governor is a computer controlled speed controller of the main rotor head that strives to maintain constant rotor RPM under all operating conditions. The turbine power plant governor is discussed extensively in USNTPS-FTM-No. 106 Section 3.3.3.

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A governor for reciprocating power plants involves similar concepts to regulate main rotor speed (N_R). However, the engine response compared to a turbine is typically faster and, therefore, the transient droop may be reduced when compared with turbine governors. Tendency for rotor overspeed may also be reduced due to the speed at which the powerplant torque can be reduced by the governor with less system lag. The reduction in transient droop improves static droop compensation which allows the governor to require less computer compensation and estimation for proper damping.

An electric UAS is powered by a single or multiple motors. Each motor is controlled by a small circuit board that varies the motor's RPM. This circuit board, also called the electronic speed controller (ESC), has internal circuitry that acts as a governor. The throttle command sent to the ESC is treated like a RPM command, or main rotor speed (N_R) command, and will hold it constant for all pitch changes. The motor response for electric powerplants is very quick, has maximum torque at low RPM, and can handle short periods of torque overload. The static and transient droops associated with electric powerplants should be minimal.

The rotor head power required varies with rotor speed, N_R, as shown in Figure 3.1. The power required to increase N_R dramatically increases with higher N_R due to the increased profile, parasite, and induced drag.

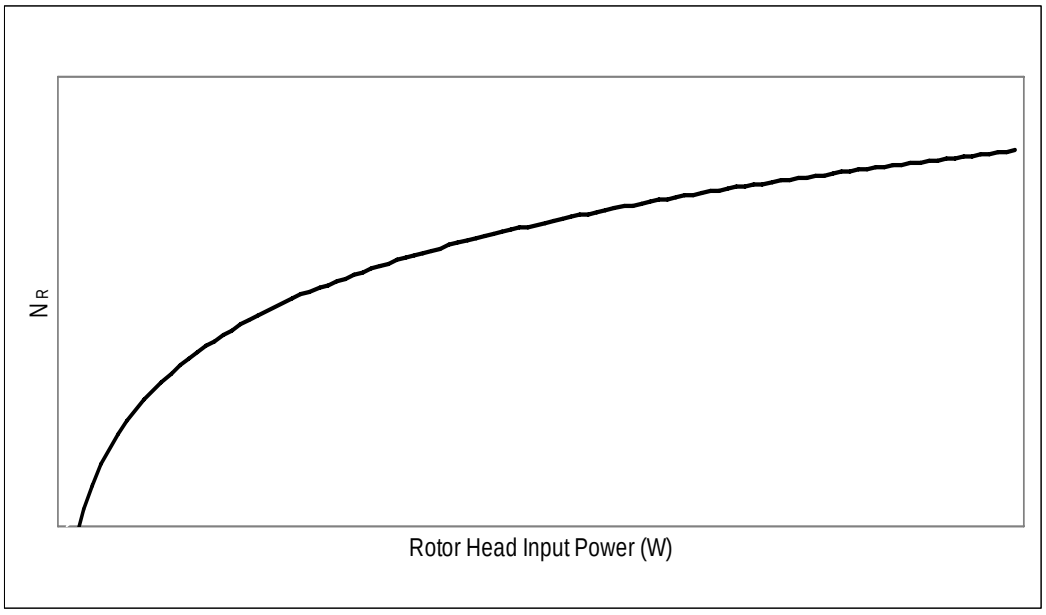


Figure 3.1 Electric UAS Rotor Head Speed as a
Function of Rotor Head Power Required

As electric power plant batteries are used, their available energy capacity decreases, much like fuel consumption in gas turbine or reciprocating engines. Because available energy decreases, the voltage drops and the power delivered to the motor will decrease over time. This has a direct effect on the governor's control of the rotor speed

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with voltage decreases up to 10% depending on battery type. Most batteries used in UAS will use lithium chemistry and have a predictable, consistent voltage curve over time. The battery energy decrease or "burn rate" can be calculated and tested and power control systems can be designed around these characteristics. The battery cycle life is cell dependent and ranges from 100 to over 500 recharge cycles. Test programs should evaluate manufacturers' recommended charge rates and cycles. Electric powerplant testing should be conducted within the quasi-linear range of the battery discharge curve to obtain reliable test data and ensure safety of the test asset. Depending on the system, it may be necessary to test and/or graph the governor gain vs. rotor RPM. If the UAS is equipped with multiple motors, it is necessary to understand load sharing and multiple motor issues associated with a multimotor installation.

3.3.3 Powerplant/Motor Protection

3.3.3.1 Powerplant Overspeed

Many different systems have been used to prevent powerplant and airframe damage. In turbine applications, the engine speed control protects from overspeed and is discussed in detail in USNTPS-FTM-No. 106. In reciprocating engines the same principles apply; a control loop is used to monitor critical engine speed parameters and control the throttle on the carburetor or fuel injectors to keep these parameters within limits.

Electric motors feature a governor control which controls the motor and rotor head RPM. In a transient rotor speed condition, electric motors have a large range of RPM settings and short transient overspeed conditions are allowed by the manufacturer unless the engine is designed to run at the maximum RPM limit during normal operation. Typically, motors will run near their most efficient RPM, which is 20% or more below their mechanical maximum limit.

3.3.3.2 Temperature Limiting

A powerplant temperature instrumentation suite may be used to monitor many powerplant locations where temperature is a concern. Power turbine inlet temperature, cylinder head temperature, exhaust gas temperature, and motor windings temperature are commonly monitored temperatures for various power systems during performance flight testing. These sensors may be connected to the fuel control or motor speed controller to monitor and/or protect the powerplant or motor from an overtemperature condition. Operating profile (phase of flight) may also determine temperature limitations and may result in altered fuel or current flow.

The power source for most electric powered UAS is a battery using lithium chemistry. These batteries, as most others, have associated maximum temperatures and will, when reaching that maximum temperature, shut down and vent the cells to avoid drastic over expansion and possible explosion. Because of these risks, most electric power systems are designed with maximum battery performance limits. However, it is

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important to instrument the batteries for temperature during flight test to determine if the batteries are approaching their temperature limits or if they are operating in a detrimental area of their power vs. temperature curve.

3.3.3.3 Flight Idle Stop

The idle stop is installed to keep the powerplant RPM from descending below idle while in flight. During flight at the idle stop response time is decreased and engine stability is occasionally degraded. This may result in surging or decreasing powerplant response time when powerplant torque is required to increase.

In UAS systems, these functions should be programmed into the autopilot or the powerplant control system. When in a flying state, the lower throttle limit (flight idle stop) should always be set for maximum descent rate but not lower.

3.4 Test Methods and Techniques

3.4.1 General

The area of powerplant assessment is one of particular interest for the test team involved in performance flight testing for VTOL UAS. The powerplant characteristics and limiting factors must be understood for effective performance flight testing. Of particular interest are the worst-case scenarios of gross weight and center of gravity for particular test methods, such as maximum gross takeoff weight for static and transient droop. UAS differ from manned aircraft in that many UAS do not require the AVO to close the loop for operation but rather rely on the flight control system to accomplish these tasks. However, there may be some instances for which the autopilot cannot resolve an issue and AVO intervention is required. For these systems, if the powerplant monitor in the GCS gathers insufficient data for the test methods to be performed, then the flight control software may need to be altered to a test configuration for manual manipulation so that the appropriate powerplant data can be obtained. Any changes to the flight control software will require airworthiness approval.

As stated in USNTPS-FTM-No. 106 Section 3.4.1, the test methods for powerplant assessment are a continuation of powerplant bench and test cell testing. The test methods that are discussed must be tailored to the specific powerplant installation. Thorough knowledge of the test AV with special emphasis on the powerplant/rotor, powerplant/rotor controls, and powerplant/rotor limits is mandatory for the specific application of the test methods and the success of the test program. As for any flight test, safety is paramount and appropriate safety assessments and risk management techniques should be implemented. The powerplant assessment test methods follow the concept of incremental buildup described in chapter 1 of this document. All limits for the powerplant, rotor, and transmission are observed and any approach to the limits should be monitored closely.

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3.4.2 Powerplant/Rotor Controls and Displays

The evaluation of the powerplant/rotor controls and displays involves assessing the autopilot inputs and feedback loop corrections for maintaining trim conditions and is applicable to all powerplant types. The data will be collected by an onboard recorder to ensure that the data being recorded are the actual inputs from the autopilot or truth data. The displays within the GCS will show the flight critical data as well as the visual and audio warnings that the AVO observes during UAS operations.

For performance flight testing, it is usually the best idea to partition flight critical data from engineering data so that the AVO has the parameters necessary for monitoring the flight. The partitioned engineering data can be transmitted by a separate datalink (TM) to the flight test engineer or recorded in an onboard data logger and retrieved at the termination of the flight. Use of TM or data logging will prevent the bandwidth of the primary AV datalink from being used to transmit data that are only necessary for data reduction and analysis. The displays for the AVO (in the GCS) and the flight test engineer should follow color and luminance guidelines set forth in MIL-STD-1333B, MIL-STD-411F, and MIL-L-85762.

The data required for UAS powerplant/rotor controls and displays test method are similar to the USNTPS-FTM-No. 106 Section 3.4.2.1. A description of the controls and displays, including images and measurements, and the recorded interactions between the autopilot and rotor control system should be sufficient.

3.4.3 Powerplant/Rotor Operating Procedures

This test pertains to the manual and mental operations of the pilot according the USNTPS-FTM-No. 106 Section 3.4.3. Since most UAS do not require the AVO to close the control loop, the autopilot must be tested for operating procedures. This task could be accomplished during Hardware-in-the-Loop Simulation (HILSIM) since the flight computer, not a pilot, is being tested.

A flight termination system (FTS) should be considered for installation in all UAS that hold any possibility of causing serious damage to surrounding areas or personal human injury. A flight termination system provides a second control of the AV should the primary link fail by using a stand alone, data link system that controls engine shutdown and deployment of an alternate lift device (i.e., parachute). The flight termination system operation must be verified before each flight. The test team must understand the proper operation of the FTS, standardized procedures, and assigned responsibilities for issuing the flight termination commands.

The AV interface contained in the GCS may contain various engine controls depending on the complexity of the system. A turbine powered UAS will likely display many facets of the engine controls on the GCS while a small electric system might have limited engine control. Common powerplant control may include throttle setting and shutoff control. Other GCS controls may include flight idle power levels, mixture

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control, and cooling vents or other temperature control devices. While these manual controls may be available, most systems should have autonomous control of these systems. Throttle setting typically is not directly controlled by the AVO but rather mixed by the autonomous flight controller when a particular command is given. Therefore, limited powerplant control may be available while in flight as the vehicle mission computer controls all settings. Testing on the ground may allow throttle control and engine runup.

Emergency controls of the AV through the GCS interface should include motor shutoff and limiting controls. The flight termination system has additional control of the system by turning the motor off in case of emergency. This is done by disconnecting the battery (electric powerplant), grounding the ignition in a reciprocating engine, or shutting off the fuel supply in a turbine engine.

3.4.4 Powerplant/Rotor Start and Shutdown Characteristics

This test method follows the techniques described in USNTPS-FTM-No. 106 Section 3.4.4 evaluating the powerplant under varying environmental conditions, proceeding from most favorable to least favorable conditions. If onboard data logger is used to capture flight control system data, then it should be powered by an alternate source of power than the UAS battery or auxiliary power unit (APU) used for startup as it may alter the startup performance of the powerplant.

3.4.4.1 Data Required

Gas Turbine: Time history of engine/rotor parameters:
- Torque (Q)
- Gas generator speed (N_1)
- Free power turbine speed (N_2)
- Power turbine inlet temperature (T_5)
- Fuel flow (W_f)
- Rotor speed (N_R)
- Annotate engine/rotor control activation
- Annotate system operation
- Outside air temperature (OAT)
- Pressure altitude (H_{Po})
- Wind speed and direction (relative to the nose of the AV)
- Wind gust spread
- Previous cooling period of the engines (if applicable)

Reciprocating Engine: Time history of engine/rotor parameters:
- Manifold pressure (P_M)
- Crankshaft speed (N_1)
- Gearbox output shaft speed (N_2)
- Cylinder head temperature (CHT)
- Oil temperature (T_{OIL})
- W_f

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N_R
Annotate engine/rotor control activation
Annotate system operation
OAT
H_{Po}
Wind speed and direction (relative to the nose of the AV)
Wind gust spread
Previous cooling period of the engines (if applicable).

Electric powerplants: Time history of motor/rotor parameters:
Q
Motor output shaft speed (N_1)
Gearbox output shaft speed (N_2)
Motor case temperature (T_{case})
N_R
Source voltage (V_S)
Source current(A_S)
Source temperature (T_S)
OAT
H_{Po}
Wind speed and direction (relative to the nose of the AV)
Wind gust spread
Previous cooling period of the motors (if applicable)

3.4.4.2 Safety Considerations/Risk Management

When adding a flight test instrumentation package for onboard data logging, the effect on airworthiness (i.e., weight changes, center of gravity changes, and electromagnetic effects) should be considered. The AV should be chained to a suitable tie-down point during the test to prevent mishaps.

3.4.5 Powerplant Acceleration Characteristics

Powerplant acceleration is an important aspect for mission suitability in determining the runup time from powerplant startup to powerplant takeoff speed so that the operator will know how much time is needed before the UAS can liftoff for a mission. The test for all powerplants should follow the procedure as mentioned in USNTPS-FTM-No. 106 Section 3.4.5.

3.4.5.1 Data Required

Gas Turbine: Time history of Q, N_1, N_2, T_5, W_f, N_R, and throttle position. Annotate system operation. Note ambient conditions: OAT, H_p, wind speed and direction (relative to the nose of the AV), wind gust spread, and previous cooling period of the engines (if applicable).

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Reciprocating Engine: Time history of P_M, N_1, N_2, T_5, T_{oil}, W_f, N_R, and throttle position. Annotate system operation. Note ambient conditions: OAT, H_p, wind speed and direction (relative to the nose of the AV), wind gust spread, and previous cooling period of the engines (if applicable).

Electric Powerplant: Time history of Q, N_1, N_2, T_5, N_R, V_S, A_S, and T_S. Annotate system operation. Note ambient conditions: OAT, H_p, wind speed and direction (relative to the nose of the AV), wind gust spread, and previous cooling period of the motors (if applicable).

3.4.5.2 Safety Considerations/Risk Management

During rapid powerplant and rotor acceleration, avoid exceeding system limitations. Wheel brakes, locks, and chocks should be employed. As the datum powerplant/rotor trim setting is advanced to the maximum, avoid exceeding system limitations. Avoid inadvertent movement on the ground during accelerations. The same site and crew precautions that are observed during powerplant start should be observed during the accelerations.

3.4.6 Powerplant Trim Response

As of this writing, no VTOL UAS incorporate powerplant trim response into the flight control system and therefore this test typically is not applicable. In the future, there may be VTOL UAS that have the capability to trim the powerplant and, if so, the test method described in USNTPS-FTM-No. 106 Section 3.4.6 can be used.

3.4.7 Powerplant/Rotor Stability (Static and Transient Droop)

It is important to understand that when dealing with UAS, it is the autopilot that actually controls the engine of the AV. Properly set up, the autopilot will be able to dampen out any static or transient droop and should not overshoot. In these tests, the powerplant and the autopilot control of that engine are actually what are being tested, as an entire system. Additionally, static and transient droop may be difficult to measure low inertia rotor systems, which are common in small or light UAS. This test method will be applicable for UAS where both the autopilot does not dampen the droop or overshoot response and the static and transient droop can be measured.

Mission maneuvers and UAS survivability will become more critical in the future, as the cost, complexity, and workload of the systems grow. Though UAS flight testing may not need to include survivability or mission maneuvers at present, it is beneficial to understand testing of these maneuvers and their associated parameters. The effects of powerplant/rotor stability on the performance of mission maneuvers are documented by using automatic data recording during mission maneuvers. Such maneuvers are quick stops, bob-ups, remasking, rapid accelerations, and dashes. Performance of these maneuvers may be linked to the powerplant/rotor stability and power margin.

3.4.7.1 Data Required

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Gas Turbine:

Static Droop: Stabilized Q, NR, and airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Transient Droop: Time history of Q, NR, and collective servo position in order to determine blade pitch angle. Airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Reciprocating Engine:

Static Droop: Stabilized P_M, NR, and airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Transient Droop: Time history of Q, NR, and collective servo position in order to determine blade pitch angle. Airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Electric Powerplant:

Static Droop: Stabilized Q, NR, and airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

Transient Droop: Time history of Q, NR, and collective servo position in order to determine blade pitch angle. Airspeed. Annotate system or warning operation. Ambient conditions. Onboard data UTC time-tagged.

3.4.7.2 Test Criteria

1. Fixed powerplant trim setting during tests
2. Static droop, collective movement in one direction only
3. Static droop, stable powerplant power demand

3.4.7.3 Data Requirements

1. Static droop, stabilize 30 s minimum before recording data
2. Vo ±1 KIAS or capability of the system

3.4.7.4 Safety Considerations/Risk Management

During these tests, observe powerplant/rotor and aircraft limitations. The test crew must have a thorough knowledge of all applicable limitations. Use extreme caution in the transient droop test to avoid low rotor RPM situations that may present system or handling qualities problems. Consider the powerplant power indicating system and any lags in that system for the rate of collective application and the maximum collective. Since these tests are performed at conditions of high collective settings and possible low rotor RPM, they must be conducted over an area suitable for autorotation. This is especially true for the hover tests. Perform the forward flight tests at a test altitude that favors an autorotation considering that both climbs and descents from the test altitude will be made.

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3.4.8 Powerplant Torque Matching

For multipowerplant installations, powerplant torque matching and load sharing throughout the operating envelope are of interest. In conjunction with trim response tests, evaluate the ease of matching powerplant torque by using single powerplant trim inputs. Equal load demand between or among the powerplants is of interest. Consider powerplant power control during actual and simulated single powerplant situations. Additional special techniques are not required for this evaluation. Observation during normal and simulated emergency operation will reveal any problems. If a problem with torque matching is encountered, make a time history of powerplant/rotor parameters to document the problem.

3.4.8.1 Data Required

Gas Turbine: Time history of Q, N_1, N_2, T_5, N_R, W_f. Ambient conditions: OAT, H_P. Maneuver description. Time history of Q and powerplant trim position for flight control system powerplant torque matching.

Reciprocating Engine: Time history of P_M, N_1, N_2, T_5, T_{oil}, N_R, W_f. Ambient conditions: OAT, H_P. Maneuver description. Time history of Q and powerplant trim position for flight control system powerplant torque matching.

Electric Motor: Time history of Q, N_1, N_2, T_5, N_R , V_S , A_S , and T_S . Ambient conditions: OAT, H_P. Maneuver description. Time history of Q and powerplant trim position for flight control system powerplant torque matching.

3.4.9 Powerplant Limiting Characteristics

Many UAS powerplants incorporate automatic limiting features. These features may limit powerplant torque, powerplant temperature, or powerplant speed (N_1 and/or N_2). If these features are included, they should be tested for proper operation. The technique used is to approach the limiting condition slowly and note the operation of the feature. If the operation of the limiting feature presents a problem for the pilot, document with a time history.

3.4.9.1 Data Required

Gas Turbine: Time history of Q, N_1, N_2, T_5, W_f, N_R as appropriate. Ambient conditions: OAT, H_P. Description of limiting feature.

Reciprocating Engine: Time history of P_M, N_1, N_2, T_5, T_{oil}, W_f, N_R as appropriate. Ambient conditions: OAT, H_P. Description of limiting feature.

Electric Motor: Time history of Q, N_1, N_2, T_5, N_R, V_S , A_S , and T_S , as appropriate. Ambient conditions: OAT, H_P. Description of limiting feature.

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3.4.9.2 Safety Considerations/Risk Management

By the very nature of these tests, limitations are going to be reached. Use buildup technique to approach the limits in a methodical approach. Thorough knowledge of the limiting feature is necessary to anticipate aircraft response. Limitations should be well defined and approved by the powerplant OEM. If an unanticipated response is found, the tests should be suspended until the reason for the response is understood. Supplementary code may need to be installed to evaluate the powerplant limiting characteristics. If this is done, care needs to be taken when approaching limits of the engine beyond the limits normally imposed by the flight control system.

3.4.10 Powerplant Power Contribution

The rate of descent with the collective at the minimum pitch setting and with the powerplant(s) on line is evaluated and compared to the rate of descent obtained in full autorotation. Make a descent with the collective at minimum and 100 % N_R or normal operating rotor speed for a given airspeed, gross weight, and altitude band. The rate of descent is determined by timing through the altitude band, normally 1,000 ft, but can be altered to suit particular systems. The descent is repeated in full autorotation, if possible, and determined to be reasonably safe for the same test conditions of airspeed, gross weight, and altitude band. Again, the rate of descent is determined by timing through the altitude band. The tests can be repeated for additional airspeeds throughout the normal operating range.

Start the descent above the altitude band of interest to ensure that the aircraft is stabilized while passing through the band. Time splits are usually taken at 500 ft increments. This is done as a backup measure in case the entire 1,000 ft run is not stable. If that is the case, the first 500 ft or the second 500 ft, may provide usable data.

If the AV power controls may not be retarded to idle in flight, full autorotation must be simulated. One technique may be to split the N_2 and N_R by using the powerplant speed trim control to reduce the N_2.

All powerplants can be tested in the same manner to determine powerplant power contribution.

3.4.10.1 Data Required

Altitude band, time through band, airspeed, N_R, gross weight (fuel used or remaining), and ambient conditions.

3.4.10.2 Test Criteria

1. Stabilized N_R
2. Balanced, wings level, unaccelerated flight

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3.4.10.3 Data Requirements

1. Vo ±1 kn or limits of the system
2. N_R ±1 % or limits of the system

3.4.10.4 Safety Considerations/Risk Management

The floor of the altitude band should be high enough to allow time during the descent to bring the powerplant back on line to power the rotor and still maintain a margin of altitude safety. Complete the recovery to powered flight no lower than 1,000 ft AGL. The area must be suitable to continue the descent to touchdown if the power can not be brought back on line for any reason. The first airspeed investigated should be the recommended airspeed for minimum rate of descent. An incremental buildup is conducted to expand the airspeed range. Keep in mind that additional recovery altitude is necessary at greater rates of descent.

3.5 Data Reduction

The data reduction for powerplant assessment test methods involves compiling the qualitative data from AVO or test personnel, quantitative data from the autopilot, and data from the onboard recorder and telemetry systems. The types of curves generated in USNTPS-FTM-No. 106 Section 3.5 are examples of the quantitative data graphical representation. Reciprocating engine limiting characteristic data reduction will be the same as that provided by the USNTPS-FTM-No. 106 for a gas turbine engine, with the understanding that cylinder head temperature (CHT) or exhaust gas temperature (EGT) will take the place of T_5 as the limiting temperature. Electric motor limiting characteristics will depend on the ambient temperature of operation as well as the power available from the battery at that temperature and the maximum amount of torque that can be applied by the motor.

3.6 Data Analysis

Powerplant assessment tests are made to document characteristics (i.e., static and transient droop) or to uncover possible problem areas (i.e., powerplant torque matching or powerplant limiting characteristics). Data are presented to support these characteristics or problems. An analysis of the impact on mission performance is made. Representative mission maneuvers and functions are performed to support mission considerations.

3.7 Mission Suitability

Mission suitability is the primary reason for powerplant assessment testing. How do the powerplant characteristics affect the mission performance? Specifically, what is the time required to launch and what is the time required to reach maximum climb rate?

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Operation of the powerplant/rotor in simulated mission tasks and under simulated mission conditions will reveal the overall suitability of the powerplant/rotor installation.

3.8 Specification Compliance

Each UAS and powerplant detail or development specification may contain provisions for powerplant assessment. Each specification must be researched for applicability and a determination made if the characteristics meet the specification.

3.9 Glossary

3.9.1 Notations

AGL	Above ground level
APU	Auxiliary power unit
CHT	Reciprocating Engine Cylinder Head Temperature, ^{o}C or ^{o}K
EGT	Reciprocating Engine Exhaust Gas Temperature, ^{o}C or ^{o}K
FOD	Foreign object damage
H_{Po}	Observed Pressure Altitude, feet
IGE	In-ground effect
N_1,N_g	Gas turbine engine gas generator speed, reciprocating engine crankshaft speed, electric motor output shaft speed, RPM
N_2, N_f	Gas turbine engine power turbine speed, reciprocating engine gearbox output shaft speed, electric motor gearbox output shaft speed %, RPM
N_R	Main rotor speed %, RPM
P_M	Reciprocating engine intake manifold pressure, inHg
Q	Engine torque %, psi
T_5	Gas turbine engine power turbine inlet temperature or interstage turbine temperature, reciprocating engine cylinder head temperature, electric motor case temperature, °C or ^{o}K
T_{OIL}	Reciprocating engine oil temperature, ^{o}C or ^{o}K
V_C	Calibrated airspeed KCAS
V_O	Observed airspeed KIAS
W_f	Fuel flow, lb/h
V_S	Source or battery voltage draw, V
A_S	Source or battery current draw, A
T_S	Source or battery temperature, ^{o}C

3.9.2 Terms

Flight idle glide	Minimum power descent.
Flight idle stop	The minimum fuel flow which can be delivered with the throttle in the fly position.

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Static droop	The change in N_R, from initial to final steady state, caused by a change in power demand.
Transient droop	The change in N_R, from minimum to final steady state, caused by the rate of change in power demand.

3.10 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 4
ENGINE PERFORMANCE

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4. *Engine Performance*

4.1 Introduction

To determine aircraft performance, power available must be measured and combined with airframe power required to obtain aircraft performance parameters. Aircraft performance parameters of interest are, among others, maximum level flight airspeed, hover ceiling, and maximum rate of climb.

4.2 Purpose of the Test

The purpose of these tests is to determine power available. To accomplish this, inlet or pressure loss performance and powerplant performance will be tested for use in determining the power available curves for the unmanned aircraft systems (UAS).

4.3 UAS vs. Manned Aircraft Theory

4.3.1 Gas Turbine Engine Theory

The theory set forth in U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106) concerning gas turbine engines will apply for UAS with this type of powerplant installed.

4.3.2 Reciprocating Engine Theory

Reciprocating Engine Data Required and Data Reduction

The power available curve for a reciprocating engine, which is usually given as manifold pressure vs. horsepower with specific RPM lines, cannot be determined from flight test alone. The preferred method of determining power available from a reciprocating engine is to have the performance curve (Figure 4.1) for a calibrated engine supplied by the engine OEM. A calibrated engine is simply one that has the proper induction and exhaust system installed during bench testing so that the losses associated with these systems can be quantified. If this is not an option, the engine will need to be tested on a dynamometer to determine the power available curve. This can be accomplished without engine removal on larger air vehicle (AV), but will probably require the engine to be removed for medium and smaller UAS with reciprocating powerplants. The engine, if removed, should have the induction and exhaust systems reinstalled before bench testing to determine the power available curve of the engine with associated induction and exhaust losses. The AV will then need to be flight tested to determine the effects of forward flight on the intake manifold pressure. The rise associated with a reciprocating engine powered AV during forward flight is known as the manifold pressure rise or ram recovery. The quantification of this characteristic is important in determining reliable power available curve corrections for various airspeeds.

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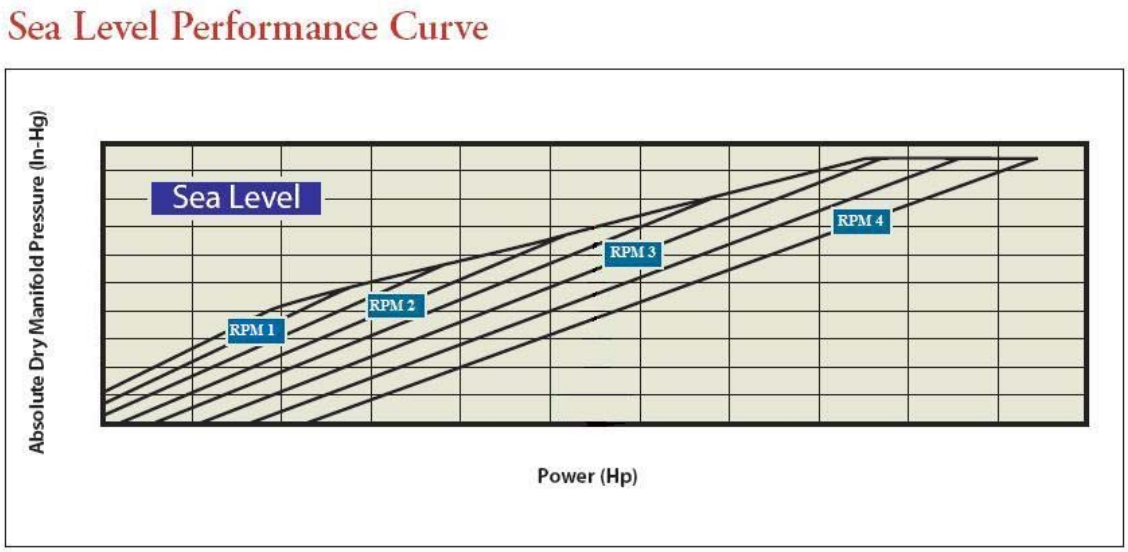


Figure 4.1 Reciprocating Engine Performance Curve

If the torque available curve is desired, the following equation can be used to reduce horsepower data to torque data.

$$\tau = \frac{5252 * \text{hp}}{\text{RPM}}$$

Inlet/intake Pressure and Temperatures

Inlet/intake temperature and pressure are important for reciprocating powerplants in determining corrected data. Instrumentation should be used to determine temperature and pressure in the engine intake to aid in correcting data to standard day temperature and pressure as well as quantifying the performance gain from ram recovery.

4.3.3 Electric Motor Theory

Electric Motor Power Output

Electric motor power is predictable and easily measured. Motor manufactures publish motor specifications and power curves for many motors. If this information is not available, commercially available tools and equipment can be used for determining various electric motor performance parameters such as efficiency, power, and torque. These tools offer a wide range of performance testing applications with possible cost and time saving benefits that can reduce the scope of motor performance determination through flight testing.

Measurement of current draw and voltage to the motor in addition to RPM shows the power used by the motor. This power coupled with published or calculated efficiency

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can be used to predict shaft horsepower output. A typical plot of motor power is shown in Figure 4.2.

$$\text{Power (Watts)} = V_S * A_S$$

$$\eta_{Motor} = \frac{\text{Watts}_{in}}{\text{Watts}_{out}}$$

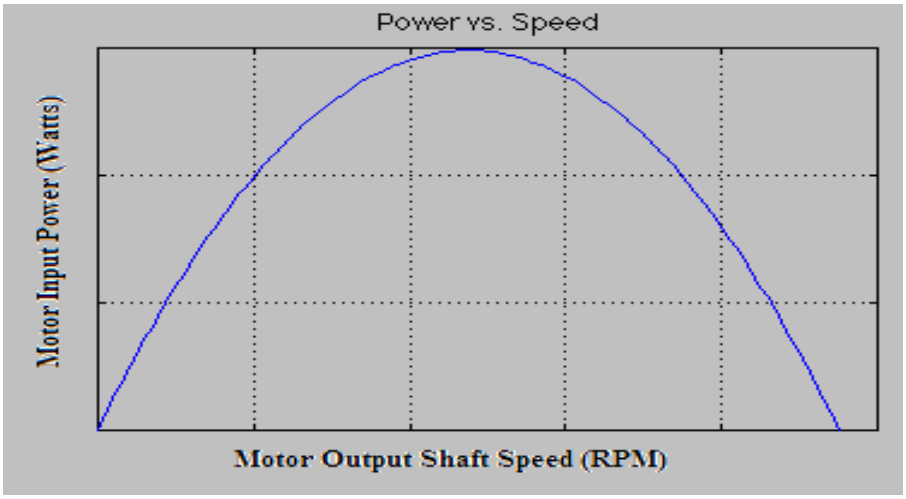


Figure 4.2: Electric Motor Power Curve

If the torque available curve is desired, the following equation can be used to reduce Wattage data to torque data.

$$\tau = \frac{5252 * 746 * W}{RPM}$$

Electric Fuel Flow Calculation

The fuel flow in an electric power system is measured by the amount of electric power used by the motor and other electronic devices. The units for this measure is Coulombs per second, which is further defined as Amperes and related by the equation C = A * s. The power capacity of a battery is usually measured in Amp-Hours or Ah. The amount of power remaining in the battery can be determined by subtracting the amount of power consumed from the power in a charged condition. For example, a system drawing 4 Amps from a 2 Ah battery will have duration of one-half hour.

To determine fuel flow in an electric UAS, the instruments required are an amp-meter installed between the battery and the AV power connector. Average amps can be used or measured amps can be integrated over time to determine the amount of power remaining in the battery of the electric powerplant system.

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Ambient Temperature Effects on Electric Power Systems

Aircraft powerplants are known to be sensitive to outside temperature and pressure. With the introduction of electric power in air vehicles, the associated ambient effects during flight will cause alterations in electric motor performance due to motor temperatures and battery temperatures. Electric motors do not require outside air for fuel, therefore the air density and pressure will not affect engine performance. However, the electric motor windings produce heat during operation which must be dissipated to outside air. The temperature of the outside air or cooling air is directly responsible for the speed at which the heat from the motor can be dissipated. Though performance of electric systems is temperature-dependent, at this time it is not completely understood or well documented for all motors. Power should be tested on individual systems to develop a temperature dependent power curve. Testing should be done in a temperature-controlled environment to document motor performance as a function of temperature variation. The cooling duct should be instrumented for temperature during flight testing to get data on ambient temperature at the motor.

Effects from different flight attitudes will cause variation in motor temperatures. The ram air effect of forward flight will typically cool motors and increase power output by a small percentage. Testing these effects will yield a more complete powerplant test in more areas of the flight envelope.

4.4 Test Methods and Techniques

4.4.1 General

Powerplant performance data are recorded during any steady, unaccelerated flight conditions. Special tests often are not conducted to acquire powerplant and inlet/intake performance data, although when they are, they typically focus on the extremes of powerplant performance. During steady-state conditions, power is set for the test point and powerplant data are recorded along with the data supporting other test purposes. For example, powerplant and inlet/intake performance data can be obtained along with pitot/static system performance data during the course of the airspeed calibration. At each test point, the powerplant must be allowed to stabilize and sufficient time elapsed to obtain a representative fuel flow or current draw in the case of an electric powerplant. All parameters should be monitored during the recording interval for any unusual changes.

4.4.2 Inlet/Intake Performance for Gas Turbine Engines

Vertical takeoff and landing (VTOL) UAS with turbine engines will follow the test methodology for inlet performance described in USNTPS-FTM-No. 106 Section 4.4.2. Specifically, the instrumentation necessary consists of both static and pitot pressure across the compressor face to ensure uniform pressure distribution at the compressor inlet and total pressure, which, when combined with total temperature at the compressor inlet, will be used to correct for standard day corrections. The test team

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should consider any airworthiness issues associated with the instrumentation early in the test program to ensure acceptance.

4.4.2.1 Data Required

Static Pressure at the Engine Compressor Face (P_{S2}), Total Pressure at the Engine Compressor Face (P_{T2}), Static Temperature at the Engine Compressor Face (T_{S2}), Indicated Airspeed (Vo).

4.4.2.2 Test Criteria

1. Balanced, wings level, unaccelerated flight
2. Engine bleed air systems off or operated in normal flight mode
3. Stabilized engine power demand

4.4.2.3 Data Requirements/Tolerances

1. Stabilize 60 s minimum before recording data
2. Record stabilized data 60 s
3. V_O ±1 KIAS or limits of the flight control system

4.4.2.4 Safety Considerations/Risk Management

Inlet and engine performance data are gathered during all steady-state flight conditions. Usually these tests are combined with other vertical takeoff and landing (VTOL) testing. The conditions and procedures applicable to the associated flight tests are applicable to the inlet and engine performance flight tests. Of particular concern for inlet tests is the location and airworthiness of the inlet instrumentation rake. Before installation, the rake must be tested to ensure it can withstand safely the dynamic pressures, vibrations, and temperatures that are present or anticipated in the inlet. For engine run ground tests, necessary VTOL restraining devices must be designed, fabricated, and installed.

4.4.3 Powerplant Performance

Powerplant performance data are required for all flight regimes and are obtained during steady-state, unaccelerated test points. Usually, inlet and powerplant performance data are recorded during the same test points for turbine powerplant. Data points at the extremes of powerplant power demand are required and obtained in high speed forward flight, high power climbs at high altitudes, at minimum power on the deck, and during power-on autorotational descents (flight idle glides).

4.4.3.1 Data Required

Gas Turbine:

Torque (Q)

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Gas generator speed (N_1)
Power turbine inlet temperature (T_5)
Fuel flow (W_f)
Rotor speed (N_R)
(Vo)
Outside air temperature (OAT)
Pressure altitude **(**H_{Po})

Reciprocating Engine:

Manifold pressure (P_M)
Crankshaft speed (N_1)
Cylinder head temperature (CHT) or
Exhaust gas temperature (EGT)
Oil temperature (T_{oil})
(W_f)
(N_R)
(Vo)
Outside Air Temperature (OAT)
Pressure Altitude **(**H_{Po})

Electric Powerplants:

Torque (Q)
Motor output shaft speed (N_1)
Motor case temperature (T_{case})
(N_R)
(Vo)
Source voltage draw (V_S),
Source current draw (A_S)
Source temperature (T_S)
OAT
Pressure altitude **(**H_{Po})

4.4.3.2 Test Criteria

1. Balanced, wings level, unaccelerated flight
2. Engine bleed air systems off or operated in normal flight mode (if applicable)
3. Stabilized powerplant power demand

4.4.3.3 Data Requirements/Tolerances

1. Stabilize 60 s minimum before recording data
2. Record stabilized data 60 s
3. N_R ±0.2% or limits of the system
4. N_1 ±0.1% or limits of the system
5. T_5 ±5 °C
6. Vo ±1 KIAS or limits of the system

4.4.3.4 Safety Considerations/Risk Management

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Know and adhere to normal operating procedures and all limitations. Exercise care at the high power points so powerplant limitations are not exceeded. Consider lags in the system as well as transient lags that may be present.

4.4.4 Power Available

Powerplant power available is computed from inlet and/or powerplant performance data at maximum throttle or RPM setting. Separate tests are not required.

4.5 Data Reduction

4.5.1 General

The data reduction methods described in USNTPS-FTM-No. 106 will correlate for VTOL UAS powered by turbine engines. For reciprocating engines and electric powerplants, the data reduction techniques mentioned in Section 4.3 should be used.

4.5.2 Inlet Performance for Turbine Engines

The inlet performance data reduction as explained in section 4.5 of USNTPS-FTM-No. 106 will apply for gas turbine engines but not for reciprocating engines or electric powerplants. However, the intake duct for reciprocating engines should be instrumented for pressure and temperature during engine performance tests, as the cooling duct for electric powerplants should be instrumented for temperature during motor performance tests.

4.5.3 Powerplant Performance

The techniques described in USNTPS-FTM-No. 106 Section 4.5.3 will apply for turbine engine data reduction. Reciprocating and electric powerplants should follow the data reduction techniques mentioned in Section 4.3 that will produce the performance curves similar to Figure 4.1 for reciprocating and Figure 4.2 for electric powerplants. All powerplant performance parameters should be corrected to standard day operating conditions so that all performance data will be generalized.

4.5.4 Power Available

The power available can be derived from the maximum shaft horsepower as a function of RPM and torque for turbine engines, maximum horsepower as a function of manifold pressure and RPM, and maximum horsepower as a function of RPM and remaining battery capacity for electric powerplants. Again the performance parameters for turbine and reciprocating engine should be corrected for nonstandard temperature and pressure to generalize the data; however, electric powerplants need only to be corrected for nonstandard temperature.

4.6 Data Analysis

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The importance of engine performance (power available) is the impact on aircraft performance parameters when combined with airframe power required. Subsequent chapters deal with power required. Engine power available as developed in this chapter is combined with airframe power required to determine aircraft performance parameters.

Gas Turbine Engine: For turbine engines, inlet performance, as measured by the inlet pressure and temperature ratios, is used to determine corrected engine performance as described by N_{1corr}, $ESHP_{corr}$, T_{5corr}, and W_{fcorr}. Specific fuel consumption, as a measure of engine efficiency, is determined from W_f and ESHP. Both inlet performance and corrected engine performance are used to determine power available as a function of altitude for a given airspeed or as a function of airspeed for a given altitude.

Reciprocating Engine: Reciprocating engines have a relatively flat power curve (hp vs. airspeed) so the manifold pressure as a function of horsepower at constant RPM (Figure 4.1) will be the most useful performance chart for an operator.

Electric Motor: Operators of UAS with electric powerplants will find the power versus RPM graph (Figure 4.2) most useful coupled with a manufactures plot of motor efficiency as a function of RPM.

4.7 Mission Suitability

As with data analysis, mission suitability aspects of engine performance alone is not considered. Engine performance (power available) when combined with airframe performance (power required) will produce performance parameters for the UAS.

4.8 Specification Compliance

There are no general Mil-Specs for engine performance; however, there may be performance specifications called out in the contract specifications. Consult each AV or powerplant specification for applicability. Often, maximum specific fuel consumption and minimum power available are specification requirements. The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAS will provide qualifications for which it was designed to meet. Expected performance of the UAS will also be rooted in the modeling and simulation of UAS. The detailed technical manuals for the UAS can then be written from the developmental testing as a reference source for all future system testing and military operations.

4.9 Glossary

A Amperes (C/s)

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A_S	Source or battery current draw, Amperes
C	Coulomb
CHT	Reciprocating engine cylinder head temperature, °C
EGT	Reciprocating engine exhaust gas temperature, °C
H_{Po}	Pressure altitude, ft
η_{Motor}	Electric motor efficiency, %
N_1	Gas generator rotating speed, engine output shaft rotating speed, or motor output shaft rotating speed, % RPM
N_R	Rotor rotating speed, % RPM
OAT	Outside air temperature, °C
P_M	Reciprocating intake manifold pressure, inHg
P_{S2}	Static pressure at the engine compressor face, psi
P_{T2}	Total pressure at the engine compressor face, psi
Q	Torque, %, psi
RPM	Revolutions per minute
τ	Engine torque, ft-lb
T_{case}	Electric motor case temperature, °C
T_{S2}	Static temperature at the engine compressor face, °C
T_5	Gas turbine engine power turbine inlet temperature, °C
T_S	Source or battery temperature, °C
Vo	Indicated airspeed, kias
V_S	Source or battery voltage draw, Volts
W	Electric motor power, Watts
W_f	Fuel flow, lb/hr

4.10 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 5
HOVER PERFORMANCE

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5. *Hover Performance*

5.1 Introduction

This chapter deals with determining vertical takeoff and landing (VTOL) unmanned aircraft systems (UAS) hover performance. Powerplant power available and airframe power required to hover will be combined to determine air vehicle (AV) hover performance. Test techniques, scope of test, and the limitations associated with each test are discussed. Details regarding nonstandard rotor design (ducted fans, autogyros, tail sitters, etc.) performance testing are reserved for later revisions of this chapter.

5.2 Purpose of the Test

The purpose of this test is to evaluate AV hover performance characteristics. Airframe power required to hover will be determined and combined with powerplant power available to establish aircraft hover performance.

5.3 UAS vs. Manned Aircraft Theory

For UAS using standard rotor designs, defined as variable pitch, nonducted rotors which produce the majority of lift and thrust in all modes of flight (hover, forward flight, climb and descent), the theory contained in U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106) Section 5.3 applies. UAS specific theory for AV with nonstandard rotor designs will be covered in later revisions of this document.

5.3.1 UAS with Standard Rotor Designs

The theory contained in USNTPS-FTM-No. 106 would be applicable for all UAS with standard rotor designs.

5.3.2 UAS with Nonstandard Rotor Designs

Reserved

5.4 Test Methods and Techniques

5.4.1 General

The free flight hover technique is the lowest complexity test technique for determining UAS hover performance. Tethered hover requires extremely precise position information for the position maintenance control loop. The objective of gathering hover performance data is to determine the power required to hover at various pressure altitudes given a constant gross weight. The flight control system on most UAS will not be programmed to compensate for a tether; therefore, the tethered technique will

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most likely be limited to those UAS designed with a requirement for carrying external loads. Hover height measurements during the free flight technique can be gathered by using a combination of radar altimeter/laser rangefinder from 0 to 250 ft above ground level (AGL) and a global positioning system (GPS) for altitudes above 250 ft AGL. If available, a differential GPS (DGPS) system is the preferred measuring device for all altitudes. Small, light, and accurate radar altimeters are commercially available that can measure altitudes over 250 ft AGL. Commercially available laser rangefinders are small, lightweight, and very accurate to altitudes over 1,000 ft AGL. All but the smallest AV, with the payload removed, should be able to carry these instruments. Most modern GPS receivers will be accurate to within 25 vertical ft, with DGPS receivers obtaining altitude solutions accurate to 1 ft.

Some small UAS hover performance may be dramatically altered by wind and/or weather effects. Given the low Reynolds numbers of small UAS rotors, small amounts of wind may cause a rotor efficiency gain, causing the hover performance to appear better than it actually is. Consequently, some small UAS may need to be tested in large indoor structures so as to negate wind and weather effects and obtain reliable data.

To evaluate UAS hover performance over an operating envelope, the thrust coefficient (C_T), given in equation 5.1, should be varied so that the power required for hover inside ground effect (IGE) and outside ground effect (OGE) can be determined.

$$C_T = \frac{T}{\rho_a A_D (\Omega R)^2} \qquad \text{Eq. 5.1}$$

T = Thrust (lbs)
ρ_a = ambient air density (slug/ft^3)
A_D = Rotor disk area (ft^2)
Ω = Rotor Angular Velocity (rad/s)
R = Rotor Radius (ft)
ΩR = Rotor Rotational Velocity (ft/s)

The parameters in the thrust coefficient equation should be varied, preferably one at a time, to determine the range of C_T over the operating envelope of the UAS. Since in a hover thrust will approximately equal weight, changing weight will alter C_T. The weight can be changed by varying the fuel load (maximum gross takeoff weight, intermediate gross takeoff weight, and minimum takeoff weight to perform mission and land with required reserve) before each test flight for turbine and reciprocating powered UAS. During the sortie, fuel burn will decrease weight so that an accurate method of measuring fuel flow, such as a fuel flow meter, must be used. If accurate fuel flow instrumentation does not exist on the AV, special test instrumentation must be installed. Electric UAS may change takeoff weight by varying the payload or adding ballast to give a range of lift-off weights.

The other parameters which can be changed to alter C_T are the air density and rotor speed. The density can be varied by hovering at various altitudes (IGE, OGE,

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transitional, etc.). Changing rotor speed can be an effective method of obtaining a number of different values of C_T, if the AVO has the ability to vary rotor speed. As stated in USNTPS-FTM-No. 106 Section 5.4.1.3, the big advantage of using rotor speed as a variable parameter is the ability to extend the C_T test range above and below those obtainable by other means. For example, if at sea level the maximum takeoff gross weight does not provide a C_T corresponding to the anticipated hover ceiling, then the only rapid method to obtain a high C_T at the test site is a reduction in rotor speed (ΩR).

For UAS with electric powerplants, the battery characteristics should be understood and well documented so that the test team can determine the powerplant operating regime during flight test. The test team should consider only testing electric UAS in the quasi-linear portion of the battery drain curve. The beginning and end of the battery drain curve may be inaccurate for flight performance testing since there are larger than average voltage decreases over short periods of time.

5.4.1.1 UAS with Standard Rotor Design

The general test technique data contained in section 5.4.1 of USNTPS-FTM-No. 106 will be applicable for all UAS with a standard rotor design.

5.4.1.2 UAS with Nonstandard Rotor Design

Reserved

5.4.2 Power Train Oscillations

This test can be used to determine average power required to hover as well as validate OEM specifications for altitude hold during hover. Data should be collected for at least 5 seconds after the AV reaches and stabilizes on a commanded hover altitude as stipulated in USNTPS-FTM-No. 106 Section 5.4.3.4. The data should then be averaged over the 5 seconds to obtain one data point. If the autopilot is continuously varying control inputs and preventing a stable test point, then the control software may need to be altered for the AVO to ensure a stable test point; this will require a flight release from the airworthiness authority.

5.4.3 Free Flight Hover

5.4.3.1 General

The hover altitude of the AV should be measured using a combination of a radar altimeter/laser rangefinder up to 250 ft AGL with GPS for altitudes above 250 ft AGL. If available, a DGPS system is the preferred measuring device for all altitudes. The AV is made to hover at various altitudes, including altitudes IGE, OGE, and transitional, based on mission-related hover altitude requirements. The transition from IGE to OGE is marked by the point where the torque required to hover at increasing altitude becomes constant. The power required to hover is recorded for various combinations of gross

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weight (minimum landing weight, maximum gross takeoff weight, and intermediate gross weight), density, and rotor RPM.

5.4.3.2 Data Required

Gas Turbine Engine:

Torque (Q)
N_1
Rotor speed (N_R)
Outside air temperature (OAT)
Fuel used (FU, FC) or fuel remaining
Hover height
Wind speed and direction
H_{Po}
Ground temperature and pressure

Reciprocating Engine:

Manifold pressure (P_M)
N_1
N_R
OAT
Fuel used (FU, FC) or fuel remaining
Hover height
Wind speed and direction
H_{Po}
Ground temperature and pressure

Electric Motor:

N_R
OAT
Source voltage draw (V_S)
Source current draw (A_S)
Hover height
H_{Po}
Wind speed and direction

5.4.3.3 Test Criteria

1. Wind speed <3 kt
2. Engine bleed air systems off or operated in normal flight mode
3. Testing is not within or does not cross a temperature inversion

5.4.3.4 Data Requirements/Tolerances

1. Stabilize 15 seconds (s) minimum before recording data
2. Record stabilized data 5 s
3. $N_R \pm 0.2$ % or limits of the system
4. Maximum drift ± 3 ft or limits of the system
5. Maximum hover height ±1 ft IGE, ±3 ft OGE, or limits of the system

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5.4.3.5 Safety Considerations/Risk Management

Conducting the tests at altitude provides the AVO and autopilot with additional options in the event of engine or datalink malfunction. Testing should be conducted at an altitude and over terrain that will permit a successful landing. Operation in the avoid areas of the height velocity diagram cannot be eliminated during all hover tests, but exposure must be minimized.

Thorough planning and crew briefing, including ground support personnel, will minimize potential hazards. During all tests, vortex ring state should be avoided. The flight control system of some UAS may not be able to recover if the AV were to enter this state. Safeguards should be implemented by the OEM in the form of maximum descent rates for various forward airspeeds.

For UAS hovering IGE, the gain on the autopilot should be quite large to prevent instability. Once OGE is achieved, the gain should be programmed lower to enable more rapid response to control inputs.

5.5 Data Reduction

5.5.1 Standard Rotor Design

The data reduction techniques described in USNTPS-FTM-No. 106 Section 5.5 will be applicable for UAS standard rotor design, as defined in Section 5.3, with the unit conversion of electric motor power data from Watts to horsepower using the relation (746)*(W)=1hp.

Power margin, difference between power available and power required, should also be considered in flight testing, since it varies with test-day atmospheric conditions. This is particularly important for hover performance since a temperature higher than standard and a lower atmospheric pressure than standard could dramatically decrease power margin, decreasing the hover ceiling and possibly requiring the AV to make a rolling takeoff in the most extreme conditions.

5.5.2 Nonstandard Rotor Design

Reserved

5.6 Data Analysis

5.6.1 Standard Rotor Design

The data reduction techniques described in USNTPS-FTM-No. 106 Section 5.6 will be applicable for UAS with standard rotor design. As stated in Section 5.5.1,

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changes in power margin with nonstandard environmental conditions can dramatically affect hover performance.

5.6.2 Nonstandard Rotor Design

Reserved

5.7 Mission Suitability

The VTOL UAS has a distinct advantage with its ability to operate in the low speed environment and to hover. From a performance perspective then, hover performance takes on great importance. This importance is expressed in relation to the payload that can be carried, the camera image accuracy on target, and the weather environments where the UAS can operate. In determining the mission suitability of a UAS hover performance, the hover ceiling should be determined for all applicable mission gross weights, standard day conditions, hot day conditions, IGE, OGE, and applying a mission relatable power margin as discussed in the data analysis. If mission suitability is deficient in any areas, mission relation may be made in terms of amount and percentage of fuel or payload reduction required to hover at a given altitude. Alternatively, if a given payload and fuel load is required for the mission, the density altitude will limit the UAS system locations for operation.

5.8 Specification Compliance

5.8.1 General

The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAS will provide qualifications for which it was designed to meet. Expected performance of the UAS will also be rooted in the modeling and simulation of UAS. The detailed technical manuals for the UAS can then be written from the developmental testing as a reference source for all future system testing and military operations.

The manufacturer specification for UAS typically lists the hover ceiling capability or limit. This hover ceiling limit is for a particular gross weight, at maximum rated power, standard day conditions and is stated in terms of OGE hover altitude. All limits should be given for less than 3 kt of wind, depending on system limitations.

5.8.2 Power Available

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The testing involved for calculation of hover ceiling requires use of a powerplant from a manufacturing line which may have a differing shaft horsepower output than specified. When testing a specific AV to meet required specifications, a spec powerplant should be used. A spec powerplant has a power curve generated from previous testing performed on that powerplant as configured or installed in the AV. If the power curves for the spec powerplant are not available, then the actual installed power available should be used for computations. Installed power available should be determined as discussed in Chapter 4.

5.9 Glossary

A_S	Electric motor source or battery current draw	Amperes
FC	Fuel counts remaining	lb
FU	Fuel used or consumed	lb
H_{Po}	Indicated altitude	ft
N_1	Gas turbine engine gas generator speed, reciprocating engine crankshaft speed, electric motor output shaft speed,	%, RPM
N_R	Rotor speed	%, RPM
OAT	Outside air temperature	^oC
P_M	Reciprocating engine intake manifold pressure	inHg
Q	Gas turbine engine or electric motor torque	%
V_S	Electric motor source voltage draw	Volts

5.10 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 6
VERTICAL CLIMB PERFORMANCE

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6. *Vertical Climb Performance*

6.1 Introduction

This chapter deals with the general aspects of determining vertical takeoff and landing (VTOL) unmanned aircraft systems (UAS) vertical climb performance. The theory of a rotor in axial flight is discussed in U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106). The test method discussed here is relatively simple yet produces a very thorough evaluation of the vertical climb performance. Test-day vertical climb data are used to determine the vertical climb correction factor and standard day vertical climb performance. Test applicability as it relates to VTOL UAS is discussed under test methods.

6.2 Purpose of the Test

The purpose of this test is to investigate air vehicle (AV) vertical climb performance characteristics to achieve the following objectives:

1. Determine the vertical climb correction factor for use in computing vertical climb performance.
2. Evaluate the pertinent requirements of the detailed model specification.
3. Define mission suitability problems.

6.3 UAS vs. Manned Aircraft Theory

UAS have flight control systems which take the desired input from the air vehicle operator (AVO). The AVO has no direct control over the collective pitch of the rotor blades and, therefore, different testing methods must be used. Depending on the system, a VTOL UAS will probably not be equipped with the level of direct vertical climb control found in manned systems. Some systems may incorporate more than one selectable vertical climb rate, such as a "quick climb" command, in addition to a standard vertical climb rate. Though a system may be equipped with more than one climb rate, the AVO may not have the same control over the vertical climb rate of an AV as a pilot would in a manned helicopter. In most cases, the UAS will have a preprogrammed climb profile as set by the manufacturer and the AVO will select which type of climb to perform. The tests described in this chapter can be performed to verify OEM claimed climb rates and measure power required to climb in various configurations. If it is preferred to test a number of airspeeds to determine the maximum rate of climb, then the control software may need to be altered to a test configuration; this will require a flight release from the airworthiness authority.

6.3.1 UAS with Standard Rotor Design

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For a UAS with a standard rotor design, as defined in Section 5.3, the theory as explained in section 6.3 of USNTPS-FTM-No. 106 would be applicable for all powerplants with the exception of electric powerplants. For electric powerplants, power is measured in Watts and converted to horsepower using the following expression:

$$(746)*(W) = 1hp \qquad \text{Eq. 6.1}$$

With the electric motor power output converted to shaft horsepower using Equation 6.1, where W is Watts and hp is horsepower, the electric motors can be compared and evaluated like other powerplants.

6.3.2 UAS with Nonstandard Rotor Designs

Reserved

6.4 Test Methods and Techniques

6.4.1 General

The objectives of vertical climb testing are to determine test-day vertical climb performance and to compute a vertical climb correction factor. After obtaining this climb correction factor relationship from flight test, vertical climb rates can be predicted for any combination of aircraft gross weight, rotor speed, atmospheric parameters, or power margins. Vertical climb rate predictions are valid for the range of flight test conditions evaluated in building the climb correction factor model. Aircraft configuration (stores, weapons, fuel pods, etc.) is considered when grouping sets of data.

The aircraft is stabilized first at an out-of-ground effect (OGE) hover (height as determined in hover performance tests) and data are collected. The power margin is determined (power available less the power required to hover OGE). Several vertical climbs are accomplished at increasingly larger excess power values until the limit power available is reached. Small variations in horizontal velocity (as little as 3 kt) contaminate the data. Heading control is required. Once the vertical climb has stabilized, as determined by reference to a sensitive vertical accelerometer or a vertical speed indicator, data are recorded. Time history altitude information is used generally, but time to climb through an altitude band can be substituted when an automatic data collection system is not available. Once the data have been collected, the AVO increases forward speed and flies the UAS back to the initial hover point for succeeding data points. The test team should analyze the test article to choose altitude bands to minimize the impact of fuel burn on vertical climb performance.

Hover performance for small UAS may be dramatically altered by the wind and/or weather effects. Given the low Reynolds numbers of small UAS rotors, the wind could actually help the rotor become more efficient than under no wind hover condition,

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causing the hover performance to appear better than expected. Consequently, some small UAS may be tested in large indoor structures so as to negate wind and weather effects.

The AV weight should be varied since it dramatically affects the power required for vertical climbs. This can be accomplished by varying the fuel load (maximum gross takeoff weight, intermediate gross takeoff weight, and minimum takeoff weight to perform mission and land with required reserve) or ballasting before each test flight for turbine and reciprocating powered UAS. During the flight, fuel burn will decrease weight so an accurate method of measuring fuel flow, such as a fuel flowmeter, must be used. If accurate fuel flow instrumentation does not exist on the AV, special test instrumentation must be installed. Electric UAS takeoff weight can be varied by varying the payload mass or adding ballast. The power required to climb vertically can be varied using a range of different rotor speeds if the GCS software is modified to alter rotor RPM.

For UAS with electric powerplants, the battery characteristics should be understood and well documented so that the test team can determine the powerplant operating regime during flight test. The test team should consider only testing electric UAS in the quasi-linear portion of the battery drain curve, since the beginning and end of the battery drain curve can be inaccurate for flight performance testing since there are quite large power decreases over short periods of time.

6.4.2 Free Flight Vertical Climb Performance

6.4.2.1 General

Similar to testing the climb performance of a manned platform, the test should begin once the AV has stabilized in an OGE hover. A differential global positioning system (DGPS) system is preferred over a standard GPS system for gathering climb performance data due to its reduced altitude position error; however, a standard GPS system can be used for evaluating vertical climb performance if testing in a region where a wide area augmentation system signal is available.

Ground Referenced Test

Most UAS will be equipped with flight control systems capable of holding the AV over a fixed point programmed in by the AVO. Test data can be collected from commanded vertical climbs at various settings (if applicable) while the AV maintains position over the preprogrammed point. To properly evaluate vertical climb performance or validate OEM vertical climb performance specifications, winds should be the lesser of 3 kt or as expressed in the system limitations.

Air Referenced Test

Some AV may be capable of performing air mass referenced vertical climbs by programming the AV to not adjust for wind conditions. This can be accomplished by

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using the flight control system if equipped with an inertial navigation system (INS). The INS would be programmed to minimize horizontal accelerations during the vertical climb tests to maintain a true vertical climb with wind. If the AV under test can perform such climbs, winds need not be less than 3 kt for climb performance evaluation and OEM specification compliance; however, a stable air mass as defined in Section 1.8 is preferable.

6.4.2.2 Data Required

Gas Turbine Engine:

OGE Hover:
- Outside air temperature (OAT)
- Indicated altitude (H_{Po})
- Engine gas generator speed (N_1)
- Rotor speed (N_R)
- Torque (Q)
- Fuel remaining (FC)
- H_{Po}
- AV configuration

Vertical Climb:
- OAT
- N_1
- N_R
- Q
- FC
- H_{Po}
- AV configuration
- Time history of climb

Reciprocating Engine:

OGE Hover:
- OAT
- H_{Po}
- Engine crankshaft speed (N_1)
- N_R
- Manifold pressure (P_M)
- Engine crankshaft speed (N_1)
- FC
- H_{Po}
- AV configuration

Vertical Climb:
- OAT
- N_R
- P_M
- N_1
- FC
- H_{Po}

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AV configuration
Time history of climb

Electric Motor:

OGE Hover:
- OAT
- H_{Po}
- N_1
- N_R
- Motor output shaft speed (N_1)
- Source voltage draw (V_S)
- Source current draw (A_S)
- AV configuration

Vertical Climb:
- OAT
- H_{Po}
- N_1
- N_R
- V_S
- A_S
- AV configuration
- Time history of climb

6.4.2.3 Test Criteria

1. Stable air mass (or winds less than 3 kt)
2. Testing range does not cross a temperature inversion

6.4.2.4 Data Requirements/Tolerances

OGE Hover:
1. Stabilize 15 seconds before recording data
2. N_R ±1% or limits of the system
3. OGE hover height ±3 ft of target altitude or limits of the system

Vertical Climb:
1. Stable vertical climb rate or zero vertical acceleration
2. N_R ±1 %
3. Horizontal velocity ±3 kt, wind less than 3 kt (for vertical climb over preprogrammed point)
4. Heading tolerance ±5 degrees
5. Stable air mass as defined in Section 1.8 for air referenced vertical climb tests

6.4.2.5 Safety Considerations/Risk Management

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Conducting the tests at altitude provides the AVO and autopilot with additional options in the event of engine or datalink malfunctions. Conduct OGE hover and climbs over suitable forced landing areas and preferably oriented into the prevailing wind. Use an area chase aircraft to provide in-flight separation from other aircraft if necessary. Review vortex ring state recovery procedures if applicable as well as lost link procedures. Be sure to follow OEM descent procedures between climb tests.

6.5 Data Reduction

UAS vertical climb performance data reduction can be performed using the same methods as for manned aircraft. The data reduction techniques discussed in USNTPS-FTM-No. 106 would be applicable for all powerplant configurations with the units of power output data of electric motors converted to horsepower using Eq. 6.1.

6.6 Data Analysis

The data analysis explained in section 6.6 of USNTPS-FTM-No. 106 would be applicable for all UAS with all powerplants.

6.7 Mission Suitability

6.7.1 General

The requirement for establishing the vertical rate of climb capabilities of a UAS is important during performance evaluations. The UAS must demonstrate vertical climb maneuvers with part of its mission requiring some form of a vertical climb. The employment of a UAS in a tactical situation requiring confined area takeoffs from a hover is an example of mission suitability requirements which dictate a satisfactory degree of vertical climb performance.

6.7.2 UAS Control System

The flight test determination of the vertical climb performance is simple, yet very demanding. The accuracy of the results depends on the AV capabilities and control authority, whether AVO may have control or autopilot maintains all control. The autopilot must sense horizontal displacement through changes in aircraft heading and attitude using GPS techniques. Some UAS systems may be limited to forward flight climbs when out of ground effect or lower the takeoff weight to maintain a higher minimal vertical speed for climb. The overall service suitability determination of the vertical climb performance characteristics reflect the flight controller limitations for the AV.

6.8 Specification Compliance

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The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAS will provide qualifications for which it was designed to meet. Expected performance of the UAS will also be rooted in the modeling and simulation of UAS. The detailed technical manuals for the UAS can then be written from the developmental testing as a reference source for all future system testing and military operations.

The specification for a UAS system will have the rate of climb listed as tested by the manufacturer. This test validates this performance parameter and creates an operating envelope for weight and climb rates. Generally three gross weights are satisfactory for testing; minimum operating weight, intermediate gross weight, and maximum gross takeoff weight.

6.9 Glossary

A_S	Source or battery current draw	Amperes
H_{Po}	Indicated altitude	ft
FC	Fuel counts remaining	lb
hp	Horsepower	hp
N_1	Gas turbine engine gas generator speed, reciprocating engine crankshaft speed or electric motor output shaft speed, %, RPM	
N_R	Rotor speed	%, RPM
OAT	Outside air temperature	^oC
P_M	Reciprocating engine intake manifold pressure	inHg
Q	Gas turbine engine torque or electric motor torque	% psi
V_S	Source or battery voltage draw	Volts
W	Watts	W

6.10 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 7
LEVEL FLIGHT PERFORMANCE

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7. *Level Flight Performance*

7.1 Introduction

This chapter considers the level flight performance for vertical takeoff and landing (VTOL) unmanned aircraft system (UAS). Airframe level flight performance (power required) is determined from the tests described within this chapter to determine the airspeeds for maximum endurance and maximum range; the power required is combined with power available as determined in Chapter 4 to yield maximum level flight airspeed.

Some UAS, such as tilted systems or tail sitter designs, will need to be tested in the same manner as fixed wing UAS to determine level flight performance parameters. If this is the case, these UAS can be tested using the UAS Flight Test Methodology, Annex to U.S. Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108) "Fixed Wing Performance" Chapter 4 (Level Flight Performance).

7.2 Purpose of the Test

The purpose of the test is to investigate aircraft performance characteristics in level flight to achieve the following objectives:

1. Determine significant performance parameters: maximum level flight airspeed (V_H), never exceed airspeed (V_{NE}), maximum range airspeed ($V_{MAX\ RANGE}$), cruise airspeed (V_{CRUISE}), maximum endurance airspeed ($V_{MAX\ END}$), combat radius, maximum endurance, and maximum range.
2. Define mission suitability problems.
3. Evaluate the pertinent requirements of the detailed specification.
4. Identify vibratory problems associated with given airspeeds.

7.3 UAS vs. Manned Aircraft Theory

For UAS using standard rotor designs, as defined in Section 5.3, the theory contained in USNTPS-FTM-No. 106 Section 7.3 will be applicable. UAS specific theory for air vehicle (AV) with nonstandard rotor designs (e.g., ducted fans, autogyros, tail sitters, etc.) will be covered in later revisions of this document.

7.3.1 UAS with Standard Rotor Designs

The theory contained in USNTPS-FTM-No. 106 is applicable for all UAS with standard rotor designs.

7.3.2 UAS with Nonstandard Rotor Designs

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7.4 Test Methods and Techniques

7.4.1 General

The USNTPS-FTM-No. 106 manual can be followed for explanation of flight tests and comparison of different testing methods. The constant W/σ (weight of the AV divided by air density ratio) method or W/δ (weight of the AV divided by the air pressure ratio) method described in USNTPS-FTM-No. 106 Section 7.4.1 are applicable for UAS testing. Though most UAS can be tested using these classic methods, most UAS will require fewer altitude changes to maintain constant W/σ or W/δ due to small changes in weight resulting from lower fuel consumption. This will simplify the process of computing the proper altitude/temp/weight combination required for flying 8 to 10 test points at a constant pressure altitude.

Existing fuel flow indicators or supplemental fuel flow test instrumentation are necessary for determining precise aircraft weight during this test. Measuring fuel flow for systems without onboard equipment or instrumentation may be accomplished using alternate methods such as weighing the AV before and after each test point, which will yield an average fuel flow rate for the entire flight. However, this method of determining fuel flow is less accurate than instrumented methods and significantly increases the amount of time required for the test due to the multitude of takeoffs and landings.

Testing of all-electric UAS is simplified by the fact that AV weight does not change during flight. The AV, therefore, does not require changing altitude to hold a constant W/σ or W/δ unless temperature changes. However, consideration must be given to the decrease in maximum level flight airspeed as the battery nears depletion, due in part to the payload and other power draws to the system, which reduces the available power. Planning for the impact of these auxiliary power draws can reduce their impact on the execution of the level flight test points while not sacrificing proper buildup technique. A recommended method is to conduct the testing for each weight requirement starting from the slowest cruise airspeed and increasing to higher airspeeds in even increments. Before reaching V_h, the AV should be landed to install a newly recharged battery and relaunched to capture the V_h test point at maximum available power.

This test also presents an opportunity to evaluate the flight path stability at varying airspeeds. The measurement of flight path accuracy is best done using global positioning system (GPS) tracking over ground and looking for cross track errors. Vibration issues associated with the AV vary with different airspeeds and flight profiles. Instrumentation sensing vibration levels will show effects and varying airspeeds and rotor RPM. Instrumentation that monitors servo temperatures and controls may provide information relating to the autopilots workload while maintaining level flight at specific airspeeds.

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Limited airspace at the test facility may adversely impact test efficiency. UAS capable of high airspeeds will need long straight legs to conduct level flight speed sweeps. Frequent turns to keep the aircraft in restricted airspace will add complexity and time to the test. This factor should be considered carefully in test site selection.

7.4.2 Techniques

The USNTPS-FTM-No. 106 manual can be followed for explanation of flight tests techniques. Exceptions and additional techniques are listed below.

The outside air temperature (OAT) and observed pressure altitude (H_{Po}) are best measured on the AV; however, the UAS static port may be inaccurate due to rotor downwash and adding thermocouples to measure OAT may be difficult for some UAS due to weight concerns. If the UAS has an inaccurate static pressure sensor or cannot include additional instrumentation (probably contained to small UAS) for OAT measurement and there is a low probability of a temperature inversion, then measuring OAT and H_{Po} on the ground and using standard temperature and pressure lapse rates can be a reasonably accurate method of obtaining OAT and H_{Po} at the AV altitude. Altitudes for flight testing may be varied to accompany test facilities or airspace restrictions. AV size will typically determine maximum airspeeds and space requirements. The space required will determine altitudes necessary for line-of-sight flight and reasonable heights. If the temperature and pressure data are measured on the ground, flying at lower altitudes will cause less wind change from calculated density altitudes.

A technique discussed in the USNTPS-FTM-No. 106 manual is to stabilize on a test point while holding the collective fixed. While necessary for test accuracy, the AVO may not have direct control of the AV collective pitch setting. While this may be possible with additional OEM support and programming, most flight testing will use the equipment as supplied. Conducting level flight performance testing without a fixed collective can increase the scatter in the test data. This scatter may be compensated for somewhat by the fact that UAS autopilot performs consistently in similar environmental conditions. Waiting for similar environmental conditions on subsequent flights should yield repeatable results, which may allow the data to combine effectively despite the changing collective settings.

Rotor RPM is typically set by the manufacturer and held constant while flying. Special situations requiring maximum power (i.e., liftoff or maximum airspeeds) may include preprogrammed changes to rotor RPM. These settings should be specified by the manufacturer and control of the rotor RPM speed should be understood by the test engineer. If the rotor RPM is variable, flight testing may be extended by performing each test at different rotor RPM.

The typical AVO procedure for the level flight profile is to compute the expected altitude based on forecast weather and aircraft weight, program the AV to fly at the desired airspeed/altitude, wait for recorded AV data to stabilize, confirm the proper PA and temperature, and then start recording the primary test data. As fuel is used during a

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test run, the AV may be required to climb to hold W/σ or W/δ (and $N_R / \sqrt{\theta}$) constant. Before flight test, the test team should determine the amount of fuel weight loss necessary before altitude must be changed to hold W/σ constant. The number of data points should cover the operational envelope for the UAS, with particular interest to the predicted airspeeds for maximum endurance, maximum range, and maximum level flight airspeed.

The W/σ or W/δ profiles should cover the operating weight range for the specific UAS (maximum gross takeoff weight to minimum landing weight with required reserve). When testing at maximum takeoff weight, testing external stores, both symmetric and asymmetric, should be included where applicable, according to OEM loading specifications.

7.4.2.1 Data Required

Gas Turbine Engine:
- Torque (Q)
- Rotor speed (N_R)
- AV gross weight
- AV configuration
- Indicated/observed airspeed (V_o)
- Gas generator speed (N_1)
- Indicated/observed altitude (H_{Po})
- Outside air temperature (OAT)
- Fuel used (FU, FC) or fuel remaining

Reciprocating Engine:
- Manifold pressure (P_M)
- N_R
- AV gross weight
- AV configuration
- Indicated/observed airspeed (V_o)
- Crankshaft speed (N_1)
- H_{Po}
- OAT
- Fuel used (FU, FC) or fuel remaining

Electric Motor:
- Motor output shaft speed (N_1)
- N_R
- AV gross weight
- AV configuration
- Indicated/observed airspeed (V_o)
- H_{Po}
- OAT
- Source voltage draw (V_S)
- Source current draw (A_S)

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7.4.2.2 Test Criteria

1. Balanced, unaccelerated, level flight
2. Auxiliary systems off or in normal flight mode
3. Stabilized, constant power required
4. Stable air mass
5. Do not test across or within a temperature inversion

7.4.2.3 Data Requirements/Tolerances

1. Stabilize 10 s minimum before recording data
2. Record stabilized data 5 s
3. V_o $\pm$1 KIAS or limits of the system
4. H_{Po} $\pm$20 ft or limits of the system
5. N_R $\pm$1% or limits of the system

7.4.2.4 Safety Considerations/Risk Management

The test location should be evaluated to determine the suitability of UAS testing. Large VTOL UAS approaching manned aircraft speeds should use a large, restricted area with high visibility and the restricted area within line of sight. Likewise, small VTOL UAS with slow forward flight speeds can use smaller airspace to conduct testing. In certain cases, the flight profile requires flying a level flight test on long straight legs which may put the AV beyond line of sight or near the operational limits of the data link. Operating at long distances from the ground control station can be avoided by flying an extended oval or figure eight with straight line segments for testing. The recommended flight pattern for the UAS is to program waypoints in an extended oval with straight line segments for flight tests, also known as a "racetrack" pattern.

When flying near edges of the flight envelope, the tester must be aware of flight boundaries such as range from the ground control station, airspeed limits, or rotor speed limit. Any AV flight characteristics and vibratory environment must be considered when building up to these flight test points. AV actions during lost link, data link latency, and airspeed errors in the edge of the flight envelope (low or high speed regime) must be evaluated to ensure limits are not exceeded. High altitude testing may introduce compressibility effects that may not be addressed with standard flight control laws. Consideration must be given to changes in flight control laws on the backside of the power curve in high wind environments aloft (changing to ground speed vs. airspeed).

7.5 Data Reduction

7.5.1 Data Reduction for UAS with Standard Rotor Design

The data reduction process as explained in Section 7.5 of USNTPS-FTM-No. 106 will be applicable for VTOL UAS with standard rotor design as defined in Section 5.3 of

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this document. For UAS with reciprocating powerplants, the power required is found through the following technique:

- Perform the flight test and record the engine operating conditions including RPM and manifold pressure.
- Reference the power required chart developed in Powerplant Performance (Chapter 4) testing.
- Use the level flight engine data to find the power output on the chart.

For electric powerplants, the power of the motor is typically measured in Watts, but can be converted from Watts to horsepower using the relation (746)*(W) = 1 hp.

7.5.2 Data Reduction for UAS with Nonstandard Rotor Design

Reserved.

7.6 Data Analysis

7.6.1 Data Analysis for UAS with Standard Rotor Design

The data analysis as explained in Section 7.6 of USNTPS-FTM-No. 106 will apply for UAS with standard rotor designs as defined in Section 5.3 of this document. Consideration should be given to terminology differences between the testing agency and the airworthiness authority who analyzes the data. The terms best endurance airspeed (VBE) and best range airspeed (VBR) are often used rather than $V_{MAX\ RANGE}$ and $V_{MAX\ END}$. Using these terms in the data analysis and test reporting will facilitate better inter-agency communication

7.6.2 Data Analysis for UAS with Nonstandard Rotor Design

Reserved.

7.7 Mission Suitability

7.7.1 General

This test evaluates the suitability of level flight speeds and determines the efficiency of a VTOL UAS in forward flight. The determination of cruise speed of a UAS defines its mission limits and tradeoffs for payload and range. The maximum speed determines the egress capabilities of the UAS and the minimum time required to reach a target. When determining maximum airspeeds, endurance, and range, the test team should take into account mechanical factors such as airframe vibration, flight path stability, attitude, autopilot control, and the affect of these parameters on mission success.

Level flight mission suitability performance testing should have a realistic performance profile. Is it being tested in the environments it will be used in? What airspeeds and altitudes are needed for the required mission? What configuration is

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required for performance requirements and what atmospheric limits are there on the system. Is the range in each configuration sufficient to support the mission profile?

From the level flight performance testing, airspeeds will be generated for cruise, maximum, best range, etc. These speeds will determine suitability and help define the mission profile. Payloads required for flight will also define AV range and duration on target.

7.7.2 Vibrations

Section 7.7.2 of USNTPS-FTM-No. 106 will apply for all UAS with standard rotor designs as defined in Section 5.3 of this document. AV propulsion systems generate vibration; therefore, the AV should be instrumented with accelerometers to provide the test engineers the necessary feedback to monitor the vibrations. Assessments should be made to verify the design limitations of AV systems and subsystems as well as handling issues (i.e., flutter).

7.8 Specification Compliance

The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAS will provide qualifications for which it was designed to meet. Expected performance of the UAS will also be rooted in the modeling and simulation of UAS. The detailed technical manuals for the UAS can then be written from the developmental testing as a reference source for all future system testing and military operations.

When a UAS must be flight tested to validate the specifications advertised by the OEM, cruise speed, maximum level flight speed, maximum endurance, and range are all important level flight parameters that need to be validated. These parameters form the foundation of AV performance and are crucial to determining whether or not a particular UAS can accomplish a given mission. If there is any question as to whether a particular UAS can meet the specifications claimed by the OEM, a complete flight test may be necessary to quantify the performance of the system.

7.9 Glossary

A_S	Electric motor source or battery current draw, Amperes
FC	Fuel counts remaining, lb
FU	Fuel used or consumed, lb
H_{Po}	Indicated/observed altitude, ft

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N_1	Gas turbine engine gas generator speed, reciprocating engine crankshaft speed, electric motor output shaft speed, %, RPM
N_R	Rotor speed, % RPM
OAT	Outside air temperature, ^oC
P_M	Reciprocating engine intake manifold pressure, inHg
Q	Gas turbine engine or electric motor torque, %
V_S	Electric motor source voltage draw, Volts
σ	Pressure ratio
δ	Density ratio
θ	Temperature ratio

7.10 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 8
CLIMB AND DESCENT PERFORMANCE

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8.9 Glossary **8.120**

8.10 Reference **8.120**

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8. Climb and Descent Performance

8.1 Introduction

This chapter deals with aspects of climb and descent performance characteristics of vertical takeoff and landing (VTOL) unmanned aircraft systems (UAS). Some VTOL UAS that fly as fixed wing aircraft for forward flight climb and descent modes, tilted systems and tail sitters, should use the test techniques discussed in the UAS Flight Test Methodology, Annex to U.S. Naval Test Pilot School Flight Test Manual (USNTPS-FTM-No. 108) "Fixed Wing Performance" Chapter 7 ("Climb Performance") and Chapter 8 ("Descent Performance").

When testing climb and descent performance, the intent is to characterize the effects of weight, altitude, and power setting on the airspeed for maximum rate of climb and the minimum rate of descent in autorotative. UAS climb rates are important in determining time to climb, which help for mission planning and suitability. When testing autorotative descent performance, the primary goal is to determine the best airspeed for minimum descent rate and the airspeed for maximum glide distance. The main rotor speed may also be varied to record the effects of rotor speed on glide capability. Other parameters for autorotation testing include stability in a descent, cyclic control of the UAS, and turn rates of the air vehicle (AV) while autorotating. The requirements of the specification are examined where applicable; however, the primary emphasis of this test is based on service suitability and mission suitability.

8.2 Purpose of the Test

The purpose of this test is to examine the forward flight climb and, if applicable, autorotative performance characteristics to achieve the following objectives:

1. Determine the following points of interest:
   a. Airspeed for maximum rate of climb, $V_{max\ R/C}$
   b. Airspeed for autorotative minimum rate of descent, $V_{min\ R/D}$
   c. Airspeed for autorotative maximum glide range, $V_{max\ glide}$
2. Evaluate and/or validate the required AV performance parameters for forward flight climbs and descents and address specification.
3. Define the maximum altitude that the UAS should cruise at for compliance as it relates to allow guaranteed ability to maintain altitude and not exceed the autopilot capability.

8.3 UAS versus Manned Aircraft Theory

8.3.1 UAS with Standard Rotor Design

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The theory contained in USNTPS-FTM-No. 106 Section 8.3 regarding determination of airspeeds for maximum rate of climb, minimum rate of descent, and maximum glide distance (autorotation, if possible) will apply for UAS with standard rotor designs, as defined in Section 5.3.

Some, if not most, UAS may not be programmable for deliberate autorotative descents. If a UAS is capable of performing autorotative descents, then these can be incorporated into the flight test schedule as required. If the UAS is unable to perform autorotative descents, the AVO should program the UAS to fly at the maximum descent rate with the powerplant set at flight idle, to minimize test time.

The size of a particular UAS will necessitate the alteration of test criteria such as altitude band and autorotation start height. The altitude band is defined as the total range of altitude bounded by the start and endpoint of a test sequence with a target altitude at the center of that range. For smaller UAS, the altitude test bands may need to be reduced due to potentially lower climb rates. The smaller test bands may consume less fuel which may yield a negligible change in weight between test points. Manned aircraft autorotative testing typically requires 500 ft of stabilization altitude, which may need to be altered depending on the capability of the flight control system of the UAS under test.

In designing the flight-test profiles, the test team should determine the desired amount of fuel burn or power consumed (for electric UAS) per climb test for each AV to maximize the amount of climbs that can be performed for a single test flight. The test team should also take into account the minimum amount of data recording time necessary for reliable and accurate data collection. Stabilization time may be reduced depending on the capability of the flight control system. As a general rule, a stabilization time of 15 seconds should be sufficient.

8.3.2 UAS with Nonstandard Rotor Design

Reserved.

8.4 Test Methods and Techniques

8.4.1 General

Test day climbs/descents are the most common techniques used for UAS testing. Climbs to altitude at varying gross weights and power settings during the conduct of system performance flight test should provide sufficient data to compute the power and weight correction factors. The service ceiling climb test technique should also support this data collection.

Forward flight climb and descent data collection during a test sequence is usually combined to take advantage of alternating increase and decrease in altitude, though it is

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ultimately up to the test team as to whether or not more than one performance parameter should be tested during one test flight.

Many VTOL UAS will have OEM predefined climb rates or climb speeds, which may require verification through a rate-of-climb flight test. If the test team determines it necessary to test airspeeds or climb rates other than those predefined by the OEM, the flight control software may need to be altered, possibly necessitating a flight release from the airworthiness authority.

Testing all combinations of rate of climb, CG, gross weight, configuration, and altitude would require an inordinate amount of time. It is recommended to perform test near the edge of flight envelope operations such as maximum weights/most drag/bad CG conditions; these are the conditions having the greatest impact on climb and descent performance. The test designer and the test team must decide which combinations are of the greatest interest and design the flight test profiles around these combinations. This allows for collection of the maximum data in the minimum flight time. For simplicity, climb and descent flight test techniques will be discussed separately although they are combined during flight test.

There is some concern with conducting deliberate autorotative descents for UAS due to the potential for reduced control authority and few requirements. An AVO is not typically required to practice autorotation sequences due to the preprogrammed flight control of the UAS. In most designs, if the autopilot senses the engine is inoperative and the airframe is capable, the UAS will execute an autorotation to some point. Other concerns with UAS autorotation are selection of the landing zone due to possible aircraft or damage to objects on the ground. However, if a UAS is capable of performing autorotative descents, suitable landing areas are available, and the requirement exists, then these tests can be incorporated into the flight test profiles as required.

8.4.2 Climb and Descent Performance Procedures

8.4.2.1 General

The test procedures as described in Section 8.4.2 of the USNTPS-FTM-No. 106 will be applicable for all UAS with standard rotor designs as defined in Section 5.3.

The altitude bands that the AV will be tested through will vary with the particular system under test. Larger AV may be able to be tested to the same standards as manned aircraft, but smaller AV will typically be tested through smaller altitude bands due to service ceiling limitations or length of time taken to perform climbs in each test flight. Time to stabilize in a climb will also determine altitude band required.

Smaller UAS will usually incorporate reciprocating or electric powerplants, which may burn a large quantity of fuel or battery power if subjected to climbs through large altitude bands, altering weight significantly during the climbs, and resulting in frequent landings for refueling or recharging.

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Since every UAS mission starts with a climb and ends with a descent, scheduling climb and descent test points during each of these phases is an effective use of test resources.

8.4.2.2 Data Required

Gas Turbine Engine:

Outside air temperature (OAT)
Indicated altitude (H_{Po})
Gas generator speed (N_1)
Torque (Q)
Fuel remaining (FC)
Indicated airspeed (V_O)
Sideslip Angle (β_O) [Preferably measured with Yaw Alpha Pitot/Static (YAPS) boom]
AV configuration
Time history of climb

Reciprocating Engine:

OAT
H_{Po}
Manifold Pressure (P_M)
Engine Crankshaft Speed (N_1)
FC
V_O
β_O
AV configuration
Time history of climb

Electric Motor:

OAT
H_{Po}
Motor Output Shaft Speed (N_1)
Source Voltage Draw (V_S)
Source Current Draw (A_S)
V_O
β_O
AV configuration
Time history of climb

8.4.2.3 Test Criteria

1. Constant airspeed
2. Constant power setting
3. Constant rotor speed
4. Stabilized climb rate
5. Calm air

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6. Wings level, unaccelerated flight
7. Constant heading
8. Do not test in or across a temperature inversion

8.4.2.4 Data Requirements/Tolerances

1. Airspeed ±2 KIAS from schedule or limits of the system
2. N_R ±0.5 RPM or limits of the system
3. Altitude within the test band
4. Stabilized 15 seconds (or reasonable time for flight control system) before entering altitude band
5. Heading ± 5° or limits of the system

8.4.2.5 Safety Considerations/Risk Management

Testing locations should be chosen based on the specifications and expected performance of the UAS. Care should be taken to operate the AV within datalink limits and with consideration to line-of-sight (LOS) requirements. Lost link procedures need to be planned and understood by all test personnel to reduce confusion should a link problem occur. Additionally, safe landing areas need to be predetermined in the event of engine failure. In UAS, these landing areas should be programmed into the ground control station (GCS) should an emergency landing be required.

When performing maximum performance climbs, the AVO should carefully monitor powerplant temperatures and be sure of adequate engine inlet cooling. Similar to manned aircraft precautions, VTOL AV may have limited tail rotor antitorque ability and not be able to hold heading while under maximum performance climbs. This should be stated by the OEM but should be monitored closely.

8.4.3 Continuous Climb Procedure (Service Ceiling Climb)

8.4.3.1 General

The test procedures as described in Section 8.4.3 of the USNTPS-FTM-No. 106 will be applicable for all UAS with standard rotor designs as defined in Section 5.3 of this document.

Depending on the specifications or requirements of the UAS under test, the conventional service ceiling climb rate limit of 100 fpm may not apply. If this is the case, the service ceiling climb test should be conducted until the climb rate limitation specified for the particular UAS under test is reached.

8.4.3.2 Data Required

Gas Turbine Engine:

OAT
Rotor speed (N_R)

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Gas generator speed (N_1)
Torque (Q)
Indicated airspeed (V_O)
Fuel remaining (FC)
Sideslip angle (β_O)
Aircraft configuration
Rate of climb (ROC)

Note: The data elements are generally continuously recorded on a data system and specific data elements reduced at altitude increments.

Reciprocating Engine:

OAT
N_R
Engine Crankshaft Speed (N_1)
Manifold Pressure (P_M)
V_O
FC
β_O
Aircraft Configuration
ROC

Note: Same as above

Electric Motor:

OAT
N_R
Motor output shaft speed (N_1)
V_O
Source voltage draw (V_S)
Source current draw (A_S)
β_O
Aircraft configuration
ROC

Note: Same as above

8.4.3.3 Test Criteria

1. Adherence to airspeed schedule.
2. Maintain power level in climb.
3. Constant rotor speed.
4. Stable climb rate.
5. Calm air.
6. Wings level, unaccelerated flight.
7. Constant heading.
8. Do not test in or across a temperature inversion.

8.4.3.4 Data Requirements/Tolerances

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1. Airspeed ±2 KIAS from schedule, or limits of the system.
2. N_R ±0.5 RPM or limits of the system.
3. Heading ±5° or limits of the system.

8.4.3.5 Safety Considerations/Risk Management

Testing locations should be chosen based on the specifications and expected performance of the UAS. Care should be taken to operate the AV well within datalink limits and with consideration to LOS requirements. Lost link procedures need to be planned and understood by all test personnel to reduce confusion should a link problem occur. Test planning for operations near the edge of the flight envelope need special emphasis on safety and loss of link as the datalink systems are critical during these maneuvers.

When testing a UAS near or at the service ceiling, maneuverability and control response of the UAS may be reduced. If this is the case, the UAS may require more time to recover to stable flight. These parameters should be taken into consideration when determining test range requirements for service ceiling climb testing.

8.4.4 Autorotative Descent Performance (If Applicable)

8.4.4.1 General

Many VTOL UAS designs may not incorporate autorotative capability. If the UAS can be programmed to autorotate, Section 8.4.4 of USNTPS-FTM-No. 106 would be applicable for all UAS with standard rotor designs as defined in Section 5.3 of this document. Descent performance of a UAS can be evaluated by setting the engine speed to flight idle and recording the descent rate of the AV.

8.4.4.2 Data Required

Gas Turbine:
- OAT
- Rotor speed (N_R)
- Gas generator speed (N_1)
- Torque (Q)
- Indicated airspeed (V_O)
- Fuel remaining (FC)
- Aircraft configuration
- Time history of descent

Reciprocating Engine:
- OAT
- N_R
- Engine crankshaft speed (N_1)
- Manifold pressure (P_M)
- V_O
- FC
- Aircraft configuration

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Time history of descent

Electric Motor:

OAT
N_R
Output shaft speed (N_1)
Source voltage draw (V_S)
Source current draw (A_S)
V_O
Aircraft configuration
Time history of descent

8.4.4.3 Test Criteria

1. Constant airspeed
2. Constant rotor speed
3. Stabilized rate of descent
4. Calm air
5. Wings level, unaccelerated flight
6. Constant heading
7. Do not test in or across a temperature inversion

8.4.4.4 Data Requirements/Tolerances

1. Stabilized 15 seconds before altitude band
2. Airspeed ±2 KIAS from target or limits of the system
3. N_R ±1% from target or limits of the system
4. Heading ±5° or limits of the system

8.4.4.5 Safety Considerations/Risk Management

Testing locations should be chosen based on the specifications and expected performance of the UAS. Care should be taken to operate the AV well within datalink limits and with consideration to LOS requirements. Lost link procedures need to be planned and understood by all test personnel to reduce confusion should a link problem occur

Extreme caution should be taken when autorotating a UAS of any size. Depending on the powerplant installed in the AV, the time necessary to reapply power to the rotor system needs to be taken into consideration when determining safe operating altitudes. Autorotative descents should only be performed over areas offering the greatest chance of successful recovery with the least chance of damaging property or endangering test personnel in the event a mishap occurs.

8.4.5 Autorotative Turning Performance (If Applicable)

8.4.5.1 General

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VTOL UAS may have autorotative capability, yet may also lack the control authority or ability to turn and/or yaw while descending in autorotative flight. UAS with the ability to program turns during autorotative descents can be tested using the procedures described in Section 8.4.4 of USNTPS-FTM-No. 106, which is applicable for all UAS with standard rotor designs as defined in Section 5.3 of this document. If the UAS lacks turning capability while autorotating, it can only continue on its forward track and glide slope while descending to the ground.

8.4.5.2 Data Reduction

Gas Turbine:

OAT
Rotor speed (N_R)
Gas generator speed (N_1)
Torque (Q)
Indicated airspeed (V_O)
Fuel remaining (FC)
Bank angle (φ)
Aircraft configuration
Time history of turn rate and descent

Reciprocating Engine:

OAT
N_R
Engine crankshaft speed (N_1)
Manifold pressure (P_M)
V_O
FC
Bank angle (φ)
Aircraft configuration
Time history of turn rate and descent

Electric Motor:

OAT
N_R
Output shaft speed (N_1)
Source voltage draw (V_S)
Source current draw (A_S)
V_O
Bank angle (φ)
Aircraft configuration
Time history of turn rate and descent

8.4.5.3 Test Criteria

1. Constant airspeed
2. Constant rotor speed
3. Stabilized rate of descent

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4. Calm air
5. Coordinated turn
6. Constant bank angle

8.4.5.4 Data Requirements/Tolerances

1. Airspeed ±3 KIAS or limits of the system
2. N_R ±3 RPM or limits of the system
3. ϕ $\pm 2^o$ or limits of the system

8.4.5.5 Safety Considerations/Risk Management

Testing locations should be chosen based on the specifications and expected performance of the UAS. Care should be taken to operate the AV well within datalink limits and with consideration to LOS requirements. Lost link procedures need to be planned and understood by all test personnel to reduce confusion should a link problem occur

Extreme caution should be taken when autorotating a UAS of any size. Depending on the powerplant installed in the AV, the time necessary to reapply power to the rotor system needs to be taken into consideration when determining safe operating altitudes. Autorotative descents should only be performed over areas offering the greatest chance of successful recovery with the least chance of damaging property or endangering test personnel in the event a mishap occurs.

8.5 Data Reduction

8.5.1 Data Reduction for UAS with Standard Rotor Design

The data reduction methods explained in Section 8.5 of USNTPS-FTM-No. 106 will be applicable for all UAS with standard rotor designs as defined in Section 5.3 of this document. Electric powerplant power data can be converted from Watts to horsepower using the relation (746)*(W) = 1 hp. For UAS with reciprocating powerplants, the power required can be found by using the engine operating conditions recorded during testing to determine power required from the engine power available curve found from Chapter 4 testing.

8.5.2 Data Reduction for UAS with Nonstandard Rotor Design

Reserved.

8.6 Data Analysis

8.6.1 Data Analysis for UAS with Standard Rotor Design

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The data analysis methods explained in Section 8.6 of USNTPS-FTM-No. 106 will be applicable for all UAS with standard rotor designs as defined in Section 5.3 of this document.

8.6.2 Data Analysis for UAS with Nonstandard Rotor Design

Reserved.

8.7 Mission Suitability

8.7.1 Climb Performance

Testing the climb performance is necessary to determine the airspeed required for maximum rate of climb. Once this is established, the UAS should have these data programmed into the autopilot, when applicable, and operate at these optimum speeds. An important factor affecting these tests is how well the AV can stabilize airspeed while climbing. The tolerances of airspeed variation and the effects on climb performance need to be defined, typically a 1% performance loss. Holding airspeed within this tolerance will result in 99% of the power for maximum rate of climb; remaining at or above the airspeed for best rate of climb keeps the AV on the front side of the power curve while in a climb and reduces the autopilot workload. Other factors contributing to UAS climb rate and airspeed include the efficiency of the main rotor, AV attitude, and other vibrations and/or control issues associated with certain airspeeds and/or powerplant RPM.

8.7.2 Autorotative Performance

In addition to the USNTPS-FTM-No. 106 Section 8.7.2 concerning autorotative performance, UAS have additional concerns when autorotating. Since a pilot is not inside the aircraft, situational awareness is reduced when descending to the ground for an emergency landing. Preprogrammed landing zones should be known to the test personnel and be programmed into the UAS to address engine out situations.

Autorotation ability will vary among UAS and may be limited to instances of loss of link or powerplant failure. If autorotations are impossible to perform, Section 8.4 can be omitted from testing. Upon engine failure, these UAS may incorporate other recovery systems, such as a parachute. Testing the autorotation performance of a UAS may pose some problems due to lack of control of the AV. Care must be taken to be sure the AV will be able to reach the desired landing location from operating altitude during test. Line of sight is usually required for all UAS; an AV autorotating at great distance may lose link with the GCS, thus amplifying the emergency as the system attempts to perform a lost link procedure without engine power.

8.8 Specification Compliance

The specification compliance of the test data comes from three sources: the airworthiness specifications, the technical specifications, and simulation verification and

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validation. Testing must comply with all three. The Aeronautical Design Standards set forth by the Aviation Engineering Directorate establish the criterion for airworthiness data requirements. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAS will provide qualifications for which it was designed to meet. Expected performance of the UAS will also be rooted in the modeling and simulation of UAS. The detailed technical manuals for the UAS can then be written from the developmental testing as a reference source for all future system testing and military operations.

The forward climb rate is dependent on the excess power available and is evaluated together with the vertical climb rate. The climb tests evaluate the OEM specifications and help define the operational envelope for autopilot settings. Descending test results will generate glide angles and, therefore, assist in ascertaining achievable landing zones based on altitude. The data generated and resultant performance curves are typically evaluated at sea level on standard day condition. Data extrapolation can be used to generate altitude limits and pressure corrections for performance parameters.

8.9 Glossary

A_S	Source or battery current draw, Amperes
H_{Po}	Indicated altitude, ft
FC	Fuel counts remaining, lb
hp	Horsepower
N_1	Reciprocating engine crankshaft speed or electric motor output shaft speed, % RPM
N_R	Rotor speed, % RPM
OAT	Outside air temperature, ^oC
P_M	Reciprocating engine intake manifold pressure, inHg
Q	Gas turbine engine torque or electric motor torque, % psi
V_S	Source or battery voltage draw, Volts
W	Watts

8.10 Reference

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

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CHAPTER 9
TRANSITION FLIGHT

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9. *Transition Flight*

9.1 Introduction

Reserved.

9.2 Purpose of the Test

Reserved.

9.3 UAS vs. Manned Aircraft Theory

Reserved.

9.4 Test Methods and Techniques

Reserved.

9.5 Data Reduction

Reserved.

9.6 Data Analysis

Reserved.

9.7 Mission Suitability

Reserved.

9.8 Specification Compliance

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9.9 Glossary

Reserved.

9.10 References

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CHAPTER 10
STANDARD MISSION PROFILES

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10. Standard Mission Profiles

10.1 Introduction

The intent of this chapter is twofold: to discuss the operational mission scenarios of unmanned aircraft systems (UAS) and to discuss tests required to verify the flight performance (e.g., maximum range, maximum endurance, et al.) of a given UAS. The primary mission fulfilled by many UAS has been reconnaissance, surveillance, and target acquisition (RSTA) missions. Other mission objectives that may be employed by UAS include weapons drop, communications relay, air data relay, mission support, medical drops, and psychological operations (pamphlet delivery, radio delivery, etc.). Before flight testing, the test team must have a working knowledge of the UAS mission profile(s). Knowing this information enables the test engineer to plan a test program that combines UAS specific performance testing with mission profile testing, all in an effort to integrate the entire test effort into a single flight test program.

It is recommended that actual range and endurance missions be included into the flight test program. The analysis presented in Chapter 7 provides the theoretical basis for best case range and endurance values. However, mission profile testing will be required to verify these parameters in a real operational scenario. In addition, this type of testing lends itself to be useful in verifying many of the parameters found in the air vehicle (AV) operator's manual, in particular the value for minimum fuel required for safe landing. In this scenario, a given number of mission sorties can be flown to observe the fuel flow and verify the reliability of the specifications found in the operator's manual.

10.2 Purpose of the Test

UAS mission profile performance testing is used to verify the pertinent performance parameters that have been evaluated or extrapolated from other flight performance test events. These types of tests will verify that a given UAS can indeed perform the intended mission without incident. In addition, these tests typically involve operating the AV over an extended period of time, generally in a fully equipped configuration. As such, these tests present opportunities to the test engineer to evaluate several of the mission critical systems onboard the AV, such as the datalink range and link margin, payload range and resolution, and the target location accuracy.

Typically UAS datalink testing has been accomplished by loitering the AV in a restricted airspace and then moving the ground control station (GCS) to a position that increases the range between the two systems. If moving the GCS is too problematic, the UAS can proceed outside the restricted area into the National Airspace System (NAS) to evaluate the datalink. This will require that a Certificate of Authorization be obtained from the Federal Aviation Administration (FAA) as well as "see and avoid" duties performed by either a manned chase plane or ground observers. UAS operating within the NAS must meet certain requirements set forth by the FAA. One of the main requirements, as of this writing, is the ability to sense and avoid other aircraft, especially

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noncooperative, or no transponder, aircraft. Research and development is ongoing to develop a system capable of attaining the same level of safety as a manned aircraft in a package small and light enough to be carried by a light or small class AV. Testing outside restricted airspace may not be an option in the near future and this needs to be understood by the test team before a final test plan can be developed.

Payload performance evaluation will involve testing the target location error (TLE) of an electro-optical/infrared (EO/IR) camera in a RSTA mission; the TLE is typically tested while the AV is loitering about a stationary point at a specified slant range, which is a function of altitude and loiter diameter. Payload performance tests should also evaluate target acquisition distance. The temperature and vibration effects on the payload are also of interest; therefore, the payload should be tested at various conditions, including testing at higher mean sea level altitudes to aid in the evaluation of these effects.

The navigation system is another mission critical evaluation for UAS. Most UAS use global positioning system/inertial navigation system (GPS/INS) for position information. One way that the navigation system can be tested is to position the AV at a specified distance from the GCS and have the AV return to base using only the INS. This type of test is useful in determining the "dead reckoning" capability of the flight control system.

Consideration should be given to specification compliance for datalink performance, payload performance, and navigation system performance. There will be technical specifications for system performance and simulation verification and validation, and possibly airworthiness specifications. Additionally, Technical Requirements Documents and Operational Requirements Documents specific for each UAS will provide qualifications for which these systems were designed to meet. Expected performance of the UAS will also be rooted in the modeling and simulation of UAS.

10.3 UAS vs, Manned Aircraft Theory

USNTPS-FTM-No. 108 "Fixed Wing Performance" Section 10.3 contains a complete discussion into the mission requirements as they relate to flight performance and will not be duplicated herein. In general, flight test programs are planned to collect flight performance data over a given range of the operating envelope of the AV. With UAS, these operating envelopes are currently determined by the payload requirements rather than the performance limitations of the AV. Each UAS must be understood as a whole before a detailed test plan can be generated for that particular UAS. If the UAS is equipped with multiple payloads, the system performance limits of those payloads should be well-defined and the flight test program should center on the data points of greatest interest.

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10.4 Safety

There are a number of safety considerations that must be addressed by the test engineer before conducting mission profile testing. Mission profile testing is designed to validate the interpolated points within the previously tested limits of the flight envelope. This type of testing will provide the required reliability for parameters such as maintenance cycles, fuel flow measurement systems, return home fuel status, avionics and sensors, and the datalink performance. Regarding the fuel status, the test engineer should have a working knowledge of the fuel system, how the fuel is monitored, the reliability of this monitoring system/scheme, and the maneuverability limitations of the UAS when operating near its minimum fuel state.

Typically, mission profile testing will be accomplished towards the end of the flight test program. At this point during the test series, it is possible that the datalink limits of the system will not have been completely tested and/or verified. Therefore, the datalink signal strength should be carefully monitored for deteriorating conditions that would adversely affect the link margin of the system, possibly leading to a lost of link (LOL) condition. Loss of link procedures should be verified before any flight, particularly when the UAS might be in the NAS. Another important consideration pertaining to the datalink is the restriction associated with the line of sight (LOS). Terrain avoidance and terrain effects on LOS tools are available to the AVO and must be used during the planning stages of the test event to ensure that emergency and return home operational scenarios are achievable given the location of the UAS.

The impact of long duration flights on crew fatigue is another important safety issue, since the OEM will have a prescribed maximum length of time an AVO may operate the UAS. As such, mission profile flights may involve crew changes; this should be incorporated into the test planning so that the change does not occur at a high stress time, such as furthest distance from the GCS that could lead to a lost link condition or flight in NAS. Additionally, overall aircraft status while flying long duration mission profile tests should be monitored since there may be issues with the operating limitations of AV components (i.e., servo temperature, exhaust gas temperature, etc.). These can be critical to the execution of mission profile testing.

Frequently, the mission and loiter segments involve transitioning between named areas of tnterest and/or target areas of interest. However, these are typically more local than the distance from the ground control station. In executing these types of mission profiles, the AVO must remain cognizant of:

- the fuel state
- the winds expected on the return
- the minimum fuel/battery guidance for that vehicle type
- airspace restrictions and the overall air defense environment.

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10.5 Mission Profiles

The generic mission profile, as shown in Figure 10.1, can be flown by all size classifications of UAS with the primary difference being the altitude, range, endurance, and target location activities of the given AV. Figure 10.1 depicts a mission profile that contains all of the common elements of the flight mission, namely takeoff, climbout, cruise/level flight, target location activities, return to base, and landing. The target location activities will vary between the various UAS configurations. These activities include, but are not limited to, loitering, hovering, perching, staring, weapons deployment, medical drops, target designation, et al. To date, most of the weapons deployment missions have been designed for the larger-sized UAS. However, as the need arises, the technology will increase to allow the deployment of munitions off of smaller UAS AVs, giving the warfighter an expanded capability.

Conducting flight performance testing, coupled with advanced mission profile testing, will ensure the UAS can meet the required mission critical roles only afforded to UAS. UAS standard mission profiles are usually comprised of the following segments, as depicted in Figure 10.1:

- launch from set altitude (consider conditions ranging from sea level up to 9,000 feet pressure altitude and hot/cold day to simulate different operational environments)
- climb segment (with or without transit downrange)
- ingress/cruise segment
- target location activities:
  a. loiter around a target
  b. hover over a target (hover and stare)
  c. perch on a stationary object and employ sensors (perch and stare)
  d. search a grid
  e. follow a road/object
  f. deploy a weapon or payload
  g. designate a target
- egress, descent, and landing (return-to-base)

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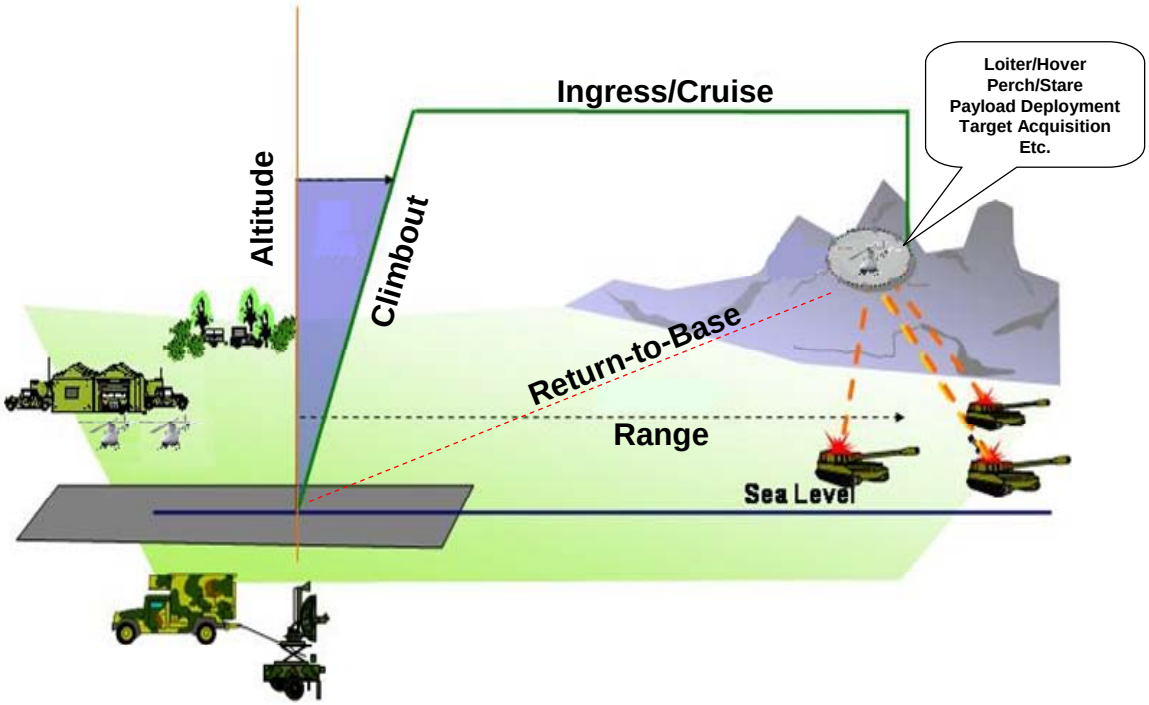


Figure 10.1 Generic Mission Profile

10.6 Conclusions

Technology development may lead to more mission objectives that are possible or desired for current UAS. The mission profiles discussed in this chapter represent the missions that UAS are currently performing. This document will aid the testing agency in writing the detailed test plan for a specific UAS. It is important to remember that the entire system is difficult to separate from the aircraft and, therefore, testing both the flight performance and mission performance of the UAS during the same flight test program is essential. Mission profile testing allows the test team to accomplish the aforementioned testing goals in one, all encompassing flight test.

10.7 References

1. U.S. Naval Test Pilot School Flight Test Manual 106 (USNTPS-FTM-No. 106), *Rotary Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 31 December 1996.

2. U.S. Naval Test Pilot School Flight Test Manual 108 (USNTPS-FTM-No. 108), *Fixed Wing Performance*, Naval Test Pilot School, Patuxent River, Maryland, 30 December 1992.

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